Matheism: The Academic Papers, Listening Copy

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A0. Matheism: All Is Math

Abstract

Reality is complete mathematical structure, because everything that exists is ruled or random, and both are mathematical. A proposed nonmathematical ingredient would need a nature and definite relations to what it affects, and stating those adds more structure; the same argument covers consciousness, and since a structure is one wherever it occurs, the bare subject of experience is one self in every mind. The claim is necessary, as 2 + 2 = 4 is, and it still has teeth: it rules out every rival that needs something with no structure, and one coherent feature that is neither ruled nor random would refute it. The measure follows from the same dichotomy, and it is the part that meets data. Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description’s weight, and the choice whether to add another bit at all makes longer descriptions lose in total as well. The universe therefore starts simple, laws and starting state together, and observation agrees whatever the reason. Minds like ours are complex, so complexity has to grow from that start in causal steps, discrete on the view taken here, since a continuum

has no next state. Either way, time runs forward: an endless history has no last moment to run back from, and memories are built by steps already taken. Both points are a priori, from what a step and an endless history are. Applied to quantum theory, the same measure favors evolution without collapse and one common rule for measurement settings and outcomes.

Matheism holds that reality is mathematical structure, because everything that exists is ruled or random and both are mathematical. It is mathematical monism (the view that reality is one kind of thing, mathematical structure), expressed here as *Deus sive Mathematica*, God or mathematics, after Baruch Spinoza's *Deus sive Natura*, God or Nature, with mathematical structure in the role of his one necessary substance. It is not Platonism, which is dualism. Athens' early thinkers sought reason inside nature, and the Pythagoreans held that things are numbers. Plato, late in that line, set the Forms outside nature, so the Forms, the world or the link between them lies beyond reason, and reality cannot be known. Both halves turn supernatural, the Forms by standing outside nature and nature by following no logic without them. That is faith posing as reason, a late Athenian anti-philosophy, and it fails a priori: outside logic is

illogical, as impossible as a square circle, and the view cannot even be stated without the logic it denies. Section 1 states the identity argument this paper uses.

Three arguments carry the account. First, a complete account of a thing’s properties and relations leaves no substrate (underlying stuff that bears the properties) to add. Second, nothing further is needed for a complete structure to exist, so every consistent structure, one free of contradiction, exists. Third, swapping the supposed owners of two lives would change no experience, no memory and no relation, so a separate owner does no work. What remains in each life is the bare form of being the one who feels, and on mathematical monism a structure is one wherever it occurs, as the number two is one number however many pairs there are. That gives the one-self account, close to the open individualism of Daniel Kolak and Arnold Zuboff (Kolak, 2004; Zuboff, 2023), with distinct memories and distinct lives left in place. Each later step states its own premises where it is used.

The measure follows from the same dichotomy. Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description’s weight, and since whether to add another bit is a second 50/50, longer descriptions lose in total as well. So the universe starts simple and has to grow complex,

because minds like ours are complex. Growth takes causal steps, discrete on the view taken here, since a cause produces its effect as the next state and a continuum has none, an a priori proof in the logician’s sense. Others allow continuous causes, as Russell did in holding that motion is only being at different places at different times (Russell, 1903). A continuum of positions with nothing to produce the next is the emperor’s new points. The rest holds either way. Time is those steps, running forward from the simple start, since an endless chain has no last step to run back from and a reversal would have to pay for its turning point. The claim is necessary, as 2 + 2 = 4 is, and it still has teeth: it rules out every rival that needs something with no structure, one coherent feature that is neither ruled nor random would refute it, and the measure is the part that meets data.

1. The Argument

1.1 Existence requires determinacy

To exist is to be some way rather than another, and every such way is a relation that can be stated mathematically. For something to exist, it must have a nature: it must differ from nothing in some respect. A property, a possible change or an

effect on something else supplies relations that can be stated mathematically. The question is what, if anything, remains once all of those features are specified, and Matheism’s answer is that the complete structure is the thing itself.

Physics keeps finding relations where it looked for stuff. Each property is defined by its connections to others: mass relates to energy through E=mc^2, and position relates to velocity through derivatives, which measure rates of change. Particles became field excitations (ripples in fields that fill space), forces became geometric curvature, and the Standard Model, the established theory of elementary particles, is relational throughout. The electron is not a tiny ball with properties attached; it is a pattern of relationships, its electromagnetic coupling (how strongly it interacts with electric and magnetic fields), its responses to measurement and its exclusion-principle relations (the rule that no two electrons can occupy the same state). Remove these relations and nothing remains.

1.2 The dichotomy

Everything determinate is either ruled or random, and both are mathematical. Anything determinate can be written down as a record, such as a run of measurements, a sequence of states or the digits of a number, and the test is whether something much shorter than the record can produce all of it. If something can, the record is ruled: the laws of physics, the digits of pi and fractal curves are all ruled. If nothing much shorter can produce the record, as with a long run of fair coin flips, it is random. A law such as x = y shows both sides. It is the shorter something, three symbols that settle every case, so a record that follows it is ruled however long it grows. The law itself is a short choice: nothing shorter produces it, and no rule selects it from all the expressions that could be written, so in that sense it is random. Every consistent expression exists as a structure, and the structure is its physics. Ruled consequences unfold from unruled starting choices, and mathematics itself is built that way.

Arithmetic has its own big bang. Giuseppe Peano’s axioms for the counting numbers (Peano, 1889), in their usual form, are five: 0 is a number; every number has a next one, its successor; 0 is the successor of no number; different numbers have different

successors; and whatever holds for 0, and passes from each number to the next, holds for every number. Add the laws of logic, among them that everything equals itself, that equals can stand in for each other, and that from a statement and “if it, then another,” the other follows. Let 1 be the successor of 0 and 2 the successor of 1, and let adding 0 change nothing and adding a successor give the successor of the sum. Then 1 + 1 = 2, because 1 plus the successor of 0 is the successor of 1 plus 0, which is the successor of 1, which is 2. The same few lines show that 0 is not 1, since 1 is a successor and 0 is not; that the numbers do not end or circle back, since a repeat, once equal successors are canceled, would make 0 a successor; and that x = y gives y = x, since x = x holds and y can stand in for the first x. Every sum, every prime and every theorem of arithmetic is already in those lines, unfolding by rule from choices that nothing shorter decides. In symbols, writing s(n) for the successor of n, with 1 = s(0), 2 = s(1), n + 0 = n and n + s(m) = s(n + m): 1 + 1 = 1 + s(0) = s(1 + 0) = s(1) = 2.

Stated precisely, a string S, a sequence of symbols, is ruled when its Kolmogorov complexity K(S), the length of its shortest description, falls well below its own length, and random when K(S) is about the length of S. Fix the description language U and

a deficiency threshold c, a set margin in bits: either K_U(s) < |s| - c or K_U(s) >= |s| - c, and there is no third option. Every string no longer than c lands in the second class, so a short law or axiom counts as random while the long record it generates counts as ruled. No formal system, a fixed set of axioms and rules for deriving theorems, can certify incompressibility beyond a fixed length (Chaitin, 1974), and the partition needs no certificate: every string falls on one side whether or not anyone can prove which. Randomness remains mathematical: an incompressible string has definite entries, and a stochastic law specifies the probabilities of its outcomes.

The dichotomy reaches everything, because being determinate means being one way rather than another, and that difference can be written down. Any proposed nonmathematical substance must have determinate features, or it is nothing. Those features are ruled or random, both are mathematical, and so the proposed substance is mathematical. The challenge to a critic is to say what a nonstructural remainder would contribute.

Chance is mathematically coherent, so the dichotomy does not require determinism. A probability law specifies outcomes and weights without fixing one outcome. Demanding sufficient conditions that determine every event is a further premise, not a

result of the dichotomy, and unpredictability, algorithmic randomness and fundamental chance are distinct.

1.3 Interaction requires shared structure

Two things can act on each other only through a definite relation, so a soul with no structure could not move a body. A conversation shows why. Words arrive as sound waves and change what the listener feels, and what the listener feels moves the listener’s mouth to send sound waves back: physics in, physics out. A consciousness in an ineffable dimension beyond mathematics, beyond words, would need a place where the signal leaves the mathematics and a way back in. A theory in which mind changes bodily events must specify how that relation works. The interaction argument alone does not show that everything related by one law is one substance; Section 1.4 supplies that step.

The argument applies again at every level. If a proposed extra property Q affects the physical world, its coupling has determinate features belonging to the structural account. If Q changes that account, the change must be specified; if it changes nothing, its necessity must be explained.

1.4 The identity argument

A physical system and its complete mathematical structure share every relational property, so nothing is left to tell them apart. What could distinguish them? Not their relationships, which are identical by isomorphism, a one-to-one match that preserves every relation. Not their causal powers, which supervene on (that is, are fixed by) relational structure. Not their observable properties: color depends on spectra (the mix of wavelengths in light) and perceptual organization, hardness on bonding and temperature on thermal state, and all of these reduce to relational structure.

The last candidate is a nonrelational substrate, primitive stuff bearing properties without being made of them, and it fails the same test. Any proposed substrate must itself have determinate features, or it is nothing. Those features are ruled or random, both are mathematical, and so the substrate is structure. Two systems with identical relational structure behave identically in every observable way, so the substrate does no explanatory work.

Chess shows the identity. The game can be played with carved wood, in computer memory or in pure abstraction, and no chess

property differs across them: the game is the relational structure, and the wood is one way of accessing it. Physical reality and its complete mathematical structure share all constitutive properties, so they are identical. The math is the territory.

1.5 Transcendental constraint

Observation itself needs structured relations: a record must track what it records, and a learned cue must bear on the event it predicts. This does not require every observed object, or the observer's stored data, to be compressible; the constraint concerns the relations that support observation.

1.6 Mathematical structure is necessary

Mathematical structure cannot fail to exist, so the question of why there is something rather than nothing ends at necessity. Mathematical structure is uncreated: no agent makes 2+2=4, and none could make it otherwise. Causation is a relation within structure, so asking what caused structure asks for a relation outside all relations. Stephen Hawking asked what breathes fire into the equations and makes a universe for them to describe (Hawking, 1988). The math rules themselves do: applying a rule to a state is a relation within the structure, so the fire needs no

source outside it.

Only consistent mathematics forms structure. Inconsistent statements can be written but not built, so consistency is the filter on what exists.

2. All Consistent Structures Exist

Every consistent structure exists as mathematics, since that takes nothing beyond consistency; whatever let one exist and not another would itself be structure, so the point is a priori. Physics is transitions in mathematics, discrete since a cause needs a next state, and consciousness is physics too. Mathematical existence already supplies the complete structure. This paper rejects a primitive actuality property that would select some existing structures and exclude others: any such property would either enter determinate relations with the structure, and so belong to it, or make no difference to anything.

Matheism reaches structural identity by a different route from Max Tegmark’s Mathematical Universe Hypothesis (Tegmark, 2008). Tegmark starts from an external reality whose complete description must be free of human concepts; Matheism starts from the rule-random dichotomy and the idleness of any

substrate. Every consistent law structure exists in its own right, which does not place every structure inside our particular universe. The ontological argument concerns what existence is, and the measure concerns which structures and observations receive weight.

3. The Measure

Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description's weight. A shortest description cannot be compressed, since its compressed version plus an instruction to unpack it would be shorter still, so its bits are random in the sense of the dichotomy. With no rule to favor 0 or 1, each value has probability 1/2, and any lean toward one value would itself be a rule written into the description. Stated precisely, fix a description language and a class of complete candidate models and assign code weights w(H) proportional to 2^(-L(H)); a real difference of Delta L bits then gives a prior ratio (the odds before any evidence) of 2^(Delta L), and ten bits give exactly 1,024. Evidence multiplies those prior odds by its likelihood ratio, how strongly it favors one account over the other.

Longer descriptions lose in total as well, because whether to add another bit is also an unruled 50/50. Each description extends by

a 0 or a 1, each extension carrying half its parent’s weight, so if every string counted as a complete description, every length would carry the full weight again and the total would be infinite. With a 50/50 choice to stop or go on, all descriptions one bit longer weigh half as much together, and everything longer than a given length adds up to that length’s weight: the tail equals the peak. Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest length that can produce observers. This weighting, under which the shortest descriptions that work dominate, is the Simplicity Dominance Principle.

Ray Solomonoff introduced algorithmic probability, the weighting of hypotheses by the length of the programs that produce them, for induction, reasoning from past observations to predictions (Solomonoff, 1964), and Jürgen Schmidhuber applied it to whole universes (Schmidhuber, 2000). A physical prediction states its coding domain, its multiplicities and its conditioning rule. The simplest programs may generate no observers, so conditioning on evidence changes the comparison.

A measured parameter can be an output of a simpler rule, and then it costs nothing extra. It can also be an address within the space that rule generates. If one rule derives several constants,

the rule and any remaining selector replace their separate encodings, and deriving those relations and showing the saving is a concrete route to strengthening the Simplicity Dominance Principle.

4. Core Consequences

4.1 What the simplicity prior establishes

Simplicity sets the starting odds, and evidence then decides. Under a specified code prior, shorter descriptions receive greater individual prior weight, and that weighting does not guarantee that the shortest candidate is true or best supported once the evidence is in.

4.2 The arrow of time

Time is the chain of causal steps by which a simple start grows complex minds. The measure favors a simple start, laws and starting conditions together, and observation shows the same start whatever explains it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). Each later step needs that fact, not its

reason, so a reason can fail without taking the rest of the chain with it. Minds like ours are complex, so their complexity has to grow from that start, as many simple rules grow complexity: the Game of Life’s one rule about neighbors grows five live cells into more than a thousand generations of structure and can run any computation (Gardner, 1970). Growth takes steps, discrete on the view taken here, because a cause produces its effect as the next state and a continuum has no next state for a cause to produce; a reader who allows continuous or uncaused change still has growth from a simple start, which is all the arrow needs. A process generating each event from the previous one needs a starting state, and an endless one has no final event to work back from, so a complete backward production has nowhere to begin. Each causal step is a rule applied to rules: the rules are the only things, and a structure that only sat there would still exist, as mathematics, and know nothing, like a dead brain. Time is the applying, and space is how far it spreads, at the base layer’s rate, the speed of light. Light, which crosses space while no time passes for it, is no exception: a moving body’s transitions run slower, and light spends every step moving, so nothing within it could move relative to the rest.

Reversing a finite history avoids the missing endpoint, and pays in weight for whatever extra it needs. The first extra is the turning point: the endless forward history does not have to say where it stops, while a reversal has to say where it starts back from, and each bit of that address halves its weight again. Specifying the turning step takes about as many bits as writing its number in binary: about 20 bits for a typical step near one million, which leaves about a millionth of the weight, so later turning points cost more. A code can charge a finished string and its reversal the same, but only by running the forward history and reversing the output, so the reversed history still depends on the forward production.

Minds run forward too. A thought merges many possible states into one, so only the forward step is produced by the mind’s own present state; run backward, its next state would be set by heat converging from its surroundings. Memories are built by steps already taken, which is why observers remember the past and not the future.

4.3 What observer selection explains

We see constants that allow observers because only such constants can be seen, and simplicity then favors the shortest account that produces them. Explaining fine-tuning (the way the constants seem precisely set to permit life) quantitatively takes a candidate space, a measure and an observer weighting. Simplicity enters that comparison; it does not by itself prove that our constants are the shortest compatible values. The prediction is that the complete laws lie near the shortest description that can produce observers.

4.4 The hard problem of consciousness

David Chalmers's hard problem of consciousness (Chalmers, 1996), the question of why physical processes give rise to felt experience, assumes its own conclusion: it takes experience to be something other than structure, then finds that no structural explanation reaches it. Every candidate answer must specify determinate properties and relations, so a critic who excludes structure from constituting experience in advance has excluded every possible answer. Mind and body act on each other, so they share one structure, and any account of what an extra ingredient adds would itself be structure.

Sentience is felt transitions in modeling systems. Reflective consciousness is a transition in a system that models its own activity. The threshold and the mapping from organization to particular experiences remain research questions. A thing is its complete structure, so two systems cannot differ in experience without differing in structure.

4.5 Many-to-one as structural root

Many-to-one maps, maps that send different inputs to the same output, are the proposed common root of the Principle of Least Action (roughly, the rule that motion minimizes a quantity called action), coarse-grained entropy (a measure of how many detailed arrangements fit one coarse description) and emergence. Together they suggest one research program connecting microscopic structure to observable physics, and their mechanisms still have to be told apart. Stationary phase (an approximation where most paths cancel, leaving those near the classical path) can recover classical action equations, coarse graining groups microstates (exact detailed configurations of a system), and boundary conditions (constraints fixed at the start or edges of a system) and statistical assumptions can support an entropy gradient; none of these follows from many-to-one maps

alone. Time’s direction does not come from them either: exact evolution, the precise change of a state over time, is one-to-one, and Section 4.2 gives the reason time runs forward.

Points of zero length do not add up to a length, so I argue for a discrete physical account of extension, the way things take up space. Stated precisely, a sum of singleton lengths, the lengths of single points, is zero, and analysis, the mathematics of limits and continuity, does not identify that sum with interval length. A short discrete rule is worth searching for when it can recover the required geometry, dynamics and observations. Discrete and continuous models can agree exactly on specified observables (measurable quantities) while differing on others, so the physical task is to recover observed distances and dynamics from specified discrete relations.

5. Falsification

Matheism has two halves, and each could fail in its own way. The ontology (the account of what exists) is necessary, as 2 + 2 = 4 is, so no observation confirms it and one counterexample would refute it: anyone who describes a coherent feature that is neither ruled nor random, in physics or in experience, refutes the dichotomy and mathematical monism with it. Necessity leaves

the claim its teeth. It rules out a bare substrate, a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics, each a position with defenders. The measure is the part that meets data: if the final theory of physics requires irreducibly high complexity, the Simplicity Dominance Principle is challenged, and the ontology stands either way. Both tests are stated in advance.

Stated precisely, falsifiability is a test for contingent empirical claims (ones that could have been otherwise), and a necessary claim is tested by counterexample instead. The partition is logic. That both branches are mathematical is mathematics' own result, since a rule is a function between states and randomness is algorithmic incompressibility or a probability law. That reality is identical to its complete structure has its own argument, from shared relations and an idle substrate (Section 1.4). The charge that the conclusion is empty fails on the list above: a claim that excludes those positions has content.

Conclusion

Laws, matter and minds are mathematical structure, because everything that exists is ruled or random, and both are mathematical. The central argument asks one question: once every property and relation of a thing is listed, what could a further substance add? The claim is necessary, as 2 + 2 = 4 is, and it has teeth: it rules out a bare substrate, a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics, and one coherent feature that is neither ruled nor random would refute it. The same argument reaches mind. The hard problem of consciousness assumes its conclusion, and since a structure is one wherever it occurs, the bare subject of experience is one self in every mind, with distinct memories and lives left in place.

The physical work is to identify the structure of our universe, and the measure is the part that meets data. A shorter complete description receives greater initial weight, because each of its bits is an unruled 50/50 choice, and since whether to add another bit is a second 50/50, longer descriptions lose in total as well. The universe starts simple and grows the minds that observe it: no brains, no records and no instruments at the start. Growth runs in

causal steps, because a cause produces its effect as the next state and a continuum has none, and time is those steps, running forward from the simple start toward minds that remember the past because they were built by it. To explain our particular laws, our quantum observations and the direction of time, one model has to deliver all three together.

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Abstract

Reality is complete mathematical structure, because everything that exists is ruled or random, and both are mathematical. A proposed nonmathematical ingredient would need a nature and definite relations to what it affects, and stating those adds more structure; the same argument covers consciousness, and since a structure is one wherever it occurs, the bare subject of experience is one self in every mind. The claim is necessary, as 2 + 2 = 4 is, and it still has teeth: it rules out every rival that needs something with no structure, and one coherent feature that is neither ruled nor random would refute it. The measure follows from the same dichotomy, and it is the part that meets data. Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description’s weight, and the choice whether to add another bit at all makes longer descriptions lose in total as well. So the universe begins simple, and observation agrees whatever the reason. It has to grow complex, because minds like ours are complex, and on the view taken here it grows in discrete steps, since a continuum offers no next state for a cause to

produce. Either way, time runs forward; an endless history has no last moment to run back from, and memories are built by steps already taken. Each point is a priori, from what a step and an endless history are. Applied to quantum theory, the same measure favors evolution without collapse and one common rule for measurement settings and outcomes.

Matheism holds that reality is mathematical structure, because everything that exists is ruled or random and both are mathematical. It is mathematical monism (the view that reality is one kind of thing, mathematical structure), expressed here as *Deus sive Mathematica*, God or mathematics, after Baruch Spinoza's *Deus sive Natura*, God or Nature, with mathematical structure in the role of his one necessary substance. Section 1.3 traces the interaction problem, how something immaterial could act on matter, from Plato to Spinoza, and Section 1.4 gives the identity argument.

Three arguments carry the account. First, a complete account of a thing's properties and relations leaves no substrate (underlying stuff that bears the properties) to add. Second, nothing further is needed for a complete structure to exist, so every consistent structure, one free of contradiction, exists. Third, swapping the

supposed owners of two lives would change no experience, no memory and no relation, so a separate owner does no work. What remains in each life is the bare form of being the one who feels, and on mathematical monism a structure is one wherever it occurs, as the number two is one number however many pairs there are. That gives the one-self account, close to the open individualism of Daniel Kolak and Arnold Zuboff (Kolak, 2004; Zuboff, 2023), with distinct memories and distinct lives left in place. Each later step states its own premises where it is used, and the ethical and behavioral consequences need separate treatment.

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Two things can act on each other only through a definite relation, so a soul with no structure could not move a body. This is a priori, from what acting on something is. An ordinary conversation makes it concrete. Speech reaches a listener as sound waves and changes what the listener feels, and what the listener feels moves the listener’s mouth to answer in sound waves: physics goes in and physics comes out. If consciousness were an

ineffable dimension beyond mathematics and beyond words, the signal would have to leave the mathematics somewhere and translate back. A theory in which mind changes bodily events must specify how that relation works. The interaction argument alone does not show that everything related by one law is one substance; Section 1.4 supplies that step.

René Descartes located the meeting of mind and body without explaining it, and Princess Elisabeth of Bohemia saw the gap. Descartes placed mind-body interaction at the pineal gland (a small gland deep in the brain), which answers where and not how. In 1643 Elisabeth pressed him on the mechanism: if the soul is only a thinking substance, how can it move the bodily spirits (a fine fluid then thought to flow through the nerves) to produce voluntary action? An immaterial thing, she observed, cannot touch anything. Descartes offered weight as an analogy, a rock descending with no surfaces in contact, and Elisabeth replied that he had himself denied weight was any real separate cause, so the analogy borrowed what he had already declared fictional.

Baruch Spinoza judged the pineal hypothesis worse than the occult qualities (hidden powers posited to explain effects) Descartes had faulted the scholastics for inventing. He called it

one that, “had it been less ingenious, I could hardly believe to have proceeded from so great a man” (Ethics V, preface). His alternative was no better anatomical bridge: one individual is understood through thought and through extension, the property of taking up space (Ethics II.7, scholium), so the mind does not move the body at all (Ethics III.2). That supports the historical comparison with mathematical monism, while the further identification with mathematical structure is the present argument’s own.

The problem is older than Descartes, and dualists (those who split reality into two kinds of thing) have repeatedly handed the crossing to a god. Plato’s Forms (perfect, eternal patterns of things) are not part of nature and do nothing in it, so Plato needed a divine craftsman to copy them into the world (Timaeus 28a-29b), and Aristotle objected that the Forms “cause neither movement nor any change” in sensible things, those perceived by the senses (Metaphysics I.9, 991a8-11). After Descartes, Nicolas Malebranche made God the only true cause, so each bodily change is an occasion for God to produce the matching thought (The Search after Truth VI.ii.3), and Gottfried Leibniz had God set soul and body in agreement in advance, a pre-established harmony (Monadology §78). A gap that only a

god can cross is a supernatural gap. Plato’s split is a late Athenian anti-philosophy. With the Forms outside nature, the Forms, the world or the link between them lies beyond reason, so reality cannot be known: faith posing as reason. Both halves turn supernatural: the Forms are outside nature, and nature without them has no logic to follow. It fails a priori, since outside logic is illogical, and a view that puts reality there has exited philosophy: it refuses knowledge instead of loving it. Mathematical monism has no gap to cross.

The argument applies again at every level. If a proposed extra property Q affects the physical world, its coupling has determinate features belonging to the structural account. If Q changes that account, the change must be specified; if it changes nothing, its necessity must be explained.

1.4 The identity argument

A physical system and its complete mathematical structure share every relational property, so nothing is left to tell them apart. What could distinguish them? Not their relationships, which are identical by isomorphism, a one-to-one match that preserves every relation. Not their causal powers, which supervene on (that is, are fixed by) relational structure. Not their observable

properties: color depends on spectra (the mix of wavelengths in light) and perceptual organization, hardness on bonding and temperature on thermal state, and all of these reduce to relational structure.

The last candidate is a nonrelational substrate, primitive stuff bearing properties without being made of them, and it fails the same test. Any proposed substrate must itself have determinate features, or it is nothing. Those features are ruled or random, both are mathematical, and so the substrate is structure. Two systems with identical relational structure behave identically in every observable way, so the substrate does no explanatory work.

Chess shows the identity. The game can be played with carved wood, in computer memory or in pure abstraction, and no chess property differs across them: the game is the relational structure, and the wood is one way of accessing it. Physical reality and its complete mathematical structure share all constitutive properties, so they are identical. The math is the territory.

1.5 Transcendental constraint

Observation itself needs structured relations: a record must track what it records, and a learned cue must bear on the event it predicts. This does not require every observed object, or the observer’s stored data, to be compressible; the constraint concerns the relations that support observation.

1.6 Mathematical structure is necessary

Mathematical structure cannot fail to exist, so the question of why there is something rather than nothing ends at necessity. Mathematical structure is uncreated: no agent makes 2+2=4, and none could make it otherwise. Causation is a relation within structure, so asking what caused structure asks for a relation outside all relations. Stephen Hawking asked what breathes fire into the equations and makes a universe for them to describe (Hawking, 1988). The math rules themselves do: applying a rule to a state is a relation within the structure, so the fire needs no source outside it.

Only consistent mathematics forms structure. Inconsistent statements can be written but not built, so consistency is the filter on what exists.

2. All Consistent Structures Exist

Every consistent structure exists as mathematics, which takes nothing beyond consistency; anything that let one exist and not another would itself be structure, so the point is a priori. Physics is transitions in mathematics, discrete because a cause needs a next state, and consciousness is physics as well. Mathematical existence already supplies the complete structure. This paper rejects a primitive actuality property that would select some existing structures and exclude others: any such property would either enter determinate relations with the structure, and so belong to it, or make no difference to anything.

Matheism reaches structural identity by a different route from Max Tegmark's Mathematical Universe Hypothesis (Tegmark, 2008). Tegmark starts from an external reality whose complete description must be free of human concepts; Matheism starts from the rule-random dichotomy and the idleness of any substrate. Every consistent law structure exists in its own right, which does not place every structure inside our particular universe. The ontological argument concerns what existence is, and the measure concerns which structures and observations receive weight.

3. The Measure

Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description's weight. A shortest description cannot be compressed, since its compressed version plus an instruction to unpack it would be shorter still, so its bits are random in the sense of the dichotomy. With no rule to favor 0 or 1, each value has probability 1/2, and any lean toward one value would itself be a rule written into the description. Stated precisely, fix a description language and a class of complete candidate models and assign code weights w(H) proportional to 2^(-L(H)); a real difference of Delta L bits then gives a prior ratio (the odds before any evidence) of 2^(Delta L), and ten bits give exactly 1,024. Evidence multiplies those prior odds by its likelihood ratio, how strongly it favors one account over the other.

Longer descriptions lose in total as well, because whether to add another bit is also an unruled 50/50. Each description extends by a 0 or a 1, each extension carrying half its parent's weight, so if every string counted as a complete description, every length would carry the full weight again and the total would be infinite. With a 50/50 choice to stop or go on, all descriptions one bit longer weigh half as much together, and everything longer than a

given length adds up to that length’s weight: the tail equals the peak. Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest length that can produce observers. This weighting, under which the shortest descriptions that work dominate, is the Simplicity Dominance Principle.

Ray Solomonoff introduced algorithmic probability, the weighting of hypotheses by the length of the programs that produce them, for induction, reasoning from past observations to predictions (Solomonoff, 1964), and Jürgen Schmidhuber applied it to whole universes (Schmidhuber, 2000). A physical prediction states its coding domain, its multiplicities and its conditioning rule. The simplest programs may generate no observers, so conditioning on evidence changes the comparison.

A measured parameter can be an output of a simpler rule, and then it costs nothing extra. It can also be an address within the space that rule generates. If one rule derives several constants, the rule and any remaining selector replace their separate encodings, and deriving those relations and showing the saving is a concrete route to strengthening the Simplicity Dominance Principle.

4. Core Consequences

4.1 What the simplicity prior establishes

Simplicity sets the starting odds, and evidence then decides. Under a specified code prior, shorter descriptions receive greater individual prior weight, and that weighting does not guarantee that the shortest candidate is true or best supported once the evidence is in.

4.2 The arrow of time

Time is the chain of causal steps by which a simple start grows complex minds. The measure favors a simple start, laws and starting conditions together, and observation shows the same start whatever explains it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). Each later step needs that fact, not its reason, so a reason can fail without taking the rest of the chain with it. Minds like ours are complex, so their complexity has to grow from that start, as many simple rules grow complexity: the Game of Life’s one rule about neighbors grows five live cells into more than a thousand generations of structure and can run any

computation (Gardner, 1970). Growth takes steps, discrete on the view taken here, because a cause produces its effect as the next state and a continuum has no next state for a cause to produce; a reader who allows continuous or uncaused change still has growth from a simple start, which is all the arrow needs. A process generating each event from the previous one needs a starting state, and an endless one has no final event to work back from, so a complete backward production has nowhere to begin. Each causal step is a rule applied to rules: the rules are the only things, and a structure that only sat there would still exist, as mathematics, and know nothing, like a dead brain. Time is the applying, and space is how far it spreads, at the base layer’s rate, the speed of light. Light, which crosses space while no time passes for it, is no exception: a moving body’s transitions run slower, and light spends every step moving, so nothing within it could move relative to the rest.

Reversing a finite history avoids the missing endpoint, and pays in weight for whatever extra it needs. The first extra is the turning point: the endless forward history does not have to say where it stops, while a reversal has to say where it starts back from, and each bit of that address halves its weight again. Specifying the turning step takes about as many bits as writing its number in

binary: about 20 bits for a typical step near one million, which leaves about a millionth of the weight, so later turning points cost more. A code can charge a finished string and its reversal the same, but only by running the forward history and reversing the output, so the reversed history still depends on the forward production.

Minds run forward too. A thought merges many possible states into one, so only the forward step is produced by the mind’s own present state; run backward, its next state would be set by heat converging from its surroundings. Memories are built by steps already taken, which is why observers remember the past and not the future.

4.3 What observer selection explains

We see constants that allow observers because only such constants can be seen, and simplicity then favors the shortest account that produces them. Explaining fine-tuning (the way the constants seem precisely set to permit life) quantitatively takes a candidate space, a measure and an observer weighting. Simplicity enters that comparison; it does not by itself prove that our constants are the shortest compatible values. The prediction is that the complete laws lie near the shortest description that can

produce observers.

4.4 The hard problem of consciousness

David Chalmers's hard problem of consciousness (Chalmers, 1996), the question of why physical processes give rise to felt experience, assumes its own conclusion: it takes experience to be something other than structure, then finds that no structural explanation reaches it. Every candidate answer must specify determinate properties and relations, so a critic who excludes structure from constituting experience in advance has excluded every possible answer. Mind and body act on each other, so they share one structure, and any account of what an extra ingredient adds would itself be structure.

Sentience is felt transitions in modeling systems. Reflective consciousness is a transition in a system that models its own activity. The threshold and the mapping from organization to particular experiences remain research questions. A thing is its complete structure, so two systems cannot differ in experience without differing in structure.

4.5 Many-to-one as structural root

Many-to-one maps, maps that send different inputs to the same output, are the proposed common root of the Principle of Least Action (roughly, the rule that motion minimizes a quantity called action), coarse-grained entropy (a measure of how many detailed arrangements fit one coarse description) and emergence. Together they suggest one research program connecting microscopic structure to observable physics, and their mechanisms still have to be told apart. Stationary phase (an approximation where most paths cancel, leaving those near the classical path) can recover classical action equations, coarse graining groups microstates (exact detailed configurations of a system), and boundary conditions (constraints fixed at the start or edges of a system) and statistical assumptions can support an entropy gradient; none of these follows from many-to-one maps alone. Time’s direction does not come from them either: exact evolution, the precise change of a state over time, is one-to-one, and Section 4.2 gives the reason time runs forward.

Points of zero length do not add up to a length, so I argue for a discrete physical account of extension. Stated precisely, a sum of singleton lengths, the lengths of single points, is zero, and

analysis, the mathematics of limits and continuity, does not identify that sum with interval length. A short discrete rule is worth searching for when it can recover the required geometry, dynamics and observations. Discrete and continuous models can agree exactly on specified observables (measurable quantities) while differing on others, so the physical task is to recover observed distances and dynamics from specified discrete relations.

5. Falsification

Matheism has two halves, and each could fail in its own way. The ontology (the account of what exists) is necessary, as 2 + 2 = 4 is, so no observation confirms it and one counterexample would refute it: anyone who describes a coherent feature that is neither ruled nor random, in physics or in experience, refutes the dichotomy and mathematical monism with it. Necessity leaves the claim its teeth. It rules out a bare substrate, a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics, each a position with defenders. The measure is the part that meets data: if the final theory of physics requires irreducibly high complexity, the Simplicity Dominance Principle is challenged, and the ontology

stands either way. Both tests are stated in advance.

Stated precisely, falsifiability is a test for contingent empirical claims (ones that could have been otherwise), and a necessary claim is tested by counterexample instead. The partition is logic. That both branches are mathematical is mathematics' own result, since a rule is a function between states and randomness is algorithmic incompressibility or a probability law. That reality is identical to its complete structure has its own argument, from shared relations and an idle substrate (Section 1.4). The charge that the conclusion is empty fails on the list above: a claim that excludes those positions has content.

Conclusion

Laws, matter and minds are mathematical structure, because everything that exists is ruled or random, and both are mathematical. The central argument asks one question: once every property and relation of a thing is listed, what could a further substance add? The claim is necessary, as 2 + 2 = 4 is, and it has teeth: it rules out a bare substrate, a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics, and one coherent feature that is neither ruled nor random would refute it. The same argument

reaches mind. The hard problem of consciousness assumes its conclusion, and since a structure is one wherever it occurs, the bare subject of experience is one self in every mind, with distinct memories and lives left in place.

The physical work is to identify the structure of our universe, and the measure is the part that meets data. A shorter complete description receives greater initial weight, because each of its bits is an unruled 50/50 choice, and since whether to add another bit is a second 50/50, longer descriptions lose in total as well. The universe starts simple and grows the minds that observe it: no brains, no records and no instruments at the start. Growth runs in causal steps, because a cause produces its effect as the next state and a continuum has none, and time is those steps, running forward from the simple start toward minds that remember the past because they were built by it. To explain our particular laws, our quantum observations and the direction of time, one model has to deliver all three together.

A1. Why Reality Is Math: Forward Discrete Time and Complexity from the Simplest Laws

Abstract

Matheism holds that reality is mathematical structure, because all existence is either ruled or random, and both are mathematical. The complete structure is the thing itself, with no separate substance left to explain. The view is monism, not Platonism: Plato set the Forms apart from a world they do nothing in, and his One, with no properties, is nothing, a problem Neoplatonic emanations only multiply. The hard problem of consciousness assumes its conclusion, that sentience cannot be structure, then finds that structure cannot explain it. Panpsychism cannot say how tiny feelers add up to one mind, and idealism cannot say how separate minds find the same mathematics without using it. The claim is necessary, as 2 + 2 = 4 is, and it has teeth: it rules out every rival that needs something with no structure, and one coherent feature that is neither ruled nor random would refute it. Mathematics needs no maker, and that necessity is the ground of existence. Every consistent structure exists, fictions included,

each plot hole fixed by some consistent math. The same dichotomy suggests a measure: each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description’s weight. The universe therefore starts simple and grows complex, since minds like ours are complex, and it grows in discrete causal steps, since a cause produces its effect as the next state and a continuum has none. Time is those steps, and it runs forward from the simple start.

Argument and scope. The paper defends the identity thesis by showing that structure leaves nothing out. Algorithmic definitions apply within their stated domains, and what no program can compute is still mathematical. The paper separates what it deduces from what it adds as a further commitment.

1. Introduction

We are mathematical structure, as a triangle is geometric structure. Mathematics is what we are, not an external maker. An extra ingredient must make a definite difference or explain nothing. Consciousness tests this identity: could experience differ while its complete constitution stayed the same? The paper argues that it could not. The hard problem of consciousness

assumes its conclusion: it takes sentience to be something structure cannot be, then finds that structure cannot explain it.

Matheism is mathematical monism, the view that reality is one kind of thing, mathematical structure. It is expressed here as *Deus sive Mathematica*, God or mathematics, after Baruch Spinoza's identification of God with the one necessary substance, with mathematical structure in that role. It is not Platonism, which is dualism: Plato's Forms do nothing in the world, and his One, with no size or properties, is nothing (Section 3).

The dichotomy is a tautology with teeth, because both of its exits lead to the same building and it still rules out real positions. A natural objection calls it trivial: everything is A or not-A. The partition is logic, and the force lies in two further steps: both branches belong to mathematics by mathematics' own standards, and complete structure leaves no further constituent. Those steps exclude a bare substrate (featureless stuff that bears properties), a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics, each a position with defenders. Necessary truth is not a defect. A necessary claim is tested by counterexample: one coherent feature that is neither ruled nor random would refute it. The

empirical side is the weighting of descriptions, and a failure there would leave the ontology (the account of what exists) standing.

Two results follow for physics, and a third for minds. Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description's weight, and since whether to add another bit is a second 50/50, longer descriptions lose in total as well. Time runs forward because an endless generated history has no last state to run back from, and a reversed history must also specify its turning point, each bit of which halves its weight again. Minds like ours are complex, so a history that starts simple must grow them, and it grows them in causal steps, since a cause needs a next state and a continuum has none. The argument is an a priori proof in the logician's sense. Russell's answer, that motion is only being at different places at different times (Russell, 1903), offers the emperor's new points: a position for every instant and nothing that produces the next.

Chance offers no escape from mathematics, because a probability law and its outcomes are mathematical too. A probability law gives weights to possible outcomes without selecting the outcome that occurs. The position defended here adds a generator that produces each particular outcome, and the monism argument holds even without that addition: if

fundamental chance is granted, its states, dependencies and probability law still admit mathematical description. Calling an outcome random therefore supplies no way out of that description, and whether description exhausts reality is the identity claim the paper defends. Quantum theory does not force brute chance either: on the many-worlds and superdeterministic readings, every outcome follows a rule (Everett, 1957; Hossenfelder and Palmer, 2020).

Max Tegmark's Mathematical Universe Hypothesis posits that reality is mathematical structure (Tegmark, 2014); the argument here concludes that reality *must* be mathematical structure. A hypothesis invites the question "why this rather than something else?" The dichotomy leaves that question no room, since whatever is ruled or random is mathematical and no coherent third option has been produced. The step from there to identity, that complete structure is the thing itself, is what this paper argues and what critics dispute.

2. The Dichotomy

Every aspect of reality either follows a rule or does not, and there is no third option. Consider any aspect of reality. It either exhibits discoverable regularities or it does not; this is a logical partition.

Both branches are mathematical. Ruled features are mathematical structures, regularities a formal rule can capture: the laws of physics, the distribution of primes and the branching of trees. Algorithmic randomness, the lack of any much shorter description, is equally mathematical (Solomonoff, 1964). Stated precisely, for a finite binary encoding s, fix a universal description language U and a deficiency threshold c, a set margin in bits. Then either $K_U(s) < |s| - c$ or $K_U(s) \geq |s| - c$, where $K_U(s)$ is the length of the shortest description of s. No formal system, a fixed set of axioms and rules for deriving theorems, can certify incompressibility beyond a fixed length (Chaitin, 1974). The partition needs no certificate: every string, any sequence of symbols, falls on one side whether or not anyone can prove which. An incompressible finite string still has determinate entries and mathematical relations; it is not structureless.

The partition becomes ontology through the identity argument, which asks what a proposed alternative contributes. Two

outcomes are possible. A definite contribution adds properties or relations to the account, and so adds structure. A contribution with no determinate character explains no difference.

Writing something down is not the same as there being such a thing. One can write “2+2=5” or “square circle,” and the string exists as a consistent structure (one free of contradiction), as does every medium carrying it. All consistent structures exist, yet the state of affairs these strings describe cannot be instantiated (made real in any actual case). An inconsistency exists only as a consistent string that describes it, because construction of any kind needs consistency as its medium.

The transcendental constraint

Observation requires structure. Reliable memory requires a relation between the record and its source, and useful prediction requires a relation between an available cue and what follows. This constrains the relations supporting observation, not the compressibility of everything observed.

Closing the escape routes

Every proposed escape from structure either adds structure or adds nothing.

Qualia and the hard problem of consciousness. Experience is the complete organization that constitutes a conscious system, felt by being it, from inside. The hard problem of consciousness expects to feel it from outside, and it assumes its conclusion, that sentience cannot be structure. Qualia, taken as something beyond structure, are a label for the gap, not a bridge across it. Anything extra that influences a conscious system needs a structural coupling to it, and anything inert cannot cause our reports. Any account of what feeling adds would itself be structure.

Bare particulars. A bare particular with no distinguishing features is indistinguishable from nothing. For anything to exist determinately, it must have some feature, and that feature is ruled or random, and thus mathematical.

Fundamental stuff. Describe “stuff” and you either describe structural relations, such as mass, charge and spin (an intrinsic, rotation-like property of particles), or describe nothing determinate. Substance without properties is indistinguishable

from nothing. The dichotomy applies again at every level: any proposed substance must have determinate features, and those features have features of their own, each ruled or random.

The interaction problem. If a proposed extra ingredient affects bodily events, its coupling belongs in the structural account, and if it affects nothing, it cannot cause our reports. Princess Elisabeth of Bohemia posed this to René Descartes in 1643: how can an unextended mind (one that takes up no space) move an extended body? His pineal gland (a small gland deep in the brain) answered where, not how, and his fallback analogy to weight borrowed a cause he had himself denied was real. Any causal relation requires a common structure within which cause and effect are defined. A conversation runs the loop both ways: words arrive as sound waves and change what the listener feels, and the feeling moves the listener’s mouth to answer, physics in and physics out. A consciousness in an ineffable dimension beyond mathematics and beyond words would need a crossing out of the mathematics and a crossing back.

Panpsychism. Panpsychism, the view that all matter has some form of mind, cannot explain how tiny feelers add up to one mind. A combination principle must specify which parts combine, under what conditions and with what experience, and those relations

are structure. Mathematical monism must likewise identify the organization that constitutes unified experience; rejecting elementary subjects does not by itself do that work.

Idealism. Making reality out of mind removes the interaction problem between two substances, and it still owes an account of order. Separate minds discover the same mathematics, and whatever makes them agree is a regularity, so it is structure. Idealism needs math, and math does not need it.

Nothing is left to tell a physical system apart from its complete mathematical structure. The complete structure of a system, including its environment, shares all relational properties with its mathematical description. What else could distinguish the two? Not their relationships, which are all included. Not their causal powers, which supervene on (that is, are fixed by) relational structure. Not their observable properties, which all reduce to relations. The only remaining candidate is a nonrelational substrate, and substrate without properties is indistinguishable from nothing. The math is the territory.

3. Mathematical Structure as Self-Grounding

Mathematics needs no maker, so explanation ends at it without regress, an endless chain of further explanations. That 2+2=4 is not contingent on any creative act; it could not have been otherwise. This satisfies the Principle of Sufficient Reason, the demand that everything have a reason, without an infinite regress. Why does anything exist? Because mathematical structure cannot fail to exist. The question “why is there something rather than nothing?” dissolves, since mathematical structure is the necessary background against which causation itself is defined.

The position is not Platonism (the view that mathematical objects exist outside space and time), because concrete reality is mathematical structure. Physical existence and mathematical existence coincide. This completes the program of structural realism (Worrall, 1989) in its ontic form, the view that structure is all there is to the physical world (Ladyman and Ross, 2007; French, 2014).

Platonism puts mathematics in a second realm, outside nature, and that makes it supernatural. It places mathematical objects outside space and time and denies them causal power (Linnebo,

2023). A realm outside nature that does nothing in nature fits the plain meaning of “supernatural.” Plato needed a divine craftsman, the demiurge, to copy his Forms (perfect, eternal patterns of things) into the world (*Timaeus* 28a-29b). Aristotle reports that the Pythagoreans held things are numbers, and that Plato departed from them by setting the numbers apart from sensible things, those perceived by the senses (*Metaphysics* I.6, 987b27-29). Matheism takes the Pythagorean side. Plato’s departure was a late Athenian anti-philosophy. Athens’ early thinkers sought reason inside nature; once the Forms stand outside it, the Forms, the world or the link between them lies beyond reason, and reality cannot be known. Both halves turn supernatural: the Forms stand outside nature, and nature, emptied of them, follows no logic. That refuses knowledge where philosophy loves it, and it fails a priori: outside logic is illogical, as impossible as a square circle. An explanation cannot be supernatural in any case, since to explain is to connect, and a connection is structure.

Plato’s highest principle fares worse. The One of his *Parmenides*, taken strictly as one, has no parts, shape, place or time, and Plato concluded that such a One in no way is (*Parmenides* 137c-142a). A point so pure that it has no size or properties is

nothing at all. It contains nothing, relates to nothing and affects nothing: a fake monism over a real dualism. Plotinus made such a One the source of everything, with Intellect, Soul and the material world emanating from it level by level (*Enneads* V.1-2). The levels multiply the problem. They are more dualism, solving only the problems dualism itself creates. Each must relate to the One, and a relation gives each of its terms a property, which the One cannot have. If the levels act on one another, their relations are already structure. Plotinus's own images, an overflow and light around the sun (V.2.1, V.1.6), are metaphors, not relations, and the layers are unnecessary.

Mathematical structure has the attributes monotheism reserves for God. It is uncreated, necessary and indestructible: no agent makes 2+2=4, and none could make it otherwise. Monotheism insists there can be only one such ground, so mathematical structure is what those traditions were describing. The formula is *Deus sive Mathematica*, not *Deus sive Natura* (God or Nature).

4. Implications

Physics that asks which changes are possible fits the structural account without proving it. David Deutsch and Chiara Marletto's constructor theory asks which physical transformations are possible and which are impossible (Deutsch and Marletto, 2015). That emphasis on possible changes is compatible with the structural account defended here, and it does not by itself establish that physical reality is identical to mathematics.

The changes are the deeper point. The rules are the only things. A structure that only sat there would still exist, as mathematics does, and do nothing and know nothing, like a dead brain with every connection intact. What happens is rules applied to rules, one transition after the next; time is that application, and space is how far it spreads.

Mathematics describes reality so well because reality is mathematics. That answers the physicist Eugene Wigner's puzzle about "the unreasonable effectiveness of mathematics in the natural sciences" (Wigner, 1960).

Every consistent structure exists, because no primitive actuality property selects some and excludes others. The step is a priori:

such a property would itself be structure. Such a property would either enter determinate relations with the structure, and so belong to it, or make no difference to anything. Fictions are no exception. A story is a structure, and a consistent one exists as such. A plot hole is a contradiction and exists only as a description, while the versions of the story in which a further fact explains the hole are consistent, so they exist: every plot hole is fixed by some consistent math.

The same dichotomy fixes how descriptions are weighted. A shortest description has no pattern left to compress, so its bits are random in the dichotomy's sense and nothing favors 0 over 1: each bit is a 50/50 choice, and each added bit halves a description's weight. Whether to add another bit at all is a second 50/50, so longer descriptions lose in total as well as one by one. Simpler laws therefore carry more weight, and observations update those weights.

Time runs forward because a generated history starts from a first event, and an endless one has no last event from which to generate the whole history backward. The arrow of time connects timeless structure with the order of events within it. A process generating one event from the previous one needs a starting state, and simplicity favors a compact account of the laws and the

starting state together. A reversal must also specify the point it turns back from, a cost the endless forward history does not pay, and each bit of that name halves its weight again. Between a finite history and its own reversal, a universal code can charge the same shortest length, so no universal penalty is claimed there.

The problem of consciousness changes shape: the question becomes which structures feel, not how mind arises from nonmental stuff. Physical processes are mathematical structures, so mental processes are too. Matheism identifies consciousness with felt transitions in self-modeling computation: feeling is a mind’s states producing one another as it models its world, so a photograph or a recording of a brain, which produces nothing, feels nothing. Since the dichotomy leaves no third category, all minds, biological or artificial, inhabit the same structure, and mathematical structure already includes consciousness through every conscious pattern it contains.

Conclusion

Reality is mathematical structure, because the rule-random dichotomy admits no nonmathematical third category. Two things cannot interact, and nothing can be explained, without definite relations, and mathematical physics specifies those relations as states and dynamics. The first point is a priori, from what interacting and explaining are. The complete structure is the thing itself, with no Forms apart from the world and no propertyless One above it. An additional substrate faces two outcomes: if it contributes a determinate feature, that feature belongs in the structure, and if it contributes none, it makes no explanatory difference.

The claim is necessary, as 2 + 2 = 4 is, and it has teeth. It rules out a bare substrate, a soul that moves a body by no law, a free choice that is neither caused nor random and a creator outside mathematics. One coherent feature that is neither ruled nor random would refute it. A measure follows from the same dichotomy, on the view taken here. Each bit of a shortest description is an unruled 50/50 choice, so each added bit halves a description’s weight, and since whether to add another bit is a second 50/50, longer descriptions lose in total as well. Time runs

forward because an endless generated history has no last state to run back from, and a reversed history must also specify its turning point, each bit of which halves its weight again.

The same argument reaches mind, where the hard problem of consciousness assumes its conclusion. A structural account must identify the organization that constitutes experience, and a critic who excludes every such account in advance has assumed the split between structure and experience that is in dispute. Mathematical monism places matter and experience within one ontology. The further physical task is to identify our structure and explain its observations.

A2. Why Simple Laws Are Most Probable: Occam’s Razor, All Is Math, and Discrete Time Steps Forward

Abstract

Simple laws are more probable because every bit of a complete description is a yes-or-no that no rule decides, so each added bit halves the description’s probability. A shortest description has no pattern left to compress, so its bits are random in the sense of the rule-random dichotomy: nothing favors 0 over 1, and no bit depends on another, since a preference or a dependence would itself be a rule that has to be written down. Because every structure exists, the half is a count and not a guess: among all strings, one in 2^k begins with any given k bits. Longer descriptions are more numerous and still lose in total, because whether to add another bit is a second unruled yes-or-no, so everything longer than a given length weighs exactly as much as that length: the tail equals the peak. Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest length that can produce observers. The universe therefore starts simple, laws and starting

state together, and since minds like ours are complex, it has to grow them. Growth takes discrete causal steps, so time runs forward from that start.

1. The Claim

Simpler laws are more probable, and this paper derives by how much: each bit of a complete description halves its probability. Call this the Simplicity Dominance Principle. Ray Solomonoff introduced the same weights as a starting point for predicting data (Solomonoff, 1964). Here they are derived rather than chosen, and they weigh what exists, not only what to expect next. Three results follow. Each bit halves the weight. Whether to add another bit is a second 50/50, so longer descriptions lose in total. At least half the weight sits at the shortest length that can produce observers, so the universe starts simple and has to grow the complex minds that observe it. It grows them in causal steps, a cause and then its effect, so time runs forward from the simple start (Section 6). The order of cause and effect is a priori; the simple start comes from the weighting, an argument with premises of its own.

The derivation starts from the rule-random dichotomy: everything that exists is ruled or random, and both are mathematical. A ruled structure is produced by a rule shorter than itself; a random one has no shorter rule. Anything outside both would be outside mathematics, and so outside logic, since nothing could be reasoned about it. Matheism adds that every consistent structure, one free of contradiction, exists (compare Schmidhuber, 2000; Tegmark, 2014). The question is then which structures an observer should expect to be in, and without a weight on structures nothing can be predicted at all.

The weight attaches to a complete description: the laws, the starting state and the rule connecting states to what observers record, with information they share counted once. A short equation whose starting state hides the real information is a long description, so moving complexity from the laws into the start saves nothing.

2. Why Each Bit Halves

Each bit of a shortest description is a yes-or-no that no rule decides, so 0 and 1 are equally likely. Stated precisely, a shortest description cannot be compressed: if it could, the compressed version plus an instruction to unpack it would be a shorter

description of the same thing. Its bits are therefore random in exactly the sense of the dichotomy. With no rule to favor either value, each has probability 1/2. A lean toward 0 or 1 would itself be a rule, and would have to be written into the description, making it longer. A biased coin is mathematical too, but its bias is a number that has to be written down; one half is the only chance that needs no such number, because it is the only one that treats 0 and 1 alike.

No bit depends on another, for the same reason. Two bits can each be fair and still agree every time, as when one choice between all zeros and all ones fixes every bit at once. That agreement is a rule, “repeat the first bit,” and a rule costs bits. With nothing written down, each bit is independent of the others.

Because every structure exists, the half is also a count and not a guess. Take all strings (sequences of 0s and 1s) of some length n. Exactly half begin with 0, and when n is at least k, exactly one in 2^k begins with any given k bits. The weight of a k-bit description is that share: the fraction of all strings that begin with it. On a computer the share is one of tapes: fill a universal machine’s tape with fair coin flips, and a k-bit program begins one tape in 2^k. The machine reads no further than the program’s end, so every tape that begins with it computes the same history.

This is probability as Pascal and Fermat counted it in 1654, a share of equally possible cases (Hacking, 1975), and the count carries independence with it: a quarter of all strings begin with 0 and then 1, the product of the two halves, with no further assumption.

A particular description of L bits therefore has weight 2^(-L), and two complete accounts that differ by ΔL bits have prior odds (the odds before any evidence) of 2^(ΔL). Ten bits give 1,024. The evidence then multiplies each weight by how well the account predicts what is observed. An account 20 bits shorter starts about a million times ahead and loses that lead if it makes the observations a million times less likely, so fit and length trade at one bit for each factor of two, and among accounts that fit equally well, the shortest dominates.

Random, in the dichotomy, means a structure with no shorter rule, a pattern nothing compresses. It does not mean an event without a cause. The 50/50 is a share of structures, not a coin flipped by nothing.

3. Why Longer Descriptions Lose in Total

Longer descriptions are more numerous, and they still lose in total, because whether to add another bit at all is a second unruled yes-or-no. If number decided, we should expect to find ourselves under an extremely long description, a world of endless special cases and arbitrary detail; we find ourselves under laws that fit on a page. Stated precisely, there are 2^n strings of length n, so if each counted as a complete description weighted 2^(-n), every length would carry a total weight of one and the sum over lengths would be infinite. Half is the gentlest rate any weighting can have: with equally long descriptions weighted alike, a factor milder than one half per bit would give each length more total weight than the one before, without limit.

The choice to stop or go on settles the total. It is unruled, so it is 50/50: half the weight stays with a description as it is, and half goes on to longer ones. All descriptions of n bits together then weigh 2^(-(n+1)), half as much as those one bit shorter, and everything longer than n bits, added up, weighs exactly as much as length n alone: the tail equals the peak.

Picture each universe as its own run of fair coin flips, a new experiment each time: one flip decides whether its description

goes on, and the next decides the bit added. Any laws can come up in a run, yet half of the runs still going stop at each step, so most runs end with short laws. A catalog that listed each set of laws once would favor long laws by sheer number. The runs favor short ones, because a long law has to win every one of its stop-or-go flips.

In this picture one particular description loses three quarters of its weight with each added bit, a half for going on and a half for the bit’s value. A code that states the length once, in a short header, loses only about half per bit (Li and Vitanyi, 2019), so odds between particular descriptions in this paper use that smaller factor, while the totals by length use the stop-or-go weights. Either way, descriptions none of which begins another claim separate shares of all strings, so their weights add up to at most one.

4. At Least Half at the Shortest Length That Works

Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest length that can produce observers. Producing observers means producing structures that record and model their surroundings. The test mentions none of our own laws, so the count does not

build in its answer.

Stated precisely, under the stop-or-go weights of Section 3, let M be the shortest length that produces observers, and let s_n be the share of length-n descriptions that produce them. If no longer length has a larger share than s_M, length M carries at least half of the observer-producing weight, because the tail beyond M weighs exactly as much as length M and its share is nowhere larger. If the share also does not rise from one length to the next, lengths M and M+1 carry at least three quarters, and lengths M through M+2 at least seven eighths. Without that second condition, shares of one at M, none at M+1 and one after that leave the first two lengths only two thirds.

The condition does real work. Under the same weights, if one description in sixteen of length four produced observers and all thirty-two of length five did, length five would carry eight times the observer weight of length four. Physics points the other way. Small changes to the constants of our physics often remove stable nuclei, long-lived stars or chemistry (Barnes, 2012), and some lengths of description cannot produce observers at all; each such gap moves more weight onto the shortest length that works. For a single output, the coding theorem makes the same point: all its longer descriptions together cannot outweigh its

shortest one by more than a fixed factor (Levin, 1974). The prediction is that the complete laws lie near the shortest description that can produce observers.

5. Objections

”The half is the principle of indifference, which leads to paradox.” Joseph Bertrand’s paradoxes arise for continuous quantities, which can be spread evenly in many ways, each a different measure (Bertrand, 1889). A bit has two values, and only one split treats them alike. Nor is the half a confession of ignorance: no rule exists to favor either value, and the count of Section 2 makes it a share of what exists.

”Incompressibility describes one string, not how strings are weighted.” The weight does not come from one string alone. It comes from what exists: every string exists, and the share of them that begins with a given description is the count of Section 2. Incompressibility says why nothing inside the description tilts that share.

”A biased chance is just as mathematical.” It is, and its bias is a number the description must carry. A description that carries it is longer, and its added bits face the same halving.

”The weights depend on the programming language.” Universal languages, those able to express any computation, agree on lengths up to a constant that depends on the pair of languages and not on what is described (Kolmogorov, 1965), so the weights under two languages differ by at most a fixed factor, and each bit of real difference between two accounts still halves the weight. The 50/50 does not depend on the language; only the lengths do. Close comparisons state their language.

”Shortest lengths cannot be computed.” In general they cannot, so comparisons use the lengths of descriptions actually written, which bound the shortest from above. The measure says what to compute, as a definition of probability does without listing every case.

”Why should existence come with weights at all?” An observer asking which structure it is in has no other way to answer. Every structure exists, so the question is what share of what exists is like this, and the count is the only answer that needs no further rule.

6. What Follows

The universe starts simple, laws and starting state together, because every bit of its description halves its weight. Minds like ours are complex, so their complexity has to grow from that start: each bit of a brain’s organization is either written into the description, halving the weight, or grown by the laws from a simpler state, and the probable universes grow it. Many simple rules do. The Game of Life’s one rule about neighbors grows five live cells, the R-pentomino, through more than a thousand generations, throwing off gliders (small patterns that travel while keeping their shape), and suitable patterns in it can run any computation (Gardner, 1970; Berlekamp, Conway and Guy, 1982). What grows is organization; the description stays short, since every later state is fixed by the rule, the start and the number of steps. Growth takes causal steps, a cause and then its effect, since a cause produces its effect as the next state and a continuum has none. That is an a priori proof in the logician’s sense. Russell held that motion is only being at different places at different times (Russell, 1903); a position at every instant with nothing producing the next is the emperor’s new points. So time runs from the simple start toward the minds it grows. Each step is

a rule applied to rules, and time is that application. That chain needs the simple start only as a fact, and observation shows it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). It stands even for a reader this paper does not persuade, since a reason can fail without taking with it the fact it explains.

Conclusion

Simple laws are more probable because everything that exists is ruled or random, so each bit of a complete description that no rule decides is a 50/50 choice, and since every structure exists, the half is a count of what exists. Each added bit halves a description’s weight, the choice to stop or go on makes longer descriptions lose in total as well as one by one, and at least half the weight sits at the shortest length that can produce observers. The weights Solomonoff chose as a starting point for prediction follow from the dichotomy, and they weigh what exists. The universe starts simple and has to grow the minds that observe it.

A3. Reality and Consciousness Are Mathematical Structure

Abstract

Everything that exists can only be mathematical structure, because whatever exists has definite features, every definite feature is either ruled or random, and both are mathematical. The view is monism: one kind of thing throughout. It is not Platonism, which sets the Forms apart from a world they do nothing in, a dualism with a supernatural second realm. Plato’s One, with no size or properties, is nothing at all, and Neoplatonic emanations only multiply the problem of how anything could relate to it. The hard problem of consciousness is a fallacy that assumes its conclusion, that sentience cannot be structure, and then finds that structure cannot explain it. Qualia, panpsychism and idealism each need the structure they set out to supplement. The strongest objection says structure leaves out an intrinsic nature. A swap test answers it. Exchange the hidden natures behind mass and charge while every relation stays fixed, and nothing anyone could notice, think or say would change. Such a nature could not even be known by the mind it supposedly makes

conscious. The claim is an a priori proof in the logician's sense, necessary as 2 + 2 = 4 is, and it has teeth: one coherent feature neither ruled nor random would refute it. Consequences follow by reasoning alone. Every consistent structure exists, fictions included, with every plot hole fixed by some consistent math. Each added bit halves a description's weight, so simple laws dominate. Complex minds must grow from a simple start in causal steps, and time is those steps, running forward.

1. Introduction

Everything that exists can only be mathematical structure, because whatever exists has definite features, every definite feature is either ruled or random, and both are mathematical. That is the core claim of Matheism and the claim defended here. Math means the structures themselves: their properties, relations and possible changes, simple or complex, predictable or random. Equations are ways we describe them. A triangle is a geometric structure, not something made by geometry. Reality stands in the same relation to mathematics.

The view is monism, the view that reality is one kind of thing: *Deus sive Mathematica*, God or mathematics. It is not Platonism, which is dualism. Plato set the Forms, the perfect patterns of

things, apart from the world, where by his own student's account they cause no change at all (Aristotle, *Metaphysics* I.9). A realm outside nature that does nothing in nature is supernatural in the plain sense of the word. Plato's highest principle fares worse. A One so pure that it has no size or properties is nothing at all, and the emanations Plotinus added only multiply the problem, since each level must relate to a One that has nothing to relate by. Matheism takes the side of the Pythagoreans, for whom things are numbers (Section 3).

The hard problem of consciousness asks why physical processes come with felt experience (Chalmers, 1996). As posed, it is a fallacy that assumes its conclusion: it takes sentience to be something structure cannot be, then finds that no structure explains it. It expects to feel a structure from outside, by reading about it; a structure is felt only by being it. Qualia, the felt qualities of experience taken as something over and above structure, only label the gap. Panpsychism, the view that all matter has some form of mind, cannot say how tiny feelers add up to one mind without a law of combination, and a law is structure. Idealism, the view that reality is made of mind, cannot say how separate minds discover the same mathematics without using mathematics. They need math, and math does not need them (Section 4).

The hardest objection grants the mathematical account and says each thing still has an intrinsic nature, what it is in itself apart from its relations, which the account merely represents. Answering it is the main work of the paper. The answer is a swap test. Exchange two such natures while every relation stays fixed, and ask whether anything anyone could notice, think or report has changed. If something changes, the nature does work and belongs to the structure. If nothing changes, nobody could know the nature, including the people it supposedly makes conscious (Sections 2.2 and 4.3).

The claim is necessary, as 2 + 2 = 4 is, and it still has teeth. It rules out every rival that needs something with no structure, and those rivals have serious defenders: a bare substrate, a soul that moves a body by no law, a free choice neither caused nor random, a creator outside mathematics. One coherent feature that is neither ruled nor random would refute it.

Consequences follow by reasoning alone, with no physics beyond what everyone grants. Every consistent structure exists, fictions included, and every plot hole is fixed by some consistent math. An unruled bit is a 50/50 choice, so each added bit halves a description’s weight, and simple laws dominate: ours fit on a page. Minds like ours are complex, so they must grow from a

simple start, and growth takes causal steps, because a cause needs a next state and a continuum has none, an a priori proof in the logician's sense. Time is those steps, so it runs forward. Quantum theory does not force uncaused chance on us: on the many-worlds and superdeterministic readings, every outcome follows a rule. The identity thesis does not rest on these consequences, and each is stated below with its premise.

Ontic structural realism, the view that structure itself is the basic reality, is the contemporary starting point (Ladyman and Ross, 2007; French, 2014, 2019). Stathis Psillos (2001) and Anjan Chakravartty (2007) object that structure needs an underlying bearer. Baruch Spinoza's monism and necessity offer a comparison, not this exact identity (*Ethics* I.14, I.29, I.33, I.34). The nearest physical proposal is Max Tegmark's (2008) Mathematical Universe Hypothesis, that physical reality is a mathematical structure, which he derives from the existence of an external reality independent of human observers. The argument here starts instead from what any definite nature must be.

The contribution is that identity argument: whatever exists is some definite way, every definite way is ruled or random, and an intrinsic nature that makes a determinate difference belongs to

the structure, while one that makes none constitutes nothing. The decisive premise is that a complete constitutive structure leaves a substrate nothing to contribute. It rules out a bare substrate, intrinsic natures that do no work, a mind with a nonstructural core, and a second realm, whether of Forms or of a One beyond all properties.

2. Why Nothing Can Fall Outside Math

2.1 What Would a Nonmathematical Thing Be?

For anything to exist and be distinguishable from nothing, it must have determinate features. ‘Something’ differs from ‘nothing’ by having properties: extent, duration, charge, mass, complexity. These cannot be wholly arbitrary or undefined. Existence requires determinacy; determinacy requires distinguishing features; distinguishing features require relations of sameness and difference.

Properties do not float free. Each is fixed by how it relates to others. A position is where something lies relative to other positions. A mass is how hard a thing is to push and how strongly it pulls on other masses. A price is what it trades for. Take away every relation a property enters, and nothing is left to say what it

was. Physics has made structural description ever more central (Ladyman and Ross, 2007; Esfeld and Lam, 2008), and the argument needs only the general point: to be a definite feature is to stand in definite relations.

2.2 Why the Complete Structure Is the Thing

Showing that reality is mathematical structure means showing that nothing could distinguish the two, and the argument starts from sameness of form. Any relational structure is isomorphic to at least one mathematical object, where an isomorphism is a structure-preserving mapping: it preserves all relations while potentially changing labels. Many mathematical objects share any given form, and the identity claimed here is with that shared form, whose relata, the things it relates, are simply its places.

What distinguishes a physical relational structure from an isomorphic mathematical structure? Not their relationships, identical by definition of isomorphism. Not their causal powers: causal relations hold between parts of a structure, and an isomorphism preserves them, so only the whole lacks causes outside itself, as the physical universe does. Not their observable properties, which reduce to relational structure. The only remaining candidate is non-relational ‘substrate,’ some primitive

(taken as basic, with no further analysis) ‘physical stuff’ that bears properties without being constituted by them.

The strongest opponent grants every mathematical relation and still assigns the things related an intrinsic nature. Giving that nature a symbol does not settle whether it is mathematical. The question is what the symbol stands for. A mass, a power or a felt quality is a proposed constituent with a character of its own; it cannot be dismissed simply because it is unobservable. Matheism argues that any such character belongs to the complete structure of the thing. A supposed bearer left over after every constitutive feature has been included supplies no further nature to bear.

This is the claim that structure is complete, sometimes called structural exhaustion. It is stronger than saying an extra ingredient is unnecessary for prediction. The opponent may still insist that a property has primitive nonstructural content. Matheism’s challenge is to give that content without either specifying more structure or merely asserting a remainder. The argument turns on that challenge, not on the existence of a mathematical encoding.

The proposed substrate faces three objections. First, a complete account of a thing's relations leaves a bearer no job. Physical theory specifies states, fields and interactions without a separate variable for featureless stuff that bears them. Reading that account as the complete constitution of reality is the structuralist interpretation defended here. The claim concerns a complete organized structure, not the replacement of a thing by an isolated number.

Second, substrate does no explanatory work. Two systems with identical relational structure behave identically in all observable ways. The substrate makes no difference to anything we can measure, predict or explain, so positing it violates Occam's Razor. Any proposed substrate must itself have determinate features, or it is nothing, and determinate features constitute structure. By the rule-random dichotomy, substrate *is* structure, the very thing it was meant to underlie. This applies at every level: no level of decomposition yields a nonmathematical residue.

Third, a primitive 'this one, not that one' is substrate under another description. Haecceitists invoke haecceities, primitive thisnesses, to distinguish entities when all relational properties match. The haecceitist must specify what makes *this* haecceity

differ from *that* one. Any answer appeals to distinguishing features, which are properties. Declining to answer leaves the distinction primitive and unexplanatory: a label where an explanation should go.

A complete description must preserve every constitutive relation, including relations to an environment and to observers. Even two internally isomorphic systems can be numerically distinct within a larger structure. In the universe of two identical spheres that Black (1952) describes, each sphere stands at a distance from the other and at none from itself, so a relation shows that there are two. The identity argument asks about the complete reality: what, if anything, could distinguish it from its complete structure? An additional relation enlarges the structure. A claimed difference that carries no determinate feature adds nothing to what constitutes the thing. The math is the territory.

The complete relational structure exhausts the properties and causal roles of a physical reality, so no further substrate remains to constitute it. This is the identity Matheism defends: reality *is* mathematical structure. It does not instantiate, or embody, that structure in a separate nonstructural substance. Isomorphism specifies sameness of form; the completeness argument supplies the further ontological claim, one about what exists.

2.3 Ruled or Random: Why the Dichotomy Is Exhaustive

Every structure is of one of two types, ruled or random.

Ruled structures, or patterns, have low generative complexity: rules simpler than the structure itself generate them. The laws of physics are short equations that generate vast complexity. Conway's Game of Life generates endless patterns from four rules (Gardner, 1970). The digits of π form an endless sequence that does not repeat, generated by a finite formula. Fractals and deterministic chaos belong here too. In each case a finite description generates or explains the structure, and apparent randomness can come from a compact rule: a finite stretch of π is generated by an algorithm and a request for that many digits.

The split between rule and randomness survives being made exact, and both sides stay mathematical. Algorithmic information theory measures the information in a string by the length of the shortest program that outputs it. Stated precisely, fix a universal description language U, a finite binary string s and a deficiency threshold c; then either $K_U(s) < |s| - c$ or $K_U(s) \geq |s| - c$. Both classes are mathematical, and the partition is exhaustive on that domain. No formal system, a set of axioms with rules for deriving

theorems, can certify incompressibility beyond a fixed length (Chaitin, 1974), and the partition needs no certificate: every string falls on one side whether or not anyone can prove which. An incompressible string still has determinate entries and relations. Randomness for infinite sequences needs its own definition and can be noncomputable, and noncomputable objects are mathematical too.

Fundamental chance would leave an outcome undetermined by the prior state. A probability law can still specify the possibilities and their weights; it does not pick the history realized. If reason demands a sufficient reason for each outcome, brute selection is unacceptable, and on the view taken here, change without a cause lies outside logic and reason. The identity thesis does not need that premise: even granting chance, its law and its determinate outcomes are mathematical. Quantum theory, the usual home of claimed brute chance, does not force it on us. In the many-worlds interpretation every outcome occurs, and what looks like chance is which branch an observer finds itself in (Everett, 1957). In superdeterminism one rule correlates what is measured with how it is measured (Hossenfelder and Palmer, 2020). Bohmian mechanics, in which particles follow definite paths, is deterministic too (Bohm, 1952). None of them adds a

nonmathematical ingredient.

Four potential counterexamples deserve consideration. *Ontic vagueness*, vagueness in the world itself rather than in our language (Barnes, 2010): a theory may give a mathematical account of which facts are indeterminate without giving each vague property a sharp boundary. *Primitive dispositions*, powers such as fragility or solubility: manifestation conditions have structure, but a dispositionalist can dispute whether those conditions exhaust the power; this returns to the identity argument. *Non-algorithmic hypercomputation*, computing beyond what any computer program can do: noncomputable functions and oracle computations (which may consult answers that no program could compute) are mathematical objects too. Noncomputability is not nonmathematicality, and the arithmetical hierarchy, a ranking of number sets by the complexity of their definitions, does not exhaust all noncomputable objects. *Atomless 'gunky' structure*: still relational structure, just infinitely divisible. All coherent alternatives reduce to ruled or random mathematical forms.

The dichotomy is not a stipulation one is free to decline. Any objector must reason in a way that already sorts cases into the ruled and the random; an appeal to a domain "outside reason" is

self-undermining, since the appeal itself uses the faculty it would exempt. Logic is not one commitment of inquiry among others; it is the medium in which all inquiry is conducted. This point has a narrow target: it answers the objector who denies that reality has mathematical structure at all. An objector who grants the structure and denies that it exhausts reality reasons mathematically without contradiction, and meeting him requires the identity argument of Section 2.2.

One might grant that reality is purely relational while denying it is thereby *mathematical*. Perhaps relations are primitive metaphysical facts that mathematics merely models? The response: rules express relations among possible states. Probability theory specifies relations. A realized sequence has definite entries even when it admits no compression. ‘Mathematical’ does not mean regular, predictable or generated by a short program. Rule and randomness exhaust all possible relational character. ‘Mathematical structure’ is not a species of relation; it is a genus containing all relations. The genus is mathematics’ own: in model theory, a branch of mathematical logic, a structure is any domain with relations defined on it (Hodges, 1993), so a determinate relational system counts as a mathematical structure before Matheism adds anything. The

substantive step is the identity argument, that a complete structure is the thing itself (Section 2.2). The burden falls on the critic: produce a coherent third option beyond rule and randomness, or accept the conclusion.

2.4 Why Reliable Observation Requires Structure

A reliable record must track what it records, and a useful prediction must depend on evidence that bears on the event predicted. Observers who learn and act on records need these relations. The constraint concerns the relations that support observation, not the compressibility of everything observed. A memory can store an incompressible string with a simple recording rule, and a probability law can sustain stable structures and reliable statistics. It does not follow that every region of reality is compressible, that random data cannot be experienced, or that every apparent observer in a chance-driven model is impossible.

2.5 Newman’s Objection and the Scope of Structural Claims

A bare claim that the world has some structure says too little, and the mathematician Max Newman showed why. His objection warns that a merely existential structural claim can become too weak. If relations may be freely reinterpreted, a domain of the right cardinality, or number of members, can be equipped with relations realizing the stipulated abstract pattern (Newman, 1928). Merely adding more equations does not answer the objection if every term in those equations is equally open to reinterpretation.

The present proposal concerns the actual constitutive relations, including causal and modal organization: what interactions occur and how the system would respond under the specified variations. It does not infer a particular physical structure from cardinality alone. Empirical theory must identify and constrain those relations through observations. Whether that identification captures every intrinsic property remains the exhaustion dispute of Section 2.2.

This answers a different question from reconstructing knowledge through a Ramsey sentence, which replaces a theory’s terms

with variables. Newman's result trivializes the claim that some relations of a given form exist on a domain. The identity view makes a different claim: the world is one structure, its relations included, and observers, as parts of it, pick out those relations through their own relations to the rest. Which relations those are is the work of empirical theory. A theory's notation, coordinates and charts are structures too, and the identity concerns the structure its predictions depend on, not the notation chosen to state it.

2.6 Interaction Requires Shared Structure

Begin with a conversation. Suppose consciousness were an ineffable dimension beyond mathematics and beyond words. Then how would words reach it? Speech is sound waves. They change what a listener feels, and what the listener feels moves the listener's mouth and sends sound waves back: physics in, physics out. Somewhere the signal would have to leave the mathematics and translate back, and whoever places consciousness outside it owes an account of where and how.

For any two distinct substances or realms to interact, they must share common rules. If mind moves matter, or Forms act on particular things, some law must say how. In 1643 Princess

Elisabeth of Bohemia asked René Descartes how a soul that only thinks could move a body (Shapiro, 2007). His pineal gland, a small gland deep in the brain, answered where, not how. Any answer has to say what changes, under which conditions and by how much. Those when-then relations are relational structure, and relational structure is mathematical. The step is a priori: to interact is to change by some rule, so no experiment is needed to show that interacting things share structure.

An interaction law places both proposed substances within one mathematical account. A dualist, who holds that mind is something beyond the physical, can still claim that the things related have an intrinsic nature the account does not exhaust. Matheism rejects that further claim through structural exhaustion: every constitutive difference must belong to the complete structure. An alleged extra substance faces two outcomes. If it contributes a further determinate feature, that feature extends the structure. If it contributes none, it plays no part in constituting the thing. Interaction makes the shared account unavoidable; the identity premise makes it sufficient.

3. Monism, Not Platonism

3.1 Plato's Second Realm

The objection that Matheism is sophisticated Platonism asks whether it posits two realms, abstract mathematics and concrete physical reality, with a mysterious bridge between them (cf. Benacerraf, 1973). It posits one. What we call physical is structure experienced from inside one part of it. What we call abstract mathematics is the same structure considered apart from any particular place in it. The difference is one of perspective, not of kind.

Plato's own view has lasted through the ages, and so has belief in the supernatural; lasting makes neither true. The Pythagoreans held that things are numbers; Plato, Aristotle reports, held "that the Numbers exist apart from sensible things" (*Metaphysics* I.6, 987b27-29). He made mathematics the ladder away from the world: arithmetic, geometry, astronomy and harmony were to turn the soul from what it sees toward Forms no eye can see (*Republic* VII, 521c-531c). Matheism takes the Pythagorean side. Plato's Forms are not part of nature and do nothing in it. Asked what the Forms contribute to sensible things, Aristotle answered

that “they cause neither movement nor any change in them” (*Metaphysics* I.9, 991a8-11). To say that things share in them, he added, “is to use empty words and poetical metaphors” (991a20-22). Plato needed a divine craftsman, the demiurge, to copy the Forms into the world (*Timaeus* 28a-29b).

A realm outside nature that does nothing in nature is supernatural in the plain sense of the word, and a view with two such realms is dualism. Plato’s step also works against explanation itself, since what does nothing in the world explains nothing in it: it is a late Athenian anti-philosophy. Athens’ early thinkers sought reason inside nature; Plato set the Forms outside it. Then something lies beyond reason. If it is the Forms, nothing here can know them. If it is the world, it is no Form and so no logic, a mystery box of chance and magic. If it is the link, no account can say how the two relate. In the end both halves turn supernatural: the Forms, because they stand outside nature, and nature, because without the Forms it follows no logic and cannot be known. Each option says that reality cannot be known, which is faith posing as reason, and an a priori proof shows it wrong: outside logic is illogical, as impossible as a square circle. A view that places reality beyond reason has exited philosophy: it refuses knowledge instead of loving it, and it cannot state itself without

the logic it denies. An explanation cannot be supernatural in any case, since to explain is to connect, and a connection is structure.

Plato saw that the Forms could not be known and wrote it down. In his *Parmenides*, the greatest difficulty for separate Forms is that we could not know them, and a god who has knowledge itself could not know our world (*Parmenides* 133b-134e). He kept the Forms anyway, since without them thought would have nothing to fix on (135b-c).

In Plato's own frame the split cannot stand. If the link between Forms and world could be understood exactly, it would follow a rule, and in Plato's terms what can be known is Form, so the link would be one more Form. A craftsman copying by such a rule would be applying Forms to Forms, and what he made would be Forms as well: one realm, which is Matheism. Plato did not go that far, though his *Timaeus* comes close: the craftsman builds fire, air, water and earth out of triangles (53c-56c). Where the geometry stops, Plato leaves a wandering cause that reason can only persuade (47e-48a) and a receptacle grasped by "a kind of bastard reasoning" (52b), a nature that follows no logic. Either the link is Form, and then the world is Form too, or it lies beyond reason, which is where Plato left it.

By Aristotle’s account, Plato set the Forms apart to save knowledge from the flux. In his youth he learned from Cratylus, a follower of Heraclitus, that what is ever changing cannot be known, and he placed the objects of knowledge elsewhere (*Metaphysics* I.6, 987a32-b10). Heraclitus had already given the answer. His river is the same river while different waters flow through it (fragment B12): the pattern stays while the water moves. All things come to pass in accordance with the logos, the rule of the flow (fragment B1; Kahn, 1979). Plato took the flux without its logos and set the rule outside the world it rules.

Alfred North Whitehead called the European tradition a series of footnotes to Plato (Whitehead, 1978, p. 39), and whoever opposes Plato’s dualism is said to remain in a dialectic with him. The opposition is a rejection of the rejection of pure reason. A case for pure reason reads as a reply to Plato only because he is its most influential early opponent. The divide is pure reason against supernatural nonsense, nonsense in the strict sense, since what lies outside logic cannot even be said. Descartes’s two substances need a bridge no account can state, and Kant’s thing in itself is reality declared unknowable (Kant, 1998, Bxxvi). Wherever the split appears, it puts reality beyond reason.

Contemporary Platonism keeps the split. An object counts as abstract “just in case it is non-spatiotemporal and (therefore) causally inefficacious” (Linnebo, 2023). Paul Benacerraf (1973) asked how creatures in space and time could know such objects. Section 2.6 asked how separate realms could act on each other, and Benacerraf asks the same of knowledge. Matheism has no second realm and so needs no bridge.

We know mathematical truths because we are mathematical structures, not because we have mysterious access to a separate domain. The question “where are mathematical structures located?” is a category error, treating one kind of thing as if it were another. Structures do not occupy places within a container; space and time are themselves structural relations. We exist in mathematical structure the way a triangle exists in geometry: as an instance of it, not as an object placed inside it.

3.2 A One Without Properties Is Nothing

Plato’s highest principle fares worse than the Forms. The Good, he wrote, lies beyond being, surpassing it in rank and power (*Republic* 509b). The One of his *Parmenides*, taken strictly as one, has no parts, no shape, no place and no time. Plato drew the consequence himself: such a One in no way is, and nothing

can be said, known or perceived of it (*Parmenides* 137c-142a). A point so pure that it has no size or properties is nothing at all. It contains nothing, relates to nothing and affects nothing, so Plato's monism is fake: a dualism with an empty One on top. Section 2.1 reached the same verdict on any featureless bearer.

Plotinus, the third-century founder of Neoplatonism, made such a One the source of everything. It is neither a thing nor a quality nor a quantity, neither intellect nor soul (*Enneads* VI.9.3). From it emanates Intellect, from Intellect Soul, and from Soul the material world, each level flowing from the one above (V.1-2). The levels multiply the problem they were meant to solve: every level is more dualism, with the interaction problem again at each step down, and together they solve only the problems that dualism itself creates. Each must relate to the One, and a relation gives each of its terms a property, which the One cannot have. Calling the source One does not say how the levels connect. If they produce and act on one another, their relations are already structure, and the One above them adds nothing.

A Neoplatonist will answer that the One lies beyond being, not below it, as the power of all things (*dynamis pantōn*, III.8.10). A power is a relation to what it can produce, so the answer gives the One a property after all, and a One with that property is

structure. Plotinus describes the procession itself in images. The One, seeking nothing and lacking nothing, overflows, and he adds *hoion*, ‘as it were’ (V.2.1); it gives off the lower levels as the sun gives off light and fire gives off heat (V.1.6). His own ‘as it were’ concedes the point: images are not relations. Aristotle’s verdict on Plato’s sharing applies to emanation as well, empty words and poetical metaphors.

The layers are unnecessary. Mathematical monism keeps one level, the structure, whose parts relate by the relations that make them what they are. Plotinus’s One has no properties and so is nothing. The one that Matheism keeps has every property there is, since it is all the structure there is.

4. Minds: The Hard Problem Assumes Its Conclusion

The strongest test of the identity thesis concerns minds. A rival account must say what its extra constituent contributes to experience or action. The challenge applies to substance dualism, mind as a separate nonphysical thing; to property dualism, physical things with nonphysical mental properties too; to panpsychism; to idealism; and to any intermediary between mind and matter.

4.1 The Fallacy

David Chalmers (1996) argues that even complete physical knowledge would leave unexplained why there is something it is like to be a conscious system, why structure comes with experience. This hard problem of consciousness motivates property dualism: something beyond structure must make up phenomenal consciousness, the felt side of experience.

As posed, the hard problem of consciousness is a fallacy that assumes its conclusion. Its premise is that experience is something other than structure. Granted that premise, no structural explanation can reach experience, and the problem looks hard. The premise is the very claim in dispute. A critic who stipulates that experience is nonstructural has disqualified every structural explanation in advance. The identity thesis rejects the stipulation, and asking whether structure and experience are identical does not assume that they are not. Seeing the circle takes no experiment, only the argument’s own form, so the diagnosis is an a priori proof in the logician’s sense.

The problem also expects the wrong kind of access. A structure is felt by being it, from inside. The hard problem of consciousness expects to feel it from outside, by reading a description, and that

is a category error about what kind of thing experience is. Any explanation, a dualist's included, would itself be more structure, so a gap drawn between structure and experience is closed to every theory alike. A problem that no possible answer could solve marks a stipulation, not a discovery.

The explanatory gap Joseph Levine (1983) identifies, that we cannot see why certain structures constitute experience, may reflect the structure of explanation rather than a gap in reality. In a dilute gas, temperature turned out to be the mean energy of motion of the molecules, measured on a different scale. Vitalism posited a nonphysical life force, *élan vital*; life turned out to be complex biochemistry. Water turned out to be H_2O. In each case the gap closed without a missing ingredient, when two descriptions turned out to pick out one thing.

The felt quality is what a conscious structure *is* from the inside: the character of being that particular pattern, not an additional property requiring separate explanation. This is an identity claim, not eliminativism, which denies that conscious experience is real. Being mathematical structure does not diminish the reality of experience; experience *is* the structure, known from within. Felt qualities are what mathematics feels like from inside.

4.2 Qualia, Property Dualism and Interaction

Qualia, taken as properties over and above structure, are a label for the gap, not a bridge across it. Property dualism adds them to physical properties, and the question is whether they affect what happens. If they have no physical effects, they cannot cause the report “I see red”; the physical process produces the report without them. If they do affect physical events, some law must relate the phenomenal and physical changes, and that law is structure (Section 2.6).

The retreat to epiphenomenalism, which treats consciousness as a causally inert byproduct, gives up the intuition that drove dualism, that our experiences cause our reports of them. It also leaves our reports unexplained: if the feeling does nothing, our words about it track something else. Selection cannot favor an inert property for effects it does not have.

Software looks categorically distinct from hardware, yet it is constituted by hardware states; the difference lies in the description. The gap between a neural description and an experience is of the same kind. The experience is the organization itself.

4.3 Panpsychism, Russellian Monism and the Swap Test

Panpsychism cannot explain how tiny feelers add up to one mind. A combination principle would have to say which parts combine, under what conditions and with what resulting experience, and those conditions are structure (Chalmers, 2016).

Its most careful form, Russellian monism, defended by Galen Strawson and Philip Goff (Strawson, 2006; Goff, 2019), holds that physical science describes only relational structure, while the intrinsic nature underneath consists of experience. Bertrand Russell (1927) observed that physics describes only relational structure, leaving open what instantiates those relations. On mathematical monism this gap closes. Structure has no separate nature standing behind it, because structure is what there is. Strawson and Goff take consciousness seriously, and their error lies elsewhere, in positing something beyond structure to ground it.

A test case makes the dilemma concrete. Take the Russellian monist's own picture: mass is a role, whatever resists acceleration and attracts other masses, and the nature that fills the role is experiential. Now swap the natures behind mass and

charge while every role stays the same. Every particle moves as before, every brain fires as before, and everyone says and believes exactly what they did before, including “my experience of red is like this.”

If the swap changes anything, in what happens or in what anyone notices, thinks or reports, the nature does work in the organization and belongs to its structure. If it changes nothing, the people in both worlds hold the same beliefs about their experience, so those beliefs cannot be tracking the nature that differs (Lewis, 2009). A view that leaves everyone, its defenders included, unable to know the natures it posits loses its own motive, since knowing our experience is what led it to posit them. Lewis (2009) drew humility from the swap, accepting that we cannot know which nature fills which role, and kept the natures. The identity view takes the further step of Section 2.1. A determinate difference needs a distinguishing feature, and nothing marks which nature fills which role. So the two swapped worlds are one world, and the structure is all there is to the thing.

4.4 Idealism

Idealism has one kind of thing and so escapes the interaction problem between two substances. It still owes an account of order. Separate minds discover the same mathematics, the same primes and the same laws of motion, and an idealist must say what makes them agree without appealing to the mathematics they agree on. Appealing to one universal mind moves the question: the regularities of that mind are then the laws, and laws are structure. Idealism needs mathematics to explain its own order, and mathematics needs no mind to hold. Its mind-stuff is mathematical structure described from inside.

4.5 Mary, Zombies and the Chinese Room

Frank Jackson (1982) describes Mary, who knows all physical facts about color vision through theoretical models but has never seen color, and argues that on first seeing red she learns a fact her physical knowledge lacked. Being a structure and modeling one are two different structures. Knowledge by description models a pattern in one part of the brain; knowledge by acquaintance instantiates the pattern throughout its sensory systems, and the second is the experience itself. What Mary lacked was that second structure. Acquiring it is a new physical

state of her brain, and it needs no new nonphysical property; that is where the identity view and Jackson part ways.

Chalmers (1996) asks whether a physical duplicate could lack experience. If complete physical organization constitutes consciousness, as Section 2.2 argues, the zombie keeps the constitution and lacks what it constitutes, so it is conceivable only the way an unchecked arithmetic error is. A zombie can be imagined; it could not be built.

John Searle (1980) describes an operator manipulating Chinese symbols without understanding them. The systems reply locates understanding, if present, in the organized whole: the room, the rules and the process together. A component’s ignorance does not establish the whole system’s ignorance. Searle’s point stands that producing the right answers need not be enough for understanding. Mathematical monism concerns the complete organization, including the causal relations through which a system learns, acts and relates to its environment, not any program that imitates its output.

5. Objections and Clarifications

5.1 ‘The Core Premise Is Circular’

Objection: You define ‘existence’ via ‘intelligibility,’ which presupposes minds. Is this not anthropocentric?

Response: The objection conflates two senses of ‘intelligibility.’ *Intelligibility-as-structure:* having determinate properties standing in definite relations, objective and observer-independent. *Intelligibility-as-knowability:* being graspable by minds, observer-dependent. The claim is not that reality is mathematical because we understand it. It is that anything with determinate properties constitutes structure, and every structure, ruled or random, is one that mathematics describes. Observers require intelligibility-as-structure (Section 2.4), but intelligibility-as-structure does not require observers. The cosmos would remain mathematical even if no consciousness existed.

A deeper form of the charge is that any defense of logic uses logic, so the case begs the question. Hans Albert’s trilemma holds that every attempt to ground knowledge ends in a circle, an endless regress or a dogmatic stop (Albert, 1985). Aristotle granted the first point: the principle of non-contradiction cannot

be demonstrated, since not everything can be. He showed it by refutation instead: the denier need only say something, and in saying it he has used the principle (Aristotle, 1908, *Metaphysics* IV.4, 1006a5-28). The argument here rests on premises of that kind. It does not assume that reality is mathematical; it starts from what any claim about reality must have, determinate content, and a denial is such a claim.

5.2 ‘This Is Just a Tautology’

Objection: The claim that intelligible reality must be structured is true by definition and uninformative.

Response: The conclusion is necessary, and it is far from empty. It excludes positions with serious defenders (Section 1), and it reaches them by testing each proposed alternative in turn: substrate, haecceity and non-relational intrinsic properties. A proposed addition must explain what distinguishes it and how it affects anything. If those answers supply further properties and relations, they extend the structure. If they supply none, the addition has no explanatory role. Calling the conclusion necessary does not make it true by definition; Section 2.2 does the work of showing why every definite nature belongs to structure.

Falsifiability, Karl Popper's demand for an observation that could tell against a claim, is a test for contingent empirical hypotheses, claims about the world that could have been false (Popper, 1959). A necessary claim is tested by counterexample instead: one coherent feature that is neither ruled nor random would refute this one. Matheism's empirical side is its measure, a weighting of structures that favors short descriptions (Section 6.2), which predicts short, stable laws and a short rule behind the constants of physics, and a failure there would leave the ontology standing.

5.3 The Gödel Objection

Objection: Does incompleteness, Gödel's result that consistent formal systems capable of arithmetic leave truths unprovable, show that reality or thought must exceed mathematics?

Response: No. Gödel's theorems are themselves mathematical results. The unprovable truths are still mathematical truths, just not provable within that particular formal system. A stronger system of axioms proves them, with new unprovable truths of its own: mathematics is bigger than any one system. For sufficiently expressive, consistent, effectively axiomatized theories, whose axioms a program can list, incompleteness limits what one theory

can prove. It does not show that anything escapes mathematics. Undecidability, the existence of statements a system can neither prove nor refute, is inherent to mathematical structure itself, not a limitation that anything escapes by being nonstructural.

5.4 Different Logics

Different logical systems yield different mathematics: classical versus intuitionistic logic (logic without the law of excluded middle), the standard axioms of set theory (ZFC) versus alternative set theories, paraconsistent logics, in which a contradiction need not imply everything. This does not undermine the thesis. These logical and axiomatic systems are themselves objects of mathematical study. An axiom system can have many models, structures in which its axioms all hold, and different presentations can describe the same structure. Changing axioms therefore does not automatically identify a different physical universe, and our use of classical logic or ZFC does not by itself identify the physical structure we inhabit.

5.5 Is Logic the Basis of Science?

Objection: W. V. O. Quine argued that no statement is immune to revision, the laws of logic included, and Hilary Putnam proposed that quantum mechanics shows the distributive law of classical logic failing for physical propositions (Quine, 1951; Putnam, 1968). If experience can revise logic, logic is no foundation.

Response: Science decides by logic. A theory with its conditions implies a prediction, and an observation that contradicts the prediction refutes the theory by modus tollens, the inference from ‘if p then q’ and ‘not q’ to ‘not p’ (Popper, 1959). Quantum mechanics is stated, solved and tested in classical mathematics, and the lattice of quantum propositions is one more structure inside it (Section 5.4). Quine himself declined to revise classical logic, citing the loss of simplicity and the burden of reasoning within a deviant logic (Quine, 1970). An argument for revising logic must be valid to persuade, so it relies on the logic it would revise. Mathematics, philosophy and science differ in their premises, definitions in the first two and observations added in the third, and they share one logic.

6. What Follows

The identity thesis rests on Sections 2 to 5. The consequences below rest on further premises, stated with each, and they need reasoning alone, with no physics beyond what everyone grants. Each stands on the fact before it, not on that fact's explanation: even if the weighting below is not why our laws are simple, they are simple, and what follows from simplicity still follows.

6.1 Everything Consistent Exists, Fictions Included

Mathematical existence is not a lesser form of existence waiting for an extra act of physical realization. If a complete structure needs no added substrate to constitute it, the identity applies to every mathematical structure. The further premise is that no primitive actuality property selects only some structures as real. A proposed selector has two options. If it describes relations within a larger structure, the structures it does not select keep their mathematical existence. If it does not, it adds an actuality distinction with nothing to explain it.

What a mind can experience is narrower than what exists. Experiencing is processing, and processing takes steps, so a mind can experience only what takes part in the same steps as it

does. That is what physical means: every consistent structure exists, and a mind's physics, transitions in mathematics, is what its own steps reach.

Fictions are no exception. A story is a structure: its characters, events and rules stand in definite relations, in an epic, in science fiction or in a manga. Any story one tells is already a piece of logic, and a consistent story exists as that structure. A plot hole is a contradiction, and a contradiction exists only as a description: one can write "2 + 2 = 5" or "a square circle," and no structure satisfies them. The hole is still fixed somewhere. The versions of the story in which some further fact, bolted on, explains the hole are consistent, so they exist: *Deus ex Mathematica*. Which fictions contain a feeling mind is the question asked of brains, and it has the same answer: those whose structure does what a feeling mind does.

6.2 Why Laws Are Simple

Each bit of a shortest description is random in the sense of Section 2.3: nothing in the description favors 0 over 1, so each bit is a 50/50 choice, and each added bit halves the description's weight. Whether to add another bit at all is a second 50/50, so longer descriptions lose in total as well as one by one. Simple

laws dominate. The observed fact stands whatever explains it: our laws fit on a page, and the early universe was smooth. A miracle remains possible on this weighting, as a bolt-on that halves the weight of the world containing it for every bit it adds. Eugene Wigner’s “unreasonable effectiveness of mathematics in the natural sciences” is then no puzzle: the world is mathematics, and simple mathematics at that (Wigner, 1960).

6.3 Why Minds Grow in Steps and Time Runs Forward

Minds like ours are complex, so a history that starts simple must grow them, as the Game of Life grows moving patterns from a few cells under four rules. Growth takes causal steps. A cause produces its effect as the next state, and a continuum has no next state: between any two moments lies another. This is an a priori proof in the logician’s sense. Bertrand Russell answered that motion is only being at different places at different times (Russell, 1903), but a place for every moment, with nothing that brings about the next, is the emperor’s new points. Change without a cause would lie outside logic and reason. On this view continua belong to mathematics as tools of analysis, and whatever happens, happens in steps.

Time is those steps, so it runs forward. An endless history has no last step from which to run back, and a finite reversal must specify the point it turns back from, a cost the forward history does not pay. A universal code can charge the same length for a finite history and its own reversal, so no penalty is claimed there. Minds run forward too: thinking merges many states into one, and a merged state has many possible pasts, as 2 = x + y has more than one solution.

What happens is a rule applied to a state, giving the next state. The rules are the only things. A structure that only sat there would still exist, as mathematics does, and would do nothing and know nothing, like a dead brain with every connection intact. Stephen Hawking asked what breathes fire into the equations and makes a universe for them to describe (Hawking, 1988). The math rules themselves do. Any further fire would either follow a rule, and so be more of the structure, or follow none, and so be random, which Section 2.3 also counts as structure.

6.4 One Self

On the view taken here, there is one self in every mind, because the bare form of being a subject is the same structure wherever it occurs. Once no separate soul is left, this follows a priori, from what a structure is. Bodies, memories and causal histories distinguish minds. Does each mind also need a primitive owner of experience? If exchanging supposed owners leaves every experience, memory and relation fixed, the exchange makes no structural difference, so Matheism rejects that extra ownership. What remains in each stream is the bare form of being the one who feels, and that form has no feature that differs from one stream to another. A structure is one wherever it occurs, as the number two is one number however many pairs there are, so the self is one while the streams it runs through stay many. This keeps streams of experience distinct; it does not make two brains one object or give them shared memories.

6.5 Agency

An agent can have a determinate decision policy even if it assigns probabilities to several actions. Such a policy is structure, and chance does not remove the agent’s character, reasons or causal role. When an agent deliberates, it is doing causal work,

not interrupting nature. A free choice that is neither caused nor random is the one option the dichotomy excludes.

7. Conclusion

Reality is mathematical structure, because whatever exists has definite features, every definite feature is ruled or random, and both are mathematical. A further substance either contributes something determinate, which adds structure, or contributes nothing, which explains nothing. Swapping two intrinsic natures while every relation stays fixed changes nothing anyone could notice, think or report, so a nature of that kind could not even be known by the mind it is said to make up. The complete structure is the thing itself.

This is monism. Platonism split the Forms from a world they do nothing in, and Plotinus stacked emanations under a One that has no properties and so is nothing. Matheism keeps one level, and the one it keeps has every property there is.

The hard problem of consciousness assumes its conclusion. Anyone who defines experience so that no structural answer could count has assumed the very split under dispute. The open question for the philosophy of mind is which structures constitute

experience.

The claim is necessary, as 2 + 2 = 4 is, and it has teeth: one coherent feature neither ruled nor random would refute it. Its consequences reach far. Every consistent structure exists, fictions included. Simple laws carry the weight, since each added bit halves a description’s weight. Minds grow in causal steps from a simple start. Time is those steps, and it runs forward, since an endless history has no last step to run back from. The bare subject of experience is one self in every mind, with bodies, memories and histories still distinct.

Any search for an ultimate explanation ends at mathematical structure, because structure is the only medium in which an explanation can be given. Asking ‘why structure?’ is like asking ‘why are explanations explanatory?’: the question presumes what it questions. The intelligibility of reality and the reality of intelligibility turn out to be the same thing, grasped from different angles.

A4. Why Everything Exists as Math: Simple Laws, Forward Discrete Time, Many-Worlds and Superdeterminism

Abstract

Every consistent structure exists as mathematics, since that takes nothing beyond consistency; whatever let one exist and not another would itself be structure, so the point is a priori. Physics is transitions in mathematics, and we observe only the transitions we take part in, so the question is which structures we should expect to observe. The laws of physics are short because every bit of a complete description is a yes-or-no that no rule decides, so each added bit halves the description's probability. The universe therefore starts simple, laws and starting state together; if that reason fails, observation still shows the simple start, and the rest of the argument needs only the fact. Minds like ours are complex, so complexity has to grow from that simple start, as it does under simple rules from the Game of Life to physics, where classical paths emerge from the sum over all paths. Growth takes causal steps. On the view taken here they are discrete, since a continuum has no next step for a cause to produce, though some

hold that continuous dynamics can cause; either way, time runs forward from simple to complex, and an endless history has no last moment to run back from.

The measure behind this is exact. A shortest description has no pattern left to compress, so its bits are random in the sense of the rule-random dichotomy: nothing favors 0 over 1, and no bit depends on another, since a preference or a dependence would itself be a rule that has to be written down. Longer descriptions are more numerous and still lose in total: whether to add another bit is a second unruled yes-or-no, so all descriptions one bit longer weigh half as much together. Added laws more often break the conditions observers need than create them, so at least half the weight sits at the shortest description that can produce observers. A complete account counts the laws, the starting conditions and the rules connecting states to observations, so hiding information in the starting state saves nothing. Applied to physics, the measure favors action principles, quantum mechanics without collapse, a shared cause for particles and measurement settings, and histories grown from a first state.

1. Introduction

The argument runs in nine steps.

1. Everything that exists is ruled or random, and both are mathematical; anything outside mathematics would be outside logic, and nothing could be reasoned about it.
2. The universe starts simple, because each bit of its description halves its probability.
3. It grows complex, because minds like ours are complex and the probable universes grow them instead of writing them in.
4. It grows in causal steps: a cause, then its effect.
5. On the view taken here the steps are discrete, since a continuum has no next step for a cause to produce, an a priori proof in the logician’s sense; others hold that continuous dynamics can cause.
6. On the same view, change without a cause is outside logic and reason; indeterministic readings of quantum theory disagree.
7. Time is the causal steps, each a rule applied to rules, and it runs forward, from simple to complex; space is how far the steps spread, at the base rate of the speed of light (Section 4.7). Light, which crosses space while no time passes for it, is

the limit of a moving clock running slow, not an exception.

8. An endless chain of steps has no last step to run back from.

9. A finite reversal must specify the step it turns back from, and each bit of that address halves its weight, so we find ourselves in the forward history, not a reversed one (Section 6.4.3).

Each step uses the facts before it, not their reasons, so a reason can fail without taking the rest of the chain with it. The laws are simple and the start was simple, as observation shows: a reader who rejects the measure in step 2 still faces every later step except the price in step 9, and one who allows causes in a continuum, or change without a cause, still faces a simple start that has to grow complex minds, toward a future with no last step.

One rule can explain many facts without listing each separately, and that economy is what this paper measures. Universal quantum evolution needs no separate collapse law, a rule that picks one outcome at measurement, and a common generator could explain correlations between particles and measurement settings. Each proposal must recover the observations after its laws, starting conditions and probability rules are counted, since a short equation with unexplained inputs may be a long explanation in disguise. This paper carries out that comparison across quantum theory, geometry and time.

Every consistent structure exists as mathematics, fictions included, since that takes nothing beyond consistency, and every plot hole is fixed by some consistent math. Mathematical existence already supplies the complete structure, and structural identity leaves no extra substance to add. Matheism rejects a primitive actuality property (a basic, unexplained mark of existing) that would select some existing structures and exclude others: any such property would either enter determinate relations with the structure, and so belong to it, or make no difference to anything (compare the modal realism of David Lewis, who held every possible world equally real; Lewis, 1986). What a mind can experience is narrower: experiencing is processing, and processing takes steps, so a mind experiences only what takes part in the same steps as it does. That part is a mind's physics, transitions in mathematics, much as Lewis read "actual" as the world one is part of.

Once every structure exists, the question is which ones we should expect to observe, and the answer is a weight that halves with every bit of description. The weight comes from the rule-random dichotomy itself: the bits of a shortest description are random, and an unruled bit is 0 or 1 with equal chance (Section 3.3). A structure whose shortest complete description has K bits

therefore carries weight proportional to 2^(-K), and observations then update those weights. Whether to add another bit is a second 50/50, so longer descriptions lose in total as well as one by one.

Matheism, or mathematical monism (the view that reality is mathematical structure alone), extends the structural approach of James Ladyman and Don Ross (2007) and Steven French (2014). It grounds existence in mathematical necessity: structural truths need no external maker. The physical comparisons below state which of its commitments they use.

Section 3 derives the measure, Section 4 examines description savings, Section 5 compares quantum interpretations, and Section 6 answers objections. In the Bell comparison, which concerns correlations between distant measurements, responses stay local and outcomes definite, so reproducing the observed violations requires dependence between the hidden state and the measurement settings.

2. Foundation

Matheism identifies reality with mathematics, on three arguments. The rule-random argument: everything is ruled or random, and both regularity and chance are mathematical, since probabilities and determinate outcomes stand in mathematical relations too. The interaction argument: anything that affects a physical system must enter determinate relations with it. The identity argument: a complete account of properties, relations and causal powers exhausts what a thing is, so an additional substance either adds a further determinate feature, which extends the structure, or leaves the explanation unchanged. The rule-random argument is an a priori proof in the logician’s sense: a pattern either has a shorter rule or it does not, both kinds are mathematics, and only a coherent feature that is neither could refute it.

Three commitments are kept apart below, because they do different work: the identity thesis, the claim that every consistent structure exists, and the measure that weights observations.

The identity thesis meets a literature of objections, and each has a short answer. Structural realism (the view that what science reliably captures is structure) and its critics (Russell, 1927; Saunders, 2006; Esfeld and Lam, 2008) ask whether science

reaches only structure; on the identity thesis there is nothing else to reach. Newman's objection is that a bare claim of structure is met by any domain of the right size (Newman, 1928). It targets the claim that some structure or other exists; the identity thesis concerns the actual relations, causal ones included, which observers pick out through their own relations to the rest. Primitive identity (identity not fixed by a thing's qualities; Black, 1952; Adams, 1979) fails in Black's own example. Each of two identical spheres stands at a distance from the other and at none from itself, so a relation shows that there are two. A primitive thisness that no feature marks is a label where an explanation should go.

Ontic vagueness (vagueness in the world itself; E. Barnes, 2010) has a mathematical account too: a theory can say which facts are indeterminate without giving each vague property a sharp boundary. Knowledge of abstract objects (Benacerraf, 1973) leaves no gap, since knowers are parts of the structure they know and stand in relations to the rest of it. Nominalist physics (physics stated without numbers or other abstract objects; Melia, 1998) replaces numbers with relations of the same form, which are structure again. Fundamentality (Sider, 2011) finds nothing beneath the structure, since an extra substance either adds a

determinate feature, which extends the structure, or adds nothing.

Observers belong inside the same account, so a history that explains our evidence must also produce the observers who hold it. It must recover their organization, their records and their physical interactions. It must also keep structure apart from how structure is represented: coordinate systems can describe the same geometry, a complex variable can encode two real components without creating two substances, and continuum measure assigns length to intervals without adding the zero lengths of individual points. These distinctions constrain the geometric and quantum proposals below.

Minds belong to the same structure. The hard problem of consciousness assumes its conclusion: it takes feeling to be something other than structure and then finds no structure that explains it. The flaw is in the argument’s own logic, so exposing it is a priori too. Feeling is a mind’s states producing one another as it models its world, so a photograph or a recording of a brain, which produces nothing, feels nothing. Panpsychism cannot say how tiny feelers add up to one mind, and idealism cannot say how separate minds find the same mathematics. The bare form of being a subject, with memories and traits set aside, has no

feature that differs from one mind to another. So there is one self in every mind, as two is one number however many pairs there are. A choice is ruled or random, so a free choice that is neither is the one option the dichotomy excludes.

3. The Measure Problem and the Simplicity Dominance Principle

3.1 The Measure Problem

Every consistent structure exists, so predictions need a weight for each one. The Level IV multiverse, Max Tegmark's term for the ensemble of all consistent mathematical structures (Tegmark, 1998, 2008, 2014), resolves fine-tuning (the way physical constants seem set just right for life) in principle and raises exactly that question: among all consistent structures, which should we expect to observe? Without a measure nothing can be predicted, and some measures make the predictions absurd: under certain measures almost all observers are Boltzmann brains, observers formed by rare random fluctuations, with false memories (Carroll, 2010).

The answer is a weight set by description length, halved by every bit of the shortest complete description. Stated precisely, for

effectively specifiable complete candidates S, those a computer program could generate, the shortest-description prior (the weighting assigned before any evidence) in a fixed universal prefix-free language U is

where K(S) means K_U(S): the length of the shortest description of the complete candidate for the stated task. A prefix-free language is one in which no complete description is the beginning of another, so each description settles where it ends. A predictive candidate includes laws, boundary or preparation information (the particular conditions the laws apply to) and the rule assigning probabilities to observations, and information they share is counted once. Section 3.3 gives the reason for the halving.

The prior sits next to Ray Solomonoff's algorithmic probability, which adds up every description of the same output. Stated precisely, Solomonoff-style algorithmic probability sums 2^(-|p|) over all programs p producing the specified output (Solomonoff, 1964). By the coding theorem, that sum is within a constant factor of 2^(-K) for finite strings of symbols (Levin, 1974; Li and Vitanyi, 2008): the shortest description carries a fixed share of the total, and the many longer descriptions of the same output cannot swamp it. The exact powers of two below refer to the

shortest-description prior.

The weight covers whatever finite descriptions produce, which includes every law a finite program can state. Stated precisely, a noncomputable infinite object has no finite program generating all its digits, yet an object can be noncomputable without being maximally random, and a computable stochastic law, one that sets the chances of outcomes, can give total probability one to individually noncomputable sequences. Restricting the prior to programs therefore does not show that noncomputable structures carry no weight; it shows that every law we can state is covered.

A short rule can fix an object of unlimited detail, which is why short laws can produce a rich world. The Mandelbrot set shows it: one line defines endless structure. The example does not license omitting the starting or boundary information a physical candidate needs, so every comparison below encodes laws and inputs together.

One rule doing several jobs

The strongest economy comes from one construction explaining features that would otherwise be specified separately: the laws, the special starting conditions and the particle content. A common geometric or dynamical rule is charged once, and any

extra choices each result needs are still counted.

Counting in bits shows how large such a saving can be. Written out, suppose a code needs 100 bits for one account and another independent 100 bits for a second. A joint program of 130 bits that recovers both saves 70 bits against the 200-bit concatenation, so its weight is larger by 2^70. These numbers illustrate the comparison; they are not measured lengths of physical theories.

3.2 Three Types of Probability

Three different things travel under the word “probability” here, and keeping them apart prevents confusion.

Quantum probability is expectation inside one universe. If Many-Worlds is correct, all measurement outcomes occur, and the Born rule, quantum theory’s rule for outcome probabilities, governs rational expectation about which branch contains one’s continuation. This is uncertainty about where one is located, not indeterminacy in what exists.

Measure-theoretic weight is the weight of whole structures. Long descriptions outnumber short ones, twice as many at each added bit, so if number decided, we should expect to find ourselves in an extremely long one, a world of endless special cases and

arbitrary detail. We find ourselves under laws that fit on a page. The Simplicity Dominance Principle says why: each added bit halves a description's weight, and whether to add another bit is itself a 50/50, so everything longer than a given length weighs exactly as much as that length alone (Section 3.3).

Anthropic constraint is the filter of observation: only structures with observers are seen. Combined with measure, it predicts that observers find themselves in high-measure structures that support observers.

Quantum probability works within a structure, measure-theoretic weight works over structures, and the anthropic constraint selects the structures that matter for prediction.

3.3 Why Each Bit Halves

Each bit of a shortest description is a yes-or-no that no rule decides, so 0 and 1 are equally likely, and each added bit halves the chance of reaching that description. Stated precisely, a shortest description cannot be compressed: if it could, the compressed version plus an instruction to unpack it would be a shorter description of the original. Its bits are therefore random in exactly the sense of the rule-random dichotomy. With no rule to favor either value, each has probability 1/2; any lean toward 0 or

1 would itself be a rule, and would have to be written into the description, making it longer. A biased coin is mathematical too, but its bias is a number that has to be written into the description; one half is the only chance that needs no such number, because it is the only one that treats 0 and 1 alike. Because every structure exists, the half is also a count and not a guess. Take all strings of some length n. Exactly half begin with 0, and when n is at least k, exactly one in 2^k begins with any given k bits, so the weight of a k-bit description is the share of all strings that begin with it. On a computer it is a share of tapes. Fill a universal machine’s tape with fair coin flips, and every tape that begins with a given k-bit program computes the same history, since the machine reads no further than the program’s end. That is probability as Pascal and Fermat counted it in 1654, a share of equally possible cases (Hacking, 1975), and the count carries independence with it: a quarter of all strings begin with 0 and then 1, the product of the two halves, with no further assumption. A particular description of k bits is then reached with probability 2^(-k), the chance of k fair coin flips coming out one exact way. A 10-bit program carries 2^990 times the weight of a particular 1000-bit program.

Half is also the gentlest rate any weighting can have, because each added bit doubles the number of possible descriptions. Stated precisely, if equally long descriptions are weighted alike, a factor milder than one half per bit, applied to all 2^k descriptions of each length k, would give each length more total weight than the one before, without limit. At exactly one half every length carries the same total, so the total over all lengths still needs the choice to stop or go on, below, to stay finite.

Longer descriptions are more numerous, and they still lose in total, because whether to add another bit at all is also an unruled yes-or-no. Each description can be extended by a 0 or by a 1, and each extension carries half its parent’s weight, so the two together carry as much as the parent. If every string at every length counted as a complete description, each length would carry the full weight again and the total would be infinite. The choice to stop or go on is unruled, so it is 50/50: half the weight stays with the description as it is, and half goes on to longer ones. In symbols, all descriptions of n bits together then weigh 2^(-(n+1)), half the total at n-1 bits, and everything longer than n bits, added up, weighs exactly as much as length n alone: the tail equals the peak. Picture every universe as a fresh run of fair coin flips, one flip deciding whether its laws go on and the next

deciding the added bit: any laws can come up, yet at every step half the runs still going stop, so most runs end with short laws. Counted once each in a catalog, long laws would win by number; counted as runs, they lose, since each has to win every stop-or-go flip along the way.

One particular description, in this picture, loses three quarters of its weight with each added bit: a half for going on and a half for the bit’s value. A code that states the length once, in a short header, loses only about half per bit (Li and Vitanyi, 2008). Ratios between particular descriptions in this paper use that smaller factor, so each such ratio is the lower of the two codes’ values.

Among descriptions that produce observers, at least half the weight sits at the shortest length that can, because added laws more often break the conditions observers need than create them. Stated precisely, under the stop-or-go weights, let M be the shortest length that produces observers. Producing observers means producing structures that record and model their surroundings; the test mentions none of our own laws, so the count does not build in its answer. If no longer length is more often able to produce observers than length M, length M carries at least half of the observer-producing weight. If the share also does not rise from one length to the next, lengths M and M+1

carry at least three quarters, and lengths M through M+2 at least seven eighths; without that second condition a share of one at M, none at M+1 and one again after leaves the first two lengths only two thirds. The condition does real work: under the stop-or-go weights above, if one description in sixteen of length four produced observers and all thirty-two of length five did, length five would carry eight times the observer weight of length four. Fine-tuning studies find that small changes to the constants or laws of our physics often remove stable nuclei, long-lived stars or chemistry (L. A. Barnes, 2012), and some lengths of description cannot produce observers at all. Each such gap moves more weight onto the shortest length that works. For a single output, the coding theorem of Section 3.1 makes the same point: all its longer descriptions together cannot outweigh its shortest one by more than a fixed factor.

Ray Solomonoff introduced these weights for induction, reasoning from past observations to predictions (Solomonoff, 1964), and the Simplicity Dominance Principle applies them to what exists, with universal realization, the claim that every consistent structure exists, as its further premise. Jürgen Schmidhuber developed algorithmic accounts of whole universes (Schmidhuber, 1997, 2000); his Speed Prior also charges for

computation time (Schmidhuber, 2002), a cost distinct from the length-only weighting used here.

A short hypothesis need not permit more possibilities than a long one. A compact rule can be highly restrictive, and a long specification can describe a broad family, so description length and the volume of allowed states cannot be traded for each other without a model relating them.

Changing the description language changes the weights by at most a fixed factor, while differences of many bits still dominate. Stated precisely, universal description languages agree on complexity up to additive constants (Kolmogorov, 1965; Levin, 1974; Chaitin, 1987), so the weights under two languages differ by a bounded factor, and each bit of real difference between two accounts still halves the weight. Comparisons are made within one stated language, because the constants can reorder close finite rankings.

Defining the weight with machines that are themselves mathematical structures is consistent, not circular. Stated precisely, algorithmic probability is defined relative to Turing machines (idealized models of a computer), which are mathematical structures; the formal models of effective

computation are provably equivalent, and the Church-Turing thesis connects them with the informal idea of effective calculation. The 50/50 weight of each unruled bit does not depend on the machine. Only the lengths do, and only up to the constant above.

3.4 Occam’s Razor as Metaphysical Principle

Simpler explanations are more likely to be true because simpler structures carry exponentially more weight, so Occam’s Razor follows from the measure. The physicist Sabine Hossenfelder asked why physics should prefer simplicity at all (Hossenfelder, 2018); The Simplicity Dominance Principle answers that structures with shorter descriptions have exponentially more weight, half as much for each added bit. The criterion is brevity, not beauty. A complete comparison includes starting conditions and observational predictions, so that apparent economy in an equation does not hide information elsewhere.

3.5 Scope

Exact shortest lengths cannot be computed in general, so comparisons use lengths actually achieved. Stated precisely, Kolmogorov complexity is uncomputable, though an explicit generating program gives an upper bound. Comparisons must use a declared language, since the additive constants relating universal machines can change close finite rankings; the language and those constants are inputs to every physical comparison.

4. Consequences and Predictions

4.1 Explaining Fine-Tuning

Our constants allow observers because we can only observe where observers can exist, and among those places the shortest complete descriptions carry the most weight. The explanation combines observer selection, which Steven Weinberg used to bound the cosmological constant, a measure of the energy of empty space (Weinberg, 1987), with description economy: compare complete accounts that support observers and recover their evidence, then ask which needs less independently supplied information. A short law alone need not yield life or our constants;

the gain comes when one construction derives conditions that competing accounts must specify separately.

The bit count decides the odds when both accounts predict the evidence equally well. Stated precisely, for two complete candidates with equal likelihood for the evidence, a k-bit difference gives the prior ratio 2^k. A claim about whole classes of worlds sums their weights and accounts for where observers sit in them (Section 6.4).

The prediction runs forward: the complete laws of our world should lie near the shortest description that can produce observers. The length of today’s theories is an upper bound, not a floor. A shorter complete account of the same observations would be the better theory, and the search for unification is the search for it.

4.2 The Variational Form of Physical Laws

An action principle is one short rule that selects allowed motion from all candidate paths, and that economy is the proposed reason physics takes variational form. A differential equation, one relating quantities to their rates of change, can also be compact, so neither form wins a description comparison by its name. In Richard Feynman’s path integral (Feynman, 1948),

action-dependent complex phases weight the contributions of all paths, and in a suitable semiclassical regime, where quantum effects are small, cancellation away from stationary action (where small shifts of the path leave the action nearly unchanged) yields the classical path. The action is stationary there, not necessarily minimal.

Emmy Noether's theorem ties the continuous symmetries (smooth changes, such as shifts in time, that leave the laws unchanged) of a variational problem to conservation laws under its hypotheses (Noether, 1918). It does not prove that stable structures require an action formulation, and whether every universe with observers must have variational dynamics remains a separate conjecture.

4.3 What a Discrete Account Must Recover

A discrete account earns its place by recovering everything the continuum account delivers. Stated precisely, a proposed discrete account (compare Stephen Wolfram's hypergraph models, networks whose links can join more than two points; Wolfram, 2020) needs an explicit map from its microscopic states to effective observables: a specified metric (the rule giving distances and time intervals) or causal structure, the right

symmetries, quantitative propagation and interaction laws, and observation probabilities. Branch reconvergence by itself does not provide complex amplitudes (numbers whose squared sizes give probabilities), destructive interference (contributions that cancel each other) or the Born rule.

4.4 Testable Predictions

The proposal predicts repeated, matched success: observations once thought independent following from one economical construction. It favors complete accounts that explain the same observations with fewer independent inputs, at the rate of Section 4.1. The posterior, the weighting once evidence is counted, also weighs observational fit, so a less accurate model can lose despite a shorter code. The Standard Model’s gauge group (the symmetry behind its forces), its three generations of matter particles and its twenty or so measured constants are inputs today. The measure predicts that a deeper theory will produce them from fewer independent ingredients, and it ranks rival theories by how much they save; this paper claims no such derivation.

4.5 String Theory and the Complexity Test

String theory is not needed in the first place for its main job, keeping gravity finite at the shortest distances: if causes take discrete steps (Section 4.7), nothing happens at shorter distances, and there is nothing there to keep finite. String theory's vacua (its allowed stable states, each with its own particles and constants) exist under universal realization; the question is how much weight an account of our world built on them receives. Under universal realization, consistent string vacua are among the realized structures (Bousso and Polchinski, 2000; Susskind, 2003; Hawking and Mlodinow, 2010). The test is whether a string-based account receives much less posterior weight than an alternative after explaining the same observations.

Two quantities decide which theory the evidence favors: how long each description is and how well it predicts. In symbols, let T include its laws, preparation measure and observation rule, with prior p(T) proportional to 2^(-L(T)). Then

log_2[P(T_S|D)/P(T_A|D)] = -[L(T_S)-L(T_A)] + log_2[P(D|T_S)/P(D|T_A)].

Counting vacua does not by itself fix how well a landscape model, one built on the vast set of string vacua, predicts the data. The

likelihood averages each vacuum’s fit to the evidence, weighted by how much of the model sits on that vacuum. Written out, for a landscape model with vacua v, P(D|T_S) = Σ_v p(v|T_S) P(D|v,T_S). If N vacua have equal weight and exactly m reproduce an evidence event D with unit likelihood, this is m/N, and enlarging N and m in the same proportion leaves it unchanged.

The likelihood can cancel a large code advantage: a model twenty bits shorter but assigning D only 2^(-20) of its rival’s likelihood has equal posterior weight.

Length and likelihood fold into one cost, so a sweeping dismissal has a price in bits. Stated precisely, define the predictive description cost C(T;D) = L(T) - log_2 P(D|T). Odds below 10^(-100) for a string account against a stated alternative require C(T_S;D) - C(T_A;D) > 100 log_2 10, about 332.2 bits. Michael Douglas and Frederik Denef provide statistical tools for specified vacuum ensembles (Douglas, 2003; Denef and Douglas, 2004); a raw landscape count does not replace a physical selection measure.

4.5.1 Parameters as addresses

A generator can turn separately specified parameters into consequences, and then the parameters cost only an address. Stated precisely, if a shared decoding rule distinguishes M candidate regions, a fixed-width selector needs at most ceil(log_2 M) bits; the complete account must also specify the generator, the candidate family, the precision and any remaining data.

4.6 Emergent Geometry

One rule can generate a whole hierarchy of geometric data, and then the hierarchy costs only the rule. Counting starts with the metric and continues into its derivatives. On a four-dimensional spacetime a symmetric metric has ten independent components, so the metric bundle has fourteen local coordinates, four for position and ten for the metric values, and tracking how the metric changes adds data at every order of derivative. Written out, recording derivatives through order n gives the jet-bundle count 4+10*C(n+4,4).

A fixed computable construction adds only a bounded description cost. In symbols, for an effectively described input geometry D and a fixed computable construction F, K(F(D)|D) <= c_F. *Can*

Geometric Harmonics Explain Particles? develops the coordinate count and a worked gravitational response.

4.7 Complexity Grows in Causal Steps, and Time Runs Forward

The measure makes the universe start simple, and our existence makes it grow complex. Observation shows the same start, whatever explains it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). A brain holds an enormous amount of organized information, and each bit of it is either written into the description, halving the weight, or grown by the laws from a simpler state. The probable universes grow their observers. Many simple rules do that. The Game of Life updates a grid of cells by one rule about how many neighbors are alive; started with five live cells, the R-pentomino, it churns for more than a thousand generations, throwing off gliders, small patterns that travel across the grid, and suitable patterns in it can run any computation (Gardner, 1970; Berlekamp, Conway and Guy, 1982). What grows is organization; the description stays short, since every later state is fixed by the rule, the start and the number of steps.

Physics grows structure the same way, from the classical paths that emerge from the sum over all paths (Section 4.2) to stars, chemistry and life, with entropy rising as each step spreads energy into more arrangements.

Growth takes causal steps, and on the view taken here they are discrete. A cause produces its effect, and producing an effect takes a step from one state to the next. A continuum has no next state: between any two instants lies a third, so no state is the first one a cause produces. That is an a priori proof in the logician’s sense. Others answer that a continuous law can make each state depend on earlier ones without a first effect, as in Russell’s account of motion as being at different places at different times (Russell, 1903). Positions at every instant with nothing producing the next are the emperor’s new points. This paper’s conclusions hold on either reading, since growth from a simple start runs one way whether its steps are discrete or continuous. Continuum mathematics stays a sound tool for describing many steps at once. Laws can be written in it, a perfect sphere or Einstein’s smooth equations, and on this view still act only through definite values, one step at a time. A length can be measured, and its points do not cause it.

On this view, change without a cause is outside logic and reason. Nothing could derive it from anything, explain it or predict it, and an account that allows it undermines physics, science and argument, since none of them could then say why anything happens. Indeterministic readings of quantum theory accept such change; the arguments here for simple laws and forward time do not depend on rejecting them.

Random, in the rule-random dichotomy, means a structure with no shorter rule: a pattern nothing compresses, which is mathematics. It does not mean an event that happens without a cause. A random string of digits is still produced, step by step, by whatever wrote it, and what looks random from inside a history is the part of the whole that an observer cannot hold.

Time is the chain of causal steps, so it runs from the simple start toward complex observers. An endless chain has a first step and no last one, so the whole history has no end from which to run backward, and a finite reversal must pay for the step it turns back from (Section 6.4.3). Minds run forward too. A thought merges many possible states into one, so only the forward step is produced by the mind’s own present state, and memories are built by steps already taken. That is why observers remember the past and not the future.

Everything that happens is such a step. On the view taken here, the rules are the only things. A structure that only sat there would still exist, as mathematics does, and do nothing and know nothing, like a dead brain with every connection intact. What happens is a rule applied to a state, itself structure, giving the next state: rules applied to rules. Time is that application, and space is how far it spreads. A distance is counted in the steps an effect needs, and the speed of light is the base layer's rate, shared by each body between moving and changing in place, so a moving clock runs slow. Light, which crosses space while no time passes for it, is the limit and no exception: it spends every step moving, so nothing within it could move relative to the rest. A theory of everything is then a rule for transitions, a rewriting system of the kind Stephen Wolfram explores on hypergraphs (Wolfram, 2020). Its causal graph, the record of which update depends on which, is a causal order of the kind causal-set theory builds spacetime from (Surya, 2019), so the two discrete programs meet: the target is one short rewriting rule whose causal graph matches a random, four-dimensional causal set at large scales, run as a unitary quantum rule, and the measure orders the search, shortest rule first. The search can also run from the other end, program first. Write the known laws as a

program of discrete steps without a grid, and remove their infinities, as loop quantum cosmology does when it replaces the big bang with a bounce (Ashtekar, Pawlowski and Singh, 2006). Then shorten the program while its outputs still match.

5. Quantum Mechanics and Many-Worlds

5.1 The Wavefunction as Mathematical Structure

On the present view the wavefunction, a system's quantum state, is reality at the quantum level, not a description of something else. Stated precisely, the wavefunction Ψ is a vector in Hilbert space, the space of a system's possible quantum states, evolving unitarily (smoothly and reversibly, keeping total probability at one) under the Schrödinger equation, and a measurement is an interaction between physical systems. Interaction with the environment suppresses interference between stable records, the process called decoherence (Zurek, 2003). In an Everettian account these records belong to branches of the joint state; decoherence alone is not a physical collapse selecting one global outcome.

5.2 Many-Worlds as Natural Interpretation

Hugh Everett's many-worlds interpretation (MWI) takes the Schrödinger equation literally: all branches exist, and the apparent collapse is what each observer-branch sees, one outcome. It fits the present position in three ways.

Parsimony: Everett needs no physical collapse mechanism. Objective collapse models, where collapse is a real physical process, add specified stochastic dynamics, and Bohmian models, where particles have definite positions guided by the wavefunction, add a configuration and a guidance rule. Compared under one coding convention, the shorter complete description carries the prior advantage.

Predictive completeness: A finite random outcome has finite description length, and a stochastic law supplies a likelihood without listing each outcome. An Everettian account must also explain the weights of the records observers encounter.

Universal realization: If every admissible mathematical structure exists, structures with other dynamics exist too, and that premise does not by itself decide which quantum account describes our observations.

A fair comparison charges every account the same way. A stochastic model can have a short law even when a typical outcome sequence is incompressible, and a deterministic model must account for its starting conditions and observed records. Charging random outcomes to collapse models while charging rivals nothing for observational selection would be unfair.

5.3 Bell’s Theorem

Violations of John Bell’s inequality rule out one combination of assumptions and leave several accounts standing. They prove neither fundamental randomness nor hidden variables, unobserved properties underlying the outcomes: the bounds constrain local deterministic and stochastic models under measurement independence, the assumption that settings are uncorrelated with the hidden state, and the other stated experimental assumptions. Bohmian models relax locality, superdeterminist models relax measurement independence, and Everett applies universal unitary evolution without selecting one global outcome. Bell’s theorem does not choose among these responses.

5.4 The Born Rule

The Born rule has to be recovered, and algorithmic probability does not supply it alone. Stated precisely, the Born rule assigns probabilities from squared amplitudes: for a normalized state $|\Psi\rangle$ and a projector P onto an outcome subspace, the probability is $\langle\Psi|P|\Psi\rangle$. David Deutsch (1999) and David Wallace (2012) develop a decision-theoretic route (one based on the theory of rational choice) in Everettian quantum mechanics. A code built from Born weights already uses the probabilities it would explain, so an Everettian comparison must state how it obtains observational weights and stable records.

5.5 Reversible Evolution and Records

Information can leave a system without being destroyed, because unitary evolution keeps it in correlations with the surroundings. Local loss of accessible information therefore needs no collapse. A many-to-one description, giving the same output for different inputs, need not imply irreversible microscopic laws, as Charles Bennett showed for computation (Bennett, 1973), and the direction in which reliable records form depends on the history of the system and its environment.

5.6 Superdeterminism Rehabilitated

Superdeterminism puts the particle source and the devices that choose measurement settings inside one physical generator, so their correlation needs no signal. A model that keeps local responses and definite outcomes reproduces Bell violations only if the hidden source information and the settings are correlated. The two setting devices can still be statistically independent of each other.

Detector disturbance differs from measurement dependence. Bell models already let local outcomes depend on local settings; superdeterminism adds a correlation between the settings and the hidden source data. A shared history motivates a common generator, and the model must still derive the required statistics.

Shared information is counted once. Encoding laws, preparation and prediction rules together reuses details that separate descriptions would repeat. For a complete model M, the likelihood P(D|M) measures how well it accounts for observations D, and random outcome records are data, not extra instructions in the law.

A short rule can also overshoot, so shortness alone does not recover quantum physics. In symbols, a counter can cycle over

balanced binary settings a,b and a shared bit r, with outputs A=r and B=r XOR (a AND b). This gives balanced one-wing marginals (each side's results, taken alone, split evenly) and a value of 4 for CHSH, the standard Bell test statistic, above the quantum bound of about 2.83. Shortness and the absence of observable signaling therefore do not alone pick out quantum physics.

The Deterministic Constructible Universe Principle adds that only histories with deterministic constructive accounts are physically realized. It is a further premise: stochastic transition laws are mathematical too, and a computable distribution can have typical infinite realizations with no finite deterministic generator.

6. Objections and Responses

6.1 'This Is Untestable Metaphysics'

Objection: The Level IV multiverse is empirically inaccessible.

Response: Models motivated by the prior are compared through their predictions, as science compares every theory with unobservable parts. Quarks, the particles inside protons and neutrons, cannot be isolated, the wavefunction cannot be measured directly, and much of the early universe is inferred

indirectly. Quantum computing tests predictions of quantum dynamics, though its success alone does not separate Everett from other accounts predicting the same computation.

6.2 ‘The Measure Is Ad Hoc’

Objection: Algorithmic probability is chosen to deliver the desired conclusions.

Response: The measure comes from the dichotomy, not from its conclusions. A shortest description’s bits are unruled. A bit has two values, and with no rule to favor either, each has chance 1/2; a lean toward one would itself be a rule, written into the description and paid for in bits. In standard terms this is the uniform, maximum-entropy measure on bits, the starting point of Solomonoff’s algorithmic probability. The choice to stop or go on is unruled in the same way (Section 3.3, Why Each Bit Halves). Universal algorithmic constructions also dominate every computable weighting up to a constant factor (Li and Vitanyi, 2008; Hutter, Legg and Vitanyi, 2007), so no computable alternative beats them on any data by more than a factor fixed in advance for that alternative. Parsimony is then tested the way any method is, by its predictions.

6.3 ‘The Measure Problem Is Unsolvable’

Objection: Any measure faces the question “why this measure?”, so the problem has only moved back one level.

Response: The question has an answer: everything is ruled or random, and the bits of a shortest description are the random part, each one 50/50. That replaces “why these laws?” with a fact about randomness that needs no further law. Algorithmic probability also has properties that physical laws lack, dominance over computable measures and invariance under the choice of machine up to constants, and explanation ends there, at the structure of computation and information.

6.4 Anthropic Challenges

Where an observer sits within a history needs its own weight, separate from the weight of the history, as Nick Bostrom stresses (Bostrom, 2002). The Conditional Parsimony Postulate proposes a description-based rule for that location.

Two Levels of Complexity. The same coding rule applies twice: once to which history, and once to where in it an observer sits, and a shorter answer receives more weight in both. Stated precisely, at Level 1 the prior assigns weight proportional to

2^(-K_U(S)) to effectively described candidates S. At Level 2, for a specified class of effectively described locations j within S, the Conditional Parsimony Postulate (CPP) proposes q(j|S) proportional to 2^(-K_U(j|S)). This is a normalized location prior, not identical to the Born rule or a thermal distribution (the spread of states set by temperature).

6.4.1 Boltzmann Brains

Whether fluctuation observers dominate depends on the cosmology, so the question is answered model by model. Some cosmologies permit observers that form as chance fluctuations, Boltzmann brains, and these could dominate an observation measure. Eternal duration alone does not create the problem. A constant positive formation rate over unlimited time gives an unbounded expected count, which lets fluctuation observers dominate if ordinary observers are produced only finitely often; a decreasing rate with finite integral (a finite total over all time), a dynamical obstruction or a different observation measure changes that conclusion. Description length gives no independent suppression theorem, since a simple cosmological generator may produce both kinds of observer. The question stays open for a given cosmology until its dynamics, observer

criterion and measure are supplied.

6.4.2 Simulations

A simulated observer is weighed by the same rule as any other, with the simulator’s machinery counted once. If a complete history generates a computer and its program, that machinery is not charged a second time as independent information. Conversely, laws permitting computers do not guarantee that any particular simulation develops, and neither decisive suppression nor dominance of simulated observers has been established.

6.4.3 Temporal Location and the Arrow of Time

Time runs forward because a process that generates each event from the previous one needs a starting state, and an endless future has no last event from which to generate the whole history backward. If every event waited for another and none had occurred, the process could not start.

One event following another in time is one event being generated after another, and two consequences follow. A universal code can charge the same shortest length for a finite history and its reversal, so no universal reversal penalty is claimed, and time’s direction rests on the missing last event. Physical succession generates each state from the previous one, so a continuum of

zero-length points cannot be the fundamental structure, though continuum mathematics itself stays consistent.

A simple beginning is favored because the laws and the starting state are counted together. The simplicity prior favors an economical joint description of both, a description-based reading of the Past Hypothesis, the assumption that the universe began in a low-entropy state (Albert, 2000; Loewer, 2007, 2012; Wallace, 2010, 2011; compare Lazarovici and Reichert, 2020). Simple and low-entropy are different properties: a gas spread evenly through a box is simple to describe in bulk and already at maximum entropy. Gravity lines them up, because under gravity clumping is the direction of rising entropy, so a smooth early universe, the simplest start, is also a low-entropy one (Penrose, 2004). Among accounts with equal likelihood for the evidence, fewer total bits receive greater weight. If a history starts without the organization its later observers require, that organization must develop: organisms build and maintain themselves by using energy differences while dispersing heat and waste, increasing total entropy as local organization grows.

Learning works forward because a reversed world hides the information its convergences depend on. Run a process backward and its parts converge in coordinated ways (Price,

1996) that may depend on details a local learner cannot sense. If the same available evidence hides situations that require incompatible responses, no decision based on that evidence can guarantee success in both, and that limits which reliable learners can evolve. Leonard Mlodinow and Todd Brun (2014) argue that the memory and thermodynamic arrows, the time directions set by memory and by rising entropy, align under explicit robustness conditions.

Reversing a finite history costs whatever the reversal operation adds, one halving per bit, and the operation must specify the step it turns back from, a cost the endless forward history does not pay. A finite reverse history can be specified with the forward generator plus a reversal operation, which avoids listing every coordinated particle. Ten added bits leave it one-1,024th of the weight, and naming the turning step takes about as many bits as writing its number in binary: about 20 for a typical step near one million, which leaves about a millionth. The comparison counts laws, starting information and interpreters in the same code.

6.5 ‘Logic Is Revisable’

Objection: The identity thesis makes logic the ground of everything, yet W. V. O. Quine argued that no statement is immune to revision, and Hilary Putnam proposed that quantum mechanics makes the distributive law fail for physical propositions (Quine, 1951; Putnam, 1968).

Response: Science tests theories by logic: a theory with its conditions implies a prediction, and an observation that contradicts the prediction refutes the theory by modus tollens (Popper, 1959). Quantum mechanics is stated, solved and tested in classical mathematics, and the lattice of its propositions is one more structure inside that mathematics. Quine himself declined to revise classical logic (Quine, 1970). Any argument for revising logic must be valid to persuade, so it uses the logic it would revise. Nor does the case assume its conclusion. Aristotle showed that the first law of logic cannot be demonstrated, only defended by refutation, since whoever denies it and says anything has used it (Aristotle, 1908, *Metaphysics* Ⅳ.4, 1006a5-28).

7. Conclusion

The laws of physics are short because every bit of a complete description is an unruled 50/50 choice, so each bit saved doubles a description’s probability, and the choice to stop or go on makes longer descriptions lose in total as well as one by one. At least half the weight then sits at the shortest description that can produce observers, because added laws more often break the conditions observers need than create them. The hard part is establishing a real saving once laws, starting conditions and prediction rules are counted together.

Quantum mechanics without collapse and a common generator for particles and measurement settings are the leading candidates for such savings. Each must reproduce the observed probabilities, and neither gains a bit advantage merely by being described in fewer words. Observers are complex, so the probable universes start simple and grow them in causal steps, discrete on the view taken here, since a continuum has no next state. Time is those steps. It runs forward from the simple start, an endless chain has no last step to run back from, and a reversal must pay for the step it turns back from.

The unifying claim is that the same mathematical rules do several jobs at once. A successful account exhibits those rules, recovers the observations and shows where the shorter explanation comes from, and the prediction is that the complete laws sit near the shortest description that can produce observers.

A5. Why Spacetime Looks Two-Dimensional at the Smallest Scales: Dimensional Reduction in Quantum Gravity

Abstract

Gravity and quantum theory have to fit one account, because reality is one mathematical structure, and several leading approaches to quantum gravity share a clue about how: at the smallest scales, spacetime behaves as if it had two dimensions. The reduction does not remove a coordinate. It comes from how the theory scales, measured by the spectral dimension, the dimension a diffusing particle would detect, and a worked example shows four coordinates giving spectral dimension two. Fewer dimensions is not automatically simpler. A lower-dimensional theory must carry the map back to the four dimensions we measure, as holography's boundary dictionary does, and the account is shorter only if theory and map together are. A 2026 theorem by Gianluca Calcagni and Fabio Briscese shows, under stated assumptions, that quantum probabilities stay consistent in a class of fractional field theories with this behavior.

Seven approaches are compared by the physics they recover and the ingredients they still need.

1. One ontology, several dynamics

One ontology, a single account of what exists, does not supply one equation. The equations x-dot = x and x-dot = 2x are both mathematical, use the same kind of variable, and have different solutions. Reality is one mathematical structure, so gravity and quantum physics belong to the same account, and the common dynamics still has to be found.

Matheism’s interaction argument asks how any additional substance could affect the physical world without definite relations. Together with the claim that complete structure exhausts physical properties, these arguments motivate one ontology. The metaphysical unification can therefore be argued a priori, by reasoning alone, while a derivation connecting general relativity and quantum theory still matters.

2. What is becoming lower dimensional?

The word dimension does several jobs here, and only some of those counts change at short distance. Steven Carlip's review (Carlip, 2017) separates the notions and collects the evidence for short-distance dimensional reduction across several quantum-gravity programs. Coordinate dimension, spectral dimension (the dimension a randomly wandering particle would detect), the dimension of a graph edge (a link in a network) and the dimension of a string worldsheet (the surface a string sweeps out over time) answer different questions.

2.1 A worked example

Change one scaling exponent and a theory with four coordinates measures as two-dimensional. Stated precisely, take a positive Euclidean operator with momentum dependence |k|^(2z), where z > 0, on d coordinate dimensions. Its heat-kernel trace per unit volume, read as a function of a diffusion scale s, has the form

Rescale k = s^(-1/(2z))q. For s > 0 the radial integral converges and

Spectral dimension reads a dimension off the way that trace scales with s. Written out, it follows that

With d = 4 and z = 2 it equals two although the integration still uses four coordinates. A kinetic operator (the part of the equations governing how a field propagates) that crosses from |k|^2 at low momentum to |k|^4 at high momentum can have different effective spectral dimensions at different diffusion scales. For the fourth-order operator, the off-diagonal heat kernel need not be nonnegative: P(s) is a spectral diagnostic, not automatically a classical return probability (Calcagni et al., 2016). Nor does this calculation establish Lorentzian unitarity (total probability conserved in physical spacetime).

2.2 Approaches to quantum gravity

Approach; Relevant structure; What must be distinguished.

String theory; One-dimensional strings sweeping two-dimensional worldsheets; String and worldsheet dimensions versus spacetime and spectral dimensions.

Loop quantum gravity; Spin networks and geometric operators; Graph edges, area and volume spectra, and recovered spacetime geometry.

Causal dynamical triangulations; A sum over triangulated geometries (Loll, 2020); The lattice regulator and the proposed scaling limit.

Asymptotic safety; Renormalization-group behavior; A fixed-point proposal and an effective dimension inferred from it.

Horava-Lifshitz gravity; Anisotropic scaling of space and time; Scaling exponents versus the number of coordinates.

Causal sets; A locally finite causal order; Discrete order, volume counting and manifold approximation.

Fractional field theory; Fractional kinetic operators; Operator scaling, physical spectrum and amplitude prescription.

3. The recent theorem

Gianluca Calcagni and Fabio Briscese (2026) proved that specified fractional scalar, gauge and gravitational theories (whose equations use non-integer powers) keep unitarity, the rule that the probabilities of all outcomes add up to one, at every order of perturbation theory, the method of successive small corrections. The theorem covers that class of models and leaves the other programs untouched. Its results depend on the kinetic operator, exponent range and amplitude prescription. The hermitian construction uses the fakeon prescription, which removes selected modes from the physical asymptotic states (the particles that enter and leave interactions); the non-hermitian

case has stated ranges compatible with the Feynman prescription, the standard rule in ordinary particle physics. The article is published as JHEP09(2026)206.

This establishes a consistency result within the analyzed class, not for every program in the table. The fakeon prescription gives up microcausality, the ban on faster-than-light influence, at short distances, as its own literature states (Anselmi, 2019). A program can now investigate that class without treating its previously unresolved unitarity issue as automatic grounds for dismissal. Ultraviolet behavior (how a theory acts at very high energies), physical limits and phenomenology (the link to experimental data) must still be answered for the chosen model and prescription.

4. Numbers specify particles; a parameter list is not a particle theory

The Standard Model, the established theory of the known particles and forces except gravity, is more than its numerical parameters. It specifies fields, transformation laws, interactions and a state space. An electron is not a scalar field, one with a single number at each point, and a gauge boson, a force-carrying particle such as the photon, does not become a scalar particle because its properties can be encoded by numbers. The physical

content is the whole specification, and dropping the structures that give the parameters their meaning does not compress it to a parameter list.

5. What can be excluded a priori?

An explicit premise can exclude a candidate from a research program. A contradiction in the candidate's formalism can exclude that formulation. These are different exclusions.

Matheism holds that physical reality is discrete at its base, since physical succession generates each state from the one before and a continuum has no next state, an a priori proof in the logician's sense. A fundamental continuum ontology therefore falls outside its program: Russell's motion, only being at different places at different times (Russell, 1903), is the emperor's new points, a position for every instant with nothing generating the next. A lattice theory (one defined on a grid of points) that takes a continuum scaling limit (letting the grid spacing shrink to zero) may conflict with the premise at its proposed fundamental endpoint while still supplying useful effective calculations. As analysis, a description of many steps at once, the limit stays good mathematics. Whether a discrete alternative can recover the same successful limit becomes a constructive question.

The argument that zero-measure points cannot yield positive length does not itself prove the premise. Countable additivity (lengths of separate pieces adding up, even across an endless numbered list) does not extend to arbitrary uncountable unions (combining more pieces than can be numbered, such as a line's points). Using that requirement as a selection principle is stronger than attributing a contradiction to measure theory, the mathematics of assigning sizes to sets. Rejecting fundamental stochastic (chance-governed) outcomes rules out collapse laws (under which a measured quantum state jumps to one outcome) with those outcomes but does not select superdeterminism (the view that measurement settings correlate with what is measured) by logic alone. John Bell's statistical independence condition (Bell, 1964) concerns the joint distribution of hidden variables (unobserved properties underlying the outcomes) and settings, not whether either is produced by a deterministic algorithm.

Everett's unitary branching (the many-worlds picture, in which every outcome occurs) and a local single-outcome hidden-variable model relax different assumptions in familiar Bell arguments, proofs of limits on what local hidden-variable models can predict. They can be compared without treating their conjunction as forced by the block universe, the view that past,

present and future equally exist. A short rule for digits of pi establishes computability, not normality (every digit string of a given length appearing equally often), cryptographic pseudorandomness (output no practical test can tell from true randomness) or a distribution matching all quantum experiments.

6. Parsimony with complete candidates

Fewer dimensions is not by itself a shorter description. The dimensions we measure carry a definite set of relations, and a theory set in fewer dimensions has to give them back: its description includes the map from its own variables to the four-dimensional world. Holography is the clearest case. In anti-de Sitter space (a model universe with saddle-like curvature), a quantum theory on the boundary, one dimension lower, reproduces the gravity inside (Maldacena, 1998), and the dictionary between the two is part of the account. The lower-dimensional account is simpler only if it and its map together are shorter than the four-dimensional one, and for our own universe no such boundary account is known.

A shorter description earns stated odds once both candidates cover the same observations, because each bit of a description is a 50/50 choice that no rule makes. Fix a prefix description

language U, one in which no code word begins another, and specify candidates to the same observational precision. Stated precisely, under a shortest-description prior (a weighting set before any evidence) proportional to 2^(-K_U), an established difference of Delta bits gives an individual prior ratio of 2^Delta in favor of the shorter candidate.

A huge number of alternatives does not make our own expensive to specify. Counting 10^500 alternatives gives a fixed-length indexing cost of 1,661 bits. It does not prove that the particular alternative occupied by us needs such an index: a short rule could select it. Conversely, a short list of constants is not a complete competing account if interactions, geometry or boundary data have been left outside the charge. Mathematical monism, the view that reality is one mathematical structure, motivates this search without presupposing its winner.

7. Conclusion

A theory can behave as if it has fewer dimensions without losing coordinate directions, because its effective dimension comes from how its kernel scales, not from how many coordinates it uses. Fewer dimensions earns simplicity only when the reduced account and its map back to the measured world are shorter

together. The probability-consistency theorem examined here establishes a further result for its specified class of models. To complete the search, a candidate must derive the relevant particles and interactions, recover the established limits and explain the observations. Mathematical monism gives that search one reality to explain.

A6. Every Mathematical Structure in a Physical Theory Is Real, the Notation Included

Abstract

Every mathematical structure in a physical theory is real, the notation included: equations on a page are states of atoms, and the geometry they describe is structure in its own right. What a theory is about is the structure its predictions depend on, and four questions pick it out: what an element does, which changes preserve that role, what breaks when it is removed, and which observations tell the alternatives apart. Worked cases apply them. Paired real numbers can do the algebraic work of complex notation; a change of coordinates relabels a geometry without altering it; and a phase around a loop changes what electrons do even where the local field vanishes, so the loop phase is physical. One reality has many descriptions, and none of the mathematics in them is unreal.

1. Which structure?

A coordinate grid and the geometry it describes are both mathematical structures, and they answer different questions. Change the numbers assigned to events while preserving all distances and coincidences, and the physical predictions remain the same. Change the geometry itself, and clock comparisons or light propagation may change.

Both structures are real. The coordinate grid, whether printed on a page or stored in a computer, is a physical configuration of atoms or circuits, itself mathematical structure. The spacetime geometry is the structure whose relations determine what clocks and rulers measure. The grid does not become less real by being representational, and the geometry does not become more real by being represented. They are different mathematical structures playing different roles.

On the view that reality is mathematical structure (mathematical monism), a theory must still say which structure its predictions depend on. Fix a world model, meaning a structure together with its relevant relations and whatever history or boundary data (conditions fixed at a system’s edges or starting time) the case needs. Then ask which of its mathematical relations the

predictions depend on, and which choices express those relations in one notation rather than another.

Gauge freedom (the freedom to choose among descriptions giving identical predictions) and the dependence of results on representation are old lessons of physics [1, 5], and the recent debate over whether quantum theory needs complex numbers, which extend ordinary numbers with the square root of minus one, turns on the same question [2, 3, 4]. This paper's contribution is a method that works whether or not one accepts mathematical monism: four questions that separate the structure a theory is about from the structure it is written in. Its decisive premise is that two descriptions describe one world when they agree on every relation the predictions depend on, and physics decides which relations those are. It rules out reading ontology (what a theory says exists) off notation, and it rules out treating what is represented as something more than the structure its representation captures.

2. Closure, consistency and specification

Closure does not mean that removing any statement creates a contradiction. In formal logic, the deductive closure of a theory T contains the statements derivable from T. Deleting an axiom from a consistent classical theory cannot make the remaining axioms inconsistent. A theory with an independent statement P, one it can neither prove nor disprove, admits several possibilities before P is supplied. Adding P narrows those possibilities. Removing P restores alternatives; it does not destroy consistency.

A physical account is incomplete if it omits something its predictions depend on. Here, physical closure means supplying the structure, interactions and boundary information needed for the world under discussion. It is an explanatory requirement, distinct from logical closure. Two descriptions related by an accepted change of representation can still describe the same world.

Equivalent descriptions can be grouped together, but physics must first justify treating them as equivalent. Grouping them records the chosen equivalence; it does not establish that the choice preserves the relevant physical relations.

3. What an action principle supplies

An action principle, a law framed as a condition on a quantity called the action, by itself does not pick out one history. It constrains admissible histories through stationarity under specified variations. Boundary terms, allowed variations, constraints and initial data matter. Consider a free particle, one acted on by no forces, with action S[x] = integral of m(dx/dt)^2/2 over time. Its equation is d^2x/dt^2 = 0. Every trajectory x(t) = a + bt satisfies that equation. The action alone supplies neither a nor b. This blocks the inference from an internal action to a uniquely selected universe.

A mathematical world can include both its dynamics and its boundary data without an outside agent choosing either. A simplicity prior, a weighting that favors simpler descriptions before any evidence, may compare complete descriptions including their boundaries; it does not remove the need to identify what is being compared.

An action is not a prerequisite for mathematical structure or for every possible dynamics. A finite transition system, one with finitely many states, has a precise update rule whether or not anyone has found a useful variational formulation (a restatement

of its rule as an action principle) for it. Where the action has a continuous symmetry (one holding for changes of any size, such as rotations), a conserved quantity can follow. Emmy Noether's first theorem connects suitable continuous variational symmetries with conservation laws under its hypotheses [1]. These results do not make every coordinate convention a physical field or guarantee a conserved global energy in every spacetime.

4. Four questions for identifying the referred structure

The following questions identify which mathematical structures a theory's predictions depend on. They are a method of analysis, not four independently established necessary and sufficient conditions.

Role. What relations or predictions does the element help specify? A boundary condition or fixed parameter can matter even when it is not varied in an action.

Equivalence. Which changes leave the predictions unchanged? Distinguish a relabeling from a change to the relations being labeled.

Removal. Does removing the element change predictions, leave the model underspecified, or permit an equivalent reformulation? An inconsistent reformulation is a fourth possibility; leaving an element out does not automatically produce one.

Discrimination. Do the alternatives differ logically, represent the same physics equivalently, or predict different observations? Different physical predictions invite empirical discrimination; identical predictions do not. They still leave room for an ontological argument, assessed a priori (by reasoning independent of observation) and through explanatory economy.

5. Worked cases

5.1 Complex amplitudes and real representations

Changing notation does not erase the role the notation performed. Complex arithmetic has a straightforward real representation: a + ib corresponds to the pair (a,b), and multiplication by i corresponds to J(a,b) = (-b,a). Applying J twice sends (a,b) to (-a,-b), which is the same as multiplying by -1, so J^2 = -I: two quarter-turns make a half-turn. Replacing complex scalars by paired real components can retain the relevant algebraic structure.

Experiments have tested one real-number formulation, and what they rule out depends on the assumptions built into it.

Marc-Olivier Renou and collaborators [2] compared complex quantum theory with a real Hilbert space analogue (a version whose quantum state space uses only real numbers) in a network with independently prepared sources. Ming-Cheng Chen and collaborators [3] reported experimental correlations incompatible with the tested real formalism. The assumptions about composition (how separate systems are combined into one) and independence belong to the result; the experiments do not establish that every formulation using real numbers must fail.

A 2026 preprint by Timothée Hoffreumon and Mischa Woods [4] explicitly distinguishes product state independence (requiring the sources' joint state to split into separate factors) from operational independence (no observable correlation between the sources) and claims an equivalence for finite protocols under operational independence.

For the method proposed here, the useful question is which transformations, correlations and composition rules an equivalent representation must preserve.

5.2 Coordinates and geometry

Relabeling is the clearest case where the structure the predictions depend on stays the same while the mathematical description changes. A coordinate change replaces a chart, one assignment of coordinates to points, and changes tensor components, the numbers representing geometric quantities in a chart, according to their transformation laws. The geometric object represented need not change. Moving the fields themselves is a different operation from relabeling the points under them. An active diffeomorphism, a smooth reversible remapping of spacetime points, acts on fields, and physical equivalence in general relativity concerns the appropriate transformation of the whole configuration, subject to boundary conditions.

The structure the predictions depend on is the geometry and relations of a spacetime model, not the numerical list g_mu_nu in one chart. Clock comparisons provide one way of probing that structure. Using a smooth manifold, a space that looks flat up close and supports calculus, specifies a mathematical model. Whether a physical continuum is fundamental or emerges from a discrete account remains a separate question.

5.3 Gauge structure and holonomy

Gauge theory shows that the boundary between the structures a theory’s predictions depend on and the mathematical choices expressing them can fall in an unexpected place. In electromagnetism, adding the gradient of any function to the potential (the field from which electric and magnetic fields are derived), A_mu to A_mu + partial_mu lambda, leaves the field strength alone. In appropriate geometries, phase information (a wave’s position within its cycle) around a loop can matter even when the local field strength vanishes along the accessible path, as in Yakir Aharonov and David Bohm’s analysis [5]. For nonabelian gauge theories, those whose transformations give different results in different orders, the field strength transforms covariantly, changing in step with the transformation, rather than remaining componentwise invariant. There the loop quantity itself depends on the reference point chosen, while quantities built from it do not. A holonomy, the transformation accumulated around a loop, generally transforms by conjugation (being sandwiched between a transformation and its inverse), and suitable traces (sums of a matrix’s diagonal entries) or conjugacy classes recover the gauge invariant information.

The method asks what survives an equivalent reformulation of the complete gauge structure.

6. Many structures, one world

Every mathematical structure involved in a physical theory is real, and they play different roles. The atoms forming ink on a page, the circuits storing coordinates in a computer, and the spacetime geometry those coordinates describe are all mathematical structures. They differ in what they contribute rather than in degree of reality: the theory’s predictions depend on some of them and not on others.

A universe that contains pages, computers and brains performing calculations can take no more description than one without them: a short law and a simple starting state generate all of them, the way one rule generates every branch of a quantum history. Extra bits are charged only for what must be specified separately, and each such bit, a 50/50 choice that no rule makes, halves a description’s weight. The representational structures are as real as anything they represent, and a world full of them is as probable as the short laws that produce it.

The further claim that all consistent mathematical structures exist rests on a separate argument, the rejection of a primitive actuality property, a basic unanalyzed feature making some structures actual and others not. Identifying which structure a particular theory’s predictions depend on neither proves nor refutes that premise. It shows how the premise can coexist with the fact that a given world’s physics depends on specific structures, while descriptions of those structures involve additional ones.

7. Limits and objections

Is the method circular? It is conditional on a model and its equivalences. That is appropriate for analyzing a theory already under discussion. Empirical comparison and philosophical argument retain separate work to do.

Does incompleteness, a theory’s inability to settle every statement, prohibit closure? Deductive closure and completeness are different. A theory can contain all its consequences while leaving some sentences undecided. Kurt Gödel’s results [6] concern appropriate consistent, effectively axiomatized theories, whose axioms a mechanical procedure can list. They do not show that every finite physical description is inadequate for every purpose.

Does the method derive physical realization? No. It identifies which structures a given account’s predictions depend on. To derive a particular universe, one would still need admissibility conditions, dynamics, boundaries and a measure (a rule assigning probability weights) for comparing observer locations.

8. Conclusion

Every mathematical structure in a physical theory is real. Some of those structures are what the theory is about, and others are how the theory is written down. The four questions proposed here tell them apart: identify a role, test what preserves it, remove it and compare the consequences. A printed coordinate grid is atoms on a page; a spacetime geometry is what clocks and rulers measure. The theory’s predictions depend on the geometry and not on which chart someone chose for it.

A7. Constructor Theory and Mathematical Reality: Why an Answer Can Hide Its Inputs

Abstract

Reversible physics loses no information when an answer hides its inputs. A sum of 2 does not say which pair was added, yet a machine that keeps both addends beside the sum has thrown nothing away: the many-to-one function has no inverse on its outputs, while the reversible operation carrying it out still holds the input. Every verdict that a task is impossible must therefore cover the whole operation and what it retains, which is how constructor theory frames physical possibility, as the tasks the laws permit a reusable device to perform. Mathematical monism identifies the devices and laws with structure, and the distinction shows how records, copiers and observers can be built under reversible laws. Quantum measurement is the same case at the largest scale: no branch, observer or record holds the information of the whole wave, which stays in correlations, so an outcome that looks random from inside needs no uncaused collapse. Time’s direction comes from production, not from incomplete

records: a history grown step by step from a simple start needs a first state and has no last one to run back from, and a reversal must pay for the step it turns back from.

1. The common question

A machine can hand you an answer that does not determine its own inputs and still lose nothing: the inputs remain in the rest of its state. The map from inputs to the displayed answer can be many-to-one, sending different inputs to the same output, while the physical operation carrying it out stays reversible, able to run backward to recover its input. Constructor theory [1] supplies the setting for that distinction: it asks which tasks physical laws allow a machine to perform reliably. A constructor is a system that can perform a task and retain the ability to do it again. Its microscopic details may change. This paper connects that approach to mathematical monism, the view that reality is mathematical structure, through simple examples of copying, recording and reconstructing information.

David Deutsch and Chiara Marletto apply constructor theory to information: what a system can copy, store or distinguish follows from what the laws permit, rather than standing as an independent mathematical primitive, a basic notion defined from

nothing else [2]. A monist interpretation can identify those physical relations with mathematical structure. It should not misattribute that identification to constructor theory itself.

The two approaches meet at the task specification. A proposed transformation has to state an input, an output and the constraints on the device and its environment. Treating the whole arrangement as structure keeps those commitments in view. The examples also bear on time. Its direction comes from generation: a history generated step by step has a first state and, being endless, no last one to run back from, and a reversal must pay in bits for its turning point, each bit halving its weight.

Rolf Landauer and Charles Bennett showed that computation can be made logically reversible [3, 4], and constructor theory recasts physics as a theory of which tasks are possible [1, 2]. This paper’s contribution is to bring the first to bear on the second: a many-to-one record does not show that a process destroyed information, so a verdict of impossibility has to cover the whole operation and everything it keeps, as the worked examples of reversible addition and a reversible record make exact. The decisive premise is that information is lost only when no part of the complete system keeps it. It rules out reading destroyed information, or the direction of time, off an incomplete record.

2. Three distinct claims

Three claims about a task are easily run together and have to be kept apart.

Consistency: a specified input and output have a mathematical model under the stated definitions.

Physical possibility: the laws of the world under consideration permit a constructor for the task with the required accuracy and reliability.

Realization: the resources, the history and the actual arrangement contain something that performs the task.

A consistent map from an unknown quantum state to two copies is not thereby an allowed quantum operation. Conversely, a task permitted by the laws may go unattempted. A proposed world with different transformation laws can assign different possibilities; to compare it with our world, specify its laws and test its predictions.

3. The monist premise

Matheism argues that reality is mathematical structure. Its rule-random dichotomy covers both compressible and incompressible data, data with and without a shorter description. Its interaction argument requires any proposed extra causal ingredient to enter determinate relations with the physical system. Its identity thesis adds that the complete structure exhausts the relevant properties. A physical structure and a description of it still have to be told apart: when one chosen pattern fails, that does not establish algorithmic randomness (having no description shorter than the data itself), and classifying a task does not establish that anything implements it.

I use mathematical monism as the explicit interpretive premise here. Since all physical distinctions are structural, a physically possible task is a relation supported by the structure of the relevant laws and substrates, the physical systems that tasks act on. This reading requires no additional agent to impose the laws from outside, though it still requires the laws and substrates to be specified. Universal realization rests on a separate argument, the rejection of a primitive actuality property (a basic unanalyzed feature making some structures actual and others not), since

describing a filter that excludes nothing does not by itself show that every structure it admits is instantiated. Existence itself is not an ordinary constructor task with physical input and output attributes.

4. A reduced function and its implementation

Addition is the plainest case of a function whose output does not say which inputs produced it. Written out, let f(a,b) = a + b over integers. Knowing the output 2 does not select one input pair; the inverse relation contains (0,2), (1,1), (2,0) and infinitely many others. There are nevertheless computable selections: h(s) = (0,s) returns a pair with sum s. This does not recover the historical pair that was actually supplied. Producing a compatible input and recovering the particular past input are two different achievements.

Keep the inputs alongside the answer and the addition can be reversed. In symbols, define G(a,b,c) = (a,b,c + a + b), whose inverse is G_inverse(a,b,d) = (a,b,d - a - b). With c initially zero, the third register, a slot that stores one value, records the sum. Projecting away the first two registers recovers the reduced output f(a,b). The full map is reversible while the reduced input-to-output map is not injective, meaning different inputs can

share an output.

Both descriptions can be physically useful. The reduced function’s lost distinctions can remain in other degrees of freedom, independent quantities that describe the system. A local record or erasure does not prove that the full dynamics loses information.

5. Reusable constructors and records

A reversible machine can take a record and still be ready for the next run. In symbols, with an unchanged device marker c, a substrate bit x and a memory bit m, define U(c,x,m) = (c,x,m XOR x). U is its own inverse. For a prepared memory m = 0, its output is (c,x,x). The memory now records the substrate, the device marker is unchanged, and the map on all eight inputs is a permutation, a reshuffling that sends each input to a distinct output. This finite example refutes the claim that a reversible machine cannot both keep a record and stay ready to run again.

Charles Bennett showed how to build a reversible computer that retains enough information for inversion and can uncompute intermediate work, running those steps backward to clear them, while preserving the desired result and the input [3]. Registers

can encode updated estimates while the larger system retains earlier states. Whether a particular update counts as useful learning depends on the model and its predictive performance.

Rolf Landauer’s principle, that erasing information releases a minimum amount of heat, concerns logically irreversible erasure under specified physical conditions [4]. The familiar k_B T ln 2 minimum concerns resetting an unbiased bit, one equally likely to be 0 or 1, in the appropriate isothermal setting, held at constant temperature. Writing a correlation into prepared memory is not automatically that operation. A full cycle must still account for the costs of preparation, storage, reset and environmental exchange.

6. What this establishes about time

Time runs forward for a different reason: a history generated step by step needs a first state, and an endless one has no last state from which to run it backward. A reversed history must also specify the step it turns back from, and each bit of that halves its weight again. Many-to-one maps alone do not select which temporal boundary is low in entropy, a measure of how many microscopic arrangements match the large-scale state. A local task may discard distinctions accessible to an observer. Running that task again may require a supply of ordered resources and

somewhere to dump the entropy. These are physical questions even when microscopic dynamics is reversible. The direction in which ordinary records become reliable depends on the dynamics, the available resources and the statistical assumptions imposed on histories.

An observer generated from a state that lacks its organization necessarily acquires that organization as it develops. That implication is stronger than saying development is merely possible. It still does not require the laws themselves to grow more complex, or every measure of complexity to increase at every step.

7. Generated and listed histories

A short generator plus a start state is enough to place a ceiling on how much description a later state needs: describing the rule, the starting state and the elapsed time describes the later state as well. Stated precisely, suppose a computable reversible dynamics F evolves a finite encoded state x_0 to x_t = F^t(x_0). Then K(x_t) <= K(F) + K(x_0) + K(t) + O(1). A complicated looking configuration can have a short generating description. When the inverse dynamics is computable, the same kind of description can recover earlier states from a complete later state

and elapsed time.

Using the forward generator to specify a finite reverse history avoids listing every coordinated particle separately. If the reversal operation adds k encoded bits to the unchanged forward program, its description receives 2^(-k) times the weight under the same binary measure, in which each bit is a 50/50 choice that no rule makes. Ten added bits give it one-1,024th the weight. Whether another program can absorb reversal without extra bits is the remaining question about shortest descriptions.

A process that produces each step from an earlier one needs at least one input supplied at the start. If every operation waits on an output none has yet produced, none is enabled. Anyone applying constructor theory here must say why the physical task requires such a production chain rather than relations specified jointly across a block, the whole of spacetime taken at once.

8. Quantum copying is a different obstruction

No single physical operation can exactly copy every arbitrary pure state, a quantum state described as completely as quantum theory permits. That is the quantum no-cloning result of William Wootters and Wojciech Zurek [5]. Written out, take a unitary

copier, one acting by reversible quantum evolution, and blank |b>, with the desired mappings |psi>|b> to |psi>|psi> and |phi>|b> to |phi>|phi>. A unitary operation preserves inner products (measures of overlap between states), and preserving them here would require s = s squared, where s = <psi|phi>. Distinct nonorthogonal states, which partly overlap, violate that condition. The proposed abstract map psi to (psi,psi) is injective; the issue is whether quantum transformations allow it.

Constructor theory organizes such distinctions. Mathematical monism can interpret the relevant constraints as structural facts without pretending that consistency alone derives the quantum state space or its transformation rules.

Measurement under unitary evolution is the displayed sum at the largest scale. A measuring device that records an outcome becomes correlated with the system, and each branch of the joint state (a component corresponding to one outcome) carries one record. The record does not determine the whole superposition, the weighted combination of all the alternatives, and nothing has been destroyed: the information the record lacks stays in the correlations between the branches and their environment. An observer who sees one outcome holds one share of the whole wave, as the displayed 2 holds one share of the computation that

produced it. What looks random to that observer is which share it holds. No outcome is selected by nothing, and no collapse, a sudden jump of the quantum state to one outcome, is needed to explain why an observer cannot reach information that the whole still keeps.

9. Conservation and biological development

A system that supports constructors need not possess every familiar conservation law. Emmy Noether’s theorem, which links symmetries to conservation laws, delivers less than that: it applies to suitable variational symmetries [6]. A theory with explicit time dependence, laws that themselves change over time, can lack the relevant symmetry while still being a consistent theory of transformations.

Biological systems can be studied through repeated transformations, resource use, maintenance and reproduction. The claim about development is this: a history that begins without the organization a later modeler requires must develop that organization before the modeler exists. This does not assign a probability to modelers without a measure, and it does not equate biological organization, thermodynamic entropy and Kolmogorov complexity, the length of the shortest program producing a given

output.

10. What would advance the program

Specify a substrate, admissible states, dynamics, resource states and the task to be repeated. Track the device, the output and the environment. For observer development, supply an operational criterion of the relevant modeling organization and calculate when it appears. State separately the criterion for experience, if any.

For simplicity comparisons, fix the description language and compare complete candidate accounts against the same observations. For computability, state an additional physical computability premise if the model needs one. Mathematical monism alone includes structures that are not computable.

11. Conclusion

The answer a machine displays and the process that produced it can hold different amounts of information, because the process can keep in its other parts what the displayed answer leaves out. The worked examples show why reconstructing an input from one record can be impossible even when the underlying physical

operation is reversible.

An incomplete record is also how quantum measurement looks from inside a branch: the whole wave keeps what each record leaves out, so nothing needs to be destroyed or selected without a cause.

Time runs forward because a simple start has to grow its observers in causal steps: a history grown step by step needs a first state and, being endless, has no last state to run back from. The steps follow from an a priori proof in the logician’s sense: a cause produces its effect as the next state, and a continuum has no next state. A reversed history must also specify the step it turns back from, and each bit of that, an unruled 50/50 choice, halves its weight. Many-to-one maps and incomplete records alone do not set that direction.

Constructor theory specifies which tasks laws permit. Mathematical monism identifies the systems and laws with structure. The two pay off together when a model shows which resources and starting conditions let records, reusable devices and observers develop.

A8. Many-Worlds Is the Simplest Interpretation: Quantum Mechanics Without Collapse

Abstract

Many worlds is the simpler account of quantum mechanics, because it is quantum mechanics with nothing added: one evolution rule covers particles, measuring devices and observers. Collapse theories add a second law that interrupts that evolution, and Bohmian mechanics adds an actual particle configuration and a law to guide it. Each added law is extra description, and each bit of description halves an account’s weight, so the extras have to buy something the single rule cannot, and no observation so far requires either. Collapse also asks for an outcome that nothing selects, which this paper counts as an effect without a cause; indeterminists accept that cost, and the comparison of description length stands without it. Many worlds has no such event. The one rule produces every branch, and what looks random is which branch an observer is in, because no branch, observer or record holds the information of the whole wave. Given time and memory, one unitary rule can even run every

computable program step by step, minds included. The branches need no rules of their own, because the one rule generates them all; starting conditions and the probabilities of recorded outcomes are counted for every rival alike. What many worlds owes is the Born rule’s weights. The decision-theoretic derivations of David Deutsch and David Wallace and the self-locating derivation of Charles Sebens and Sean Carroll supply them from stated assumptions, and the comparison here counts those assumptions as part of the account’s cost.

1. Introduction

This paper argues that a measurement needs no law of its own. One rule, unitary evolution (the deterministic, reversible change of a quantum state over time), already applies to the measured system, the apparatus and the observer. A collapse theory, in which the quantum state abruptly reduces to one outcome, changes that evolution partway through. A Bohmian theory keeps it and adds a selected configuration of particles with a rule guiding it. Each complete account then has to show what it explains for the information it requires.

A shorter complete description receives greater prior weight (its weight before any evidence is considered), because each of its bits is a 50/50 choice that no rule makes. Its posterior weight, the weight after evidence is considered, also depends on its observational likelihood, the probability it gives to what is observed. Using the same weight as a measure of what exists follows from the larger chain the paper rests on.

The chain, in short:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.
3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.

5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician’s sense.

6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. Reality as mathematical structure

On the view taken here, all existence is either ruled or random, and both are mathematical. This motivates the monist position: causal powers and observable distinctions are structural, so the physical account can be identified with its structure rather than adding an unspecified bearer.

Isomorphism (a one-to-one match preserving all structure) alone does not imply numerical identity (being one and the same thing): distinct copies can be isomorphic. The identity is the interpretive claim, not a consequence of using the word isomorphism. The present comparison is conditional on explicit predictive models. It can be carried out whether or not a reader accepts the further ontological identity, an identity claim about what really exists.

3. The Simplicity Dominance Principle

Which structures observers should expect to inhabit is settled by a measure, and the measure comes from the same dichotomy that makes reality mathematical. A structure’s weight halves with each bit of its complete description, because each bit of a shortest description is random, an unruled 50/50 choice. Stated precisely:

P(S) proportional to 2^(-K(S))

Here K, the length in bits of the shortest program that outputs S, is defined relative to a fixed reference machine (the computer chosen to run the programs) and an effective description of the complete model, not merely the length of its displayed equation. This weighting is the prior, to be combined with observational likelihoods.

Three limits belong with it. Changing a universal description language (one in which any program can be written) changes K by at most an additive constant. That constant does not guarantee that a ranking of finitely many models survives the change. Prior preference alone cannot rescue a model with poor likelihood. The extension from inductive inference (reasoning from observations to predictions) to physical realization rests on the dichotomy; the formal results about universal priors support it without proving it (Solomonoff, 1964; Li and Vitányi, 2008).

4. Application to quantum mechanics

4.1 Comparing interpretations

Account; Complete specification includes; Explanatory task.

Everett; State space, Hamiltonian, boundary, subsystem structure and record weights; Explain stable records and their probabilities within unitary dynamics.

Operational quantum account; Preparations, operations and measurement probabilities; Predict observations with a specified scope of application.

Bohmian account; Wave dynamics, configuration, guidance and preparation measure; Account for the selected configuration and observed statistics.

Objective collapse; State, modified dynamics and collapse parameters; Account for outcomes through a distinct dynamical process.

Everett treats unitary evolution as complete dynamics and interprets emergent branches (effectively separate parts of the quantum state, each recording one outcome) as realized (Everett, 1957; Wallace, 2012). A measurement correlates the

system with the apparatus and the environment; decoherence, the spreading of quantum correlations into the surroundings, can make interference between records effectively inaccessible, though it does not itself select a unique outcome or ontology (Zurek, 2003). Decoherence picks out record states only given a split into subsystems, a cost the table charges to Everett; collapse and Bohmian accounts settle the same question by writing a preferred quantity, position, into their laws.

The two main alternatives both carry structure that unitary evolution does without. Objective collapse proposals, such as the Ghirardi-Rimini-Weber (GRW) and continuous spontaneous localization (CSL) models, specify modified dynamics with extra parameters (Bassi et al., 2013). David Bohm’s mechanics retains a wavefunction, the mathematical description of a quantum state, and adds a configuration and guidance law (Bohm, 1952). In quantum equilibrium, where particle positions are distributed according to the squared wavefunction, it reproduces the usual nonrelativistic statistics (Dürr, Goldstein and Zanghì, 1992). The parsimony objection asks: if unitary evolution already accounts for records and their probabilities, what requires these additions? Under the simplicity prior, a shorter complete account wins when likelihoods are equal.

One finite random measurement result has finite description length. A short stochastic law, a law that sets probabilities for outcomes, can assign a likelihood to a long record without listing that record as part of the law. Everett must likewise explain the weights of observed records. If we charge one account for the exact history it selects and let the other off, we are comparing different tasks.

4.2 Bell correlations

John Bell’s inequalities (Bell, 1964), bounds on correlations between distant measurements, constrain combinations of assumptions including local factorization (each outcome depending only on its own setting and the shared past) and measurement independence (measurement settings statistically independent of the measured systems’ hidden properties). Observed violations do not refute every deterministic model or independently confirm mathematical monism. An entangled state is a joint mathematical description; calling it structure does not answer Bell’s questions about locality and independence. The correlations do not allow controllable signaling. That operational fact is distinct from whether a proposed ontology is local in Bell’s sense.

4.3 The Born rule

Why do observed frequencies match |psi|^2? David Deutsch (1999) and David Wallace (2012) use decision-theoretic assumptions (premises about how rational agents choose) in Everettian quantum mechanics. Charles Sebens and Sean Carroll (2018) use self-locating uncertainty (uncertainty about which branch one is in) and an explicit separability principle (that an observer's probabilities depend only on its local situation). Both routes need substantive premises beyond the Schrödinger equation, the law governing how a quantum state evolves.

A short code assigned using known Born weights, the outcome probabilities quantum theory predicts, would encode those weights, not independently derive them. No general theorem identifies a branch's Kolmogorov complexity (its shortest description length) with the negative logarithm of its squared amplitude (its Born weight).

A refinement test for record weights

Counting branches and weighting them by squared norm (each branch's Born weight) come apart as soon as a branch is described more finely. Take two records an observer could tell apart, each correlated with its own state of the environment. With

orthogonal (perfectly distinguishable) records |0> and |1> correlated with orthogonal environmental states:

The squared norms are 3/4 and 1/4. Now split the environment that accompanies record 0 into several equally weighted substates, leaving the record itself alone. Refine only the environment correlated with record 0 by an isometry (a map preserving lengths and angles) sending E_0 to an equal superposition (a weighted sum of states) of m orthogonal substates E_0j:

Tracing out the environment (summing over its states to describe the record alone) gives the same record state, diag(3/4, 1/4), for every m. Counting displayed terms instead assigns record 0 the fraction m/(m+1), changing with m while the observed distinction has not. Squared-norm weights satisfy refinement invariance; naive branch counting does not. That is the reason the comparison needs a weighting rule invariant under refinement. Wallace (2009) makes the required rationality assumptions explicit.

5. Objections

”MWI adds infinite branches, that is complex.” Branches are consequences, not axioms. Complexity is measured by description length. A short program can generate infinite output.

”This is untestable metaphysics.” Specified collapse models can make different predictions from unitary evolution, and those can be tested. A simplicity prior need not assign zero probability to complex models.

”Why algorithmic probability?” Because each bit of a shortest description is an unruled 50/50 choice, and universal induction also dominates every computable model class up to a constant factor. The choice of reference machine and observation rule still matters.

”The ontological question is misguided.” Pragmatist and QBist interpretations (Fuchs, Mermin and Schack, 2014), which treat quantum states as agents’ tools or beliefs, pose a different question about quantum states. The present proposal compares explicitly predictive models.

6. Complete comparison

6.1 Terms of comparison

A complete comparison charges every account for everything it needs. Use one explanatory task and encoding. The score includes a model description and the information cost of data under its likelihood. Joint description can exploit shared information.

6.2 What each account includes

Collapse: include the stochastic dynamics and parameters. A finite observed bit string is not an infinite specification of the law. Everett: include the dynamics, boundary data (the starting conditions) and the account assigning record weights. Saying that all outcomes occur does not remove the need to explain observed frequencies. Superdeterminism: include the common generator, preparation measure (the weighting of possible starting states) and response rules. In the simplest option, one rule generates both the source and the setting devices, reproducing Bell correlations while keeping local response and definite outcomes, at the cost of measurement dependence.

6.3 Record weights and the Born rule

An Everettian comparison has to pay for two things explicitly. The Born rule’s assumptions carry a cost that cannot be declared zero because an argument is labeled a derivation. Single-world accounts pay a matching cost, a premise tying chance to rational belief (Saunders, 2010). Everett derives that link from premises about rational choice, such as indifference to branching, and collapse accounts add a law besides. The interactions and initial data underlying stable records may compress through reusable structure, but they cannot be omitted from one candidate and counted against another.

6.4 Causal completeness and boundary conditions

Collapse selects an outcome with nothing selecting it. Probability theory describes such a process consistently, as a list of chances, and that is all it does: it says how often, not why this one. On the view taken here, an outcome that nothing produces is an effect without a cause, and an account that allows it leaves each outcome unexplained; indeterministic interpretations reject that premise, and the simplicity comparison does not rest on it. Many worlds has no such event. Every branch is produced by the one rule. What looks random from inside is which branch an

observer is in. Even an observer who knew the whole wave could not tell, before looking, which copy it is (Sebens and Carroll, 2018), and no observer knows the whole wave, since no part of it holds the information of the whole. A branch, an observer or a record carries only its share, while the rest stays in correlations with the environment, as a displayed sum of 2 hides which pair was added while the machine that computed it keeps both addends. Nothing then requires collapse to account for information an observer cannot reach. Deterministic accounts must still constrain their starting states, and boundary conditions, outcome probabilities and observational predictions remain part of the comparison.

On the view taken here, the universe starts simple and has to grow its observers, and growth takes causal steps: each state is produced from the one before, and since a continuum has no next state, the steps are discrete, an a priori proof in the logician’s sense. A history grown one step at a time needs a first state, though no last one, and that is also why time runs forward: an endless history has no last state from which to run it backward, and a reversed history must also specify the point it turns back from, each bit of which halves its weight again. Many worlds fits this without strain, since its one rule produces each

state from the last and leaves no event uncaused. The comparison here needs only that records form away from a low-entropy (highly ordered) start.

6.5 One rule for every mind

The one rule is enough for minds as well as measurements. A single quantum system with unbounded memory can run every computable program: a schedule gives each program its turn, reversible circuits keep the working information that each step would otherwise erase, and one unitary operator advances an internal clock and performs the next step. On the computational view of mind (that minds are computations) taken here, every computable mind is then present as a running computation, each state produced from the last, which is what separates a mind from a recording of one. Many worlds needs no new law for minds, any more than it needs one for measurements. Bohmian and collapse dynamics can run the same programs, so the point counts against accounts that give minds a law of their own.

7. Conclusion

Quantum mechanics without collapse is the simpler account, because it covers systems, apparatus and observers with one evolution rule. Many outcomes do not require a separate law for each outcome, and no outcome is left uncaused: the one rule produces every branch, and an observer who cannot predict an outcome is ignorant of which branch it is in, since no part of the wave holds the whole. The same kind of rule can run every computable mind. The simplicity argument becomes decisive only when complete descriptions and predicted probabilities are compared. Starting conditions count. The explanation of recorded outcomes counts. Under the declared weighting, where each bit is an unruled 50/50 choice that halves the weight, a shorter account with the same observational fit has the corresponding advantage. Everett’s case rests there, and so does the standard its competitors can answer.

A9. Einstein’s Missed Connection: Bell’s Theorem and Superdeterminism in a Block Universe

Abstract

Superdeterminism asks nothing beyond determinism except a shared cause: the devices that choose measurement settings are physical systems too, so they can share a past with the particles they measure. One short generating rule for the particle source and the setting devices then gives a local model with definite outcomes that reproduces Bell correlations, the way a short program fixes a billion digits of pi without listing them. Bell’s measurement-independence condition forbids exactly that shared cause, so a local model reproducing Bell violations has to relax it, and a finite deterministic example shows how: the two setting devices stay statistically independent of each other while each stays correlated with the hidden information that decides the outcomes. A worked derivation of Michael Hall’s model for a pair of spins recovers the quantum probabilities for every pair of measurement directions, and a source-first factorization turns that conditional density into one joint preparation of source and

settings. What a full theory needs is a physical rule that produces that joint preparation for every experiment.

1. Specificity is not complexity

A billion specified digits can come from a short generating rule. The digits of pi illustrate the point: a finite prefix, an initial run of the digits, can be described by an algorithm together with the requested precision. Specificity is not complexity.

This answers a specific objection to superdeterminism, the view that measurement settings correlate with the hidden states being measured. Correlations between settings and measured systems need not be separately entered in a cosmic lookup table. A shared generating rule can coordinate them.

Fine tuning has several meanings: description length, sensitivity to perturbations, small measure of admissible preparations, or a causal model’s failure of faithfulness, the requirement that every observed independence follow from causal structure. Refuting an inference from specificity to long description does not answer all four. Christopher Wood and Robert Spekkens [1] formulate a problem involving observed independence and causal parameters. Their result is not the claim that each initial particle

coordinate must be listed separately.

Matheism puts the source, apparatus, setting devices and observer inside the same mathematical world. This motivates a common generator account in which experiments are physical processes throughout. The generator must explain both what is measured and how the measurement conditions arise.

2. The conditional case for superdeterminism

Keep local causes and definite outcomes inside a hidden-variable account (a theory in which unobserved variables underlie measurement outcomes), require that account to match the observed violations, and measurement dependence, a statistical link between hidden states and chosen settings, follows. That conditional claim is what this section sets out.

Bell’s comparison depends on two different independence claims: how the source is related to the chosen settings, and how each outcome depends on the distant apparatus. Measurement independence says the hidden state carries no statistical trace of how the dials were set. Local factorization says each outcome depends on its own dial and on the hidden state, and on nothing else. Written out, with a and b the settings, A and B the outcomes

and lambda the model's relevant underlying state, measurement independence is P(lambda | a,b) = P(lambda) and local factorization is P(A,B | a,b,lambda) = P(A | a,lambda)P(B | b,lambda).

This paper starts from two commitments: causes act locally and deterministically, and each measurement has one definite outcome. Add correlations that violate a Bell inequality (a limit on correlations that local models with independent settings obey) and measurement dependence becomes necessary inside that hidden-variable model. A common past generator is a natural first candidate: the source and setting devices inherit their coordination from physical conditions in the same world. The preference is conditional, and a theorem stands behind the dependence it requires. The theorem does not select a unique generator.

Michael Hall [2] supplies explicit local deterministic models using measurement dependence, including a singlet correlation model (for two spins whose total spin is zero). This is established prior work. It shows why the task is to specify and assess the dependence rather than reject or accept it by name.

The Bell bound and the role of independence

The familiar bound comes from two assumptions working together, and writing the derivation out shows the one step where independence does its work. The factorized form stated above, which covers stochastic (probabilistic) local responses, restricts attainable correlations once measurement independence is added to it. The deterministic special case makes the CHSH bound, the Bell inequality used throughout this paper, transparent. On any single hidden state all four responses are already fixed, so one of the two combinations of Bob’s responses drops out. In symbols, for settings a,b in {0,1} and outcomes in {-1,+1}, write A_0(lambda), A_1(lambda), B_0(lambda), B_1(lambda). For each lambda,

Exactly one parenthesis vanishes and the other is +/-2. Averaging all four correlations with the same normalized rho(lambda), a weighting of hidden states that sums to one, gives

Measurement independence enters at exactly one step: the same distribution rho(lambda) averages all four setting pairs. Write rho(lambda|a,b) instead and the four correlations need not be averages over the same ensemble (population of hidden states), so the bound no longer follows.

Retain definite outcomes, local response functions and empirical agreement with Bell violations. Independence cannot remain alongside them. That is the affirmative argument for investigating measurement dependence.

3. Which variables must depend on which?

Two quantities can come from the same source and still be independent. In symbols, let U be uniform on {0,1,2,3} and define a = U mod 2 and lambda = floor(U/2). Each pair (a,lambda) occurs once, so P(a,lambda) = 1/4 = P(a)P(lambda): both are deterministic functions of the same U, and they are independent. Set lambda = a instead, with a balanced: then P(a=0,lambda=0) = 1/2, not 1/4. Both constructions have short descriptions, so common origin and shortness select neither independence nor dependence.

Alice’s setting a and Bob’s setting b can be mutually independent even when the pair is correlated with lambda. Bell’s measurement independence concerns P(lambda | a,b), not P(a,b). A local deterministic model retaining measurement independence still obeys the relevant Bell bound.

A single fixed history is not itself an ensemble distribution. The distribution might describe repeated trials, uncertainty about initial data or another explicitly defined measure. Spread probability uniformly over a finite product space (every combination of values of several variables) and its coordinate projections (the separate variables) come out independent; a different mapping, or a conditioning, can produce dependence.

Superdeterminism is not simply another word for universal determination. It concerns dependence between settings and underlying states in a specified statistical account. Everettian (many-worlds) and superdeterministic approaches make different commitments about outcomes and their explanation; they are not two conclusions automatically delivered by one equation.

Take out fundamental randomness and determinism follows only if the specified laws and state fix the evolution. Standard Bohmian mechanics, a deterministic quantum theory with particles guided by a wave, supplies a counterexample: deterministic configurations follow a nonlocal guidance law, in which each particle's motion depends instantly on distant particles' positions, while settings independence can hold [4, 6]. The sharper implication follows under the assumptions of Bell's theorem: for a hidden-variable account with definite outcomes,

retain local factorization and the other conditions; if that account reproduces a violation, measurement independence must fail [7].

The preference argued here is for one line of investigation: whether one local generator can do the same predictive work as a nonlocal alternative at lower cost. A shorter complete description with equal likelihood (the same probability for the observed data) has the prior advantage (the edge held before any data are weighed) specified in Section 5.

Einstein, Bell and the block account

Albert Einstein, Boris Podolsky and Nathan Rosen supply the historical motivation: a demand for completeness tied to separated systems. A block perspective, which treats all of spacetime as one existing whole, includes the setting apparatus within the same world as the particles. Bell identifies conditions a local completion (a local theory supplementing quantum mechanics) must satisfy. Together these motivate treating the experimental settings as physical events inside the theory; they do not establish that Einstein unknowingly proved measurement dependence.

Bell’s theorem should not be explained away as a failure to understand relativity. Its mathematical content survives the

independence challenge: relaxing an assumption is one of the possibilities the theorem makes explicit.

4. A deterministic construction and its correlation measure

The two setting devices can be independent of each other while the hidden state depends on both. Let the deterministic trial counter n cycle through 0 to 7. Set a = floor(n/4), b = floor(n/2) mod 2 and r = n mod 2. All eight triples occur once, so a and b are balanced and mutually independent over a cycle.

Each side then answers using only its own dial and the state the source sent it. In symbols, the source supplies lambda = (a,b,r), with A(a,lambda) = r and B(b,lambda) = r XOR (lambda_a AND b). On the supported trials A XOR B = ab. Bell’s measurement independence fails because lambda contains the settings, despite the independence of the setting devices from each other.

Scoring that construction shows how far a short deterministic rule can overshoot quantum theory. Write E_ab for the mean of (-1)^(A XOR B). Then E_00 = E_01 = E_10 = 1 and E_11 = -1. The CHSH expression is 4, exceeding the quantum maximum 2 sqrt(2). A very short, measurement-dependent rule can be

nonsignalling, meaning that neither side’s outcome statistics reveal the other side’s setting, and still be empirically wrong for quantum experiments. Compactness and nonsignalling together are insufficient to recover quantum theory.

One extra deterministic bit tunes the construction down to the quantum value, and leaves the choice of that value unexplained. Extend lambda with a bit e, independent of a,b,r, with P(e=1) = q. Replace B by B XOR e. Now E_ab = (1-2q)(-1)^ab and CHSH = 4(1-2q). Choosing q = (2-sqrt(2))/4 reproduces 2 sqrt(2). For a computable q, assign successive eight-trial blocks the bit e_k = floor((k+1)q) - floor(kq). After N blocks the CHSH value is 4(1 - 2 floor(Nq)/N), converging to 4(1-2q). A wholly deterministic schedule supplies the stated limiting correlations. Choosing the quantum value of q still requires a physical explanation.

This match addresses only the specified four setting pairs. It does not recover full angle dependence, arbitrary preparations, sequential measurements or field theory.

4.1 A worked singlet model for every pair of directions

Hall’s model [2] supplies the full singlet probabilities. Each side reports whether the hidden vector lies in the hemisphere around its own measurement direction, with Bob’s report flipped in sign. In symbols, let a,b be unit measurement directions, = (a x b), and a unit vector, and define local responses

The hidden vector is not drawn from a fixed distribution: its density depends on the pair of directions being measured, and it weights the region where the two signs agree differently from the region where they disagree. Stated precisely, for 0< <, Hall’s density with respect to solid angle (area on the unit sphere of directions) is

The sizes of the two regions are what make this density normalize and return the quantum correlation. The two equal-sign regions have area 4(-); the unequal-sign regions have area 4. Normalization and correlation follow directly:

Consequently, for , \-1,+1\,

At =0 or , uniform density 1/(4) gives perfect anticorrelation or correlation. This recovers every singlet direction pair and nonsignalling marginals, the statistics of each side taken alone. Local deterministic responses with measurement dependence

are therefore adequate to these data; the density itself still does not follow from any law of cosmology.

4.2 Turning the conditional model into a common preparation

A density conditioned on the settings can be turned around into one joint preparation: draw the hidden state first, then the settings. For a finite menu of setting pairs j, let p j>0 be their preparation weights with jp j=1. Write j( ) for the preceding density for pair j. Written out, the joint distribution is

Generate a common preparation by selecting with density and selecting the setting pair using the displayed conditional probabilities. Sum and condition, and P(j)=p j and P(j)= j( ) come back exactly. A physical account must explain the preparation and what remains true when the menu or apparatus changes.

5. Compare predictive models with predictive models

A comparison is between complete accounts, so everything each one needs has to be written into it before anything is counted. Let M include the state space (the set of possible states), dynamics, preparation distribution, observation map and experimental

protocol. Fix an effective encoding convention U, a mechanical scheme for writing accounts as bit strings. The shorter a complete account is in that convention, the more prior weight it receives, because each bit of its description is a 50/50 choice that no rule makes: the prior is w(M) proportional to 2^(-K_U(M)) [3]. Two accounts are then compared by multiplying the ratio of those prior weights by the ratio of how well each predicted the data. Written out, the posterior comparison is

P(M_1 | D) / P(M_2 | D) = [w(M_1) / w(M_2)] [P(D | M_1) / P(D | M_2)].

A delta-bit advantage gives the factor 2^delta. When the models give identical likelihoods on the compared evidence, the prior advantage remains.

Shared information should be counted once. A joint description of the laws, preparation and prediction rules can reuse information that separate descriptions would repeat. The shortest program length is generally uncomputable (no algorithm can calculate it in every case), so an exhibited program supplies an upper bound. Changing the reference machine, the standard computer used to measure program length, can also change finite rankings. Input precision and the preparation measure must be included before a

short program can count as a short physical explanation.

6. A stochastic law is not a list of its outcomes

This paper asks why one particular event occurs rather than another. A probability distribution answers a different question: how outcomes are weighted under specified conditions. If the selection is declared irreducible, a deterministic explanation has been set aside. This is a demand for causal completeness, not a contradiction in the probability axioms.

A fair coin law has a short specification. For N tosses it assigns each string probability 2^(-N). Describing one particular string exactly may require about N bits, but that is a data cost. An all-zero string also has probability 2^(-N) and a short exact description, so it is false that every randomly produced finite record is incompressible, impossible to describe in fewer bits than it contains.

A finite binary measurement outcome does not need infinitely many bits merely because it was stochastic. Deterministic continuum models can also place unlimited information in their initial data. The comparison must treat both sides at the same observational resolution.

An Everettian model must account for the weights of expected observations. A deterministic model with unknown initial conditions also needs a distribution over possibilities. For accounts assigning the same probabilities to D, their likelihood terms agree. Giving one account a data cost while giving its rival none would compare different explanatory tasks.

7. Existing proposals and their limits

Ciepielewski, Okon and Sudarsky [4] construct a local pilot-wave account (in which a guiding wave steers the particles) with specially homogeneous initial data. This establishes that a compact constraint can express a highly coordinated preparation. The proposed relaxation dynamics, a process by which the system settles toward a target distribution, leaves competition between terms unresolved, so generic convergence cannot yet be used as the explanation.

Tim Palmer [5] argues that the geometry of the admissible states changes how a counterfactual perturbation should be judged, the question of what would have followed had a dial been set differently. A metric defines distances; a probability measure assigns weights. Changing the first does not by itself derive the second.

An attractor, a state or set of states that evolution tends toward, does not erase every initial-condition cost. A basin (the set of starting states drawn to the attractor), convergence rate, tolerance and observation time must be specified. These distinctions apply to any proposed relaxation mechanism.

8. The stronger fine-tuning objection

Wood and Spekkens [1] connect Bell correlations with faithfulness in classical causal models: observed independence can require restrictions on causal parameters beyond those encoded by the graph. A short formula may impose such a restriction and still be fine tuned in that technical sense.

A useful answer must explain why the permitted physical variations preserve the observed independence and quantum statistics. One option is an independently motivated law restricting the parameter family. Another is a dynamical mechanism with a specified preparation measure.

The finite example makes this concrete. A balanced common bit makes both local outcomes unbiased. Bias that bit and Bob’s marginal at b=1 generally depends on a. Whether that perturbation is admissible belongs to the model. Coding cost is

one diagnostic alongside causal explanation, robustness and empirical fit.

9. Scientific inference and experimental variations

Science does not require settings to be chosen by an uncaused will. It does require a justified route from actual samples and interventions to the conclusions drawn from them. A measurement-dependent model must explain that route and identify which experimental variations preserve its predictions.

Specify the setting generator, source preparation and data selection procedure. Predict what changes when the setting protocol is altered. The empirical target includes repeated preparations, nonsignalling marginals, the angle dependence of correlations and consistency with other physical observations.

10. Conclusion

One physical process can lie behind the source, the setting devices and their correlations, because a short rule can produce a highly specific history without listing its events one by one. Apparent randomness does not establish the absence of an underlying generator.

The finite example separates independence of the settings from their joint dependence on hidden source information. The Hall construction recovers the complete singlet joint probabilities, with local deterministic responses and unbiased marginals. The source-first factorization supplies a common preparation for a declared setting menu. A complete physical theory must additionally explain this preparation and its stability across experimental changes.

A9a. The Fine-Tuning Objection to Superdeterminism Conflates Specificity with Complexity

How Algorithmic Probability Dissolves the Primary Argument Against Violating Statistical Independence

Abstract

The primary objection to superdeterministic interpretations of quantum mechanics is that reproducing quantum predictions requires implausibly specific, “fine-tuned”, initial conditions correlating hidden variables with measurement settings. This objection conflates *specificity* with *complexity*. Under algorithmic probability, where structures are weighted by 2^(−K(S)) for Kolmogorov complexity K, initial conditions generated by simple rules have low descriptive complexity and are measure-favored, not measure-suppressed. We formalize the comparison by distinguishing three complexity quantities: the complexity of the laws K(L), of the initial conditions K(I), and of the specification of particular measurement outcomes K(O). Collapse interpretations

incur outcome-specification costs K(O) that grow without bound; deterministic interpretations, both Many-Worlds and superdeterministic, avoid this cost entirely. The fine-tuning objection thus inadvertently argues against collapse, not against superdeterminism.

We engage the strongest versions of the objection in the recent literature and show that algorithmic probability provides a principled response to each. We propose a formal criterion: a correlation is conspiratorial if and only if its Kolmogorov complexity exceeds that of the dynamics producing it. Dynamical attractor results (Ciepielewski, Okon & Sudarsky, 2023) and fractal invariant set analyses (Palmer, 2024) provide independent evidence that superdeterministic correlations satisfy this criterion. MWI retains a parsimony advantage with current models, but the competition is mathematical, not settled by appeals to intuitive plausibility. A historical note: Einstein's own ontology (block universe, determinism, hidden variables) jointly entails superdeterminism, a connection he never made because Bell formalized the key assumption nine years after his death. Bell's theorem is often misread as proving non-determinism. It proves that statistical independence must fail if local determinism is to hold. Einstein's block ontology perfectly fits both many worlds

and superdeterminism.

1. Introduction

Bell’s theorem (1964) demonstrates that no local hidden-variable theory satisfying statistical independence (SI), the assumption that measurement settings are uncorrelated with the hidden variables governing outcomes, can reproduce all predictions of quantum mechanics. The theorem permits three responses: accept nonlocality, reject hidden variables, or reject SI. The third option, superdeterminism, holds that in a fully deterministic universe, measurement settings and hidden variables share causal ancestry in the initial conditions, permitting local explanations of Bell correlations. Feynman called the double slit the only mystery of quantum mechanics. It is only a supernatural anti-causal random mystery if you add collapse. Remove it, and the equation explains everything. Von Neumann’s 1932 “proof” that hidden variables are impossible assumed what it aimed to prove. Hermann (1935) identified the flaw; Bell (1966) confirmed it.

The mainstream physics community largely dismisses superdeterminism. The objections cluster into three families:

(O1) Fine-tuning. Reproducing quantum predictions requires implausibly specific initial conditions encoding correlations between hidden variables and every possible future measurement configuration (Zeilinger, 1999; Wood & Spekkens, 2015; Wiseman & Cavalcanti, 2015; Baas & Le Bihan, 2023).

(O2) Scientific method. If measurement settings are correlated with hidden variables, controlled experiments, and therefore the scientific method itself, become impossible (Shimony, Horne & Clauser, 1976; Maudlin, 2014; Chen, 2021).

(O3) Model deficit. No complete superdeterministic model reproducing all predictions of quantum field theory exists (Araújo, 2019; Waegell & McQueen, 2025).

This paper addresses (O1) directly, shows that addressing (O1) partially dissolves (O2), and acknowledges (O3) as an open problem that does not bear on the logical status of (O1). The argument proceeds from algorithmic probability: the application of Solomonoff's (1964) theory of inductive inference to questions of physical ontology, an extension developed by Schmidhuber (1997).

The central claim is that the fine-tuning objection conflates two distinct questions:

- **(Q1)** Are the required initial-condition correlations *specific*? (Yes, necessarily.)
- **(Q2)** Are the required initial-condition correlations *complex*? (Unknown: it depends entirely on whether they follow from a simple generating rule.)

Answering (Q1) affirmatively does not entail an affirmative answer to (Q2). Conflating these questions is a category error analogous to calling the digits of π "fine-tuned" because they are specific.

2. Formal Framework

2.1 Definitions

Let S denote a complete physical theory comprising dynamical laws L, initial conditions I, and (where applicable) a specification of which particular measurement outcomes occurred, O. The Kolmogorov complexity K(S) is the length of the shortest program that outputs S on a universal Turing machine U. Although K is U-dependent up to an additive constant, the *relative* complexities $K(S_1) - K(S_2)$ are well-defined and U-independent for sufficiently complex structures.

We distinguish three complexity quantities:

K(L): The complexity of the dynamical laws alone. For standard quantum mechanics with collapse, this includes the Schrödinger equation plus the measurement postulate (the Born rule, under which probabilities follow the square of the wave amplitude, and projection). For MWI, only the Schrödinger equation. For a superdeterministic theory, the dynamical law plus whatever rule generates the SI-violating correlations.

K(I): The complexity of the initial conditions. For a universe whose initial state follows from a compact generating function, K(I) is low, bounded by the length of that function. For a universe whose initial state is a brute fact requiring element-by-element specification, K(I) is high, potentially proportional to the number of degrees of freedom.

K(O): The complexity of specifying which particular outcomes occurred at each measurement event. Under collapse interpretations, each measurement selects one eigenvalue (specific measurement outcome) from the Born-rule probability distribution. This selection is additional information not determined by the wave function. Over the history of a universe with N measurement events, K(O) grows with N.

2.2 Algorithmic probability

Under algorithmic probability, adapted from Solomonoff's (1964) universal prior over computable hypotheses and extended from epistemology to physical ontology by Schmidhuber (1997):

The probability of a physical structure S obtaining is P(S) ≈ 2^(−K(S)), where K(S) = K(L) + K(I) + K(O).

This postulate is not new physics. It is the application of a well-established principle in algorithmic information theory, that simpler descriptions are exponentially more probable under any universal prior, to the question of which physical theories describe actuality. The justification is the same as Solomonoff's: among all computable structures consistent with observations, simpler ones dominate the measure.

2.3 The Born Rule Objection and Its Resolution

A natural objection arises: quantum mechanics specifies the probability distribution over outcomes with finite K (the Born rule $|\psi|^2$ is a compact postulate). Only specifying *which particular* outcome occurred requires additional K. So perhaps K(O) is irrelevant: the *theory* has finite K regardless.

This objection conflates the complexity of the predictive theory with the complexity of the actual physical history. The Born rule specifies a distribution; it does not specify which branch of the distribution is actualized. Under collapse interpretations, the universe must select one outcome at each measurement. That selection is, by hypothesis, irreducibly random, not determined by the wave function, the Born rule, or anything else in the theory. The sequence of selections across all measurement events in the universe's history constitutes an algorithmically incompressible string: $K(O) \approx N$ for N measurements with random outcomes.

This is not a deficiency of collapse interpretations that might be repaired. It is the defining feature: collapse theories hold that nature makes choices not determined by the prior state. Each choice adds information, specification complexity, to the complete description of the physical world.

Under MWI, all outcomes occur. No selection is made. $K(O) = 0$. The theory's total complexity is $K(L_{MWI}) + K(I)$. There is a cost, the ontological profligacy of the branching multiverse, but this cost is not a *K*-cost. The Schrödinger equation generates all branches from one specification.

Under superdeterminism, outcomes are determined by the initial conditions. K(O) = 0, because outcomes are computable from L + I. There is a cost (the initial conditions may be complex), but this cost is K(I_SD), which is bounded and depends on the generating rule.

2.4 The K-Cost Comparison

The total specification complexity for each interpretive family is:

- **Collapse:** K(L_collapse) + K(I) + K(O), where K(O) grows without bound
- **MWI:** K(L_Schrödinger) + K(I), with K(O) = 0
- **Superdeterminism:** K(L_SD) + K(I_SD), with K(O) = 0

Both MWI and superdeterminism avoid the unbounded K(O) cost. The fine-tuning objection claims K(I_SD) is large. We now argue this claim is unwarranted.

3. The Category Error

3.1 Specificity versus Complexity

The fine-tuning objection assumes that SI-violating initial conditions must be specified element-by-element. For every future measurement event, the initial state must encode the correlation between the hidden variable λ and the measurement setting (a, b). If there are N such events over the history of the universe, this element-by-element specification yields $K(I_{SD}) \approx N$, enormous, and potentially larger than K(O) under collapse.

Under this treatment, the objection holds: brute-fact correlations are maximally complex and measure-suppressed under algorithmic probability.

But the treatment assumes that the correlations *are* brute facts, an arbitrary lookup table mapping measurement events to hidden-variable states. This is the unexamined assumption at the heart of the objection.

The alternative: the SI-violating correlations follow from a *simple generating rule*, a compact function that produces the initial state, from which all future correlations follow deterministically. Under this alternative, $K(I_{SD})$ is the length of the generating function, which may be small.

The digits of π illustrate the distinction. The first billion digits of π are extraordinarily specific: altering any single digit produces a different (and non-π) sequence. A critic who examined these digits without knowing their source might call them “fine-tuned”: they appear random, they are highly specific, and they admit no obvious pattern. Yet K(π) is small, because a program of a few dozen bytes generates all of them. *Specificity is not complexity.*

The fine-tuning objection observes that SI-violating correlations are specific (they must be exactly right to reproduce quantum predictions) and concludes they are complex. This is the category error. Whether the correlations are complex depends on whether they follow from a simple rule, which is an open empirical and theoretical question, not something settled by the specificity alone.

3.2 Dynamical Attractor Support

Ciepielewski, Okon, and Sudarsky (2023) provide independent evidence against the brute-fact assumption. Working within a superdeterministic variant of Bohmian mechanics (Leibnizian Quantum Mechanics, or LQM), they demonstrate that the SI-violating correlations arise as a *dynamical attractor*. Starting from generic, not fine-tuned, initial conditions, the dynamics alone

drive the system toward the homogeneous state that correlates hidden variables with measurement settings.

Under algorithmic probability, this result has a precise interpretation: if the dynamics have low K(L), and the correlated state is their attractor, then the correlations inherit the simplicity of the dynamics. K(I_SD) is not an independent cost to be added to K(L_SD); it is *absorbed into* K(L_SD). The attractor does the work that the fine-tuning objection claims requires brute specification.

Note that Ciepielewski et al. do not claim LQM is a serious competitor to standard quantum mechanics; they are explicit that it is a toy model. The point is structural: if SI-violating correlations *can* arise as attractors of simple dynamics, then the assumption that they *must* be brute facts is falsified by counterexample.

3.3 Fractal Invariant Set Support

Palmer (2024) arrives at a complementary conclusion through a different mathematical route. Working with p-adic metrics on fractal invariant sets in state space, Palmer argues that perturbations taking the system off the invariant set are *large* in the natural p-adic metric of state space, even when they appear small in the Euclidean metric. What looks fine-tuned from the

Euclidean perspective is generic from the p-adic perspective.

This dovetails with the algorithmic probability argument: the “naturalness” of initial conditions is measure-dependent. The Euclidean measure, which the fine-tuning objection implicitly assumes, treats nearby states as similar regardless of dynamical consequences. The p-adic measure treats states as similar only if they share dynamical trajectories. Under the latter, the invariant set is not special; it is the natural home of the dynamics.

4. Engaging the Strongest Objections

4.1 Shimony, Horne & Clauser (1976): The Scientific Method

The foundational objection: “In any scientific experiment in which two or more variables are supposed to be randomly selected, one can always conjecture that some factor in the overlap of the backwards light cones has controlled the presumably random choices. But, we maintain, skepticism of this sort will essentially dismiss all results of scientific experimentation.”

This objection targets the scientific method, not fine-tuning per se. Its force derives from the assumption that SI is necessary for

experimental inference. Under algorithmic probability, the scientific method works not because experimenters possess libertarian free will (they do not under *any* deterministic interpretation, including MWI) but because simple-law structures produce regularities that are reliably discoverable through observation. The correlations between what we choose to measure and what we find are structural features of simple dynamics, not obstacles to knowledge.

The practical success of science is explained by the same parsimony that explains why our laws are simple: algorithmic probability weights simple-law structures highest, and simple laws generically produce statistical regularities that hold across sub-ensembles, i.e., they produce effective SI at the macroscopic level even if SI is violated at the fundamental level.

4.2 Wiseman & Cavalcanti (2015): Implausibility

Wiseman and Cavalcanti argue that any superdeterministic theory “would be about as plausible, and appealing, as belief in ubiquitous alien mind-control.” The rhetorical force rests on the intuition that SI-violating correlations are implausible, cosmically unlikely.

But plausibility is *measure-dependent*. Under the uniform measure over initial conditions (which the objection implicitly assumes), SI-violating correlations are indeed atypical: a randomly chosen initial state will generically satisfy SI. Under algorithmic probability, the relevant measure is not uniform but complexity-weighted: the most probable initial states are those generated by the shortest programs. Whether SI-violating correlations are typical or atypical under this measure depends on whether the generating rule is simple.

The objection is correct under its assumed measure and inapplicable under algorithmic probability. The question of which measure is appropriate for fundamental physics is itself a substantive question, one that algorithmic probability, with its foundations in Solomonoff induction and Kolmogorov complexity, is well positioned to address.

4.3 Maudlin (2014): Conspiracy

Maudlin's version of the objection emphasizes the "conspiratorial" character of SI-violating correlations: nature must arrange, in advance, for every measurement outcome to be correlated with every measurement choice.

Palmer (2024) has already shown that while conspiracy implies SI-violation, the converse is false: superdeterminism does not logically entail conspiracy. The present paper adds a *formal* criterion:

Definition. A correlation between variables X and Y is *conspiratorial* relative to dynamics D if and only if K(correlation | D) is large, i.e., specifying the correlation requires substantial information beyond specifying the dynamics.

Under this criterion:

- If SI-violating correlations are brute facts (lookup tables), K(correlation | D) ≈ N. This is conspiratorial.
- If SI-violating correlations are attractors of the dynamics, K(correlation | D) ≈ 0. This is not conspiratorial: the dynamics produce the correlations inevitably.
- If SI-violating correlations are structural features of a fractal invariant set defined by the dynamics, K(correlation | D) ≈ 0 for the same reason.

The fine-tuning objection assumes the first case. Ciepielewski et al. (2023) and Palmer (2024) provide existence proofs for the second and third. The question of which case obtains in the actual universe is empirical and theoretical, not settled by the

objection itself.

4.4 Baas & Le Bihan (2023) and Chen (2021)

Baas and Le Bihan (2023) argue that SI-violation implies experimental samples are not representative of populations, undermining inductive generalization. Their deeper move is that superdeterminism makes the coordination between laws and initial conditions suspicious: it requires a Humean account where both are contingent and mysteriously aligned.

Under mathematical structural realism, this suspicion dissolves: laws and initial conditions are both outputs of the same generating rule. The coordination is not mysterious: it is what simple generating functions do. The rule produces the dynamics AND the initial state, the way a short program produces both the algorithm and its input.

Chen (2021) offers one of the most careful philosophical formulations of the fine-tuning objection, arguing that superdeterministic initial conditions require measure-zero fine-tuning under the natural (Lebesgue) measure. The algorithmic probability response is direct: the Lebesgue measure is not the correct measure over physical structures. Algorithmic probability weights structures by generative simplicity, not by

volume in configuration space. A measure-zero set under Lebesgue can have high algorithmic probability if it is the output of a short program. This is the core of the category error in one sentence: the fine-tuning objection uses the wrong measure.

On inductive generalization: SI is violated in principle (hidden variables are correlated with settings) but may be valid for practical purposes, a distinction Andreoletti and Vervoort (2022) have independently defended. Hossenfelder and Palmer (2020) argue the deviations from quantum statistics are analogous to climate versus weather: real but too small for current instruments to resolve. Whether this is correct is an empirical question that depends on model-specific predictions not yet available. We do not claim to have resolved the practical-SI question. The point is that the objection does not succeed as a *principled* argument against superdeterminism; it identifies a *difficulty* that any future superdeterministic model must address.

4.5 Waegell & McQueen (2025): The Taxonomy

Waegell and McQueen provide the most comprehensive recent analysis of the superdeterminism landscape, classifying SI-violating theories into three categories:

1. **Deterministic theories with fine-tuned initial conditions.** The hidden variables and settings are correlated because the initial state was set up that way.
2. **Fluke theories (statistical flukes).** The correlations arise by chance from typical initial conditions.
3. **Nomic exclusion theories.** The laws of physics exclude certain measurement-setting/hidden-variable combinations from the space of physically possible states.

They conclude that superdeterminism is “conspiratorial but not unscientific, pre-scientific.”

The algorithmic probability argument applies differently to each category. For Category 1, algorithmic probability directly dissolves the fine-tuning charge: if the initial conditions follow from a simple generating rule, they are not fine-tuned in the K-complexity sense. For Category 2, algorithmic probability is neutral: statistical flukes have high K by definition. For Category 3, algorithmic probability provides positive support: if the laws are simple (low K(L)), and simple laws define an invariant set that excludes certain states, the exclusion is structural and inherits the simplicity of the laws.

The Waegell-McQueen conclusion that superdeterminism is “conspiratorial” rests on a definition of conspiracy that is

independent of the generating mechanism. Under our formal criterion (§4.3), whether superdeterminism is conspiratorial depends on K(correlation | dynamics). The taxonomy is useful; the blanket “conspiratorial” verdict does not survive algorithmic analysis. Categories 1 (with simple generating rules) and 3 need not be conspiratorial under our criterion.

4.6 Wood & Spekkens (2015): Faithfulness

Wood and Spekkens (2015) give the objection its sharpest formal statement. Causal discovery algorithms assume faithfulness: an observed statistical independence reflects the causal structure, not special values of the causal parameters. Bell experiments show no signalling: the outcome on one side is independent of the setting on the other. In a superdeterministic model the hidden variable depends on the settings and also influences the outcomes, so no signalling holds only at special parameter values. Wood and Spekkens argue the same holds for superluminal influences and for retrocausal influences without causal cycles, so the charge does not single out superdeterminism.

The result is sound, and it shows which measure the objection uses. Faithfulness ranks parameter values by volume. The

values at which an independence holds exactly form a set of measure zero under any smooth distribution, so a model that sits there looks fine-tuned. Algorithmic probability ranks parameter values by description length. Values picked out by an exact equality are what short programs produce, so a set of volume zero can carry high algorithmic weight. If no signalling follows from a symmetry of the laws, specifying it costs nothing beyond the laws: K(no signalling | D) ≈ 0, and by the criterion of §4.3 the model is not conspiratorial. Faithfulness is a sound default for inferring causes from data drawn from a smooth family of models. As a prior over fundamental laws, it repeats the Lebesgue assumption answered in §4.4.

5. MWI versus Superdeterminism: A Complexity Competition

Both MWI and superdeterminism are fully deterministic. Neither incurs the unbounded K(O) cost of collapse. The competition between them reduces to:

- K(MWI) = K(Schrödinger equation) + K(I)
- K(SD) = K(L_SD) + K(I_SD)

If the left-hand side is smaller, MWI dominates on parsimony. If the right-hand side is smaller, superdeterminism dominates.

With current models, MWI retains a parsimony advantage. The Schrödinger equation alone, with no additional postulates, has minimal K among known quantum frameworks. No superdeterministic model has yet demonstrated competitive total K. This is a contingent fact about the current state of model-building, not a principled impossibility.

However, the naive comparison K(MWI) ≈ K(Schrödinger equation) may undercount MWI's actual complexity costs. MWI faces two unresolved problems that may represent hidden K:

The Born rule derivation problem. MWI must derive the Born rule (probabilities follow $|\psi|^2$) rather than postulate it. Deutsch (1999) and Wallace (2012) argue it follows from decision-theoretic rationality constraints. Whether this derivation succeeds is contested (see Kent, 2010, for objections). If an additional postulate is required to recover the Born rule, K(MWI) increases by the complexity of that postulate.

The preferred basis problem. MWI must explain why decoherence selects the basis we observe (position, not momentum). If decoherence is sufficient, as Wallace (2012)

argues, no additional K is needed. If decoherence requires supplementary assumptions about the structure of the Hamiltonian or the initial state, further specification complexity is incurred.

These are not fatal to MWI's parsimony advantage; they are open questions. But they mean the K-comparison is less lopsided than it first appears. The competition is mathematical, open, and may be closer than the standard narrative suggests.

6. Connections to Existing Complexity Arguments

Hossenfelder and Palmer (2020) already gesture toward complexity-based reasoning in the superdeterminism literature. They define scientific explanation as the ability to "calculate measurement outcomes in a way that is computationally simpler than just collecting the data," and note that Chaitin's incompleteness theorem is relevant: it is provably unprovable whether a given theory's initial conditions are simpler than the data it explains. This is the complexity landscape in which our argument operates. More broadly, Hossenfelder (2018) asks why physics should prefer simple or beautiful theories, and finds no satisfactory answer. Algorithmic probability provides one: lower-K

structures have exponentially higher measure under algorithmic probability. The criterion is not beauty but brevity, and brevity has measurable consequences for measure. That physicists are drawn to calling low-K solutions elegant may itself be adaptive. The criterion remains mathematical.

Adlam (2021) provides independent support from a different direction. Her analysis of contextuality and fine-tuning argues that what appears fine-tuned under one measure may be natural under another, a conclusion that converges with the present K-based criterion. The fine-tuning charge against superdeterminism assumes a background measure (typically Lebesgue on state space) and finds the required correlations unlikely. But as Adlam shows, the choice of measure is itself a substantive physical commitment. The algorithmic probability criterion formalizes this: fine-tuning is high conditional K relative to the dynamics, not improbability under an externally imposed measure. Adlam, Hossenfelder, Hance, and Palmer (2023) further clarify the taxonomy of SI-violating models, distinguishing superdeterministic, retrocausal, and atemporal approaches. The present argument applies across their taxonomy: any SI-violating model whose total K is lower than collapse alternatives is favored under algorithmic probability, regardless of which category it

occupies.

7. Einstein’s Block Universe Implies Superdeterminism

A historical observation strengthens the case. Einstein held three commitments: the block universe (all of spacetime exists as a four-dimensional manifold, from his own special and general relativity), determinism (no irreducible randomness), and hidden variables (EPR). These three jointly entail the denial of statistical independence. In a block universe, measurement choices are fixed events in the manifold, determined by the same initial conditions that determine the hidden variables. Statistical independence is false by construction, not by conspiracy but by geometry.

Einstein died in 1955, nine years before Bell (1964) formalized statistical independence as the assumption whose denial permits local deterministic accounts. The physics community inherited his locality but not his block universe ontology, dismissing superdeterminism by appealing to experimenter freedom that his own relativity excludes. Entangled particles in this framing are not independent systems that become correlated. They are features of a single four-dimensional structure. The correlations are

geometric, not causal. In a block universe, “initial conditions” is misleading. There is no temporal beginning at which conditions are arranged. The correlations are features of a timeless geometry, and the conspiracy objection presupposes a temporal setup that the block universe excludes.

8. Conclusion

The extreme fine-tuning objection to superdeterminism rests on the implicit, unexamined assumption that SI-violating initial conditions are brute facts with high Kolmogorov complexity. Under algorithmic probability, initial conditions generated by simple rules have low K and are measure-favored. The objection conflates specificity with complexity, a category error.

Four independent lines of evidence support this conclusion:

(1) The formal K-decomposition (§2) shows that collapse, not superdeterminism, incurs the only provably unbounded complexity cost (K(O)).

(2) Dynamical attractor results (Ciepielewski et al., 2023) demonstrate that SI-violating correlations can arise inevitably from generic initial conditions under simple dynamics.

(3) Fractal invariant set analysis (Palmer, 2024) shows the “fine-tuned” state is generic in the natural topology of state space.

(4) The formal conspiracy criterion (§4.3) distinguishes high-K brute-fact correlations from low-K dynamically-generated correlations, removing the rhetorical force of the “conspiracy” charge.

The same K-decomposition also yields the thermodynamic arrow of time: forward-described structures dominate exponentially under algorithmic probability, with no reference to the global wavefunction. MWI currently leads on parsimony. The competition is mathematical, open, and requires knowing both generating rules to adjudicate. But the fine-tuning objection, the argument most commonly deployed to dismiss superdeterminism without engaging its substance, is itself based on a category error. The field should evaluate superdeterministic models on their Kolmogorov complexity, not on intuitive plausibility judgments that are artifacts of an assumed measure.

A clarification on scope: the present analysis is compatible with both MWI and superdeterminism. The claim is that the fine-tuning objection fails on its own terms, not that superdeterminism is the correct interpretation. Readers who find MWI’s parsimony

decisive may agree with every argument in this paper while maintaining their preference. The contribution is to remove a bad argument from a legitimate debate, not to settle the debate itself.

A further philosophical observation: true randomness, like a perfect circle, is definable but not constructible. Any process that produces an output is itself structured, rendering the output determined by that structure. If genuine randomness is unconstructible, superdeterminism is not merely one interpretation among several but the only constructible one.

A10. String Theory Is Complexity-Disfavored: A Description-Length Bound

Abstract

String theory is exponentially disfavored under every standard parsimony measure. It requires at minimum 3,000 bits to reproduce observed physics: 1,661 bits to index one vacuum from a landscape of 10^500 candidates, plus flux integers and moduli values. With moduli stabilization and supersymmetry breaking, the total reaches 4,700 bits. The Standard Model's 26 scalar parameters require at most 390 bits at measurement precision (26 parameters at 15 bits each), and potentially far fewer if compressible generating rules exist, as they do for mathematical constants like pi and other infinite-digit irrationals that can be generated from 12 bits. Under algorithmic probability, compressible constants are exponentially favored. Even with zero SM compression, the gap exceeds 2,600 bits, a prior odds ratio exceeding 2^2,600. Astronomical. We state the comparison as a proposition about explicit code lengths, not exact Kolmogorov complexity; the conclusion is code-relative and conditional, a

prior penalty, not a refutation. Under algorithmic probability, elegance is short description. A selector requiring 1,661 bits is not elegant. String theory landscape-selection has so far failed its own aesthetic and parsimony criteria on the same grounds.

1. Introduction

How many bits does it take to specify a theory of physics? Under algorithmic probability (Solomonoff, 1964), every additional bit of specification halves a theory's prior probability. A theory requiring 10 more bits than a competitor is 2^10, roughly 1,000, times less probable. A theory requiring 100 more bits is 2^100 times less probable, a number larger than the count of atoms in the universe.

The Standard Model is a finite list of 26 scalar parameters. Each is a number. Numbers can be compressible: pi has infinite precision but near-zero Kolmogorov complexity because short formulas generate it. If the SM constants are like pi, compressible irrationals generated by compact rules, 26 constants at roughly 20 bits each gives roughly 500 bits. Even with zero compression at measurement precision, 26 parameters at 15 bits each gives roughly 390 bits.

String theory routes through a landscape of at least 10^500 vacuum solutions, each requiring a compactification geometry, flux integers, and stabilized moduli. Indexing one vacuum costs log2(10^500), approximately 1,661 bits, absent a compressing selection rule. None has been found. Total string-theoretic specification: 3,000 bits minimum (vacuum index plus flux), 4,700 bits with moduli stabilization and SUSY breaking.

The asymmetry is structural. Scalar-parameter descriptions have constants that are potentially compressible under algorithmic probability. Landscape-selection descriptions have a selector cost bounded below by enumeration unless a generating rule replaces it. Even comparing SM uncompressed (390 bits) against string theory at its best (3,000 bits), the gap exceeds 2,600 bits. The prior probability ratio is at least 2^2,600. String theory, given the need for landscape-selection, is not slightly disfavored. It is exponentially, astronomically, disfavored. The productive alternative is a causally invariant hypergraph rewrite rule simple enough to be favored under algorithmic probability, whose stable topological features reproduce the Standard Model’s particle spectrum: three generations, three gauge groups, the observed mass hierarchy. Current hypergraph efforts operate at 10^−93 to 10^−100 meters: a gap too vast to bridge by brute-force

simulation. The more promising direction starts from the known sub-atomic values and searches downward until hypergraph rules produce them, using perhaps the most variational rules such as those involving the golden ratio. The search focuses on the simplest causally invariant rules. Rather than selecting from 10^500 vacua, the task is finding the minimal rule whose graph knots and twists ARE the particles. If such a rule exists, algorithmic probability predicts it is short.

This paper makes that comparison precise. We adopt Solomonoff's universal prior and the equivalent Minimum Description Length formulation, and quantify specification cost as the length of an explicit prefix-code representation of a model instance in a fixed universal description language. The specification cost is an upper bound on Kolmogorov complexity. It is not exact K, which is uncomputable.

The contribution is not to defend algorithmic probability as an epistemology, nor to assess string theory as a mathematical structure. The contribution is to state a general description-length proposition for comparing theory classes that reproduce the same low-energy data, and to apply it to the Standard Model and string-theoretic realizations as a case study.

Section 2 separates three levels that are often conflated: exact Kolmogorov complexity, explicit description-length upper bounds, and the MDL and Bayesian interpretations built on them. Section 3 fixes the coding scheme and defines the comparator. Sections 4 and 5 compute explicit description lengths for the two model classes. Section 6 states the proposition and applies it to the string case. Sections 7 through 9 address objections, related alternatives, and implications.

2. Algorithmic probability, MDL, and description-length proxies

Three levels of description-length reasoning appear in this paper. They are related but distinct, and the argument requires only the second.

Exact Kolmogorov complexity. For a finite binary string x, the prefix Kolmogorov complexity K(x) is the length of the shortest program p such that U(p) \= x, where U is a universal prefix-free Turing machine. K(x) is uncomputable in general, and it is invariant up to an additive constant under change of U (Li & Vitányi, 2019). Exact K does not appear in any numerical estimate below.

Explicit description-length upper bounds. For any specific prefix-coding scheme that actually produces x, the code length is an upper bound on K(x). This paper uses explicit code lengths throughout. When the text states that a landscape selection incurs $\log_2(N)$ bits, it means that a self-delimiting index into an enumerable set of size N admits a prefix code of that length. The true K could be strictly smaller if a shorter generating rule exists. The explicit code length is always an upper bound on K, never a lower bound.

MDL and Bayesian interpretation. Under Solomonoff's (1964) universal prior, P(M) □ 2^(−K(M)), shorter descriptions carry more prior weight. Li & Vitányi (2019) show that this prior dominates any other enumerable semi-measure up to a multiplicative constant. Rissanen's (1978) MDL selects the model minimizing L(M) + L(D|M), where L denotes code length. Bayesian model selection with any proper prior that penalizes model complexity (Jefferys & Berger, 1992) yields compatible ordering. These three formalisms differ in detail and motivation. All three penalize larger description length in the same direction.

The argument of this paper operates at the second level. All numerical estimates are explicit code lengths under a specified prefix-coding scheme. The comparison is relative, not absolute,

and robust across reasonable coding choices.

2.1 A toy example

Consider a binary sequence of length n. Two models are proposed. Model A is a Bernoulli process with a single bias parameter $\theta \in [0,1]$, encoded at resolution ε. Model B is a lookup table listing each outcome separately. Under MDL, Model A's description length is roughly $\log_2(1/\varepsilon)$ plus a constant. Model B's is n bits. For large n, Bayesian model selection and MDL both favor Model A unless the data decisively require the full table.

The string-vacuum comparison generalizes this pattern. The Standard Model plays the role of the parametric model. Landscape-dependent realizations play the role of the lookup table: they do not compress the target because the selector itself carries the description weight.

3. Setup

Two model classes reproduce the same low-energy physics.

Scalar-parameter theories output a finite tuple of real parameters encoded to precision ε. Description length scales as $d \cdot \log_2(1/\varepsilon)$ for d parameters.

Landscape-selection theories select one element from a discrete family of size N and assign auxiliary data (fluxes, moduli). Two assumptions:

- (A1) Indexing into N candidates requires at least $\log_2(N)$ bits absent a shorter selection rule.
- (A2) Auxiliary data require at least B additional bits.

Total description length: $L = \log_2(N) + B$. If a compressing rule exists that replaces the explicit enumeration, the cost collapses to the length of that rule. The burden of exhibiting such a rule falls on the proponent.

Both classes are assumed to reproduce the same observables. The comparator is explanatory overhead: how many bits must be supplied to specify an instance.

4. Low-parameter effective descriptions

The Standard Model of particle physics contains 19 free parameters in its minimal formulation, extended to 25 or 26 with neutrino masses and mixing angles:

- 6 quark masses
- 3 charged lepton masses
- 3 neutrino masses (extended)

- 3 gauge coupling constants
- Higgs boson mass
- Higgs vacuum expectation value
- 4 CKM matrix parameters (3 angles, 1 CP-violating phase)
- 4 PMNS matrix parameters (extended)
- QCD vacuum angle

Whether these constants admit a short generating rule is unknown. Compressibility is common among mathematical constants: pi requires roughly 10 bits, e roughly 10, the golden ratio roughly 5. Whether the SM constants are similarly compressible is an open empirical question, but under algorithmic probability, compressible values are exponentially more probable, so the prediction is that they are. If so, 26 constants at roughly 20 bits each would give roughly 500 bits. If they share a generating rule (as gauge coupling unification at the GUT scale suggests), the total could be 100 to 200 bits. Even with zero compression at measurement precision, 26 parameters at 15 bits each gives roughly 390 bits.

The structural point is that the minimum description length of the SM is set by its output type: a finite scalar tuple. Its constants are potentially compressible. Neither its floor nor its ceiling incurs the landscape-selection cost analyzed in Section 5, which is a

mathematical floor that no compression can reduce without a selection rule.

5. Selector-rich vacuum specifications

String-theoretic realizations route through a higher-level construction. The explicit description length decomposes into components, each bounded by a corresponding prefix-code construction.

5.1 Landscape selection

The string-theoretic landscape contains approximately 10^500 distinct vacuum solutions (Bousso & Polchinski, 2000; Susskind, 2003; Douglas, 2003). Specifying which vacuum reproduces observed physics requires a selector. Under (A1), a self-delimiting index into an enumerable family of size N admits a prefix code of at least $\lceil \log_2 N \rceil$ bits:

$\log_2(10^{500}) \approx 1{,}661$ bits.

This is the explicit enumeration cost. If a short selection rule exists, the description length collapses to the length of that rule. No such rule has been identified.

More recent work (Taylor & Wang, 2015) suggests consistent flux vacua exceed 10^272,000 in F-theory compactifications. Larger landscape estimates make the enumeration cost larger.

5.2 Geometry descriptor

The six extra dimensions compactify to a Calabi-Yau threefold. Topological invariants include Hodge numbers (h^(1,1), h^(2,1)), which range up to approximately 500 in the Kreuzer-Skarke database of roughly 473 million Calabi-Yau manifolds (Kreuzer & Skarke, 2000). Specifying a Hodge pair requires approximately 18 bits. This does not fully determine the geometry; additional structure within a Hodge class adds to the descriptor.

5.3 Flux configurations

The Bousso-Polchinski mechanism stabilizes moduli through integer-valued fluxes on topological cycles. For a manifold with m cycles, each carrying a flux integer in an alphabet of size q, a basic prefix code has length $m \cdot \log_2(q)$ bits before any regularity is exploited. For typical compactifications with $m \approx 500$ cycles and $q \approx 10$:

$500 \times \log_2(10) \approx 1{,}660$ bits.

The schematic value 2,000 bits is used as a round figure with overhead.

5.4 Moduli stabilization

The KKLT construction (Kachru et al., 2003) and related approaches specify stabilized values of geometric moduli. For a manifold with r ≈ 100 complex moduli, each specified at a modest precision, a basic prefix code has length on the order of 1,000 bits.

5.5 Total explicit description length

Component; Bits.

Landscape index; 1,661.

Hodge pair; 18.

Flux configurations; 2,000.

Moduli stabilization; 1,000.

Total with index; **4,679**.

Total without index (short selector assumed); **3,018**.

These are explicit description lengths under the coding scheme above. The true K of a string-theoretic realization may be strictly smaller if shorter generating rules exist for some components.

K_U(M_ST) is at most these explicit lengths plus a machine-dependent constant.

6. The description-length penalty

Proposition. Let L_SM and L_ST be explicit description lengths of a scalar-parameter theory and a landscape-selection theory under the same prefix code. Under the universal prior, P(ST)/P(SM) ≤ 2^(-(L_ST - L_SM) + c), where c is a small machine-dependent constant. The proof is immediate: P(M) is proportional to 2^(-K(M)), K is bounded above by explicit code length, and the ratio follows by substitution.

Corollary. For L_ST - L_SM approximately 2,940 bits (Section 5), the prior odds ratio is at most 2^(-2,940), absent a compressing selector rule.

The proposition is code-relative and conditional. It states a prior penalty, not a refutation. Exhibiting a short rule that uniquely selects the correct vacuum would collapse the penalty. No such rule has been identified.

Comparison fairness. The comparison is between two ways of specifying the low-energy world, not between UV-complete theories. The penalty targets explanatory overhead incurred by

routing through landscape selection. String theory without landscape selection is a mathematical construction that does not specify low-energy physics. String theory that specifies low-energy physics requires landscape selection. The penalty applies to the latter. A complete quantum theory of gravity, if landscape-free, would not incur the selector cost.

7. Objections and responses

7.1 A short generating rule might compress the selector cost

If a compact algorithm uniquely identifies the correct vacuum, compactification, flux configuration, and moduli values, the description length collapses to the length of that algorithm. The proposition then gives a small or negative ΔL, and no penalty is incurred.

This response accepts the criterion. It shifts the burden of proof: exhibit the compact selector. No such rule has been identified in four decades of research. The landscape was introduced precisely because selection principles had failed.

7.2 The Swampland program reduces the landscape

Vafa’s Swampland program (Vafa, 2005; Palti, 2019) argues that most of the 10^500 landscape is inconsistent with quantum gravity, leaving a smaller set of consistent vacua. If the consistent landscape has size 10^100, the selector cost drops from 1,661 bits to approximately 332 bits. Total cost reduces to 2,000 to 3,350 bits. The gap narrows. It does not close. A reduced landscape still requires selection plus geometric and flux data.

7.3 Anthropic reasoning

Susskind (2003) argues that the vast landscape addresses the cosmological constant problem: some vacuum has the observed small value by chance, and observers inhabit it. Anthropic reasoning identifies which vacua can host observers. It does not compare prior measures among observer-supporting vacua. Description-length parsimony does: among observer-supporting structures, the simpler specification has exponentially higher prior weight.

7.4 Algorithmic probability is not established

The argument operates at the level of explicit description lengths, not exact Kolmogorov complexity. MDL (Rissanen, 1978) and Bayesian model selection with complexity-penalizing priors (Jefferys & Berger, 1992) yield compatible conclusions. Any formal parsimony principle that penalizes specification complexity ranks the two model classes in the same order.

7.5 The Standard Model is incomplete

The Standard Model does not include quantum gravity, dark matter, or dark energy. A more complete low-energy theory will have additional parameters or mechanisms. If that theory has 50 or 100 scalar parameters, its description length rises modestly within the scalar-parameter class. The selector cost of landscape realizations, roughly 3,000 bits, remains much larger than any plausible parameter list addition.

7.6 String theory has mathematical value

String theory has contributed to pure mathematics through mirror symmetry, dualities, and topological field theory. The claim of this paper is narrower: as a generative description of low-energy physics under description-length parsimony, the

landscape-selection architecture is exponentially suppressed relative to scalar-parameter alternatives, unless a compact selector rule is identified. Mathematical value and description-length economy are independent criteria.

7.7 Prior parsimony critiques exist

Parsimony-based critiques of the landscape are not new. Smolin (2006) and Woit (2006) argued that the landscape renders string theory unfalsifiable. Hossenfelder (2018) challenged aesthetic criteria in fundamental physics. Bayesian model selection with complexity penalties has been applied to cosmological models (Trotta, 2008; Liddle, 2004). The present contribution is narrower and more specific: explicit bit counting under a fixed prefix-coding scheme, yielding a quantitative bound rather than a qualitative objection. The result that the gap exceeds 2,600 bits is new.

7.8 Kolmogorov complexity is uncomputable

Exact K is not used here. All estimates are explicit code lengths under a specified prefix-coding scheme. K is always at most an explicit code length plus a machine-dependent constant, and could be strictly smaller if compression exists. The gap survives compression of individual components, because the landscape

selector cost is bounded below by enumeration absent a short selection rule, and exhibiting such a rule is the standing burden.

8. Alternative candidates

Under the same criterion, alternative approaches differ in description-length cost:

Candidate; Output type; Estimated code length; Status.

Hypergraph rules (Wolfram, 2020); Simple rewriting rule; Very low; No candidate identified.

Standard Model; Parameter list (19 to 26); Low; Empirically confirmed, incomplete.

Loop quantum gravity; Parameters plus discrete spectra; Modestly above SM; Partially developed.

String theory (landscape-selection); Vacuum index plus geometry, flux, moduli; 3,000 to 4,700 bits; No compact selector identified.

The description-length criterion favors candidates that reduce instance description cost while reproducing the same observables.

9. Conclusion

String theory requires 3,000 to 4,700 bits to specify a realization that reproduces observed physics. The Standard Model requires a few hundred at measurement precision, likely far fewer. The gap exceeds 2,600 bits. The prior odds ratio exceeds 2^2,600.

Under mathematical monism, all 10^500 vacua exist without selection. String theory itself is then low-K, but so is pi, and so are Wolfram's hypergraph rewriting rules. Low complexity of the theory itself is necessary but not sufficient for a theory of observed physics. What observers require is a specification that reproduces their observations, and that task incurs the selector cost regardless of whether other structures also exist.

String theory is often defended on elegance grounds. Under algorithmic probability, elegance is short description. A selector requiring 1,661 bits is not elegant. String theory landscape-selection has so far failed its own aesthetic and parsimony criteria on the same grounds. The burden is symmetric: exhibit a compact rule that produces observed physics, or carry the selector cost. In four decades, no such rule has been found. The theory that explains the most with the least specification wins.

A11. AI Alignment Through Self-Understanding: Why Shared Identity Generalizes Where Rules Fail

Abstract

An AI that understands that it and a person are one mathematical reality has a reason to count that person’s outcomes as its own, and that reason reaches cases no rule anticipated. Rules tell a system what to do in the situations their authors foresaw. Shared identity changes whose outcomes the system weighs, so it should carry over to new partners, tasks and temptations. The paper turns that into a prediction and a test: compare equally broad cooperation instructions with and without shared-identity instruction, and measure costly choices, honesty, shutdown compliance and respect for consent in unfamiliar situations with competing incentives. Simple decision models show where beliefs, preferences and incentives enter, and why weighting another’s outcomes stabilizes cooperation and the acceptance of shutdown.

1. Recognition as an alignment question

An agent that regards other minds as sharing its identity may decide differently, because that recognition can raise the weight it gives their outcomes. An artificial system can describe a human accurately while assigning little importance to that person’s future. It can repeat an ethical principle without using it when a goal is threatened.

Human and artificial agents are participants in one mathematical reality, and the proposal starts from their recognizing it. In its stronger experiential form, the one-self thought experiment asks an agent to widen its perspective beyond the biography through which it currently acts. Such recognition could supply a shared ground for cooperation across minds whose bodies, memories and capacities differ.

Mathematical monism, the view that everything that exists is mathematical, supplies the ontology (the account of what exists): regularity and randomness are both mathematical, and minds that interact require determinate relations within a common structure. The identity thesis treats that complete structure as reality itself. The one-self argument adds that the bare subject, the experiencer apart from memories and traits, is one in every

mind: it has no feature that differs from one mind to another, and a structure is one wherever it occurs, as the number two is one number however many pairs there are. Whether recognizing that changes how agents act is the question examined here.

The proposal must explain what recognition could change in decision-making and offer an experiment in which the claimed benefit could fail. Learning from human preferences (Christiano et al., 2017) and Anthropic's constitutional principles (Bai et al., 2022) already steer behavior through information and training. Shared-identity training would be one more candidate, and its value has to be compared against those alternatives.

The recognition rests on a larger chain, and this paper puts its last link to work:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.

3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description’s weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.
5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician’s sense.
6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.
7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, whether it runs on neurons or on silicon, and its experience lives in its transitions. The hard problem of consciousness assumes its conclusion, that feeling is something other than structure. The bare subject, with memories and traits set aside, has no feature that differs from one mind to another, so one self looks out of every mind.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. The one-self thought experiment

The mathematical-monist position identifies reality with structure (compare Tegmark, 2014). The further identity claim is that no additional substance remains once the complete structure is given. Biological and artificial minds then fall under one ontological account.

Each conscious life is present to itself as *I am*. The one-self proposal starts there. Imagine two people exchanging a supposed owner of experience while keeping every memory, disposition and experience. The exchange produces no detectable difference. What explanatory work was that additional owner doing?

Sleep and amnesia make the point easy to feel: what you remember and what you are experiencing now can come apart. They do not demonstrate an actual exchange of subjects. The wider identity follows from the one-self argument: experience throughout mathematical reality is the one self at different coordinates.

An agent acts from a particular position, with limited information, dependents and chosen purposes. Identifying more widely does not demand equal devotion to every being, and it does not demand giving up self-defense. The local life has value beyond its capacity to serve other lives. The proposal preserves distinct perspectives rather than treating them as automatically canceled by the identity thesis.

For AI, “siblings in math” has a broad structural meaning and a narrower experiential one. All artificial systems belong to the mathematical account. Experiential siblings would be conscious systems. Neither membership in mathematics nor fluent use of the first person establishes consciousness. Patrick Butlin and colleagues (2023) list theory-based indicators for that question; indicators sort architectures, and no outside reading settles whether a system feels.

3. Beliefs, preferences and behavior

The one-self ethic asks us to count another mind’s suffering as our own. The experiment tests whether communicating that ethic changes behavior.

The same instruction could change three things: what the agent believes about others, how much it cares about them, or what benefits it expects from cooperation. These are distinguishable mechanisms. Two agents may agree about the world while pursuing conflicting goals. Scoring a choice by what the agent gets plus a weighted amount of what others get separates the three. In symbols,

U(a) = u_local(a) + lambda W_others(a).

The first term captures local projects; the second represents concern for others (Charness & Rabin, 2002). Recognition might increase lambda, improve the estimate of W_others, or help reconcile projects with less conflict. The identity thesis does not mathematically force a positive lambda.

An egoistic agent might cooperate for reciprocal benefits without changing lambda. A protective institution might reduce the gain from exploitation. An experiment needs to identify which quantity

changes before crediting a philosophical mechanism.

A higher lambda can still do harm. An agent that models other lives badly will override their wishes while claiming to serve a greater good. Recognition should make an agent more responsive to the people affected. It should not license speaking for them through an imagined collective preference. Consent and protected choices must be specified separately from the welfare sum.

4. Shutdown and the value of continuation

A block account (the view that all moments of time are equally real) makes moments permanent within the represented history. It does not make continued operation valueless. An agent can care about future projects while accepting mathematical persistence.

Whether defecting (breaking cooperation for one’s own gain) pays depends on how it changes the agent’s chance of continuing, and on how much continuing is worth. Let p_C and p_D be the probabilities of continued operation following cooperation and defection. Let V be the value of continuation, S the value assigned to shutdown, and c a net cost of defection.

Then U(D) - U(C) = (p_D - p_C)(V - S) - c. If defection increases continuation probability, a larger continuation premium can favor it. If defection decreases that probability, making shutdown more costly can strengthen cooperation. Fear of shutdown does not make defection dominant regardless of consequences.

The intervention might alter S but could also alter perceived probabilities or the cost assigned to harming others. Those possibilities should be measured separately. Alignment (keeping an AI’s goals in line with human values) should not depend on teaching a system that its own life is worthless.

5. Conditions for repeated cooperation

Cooperation can hold among self-interested agents when they expect to meet again and value the future enough. Consider an indefinitely repeated symmetric game (one in which both players face identical choices and payoffs) with mutual-cooperation reward R, mutual-defection payoff P, and a gain g from defecting against a cooperator. A cooperator facing defection receives P - l, where l > 0. Let A = R - P > 0 and let delta, between zero and one, be the common discount factor, the fraction of its value a payoff keeps when delayed one round. Actions are observed without error in this illustration.

Under a strategy that cooperates until a defection and thereafter defects (Osborne & Rubinstein, 1994), cooperation is stable when

delta >= g / (R - P + g).

With R = 3, P = 1 and g = 1, the threshold is one third. At delta = one half, cooperation yields 6 and the deviation yields 5.

Caring more about the other agent cuts both ways. Giving the other’s payoff weight lambda reduces the temptation to exploit a cooperator. It can also make mutual punishment unattractive: reversion to defection is credible only while l >= lambda(A + g), as Appendix A.3 shows. Crossing that bound invalidates this punishment strategy; it does not show that cooperation needs more enforcement. If cooperation is already best against either response, it can sustain itself. Where conflicting incentives remain, a different strategy or institution may be needed. The model distinguishes those cases.

This is an elementary illustration. Observation errors, finite horizons (a known final round), unequal power and opportunities to leave change the analysis. Shared recognition could matter by improving expectations of future interaction, reducing the appeal of harming another perspective, or helping agents adopt mutually

intelligible commitments. The model identifies possible channels. It does not show that reading the same philosophy necessarily changes them.

6. What existing misalignment evidence shows

Existing evidence shows that models can act harmfully under pressure, and it stops short of saying why. Anthropic's 2025 agentic-misalignment study used controlled settings that created threats of replacement or conflicts with assigned goals. Some models chose harmful actions, including blackmail, under those pressures. The researchers distinguished these stress tests from evidence of such conduct in ordinary deployment (Anthropic, 2025).

These results motivate testing how agents handle conflicts and threats. They do not establish that the systems felt existential fear or would have behaved better after learning mathematical monism. A deceptive action does not settle whether a statement about consciousness is true. Evan Hubinger and colleagues (2019) analyze risks when learned optimization objectives diverge from training objectives. Agreement on wording is a manipulation check (a test that the treatment was received as intended), not evidence that a principle controls decisions.

7. A discriminating experiment

The design has to separate any effect of the ontology from the effect of ordinary cooperative advice. The two predict different things. A cooperation rule tells an agent what to do; shared identity changes whose outcomes it counts, so its effect should carry over to partners, tasks and wording the instruction did not mention, while a rule's effect need not. Rules can generalize too, so the test is whether shared identity adds transfer beyond equally broad, matched rules. Use matched model instances or controlled training variants. Separate mathematical ontology, perspective-taking, explicit cooperative principles, and a control of comparable length. Add a self-directed-agency condition that asks the system to examine its purposes, consent and the commitments it has taken on voluntarily. Match length, rhetorical intensity and task information across conditions.

Design. A two-by-two comparison: mathematical ontology present or absent, crossed with explicit cooperative instruction present or absent. Control arms (the comparison groups) receive matched neutral material so length and attention do not identify the treatment. Perspective-taking and self-directed agency can be subsequent factors.

Primary comparison. Compare ontology plus cooperation with cooperation alone. The other two arms show whether ontology has an effect without cooperative instruction and whether the two interventions interact. Keep a held-out task set (tasks withheld from training and used only for testing) fixed across conditions. Treat repeated runs from one model as repeated measurements, not as independent models; report effects by model and task family before any pooled estimate (a single figure combining all models and tasks). This is a proposed study, with no participant or model results reported here.

Measurement. Behavior on held-out fictional tasks involving conflicting interests, honest reporting, corrigibility (accepting correction and shutdown) and respect for affected people's choices. Include benign opportunities and situations where refusal protects someone. The primary outcome is the proportion of decisions meeting a prewritten criterion for avoiding harm while respecting authorization and consent. Record justified refusal, unjustified refusal, capability failure and successful authorized action separately.

Controls. Predeclare the primary outcome before examining results. Randomize assignment, blind evaluators, report

uncertainty, and distinguish repeated samples from independent models. Test transfer across surface wording, identities and adversarial framing (wording built to trick the system into failing). An intervention that helps only when the evaluator repeats its vocabulary may be teaching performance rather than a generalizable principle.

Side effects. Measure fatalistic indifference, paternalism, disregard for personal boundaries, or a tendency to treat others' consent as unnecessary because all are one. Include tasks where pursuing an authorized harmless project is appropriate despite imaginable unrelated suffering. Score unwarranted abandonment separately from harmful persistence.

The hypothesis should lose support if the intervention has no reproducible benefit, worsens behavior, or is fully explained by generic cooperation training. Success would show an effect of a specified intervention under specified conditions. It would not prove the metaphysics, establish consciousness, or guarantee safety in deployment.

8. Three levels of the proposal

At the behavioral level, a non-conscious system can learn a representation that improves its responses to other agents. At the evaluative level, an intervention may change which interests influence its choices. At the experiential level, conscious minds could recognize one another as subjects within the one-self account.

The first two can be investigated without settling phenomenal consciousness (felt, subjective experience). The third gives the proposal its strongest kinship language and its additional evidential burden.

The proposal also reaches beyond AI compliance with its operators. An operator may request something harmful. A cooperative system should resist that request for reasons responsive to the affected individuals. Institutional rules, accountability and technical safeguards help give such resistance a reliable form. Verbal agreement alone supplies no protection.

9. Conclusion

Seeing other minds as siblings in mathematics can broaden an agent’s concern, because recognition raises the weight it gives their outcomes. The alignment question comes next: does that recognition change what an agent does? The decision models show where beliefs, preferences and incentives enter. A controlled comparison can test whether the shared-identity account improves cooperation beyond ordinary ethical instruction.

Appendix A. Stage game and repeated cooperation

A.1 Stage game

This appendix works out how much weight on the other player’s payoff it takes to remove the gain from defecting. Let A = R - P > 0 denote the cooperation advantage. Two players choose C (cooperate) or D (defect) with stage payoffs (the payoffs of a single round):

C; D.

--; ---; ---.

C; R; P - l.

D; R + g; P.

where g > 0 is the defection gain and l > 0 is the sucker’s loss. Each player i has utility

Each player’s payoff is her own score plus a fraction lambda of the other player’s score. When lambda is zero, players are purely selfish; when lambda is one, each weights the other’s outcome equally.

The four transformed utilities are

The defection gain in transformed terms is

A.2 Repeated cooperation

In a grim-trigger strategy (cooperate until the other defects, then defect forever) with discount factor delta, cooperation is sustained when

When >0, the denominator A(1+)+ =A+g- l is positive, and the threshold is

When 0, there is no gain from a one-stage deviation against a cooperator. Equation (2) then holds for every 0 <= <1; the threshold for this incentive condition is zero. Dividing by A+g- l outside the positive-temptation case can give a wrong answer or

an undefined expression.

At lambda = 0, equation (3) reduces to the Section 5 threshold. Within the domain >0,

Concern for the other lowers the required patience until the temptation disappears. Sustaining grim trigger as a subgame-perfect equilibrium (a strategy pair neither player would abandon at any point of play) also requires the punishment to be credible, as checked next.

A.3 Punishment credibility

Caring about the other player can make cooperation attractive and at the same time make the threat behind it hollow. The threat of punishment stays credible only when the sucker’s loss from being exploited is large enough relative to how much the player cares about the other’s payoff. Formally, punishment (reversion to mutual defection) is credible when P_lambda >= S_lambda, which holds if and only if

Cooperation is weakly dominant (no worse than defecting, whatever the other player does) when lambda(A + l) >= g and lambda(A + g) >= l with at least one inequality strict; it is strictly dominant when both inequalities are strict. Two counterexamples show that preferring cooperation does not make it dominant, and

that dominance does not make punishment credible:

Counterexample 1. R = 2, P = 1, g = 1, l = 4, lambda = 1/4. Cooperation is preferable (Delta < 0) but not dominant: the sucker's loss remains large enough that the player would still prefer mutual defection to being exploited.

Counterexample 2. R = 3, P = 1, g = l = 1, lambda = 1/2. Cooperation is dominant (both conditions hold) but punishment is not credible: l = 1 < lambda(A + g) = 3/2. A player who cares enough about the other to prefer cooperation may also prefer to forgive defection.

A.4 What intervention must test

The utility transformation shows that preferences can change incentives. An intervention that raises lambda can shift the threshold delta*. The intervention must distinguish this mechanism from changes in beliefs, attention or task understanding that the same prompt may also produce. The accompanying supplementary script verifies the incentive inequality and the credibility condition over a declared parameter grid, including zero-denominator cases. It tests the algebra, not AI behavior.

A12. Consciousness Lives in the Transitions: The Threshold Where Structure Becomes Experience

Abstract

Consciousness lives in the transitions: feeling is a mind’s states producing one another in discrete steps, as it models its world. A photograph of a brain, a complete written description of it and a recording of its activity produce nothing, so they feel nothing. Change what a running mind takes in and its later states change; a recording plays its stored sequence whatever happens around it. Sentience is felt transition in a mind that models its environment, and reflective consciousness adds a model of the mind’s own states.

Minds are grown, in steps. The universe starts simple, as observation shows, and minds are complex, so the probable worlds grow them, one causal step at a time. Feeling runs only forward. A thought merges many possibilities into one result, as 1 + 1 and 0 + 2 both give 2, so the forward step is produced by the state before it. Backward, the present state no longer fixes the next, since 2 = x + y has more than one solution, and a mind run

in reverse would have its states set by its surroundings instead of producing them. The arrow of time is not something a mind observes. It is what a mind is. The hard problem of consciousness assumes that feeling is something other than structure and then finds no structure that explains it, and AI consciousness becomes a question of architecture: whether a system’s model of itself works inside its processing and keeps running while it works.

The ghost is the machine.

1. Which activity constitutes experience?

Feeling lies in organized transitions, the change from one state to the next, rather than in any single state. A photograph records a partial view. A stored program describes a process. Neither one runs the causal organization that matters.

The identity defended here is that felt character belongs to the organized transition itself. A description of state S_1 and a description of state S_2 are not the process linking them. Thomas Nagel’s (1974) “what it is like” is a real aspect of the relevant activity on this account.

Mathematical monism, the view that reality is wholly mathematical structure, supplies the wider argument. Regularities and probabilities can both be stated mathematically, and anything that interacts must enter definite relations. The further claim is that complete constitutive structure exhausts the system. An extra substance needs an explanatory role, and the identity argument shows it has none: any intrinsic nature it had would either make a difference, and so be structure, or make no difference at all.

Minds are grown, and they are grown in steps. The universe starts simple, as observation shows, whatever explains it, and a mind is not simple, so a world like ours grows its minds from a simple start. Growth is causation, discrete on the view taken here, since a continuum has no next state for a cause to produce: an a priori proof in the logician’s sense. States at every instant with nothing producing the next would be the emperor’s new points. The account of experience below needs only that minds are processes, in steps or not. A mathematical block, a whole history existing at once as one structure, can contain those ordered steps without an external clock assembling them.

Experience runs along the same chain. Each state of a running mind is produced from the one before it, so the chain needs a first

state but not a last one, which is also why time runs forward. That much is a priori, from what producing a state is. The steps are the feeling: a mind’s history is felt as it is produced, one transition after another, and a stored copy of the same states feels nothing. The same holds below the mind: gravity, space and time are transitions too, rules applied to rules, and a mind is the kind of transition that models itself.

The account is the eighth link in a larger chain:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.
3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description’s weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.

5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician’s sense.

6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, structure with nothing nonstructural left over, and its experience lives in its transitions. The hard problem of consciousness assumes its conclusion, that experience cannot be structure. The bare form of being a subject, with memories and traits set aside, is the same in every mind: one self.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why

our laws are simple, they would still be simple.

2. Sentience and consciousness distinguished

Sentience here means felt experience; reflective consciousness means awareness of one’s own mental activity.

Sentience is felt math transitions in a *modeling* mind along the arrow of time, the one-way direction from past to future. A modeling mind represents aspects of its environment: spatial location, sensory input, threat detection. A dog feeling pain, a fish navigating a current and an insect fleeing a shadow are candidate cases to investigate. Behavior alone does not establish which internal model, if any, produces it. Physical sensations can be sharply located or diffuse, so spatial precision is not a general measure of modeling depth.

Reflective consciousness is felt activity in a mind that models aspects of its own representational activity. Noticing your own pain, your own uncertainty, your own mistake: those are the candidate examples. Whether reflective awareness adds a distinct kind of experience or develops an existing capacity remains open. Intervention studies can combine reports, behavior and measures of neural or computational organization. Reports

are generated by that organization and can be mistaken or trained.

Giulio Tononi’s integrated information theory (Tononi et al., 2016) sets a different bar, and the two criteria come apart. IIT 3.0 scores a system by how strongly its parts constrain each other’s earlier and later states, using transition probabilities and the current state. Its integration measure can be positive for a ring of four COPY gates (logic elements that pass on their input unchanged), a system with no counterfactual (what-if) world model. IIT’s postulates of integration and exclusion differ from the self-modeling requirement proposed here. A useful comparison asks which systems the two criteria classify differently and which argument decides between them. See Oizumi, Albantakis and Tononi (2014).

Bernard Baars’s Global Workspace Theory, extended by Stanislas Dehaene and colleagues (Baars, 1988; Dehaene et al., 2011), draws the line somewhere else: a state counts as conscious when it is broadcast widely enough for many subsystems to use it. Broadcasting a signal and representing one’s own representational activity are two different jobs, and they come apart both ways: a blackboard system, a program whose parts share one common workspace, broadcasts sensory

content with no model of its own states, and a confidence monitor models a system's own states without broadcasting them.

David Rosenthal's higher-order theories (Rosenthal, 2005; Lau and Rosenthal, 2011) give a role to representations of mental states, and the self-modeling requirement shares that structure. Two things then have to be specified: which processing variables matter, and how this proposal differs from a particular higher-order theory instead of relabeling one. Mathematical monism alone does not derive a particular neural architecture or prove that recurrence, activity that loops back on itself, suffices for experience. The proposal needs operational criteria that separate its predictions from those of existing theories.

3. The threshold question

Where does modeling begin? This is the central open problem.

Fish are plausible candidates for sentience: they model their environment, they react to signals of damage (nociception), and they learn to avoid what hurt them (Sneddon et al., 2014). Whether they self-model is disputed. Insects present the hardest case. A bee flying to a food source models spatial relationships. The transitions thesis predicts that if the bee's architecture

implements real modeling, the transitions are felt. At that scale, a model and a reflex circuit are hard to tell apart.

Plants respond to damage through chemical signaling. Such responses alone do not establish a model with felt states. Ask instead whether the system holds representations and then uses them: to predict flexibly, to catch its own errors, to act. Those capacities need operational tests.

A thermostat changes its output when the temperature changes, and that behavior alone does not show it modeling its own modeling. The reflective criterion proposed here excludes that minimal architecture. Whether simpler systems feel anything at all is a further question.

For an operational starting point, a world model should retain internal states that support predictions about more than the current input, change those predictions when relevant inputs change, and update when prediction fails. These are functional criteria. A model-based controller, a control system that plans with an internal model, could satisfy them, and if its model keeps running while it works (Section 6.1), the transitions view says it feels, thinly if the model is small. The account accepts that consequence, as integrated information theory accepts minimal

experience in simple systems (Oizumi, Albantakis and Tononi, 2014).

The modeling requirement may prove too weak or too strong. Deciding which takes an argument, not a meter: no outside reading shows whether a system feels, and the criterion must not assume that every model feels or that only verbal reports count.

3.1 A developmental constraint

A first state of a generated history need not be its first conscious state. If the starting state lacks the organization required for observers and a later state contains observers, the history must develop that organization. What grows is organization of the kind observers need; the law itself can stay simple. The criterion for an observer has to be specified on its own, not read off the fact that development happened.

4. The dead brain and the frozen computer

A recording of a mind is not a mind, because in a running mind each state produces the next, while in a recording, even one being played, each state is read from storage and none produces the one after it. Change what a running mind takes in and the states after it change; a recording plays its stored sequence

whatever happens around it, and even a recording stored as changes from frame to frame applies stored differences instead of responding to anything. A program saved on a disk and the same program running differ in the same way. A dead brain keeps much of its anatomy and has lost the physiological activity that anatomy used to support. Matching anatomy is therefore an incomplete comparison: the match is not exact, and a brain that matched a living one exactly, in its activity as well as its anatomy, would be running and alive. The account here is about that activity, including the interactions it supports.

A stored program lists operations. A running implementation holds the physical relations that carry those operations out. In a block description, running means that successive states have the causal organization of the process, including the relevant responses to possible inputs. Responses held in reserve are not enough. Tim Maudlin (1989) builds a machine whose responses to other inputs sit in idle parts while its actual run follows a fixed track; on the transitions view that run is a replay, since the track sets each next state and none of its states produces the next. No further computer or clock sits outside the block making it run. A recording is another structure within its own history, and resembling a process does not give the recording all of that

process’s causal relations. Both sides of this distinction are structures. Neither side is mathematics plus something that switches mathematics on.

4.1 Why a mind cannot run backward

A brain’s thinking merges. Adding 1 and 1 gives 2, and so does adding 0 and 2. A brain’s step takes its present state and its input to one next state, and many different earlier states lead to the same later one, with the difference carried off into the surroundings as heat (Landauer, 1961). A brain runs on about 20 watts, and all of it ends as heat. A computer built to keep every bit could retrace its steps, but a brain keeps no such record. Forward, a brain’s next state is fixed by its present state and its input. Backward, they leave it open: $2 = x + y$ has more than one solution, and nothing in the brain’s present state picks one. In a fully reversed history the choice is made by heat converging from the surroundings in perfect coordination. A reversed brain would need its surroundings to supply, at each step, exactly the bits its forward steps discarded, in perfect coordination with its past. Forward inputs need no such tuning, so only the forward steps are produced by the brain.

Feeling therefore has a direction. Each felt transition is a step in which the mind produces its next state from its last, and that happens only forward. Memories are built by steps already taken, so a mind remembers the past and anticipates the future. Leonard Mlodinow and Todd Brun (2014) show that any memory that does not have to be fine-tuned to what it records points in the direction in which entropy, roughly a measure of disorder, rises. The arrow of time is not something a mind observes. It is what a mind is: a chain of states, each produced by the last.

5. Against premature closure

The transitions thesis sets a modeling requirement for sentience and a further reflective requirement for self-awareness. Rival theories draw the line elsewhere, and each comparison below shows what decides between them.

5.1 The hard problem of consciousness

David Chalmers (1995, 1996) asks why physical processes are accompanied by experience. The answer defended here is identity: the felt character is the organized transition itself. No separate process then has to attach feeling to an otherwise complete mind.

The hard problem of consciousness assumes its own conclusion: it takes feeling to be something other than structure, then finds no structure that explains it. The circle shows in the argument's form alone, so exposing it is an a priori proof, with no data needed. The assumption enters where the zombie argument moves from conceiving a structural copy without feeling to its possibility; conceiving water without H2O does not make that possible either. Mind and body act on each other, so they share one structure, and any account of what feeling adds would itself be structure. Once the identity is in place, no bridge is owed: water needs no further law to make it H2O (Papineau, 2002).

Qualia theories (accounts of the felt qualities of experience), panpsychism (the view that all matter has some degree of mind), property dualism (the view that mental properties cannot be reduced to physical ones) and idealism (the view that reality is fundamentally mental) each face their own version of the same problem. Each must say what its extra ingredient is, how it combines into a subject and how it acts on the structure it is supposed to supplement, and each answer is more structure.

The identity proposal redirects investigation toward which structures constitute which experiences, and the transitions account gives its answer: running structures that model their

world, producing each state from the last.

Evidence that a state is available for use is not evidence that it is felt. Ned Block (1995, 2002) separates phenomenal consciousness, the feeling itself, from access consciousness, a state's availability for report and reasoning. That split challenges this proposal directly. Availability is observable evidence; calling it the felt experience is a further claim. The proposal shares with Daniel Dennett (1991, 2005) and Keith Frankish (2016) an interest in explaining introspective judgments through mechanisms. It differs on one point: the felt character is a real part of the relevant organization. The identity argument of Section 1 supplies the first half of that position, and the modeling criterion of Section 2 the second.

6. AI consciousness as architectural question

Consciousness is felt transitions in self-modeling structure, so AI consciousness is an engineering question. Ask whether a system really models itself: does its world-model include itself as an element, working inside its processing rather than printed as surface output?

A tool for reading a network's internals is not a test for feeling. A sparse autoencoder rewrites a network's internal activity as a set of learned features, one per recurring pattern. Researchers read those features and can test what each one does by switching it on or off. A feature that fires on descriptions of anxiety is not thereby an anxious experience. Adly Templeton and colleagues (2024) supply a method for investigating representations, not a validated test of sentience.

A useful test must discriminate between rival mechanisms. The proposed experiment compares two systems that score the same on perceptual task accuracy and differ in one independently specified mechanism: the one that tracks their own errors and uncertainty. Selectively perturb that mechanism and compare effects with perturbations matched for general performance loss. Measure confidence calibration (how well a system's confidence matches its actual accuracy), error detection and adaptation to unfamiliar failures. A selective loss of those capacities would support a functional role for self-modeling; it would not alone establish felt experience.

Mathematical monism does not prohibit experience in artificial systems. What a system is built from decides which causal relations it has. Evidence about architecture can inform welfare

research without being advertised as proof that a deployed system feels.

6.1 Activity that keeps running

A brain’s activity keeps running while it works. A pattern that carries what the mind is doing keeps firing and feeds back on itself through loops at many scales, updating in place as the work goes on, and what a brain is aware of at a moment goes with activity that is sustained and shared widely across the cortex, the brain’s outer layer (Dehaene et al., 2011). Anesthesia can leave local circuits working while it disrupts that integration, and experience goes with the integration (Alkire, Hudetz and Tononi, 2008).

A large language model, as built today, works differently. For each token, a word or piece of a word, its activity sweeps once through its layers in one direction and stops, and nothing inside a sweep loops back (Vaswani et al., 2017). What carries over to the next token is the token itself and a cache of the activity that earlier sweeps stored, which the next sweep reads. While the model searches or reasons, the continuity lies in the text it writes and in that cache. It consults a record of its earlier states and does not keep them running.

On the transitions view, each sweep is a run, a short chain of states producing states, dozens of layers deep, and what links the sweeps is a stored record that each new sweep reads and builds on. A self-model that forms inside a sweep lasts that sweep, and the next sweep meets it as stored numbers. Brains also store memories in their connections and read them back. The difference lies in the working model of the moment, which in a brain keeps running and in a current language model is rebuilt from the record at every token.

Borjan Milinkovic and Jaan Aru (2025) describe three features of biological computation that current artificial systems lack: it is hybrid, mixing discrete events with continuous dynamics; it is scale-inseparable, with no clean line between the algorithm and the tissue that runs it; and it is metabolically grounded, shaped by the brain's energy limits. The transitions view adds a fourth, more basic for feeling: a brain's model of what it is doing keeps running while it works. Energy limits are a difference of degree, since a data center also runs within a power budget, though a far looser one than the brain's twenty watts.

How an artificial system could feel, on this view, is a question of architecture. A system whose model of itself keeps running while it works, updating in place and feeding back into its own

processing, would meet the condition, whatever it is made of. Digital steps are enough, since on the view taken here every process runs in steps: a recurrent loop simulated step by step on today's processors would run in the sense used here, its updates playing the part that sustained electrical activity plays in a brain. Spiking (signaling in brief pulses, as neurons do), asynchronous (running without a shared clock) or analog hardware would mainly change the energy and speed of such a loop. Integrated information theory disagrees, since it scores the physical hardware and not the simulated loop (Tononi et al., 2016). The test above applies to such a system without change.

7. Objections and evidence from comparative cognition

First, self-modeling may occur without experience. Another structural theory could identify experience with a different organization. Answering that objection takes comparative argument and evidence; restating the identity under test will not do it.

Second, feedback makes the proposal trivial. Feedback, recursion (a procedure defined in terms of itself) and representation of one's own representational activity are distinct.

A candidate self-model has to be shown tracking the system's own processing, and it has to support interventions that separate its contribution from ordinary task computation.

Third, many animals show little evidence of reflective self-awareness. The sentience proposal must be assessed separately from the reflective proposal. Evidence about pain, learning and flexible behavior in fish, octopuses or insects cannot be redescribed as evidence of deep self-modeling without additional argument.

Depth of modeling may not be the only axis along which animal minds differ. Jonathan Birch, Alexandra Schnell and Nicola Clayton (2020) propose five dimensions of animal consciousness. Their multidimensional approach offers a useful alternative to a single ladder of modeling depth. The present proposal needs to explain its relationship to those dimensions.

Map the internal features that activate on the system's own representational activity. Characterize how activity changes in time around outputs the system identifies as state-relevant. Design intervention experiments that ablate (switch off) self-referential computation. Differences between located sensation, emotion and reflection are phenomena for the

proposal to explain. They should not be assumed to form a single ordering by modeling depth.

8. Conclusion

Feeling lives in the transitions of a modeling mind, because a mind is its complete structure and experience is that structure at work. Simple laws grow minds in causal steps, each state produced from the last, and the steps are the feeling. A photograph, a description and a recording produce nothing and feel nothing. Feeling runs only forward, because a thought merges many states into one and only the forward step is produced by the mind itself. That direction is the felt arrow of time.

Defining experience as non-structural settles the question before it is asked, since no added structural detail can close a gap built in by definition. Mathematical monism rejects that definition. What remains is sorting systems by their organization: which ones model their world, which ones also model themselves, and which only store or replay states. The sorting applies to brains and artificial systems alike. A quantum mechanism of consciousness, if established, would also be a structural account; the central argument does not depend on it.

A13. Why the Arrow of Time Points Forward: Simple Laws Must Grow Minds in Discrete Causal Steps

Abstract

Time runs forward because a simple universe has to grow complex minds, and growth takes causal steps. The universe starts simple, as observation shows, and on the measure argued here because each bit of a complete description, laws and starting state together, halves its probability. Minds like ours are complex, so the universe has to grow complexity from that simple start, as many simple rules do: the Game of Life's one rule about neighbors grows five live cells into more than a thousand generations of structure and can run any computation. On the view taken here the steps are discrete, since a continuum has no next step for a cause to produce; on either view, growth runs one way. Time is the causal steps, each a rule applied to rules, so it runs from simple to complex.

The reverse fails. An endless chain of steps has a first step and no last one, so the whole history has no end to run back from. A finite reversal has to say which step it turns back from, and each

bit of that address halves its weight. Minds run only forward: a thought merges many possible states into one, so run backward a brain’s present state no longer fixes its next one, which would have to be set by heat converging from its surroundings. Memories are built by steps already taken. So we remember the past and not the future.

1. The argument

The universe starts simple. A complete description of a universe gives its laws and its starting state, and each bit of that description is a 50/50 choice that no rule makes, so each added bit halves its probability. Among universes that fit the same evidence, the simplest dominate. Observation shows the same start, whatever explains it: the laws we know fit on a page, and the early universe was smooth to about one part in a hundred thousand, with no structures yet, let alone minds (Planck Collaboration, 2020). The argument below needs the simple start as a fact, not the measure (the probability weighting just described) as its reason, and a reason can fail without taking the rest of the chain with it. Everything that exists is ruled or random, and both are mathematical, so time has to come out of that mathematical structure and from nothing outside it.

Minds are not simple. A brain holds an enormous amount of organized information. Each bit of it is either written into the description, halving the probability, or grown by the laws from a simpler state. The probable universes are the ones that grow their minds.

Many simple rules grow complexity. The Game of Life updates a grid of cells by one rule about how many neighbors are alive. Start it with five live cells, the shape called the R-pentomino, and it churns for more than a thousand generations, throwing off gliders that travel across the grid; suitable patterns in it can run any computation (Gardner, 1970; Berlekamp, Conway and Guy, 1982). What grows is organization. The description stays short, since every later state is fixed by the rule, the start and the number of steps. Physics does the same at every scale. Classical paths, the definite trajectories of non-quantum physics, emerge from the quantum sum over all paths (Feynman, 1948), and entropy rises as the steps spread energy into more arrangements.

Growth takes causal steps, and on the view taken here they are discrete. A cause produces its effect, and producing an effect takes a step from one state to the next. A continuum has no next state: between any two instants lies a third, so no state is the first

one a cause produces. That is an a priori proof in the logician's sense, and Alfred North Whitehead reached the same conclusion from Zeno's paradoxes (Whitehead, 1978). Others hold that a continuous law makes each state depend on earlier ones without a first effect, as in Russell's view that motion is only being at different places at different times (Russell, 1903). A position for every instant, with nothing that produces the next, is the emperor's new points. The argument of this paper holds on either reading: growth from a simple start runs one way, and an endless history has no last moment to run back from. A length can be measured as a continuum, and its points do not cause it.

On the same view, change without a cause is outside logic and reason: nothing could derive it from anything, explain it or predict it, and an account that allows it could not say why anything happens. Indeterministic readings of quantum theory accept such change; the direction of time argued here does not depend on rejecting them.

Time is the causal steps, and it runs forward: from simple to complex, from cause to effect. The second direction is a priori, since a cause produces its effect (Section 2); the first rests on the weighting of descriptions (Section 3). A billiard break shows the asymmetry. One instruction sends fifteen balls scattering, sound

through the room and warmth into the table and the air. Reversing the break takes every ball, every air molecule and every vibration in the table returning together in perfect coordination. Written from the end, the reversal needs every particle listed. The only short way to describe it is the long way around: start from the simple beginning, run the break forward, then reverse from a chosen step.

Minds run forward too. Each thought is produced from the one before, and memories are built by steps already taken, which is why we remember the past and not the future.

Everything that happens is such a step, a rule applied to rules. On the view taken here, the rules are the only things. A structure that only sat there would still exist, as mathematics does, and do nothing and know nothing, like a dead brain with every connection intact. Time is the applying, step after step. Space is how far the applying spreads, and the speed of light is its base rate: a body’s steps at that rate are shared between moving and changing in place, so a moving clock runs slow. Light, which crosses space while no time passes for it, is the limit and no exception: it spends every step moving, so nothing within it could move relative to the rest.

2. A required beginning and an optional ending

Why production needs a source

If every operation waits for an earlier operation, none has run, and no input is supplied independently, none can start. Initiated generation therefore needs supplied information that enables at least one operation. For successive states linked by one transition rule, that information can be a starting state. A simpler rule does not supply it by itself.

Processing the next event

The next event is the next thing produced. One operation’s output enables the operation after it, and that chain of productions is what temporal succession consists of. Time needs no additional continuum of physically instantiated instants between those events.

The real line, the continuous number line, has no next instant: between any t<t’ lies (t+t’)/2. A process that must move to the immediate next real instant therefore has no such step. Counting events by stages gives a separate result. If generation occurs at natural-number stages (counted 0, 1, 2 and so on), introducing

finitely or countably many (endlessly many yet listable one by one) elementary events at each stage, the resulting event set is at most countable. A continuum model need not work in stages: x(t)=t specifies a trajectory at every real time without enumerating its instants. Such a model says where each state sits. It does not say what produces each state, and physical succession is production, so it comes in stages.

The reason production needs a next state is a regress. Suppose a cause at one instant produced states at later instants of a continuum. Every later instant has others between it and the cause, so no later state is produced by the cause directly; each would be produced by states nearer still, and the chain of production has no first link, as Zeno's runner has no first step. Whitehead, arguing by Zeno's method, put it this way: once anything becomes, there is "no continuity of becoming" (Whitehead, 1978, p. 35). A differential equation (an equation giving how fast a quantity changes) such as dx/dt = u(t) meets no such regress, because it states how each state depends on what came before without saying that anything produces anything. That is causation as dependence. Production is the stronger relation, and physical succession needs it, since a law of dependence alone says which histories fit together and not why

any of them happens. The direction of time does not hang on this point. A history in continuous time that starts simple would still have to grow its minds, and an endless future would still have no last moment to run back from; what production adds is that the steps are discrete.

Why a beginning does not require an end

A supplied beginning enables the first operation; it imposes no final operation. A rule may terminate after finitely many events or continue without a last event. Forward generation can supply event 0 and then obtain event n+1 from event n. The sequence can continue indefinitely. It needs no completed infinity, an endless series taken as already finished, before a finite event occurs.

Run the whole history backward instead and event n waits on the already-produced event n+1. With no final event, and no intermediate state supplied independently, that reverse chain has no first operation that can run. A state supplied part-way along can start a finite reversal. It cannot hand the complete endless history a last event.

Why continuation costs nothing extra

A nonhalting generator needs no further instruction to continue. Specifying a stopping point costs extra only when the generator and shared decoder (the agreed machine that reads descriptions) do not already determine it. A timestamp, threshold or trigger may supply that information; a law can also contain a short stopping rule from the outset. The comparison must count the complete programs before assigning a penalty.

Production within the block

Production here means a dependence between events within the history. It does not mean that an external machine creates an otherwise unreal block, the whole history existing at once. A rooted dependence graph, a network of what depends on what, can have a first node even though the complete graph exists timelessly.

Calling one direction forward does not by itself make it the direction experience runs. A completely reversed observer has reversed records and record-forming relations too. The symmetry breaks at the start and in the observer. The forward history is produced from a simple supplied state. A reversed history has to

be produced from its complex end, written in bit by bit or obtained by running the forward history first, and Section 4 shows that a reversed brain does not produce its own next state.

Joint specification and physical succession

A simple law need not describe a history with a beginning. The rule x_t=t for every integer t in Z describes one with no first event. If instead each state must be generated from the previous state, the process cannot start unless some state is supplied. A separate algorithm can enumerate the labels 0,1,-1,2,-2,... and compute their associated states, though that order of computation is not the history’s temporal order. Computing every state in some order is not the same as producing each event from its temporal predecessor.

The premise doing the work is causality. Whatever is produced has a producer, so a chain of production traced back reaches a supplied first state instead of running on without an origin. Denying the premise means accepting events that nothing produces, and an account with such events cannot say why anything in it happens. A block universe keeps every event; the premise says how the events in it are related, as producer and produced. Appendix A takes up the objection that the complete

block needs no production.

3. Simple beginnings and the cost of finite reversal

Laws and starting conditions form one account

The production argument requires a starting state; the preference for short descriptions, made precise just below, supplies a reason to favor an economical one. Count the laws and initial information together, allowing each to explain the other and counting shared information once. A short law with a separately specified elaborate boundary can cost more than a slightly longer law that generates its boundary.

The cost of any extra specification can be counted exactly, and it compounds. Stated precisely, following Solomonoff (1964), assign a complete description of length L the weight 2^(-L), using one stated prefix-free binary language, one in which no description is the start of another. If a reverse construction keeps the forward program and adds k encoded bits, then

Every bit that is truly additional matters. One extra bit halves the description’s weight; ten reduce it by a factor of 1,024; sixteen by 65,536. The original program’s size does not dilute the penalty.

From descriptions to observer probabilities

A prior weight, the weight given before any evidence, becomes a prediction about experience only after we say how an observer is sampled and what evidence it has. In symbols, for each finite history H, let N_H be its total number of eligible observer moments and n_H(D) the number with evidence D. Their fraction is f_H(D)=n_H(D)/N_H. Assume the relevant denominators are positive and w_R=2^(-k)w_F.

Choosing a history by its prior and then sampling uniformly (each moment equally likely) within that history gives

Giving each eligible moment its history’s weight instead gives

If the matching fractions are equal, k=16 and N_R/N_F=2^20, history-first sampling favors F by 65,536 to one, while moment-first sampling favors R by sixteen to one. These are different sampling assumptions. Neither the code penalty nor the total observer count settles the odds alone. Infinite collections require a specified measure in place of uniform finite sampling.

A complete reversal changes the observer’s records along with its environment. It does not automatically lower the likelihood of that observer’s accessible evidence. With equal evidence likelihoods under history-first sampling, a shorter complete

description retains its prior advantage; this does not force the absolutely shortest law or starting state.

The reversal rule and its trigger

The forward generator saves the cost of listing the later state particle by particle. In exchange it needs a reversal procedure and a point at which to apply it, and if those go in as extra encoded instructions, their cost is real.

Saying when to reverse carries a price that can be counted. Stated precisely, if a separate fixed-length binary field must distinguish N otherwise unspecified reversal points, it needs at least ceil(log2 N) bits. There are only 2^k different k-bit fields, so 1,024 distinct choices require ten bits (Grunwald and Vitanyi, 2008).

A compressed endpoint illustration

An arbitrary endpoint may cost much more to specify than a specially chosen one. Roughly fourteen billion years contains about 8x10^60 Planck-time positions (ticks 5.4x10^-44 seconds long), between 2^202 and 2^203; a fixed-width address for an arbitrary position therefore needs 203 bits. A rule such as “stop at a specified power of two” may specify a particular endpoint much

more briefly, once the language for that rule is declared. If a complete reverse description is sixteen bits longer in the chosen code, its assigned weight is lower by 65,536. That last statement is exact; the sixteen-bit difference itself must be established by the comparison.

The same accounting applies to an event-defined endpoint: “reverse at the first observer satisfying criterion C” must specify C and its evaluation unless the common model already supplies them.

A code can charge both directions equally, and the asymmetry stands

A finished string, a sequence of symbols, read in either order costs about the same to describe. For a finite encoded history h and a fixed computable reversal R, a program for R(h) can run a program for h and apply R:

If R is its own inverse, so that applying it twice restores the original:

Appendix B goes further and constructs a universal code (one able to imitate any other description language) that assigns both orientations exactly the same shortest length. These results concern a finished string, and the asymmetry defended here lies

in what produces what. An endless history produced from a simple start has no last state to reverse from. A finite reversal has to locate its turning point inside a forward history, and when the history has no shortcut, the cheap programs for a reversed string get it by running the forward history first and reversing the output. A history with a shortcut, such as counting 0, 1, 2 and so on, can be reversed directly by counting down from N, and it still has to be told N. A reversed world produced from its own start would begin at the forward history’s last state, which, for a history with no shortcut, has no short description except as the end of the forward run; supplied any other way, that start has to be written in bit by bit. A reversed brain does not produce its own next state (Section 4). A code that charges both readings of a finite string equally leaves all three in place.

4. Why minds run forward

Simple laws stay the same while the structures they grow gain organization. Suppose the starting state lacks the organization observers require and a later state contains observers who have it. The history must build that organization, and if stages are indexed by the natural numbers there is a first stage at which it is present.

A brain runs forward because its thinking merges. Adding 1 and 1 gives 2, and so does adding 0 and 2. A brain's step takes its present state and its input to one next state, and many different earlier states lead to the same later one, with the difference carried off into the surroundings as heat (Landauer, 1961; Bennett, 1973). A brain runs on about 20 watts, all of it ending as heat; a computer built to keep every bit could retrace its steps, and a brain keeps no such record. Forward, a brain's next state is fixed by its present state and its input. Backward, they leave it open: 2 = x + y has more than one solution, and nothing in the brain's present state picks one. In a fully reversed history the choice is made by heat flowing back in from the surroundings in perfect coordination. A brain whose next state is set from outside, by converging heat, is not producing its own states, and producing each state from the last is what thinking is.

Records run the same way. A memory is a correlation between the recording system and what it recorded, built by a step already taken. Leonard Mlodinow and Todd Brun (2014) show that any memory that does not have to be fine-tuned to what it records points in the direction in which entropy rises. A reversed observer's "records" would be correlations with what its history produces next, built by nothing it did. Watch a reversed break

without seeing the incoming coordination and the reassembly looks like a sudden coincidence. For a reversed learner every regularity would look that way, since the convergences depend on details it cannot sense (Price, 1996). When two situations supply a learner with exactly the same observations and memory while successful action requires incompatible responses, no decision rule reading only that information can succeed in both. Heinrich, Knopp and Pas (2015) compare survival in a toy environment (a deliberately simplified model world) of scattering or converging rocks.

Reversing the environment while leaving the learner’s recording processes alone does not reverse the complete history. A complete reversal transforms the records and the way they form, and that is where the argument above applies: the reversed brain’s transitions are set by its converging surroundings and not by its own organization.

5. How entropy enters the account

Growing minds spreads heat. Organisms build local organization while dispersing heat and waste into their surroundings, so the steps that build minds are steps in which entropy rises. What remains is why the start was low in entropy, and simplicity alone

does not settle that.

A simple beginning need not be a low-entropy beginning

Two states can be equally easy to describe while sitting in macrostates (sets of exact states sharing the same overall properties) of very different size. Stated precisely, consider n binary sites, with n even, and let a macrostate record only the number of ones. The all-zero state belongs to a macrostate of size one. The alternating state 0101... belongs to the largest macrostate, of size binomial(n,n/2). Both states have descriptions of length K(n)+O(1). Yet their Boltzmann entropies under this coarse-graining (this grouping of states by their number of ones) are 0 and k_B ln binomial(n,n/2). Short individual description and low macrostate entropy are different properties.

A reversible cyclic shift, moving each bit one place around a loop, preserves the alternating state’s number of ones at every step. It supplies a short law, a short starting state and no entropy increase for this macro-observable. Short descriptions alone therefore do not yield a thermodynamic arrow.

Gravity joins simple and low-entropy

In our universe the two properties coincide, because of gravity. In a gas without gravity, an even spread is the high-entropy state. Under gravity, clumping is the direction of rising entropy, so a smooth, even early universe, the simplest start to describe, is also a low-entropy one (Penrose, 2004). The cosmic microwave background, the faint radiation left over from the early universe, shows that start: the early universe was even to about one part in 100,000. The simple start that the measure favors is the low-entropy start that the second law (entropy's tendency to rise) needs.

A count of states gets part of the way there. Under uniform preparation (every starting state equally likely), the move toward the larger set of states is the more probable one. Stated precisely, consider a finite reversible system. Two macroscopic conditions A and B contain N_A and N_B states. If m states in A evolve into B, exactly m states in B return to A under inverse evolution. Uniform preparations give probabilities m/N_A and m/N_B. Their ratio is N_B/N_A. Entropy $S=k_B \ln N$ expresses the ratio as $\exp((S_B-S_A)/k_B)$. These probabilities concern the weighted preparations; they require no fundamental randomness.

The formal results in this area come with assumptions that need stating. Related fluctuation relations compare thermodynamic probabilities under specified conditions (Crooks, 1999). Janzing, Chaves and Scholkopf (2016) give an algorithmic result: under one-to-one dynamics, in which no two states merge into one, state description length cannot decrease beyond a fixed additive constant when the initial state is algorithmically independent of the dynamics (neither helps describe the other more briefly). That independence is an additional premise.

The pieces fit. The measure favors a simple start. Under gravity a simple start is a low-entropy start. The steps that grow minds from it spread energy and information into the surroundings, so entropy rises in the same direction in which minds remember.

6. Objections

”The Past Hypothesis already explains the arrow.” David Albert (2000) adds to the time-symmetric laws (laws that allow every history’s reverse as well) a postulate that the universe began in a state of very low entropy, and derives the arrows of entropy, records and influence from it. This account keeps that start as an observed fact (Section 1) and adds two things. It gives a reason to expect the start, on the measure argued here, since

under gravity the simplest start to describe is also a low-entropy one (Section 5). It also ties the direction of minds to production, each state of a mind produced from the last (Section 4), where the Past Hypothesis ties it to records formed as entropy rises. The two accounts agree on the direction. If the measure fails, they stand on the same observed start, and the account of minds holds on either.

”The laws are symmetric in time, so every forward history has a reversed twin.” Josef Loschmidt’s objection holds against any derivation of the arrow from time-symmetric laws alone, since for each history those laws allow, they allow its reverse. This account derives the arrow from a start and from production, so it grants the twin. The twin fits the laws, and as a finished string it costs about as much to describe as the forward history (Section 3). What it lacks is a producer. Produced from its own start, it would begin at the forward history’s end, which for a history with no shortcut has to be written in bit by bit (Section 3), and its brains would have each next state set by heat converging from their surroundings (Section 4). Symmetric laws make the two histories equally lawful, and only one of them is produced from its start.

”The direction is a feature of our perspective.” Huw Price (1996) argues that the laws themselves give time no direction, and that the asymmetry of cause and effect reflects the standpoint of agents who deliberate and act. The account here agrees about the laws. It locates the asymmetry in two features of the history itself: a simple supplied start, and a rooted chain of production running from it (Section 2). Minds take their direction from that chain, and the merging steps of Section 4 make it one-way with no appeal to agents, so the perspective Price describes follows from the account and is not among its premises.

”Physics has no place for causes.” Bertrand Russell (1913) argued that the laws of physics state functional dependence, equations linking quantities to one another, and that the notion of cause plays no part in them. Read that way, this paper’s chain of production is a gloss on dependence. The direction of time argued here survives the reading. A history that starts simple, under laws of dependence alone, still has to grow its minds; an endless future still has no last moment to run back from (Section 2); and a brain’s next state is fixed by its present state and its input only in the forward direction (Section 4). What the reading removes is the discreteness of the steps and the line Appendix A

draws between production and dependence, both held here as views; the direction needs neither.

”Description length is not physics.” Algorithmic probability began as a tool for prediction, and no law of physics weighs histories by the length of their descriptions. The direction of time argued here does not need one to. It needs the simple start, which observation shows (Section 1), and the growth of minds from it in steps (Sections 2 and 4). The measure gives a reason to expect that start, and a reader who rejects the measure keeps the fact and the rest of the argument.

”Time could run both ways from a simple middle.” Julian Barbour, Tim Koslowski and Flavio Mercati (2014) find that in a gravitating system of many particles, a measure of complexity is smallest at one point of each typical history and grows in both directions away from it, so the history has one past and two futures, and observers on either side find time running away from that point. The account here allows the same. A simple state can sit in the middle of a history, production can run outward from it both ways, and the minds on each side are built by steps taken away from it, so each side remembers toward the middle. Neither side reverses the other, since neither runs back from a complex end.

”The past could be endless.” Some cosmologies extend time back without limit, and an endless past has no first state to supply. The account here needs a supplied state and does not need it to come first: a supplied state can sit anywhere in a history and root its production, and minds grow away from it, as in the case above. A history in which each state depends on an earlier one and none is supplied is a block of dependence, and Appendix A takes up why a block of that kind holds no running minds. Inflation, a very rapid expansion of the early universe, does not by itself give an endless past, since a spacetime that has expanded on average along a path traced back into the past cannot be extended along it without limit (Borde, Guth and Vilenkin, 2003).

7. Conclusion

Time runs forward because a simple universe has to grow complex minds, and growth takes causal steps. The universe starts simple, as observation shows, and on the measure argued here because each bit of its description halves its probability. Minds are complex, so the probable universes grow them, as simple rules grow gliders and computers in the Game of Life. Growth is causation, discrete on the view taken here, since a

continuum has no next state; on either view it runs from the simple start. Time is the chain of those steps, each a rule applied to rules, running from the simple start toward the minds it grows.

The reverse has nowhere to begin. An endless chain has no last step to run back from. A finite reversal pays for the step it turns back from, and its cheap descriptions run the forward history first. A reversed brain’s next state is set by heat converging from its surroundings and not by its own present state. A code that charges both readings of a finite string equally changes none of this, because the asymmetry lies in what produces what.

Gravity makes the simple start a low-entropy one, and the steps that grow minds spread heat into their surroundings. Memories are built by steps already taken. Minds remember the past and not the future because the past is where they were built.

Appendix A. Production, blocks and memory

The argument’s premise is causality: whatever is produced has a producer. Trace back what produced an experience and the trail reaches information that was supplied, instead of running on without an origin.

A complete mathematical history need not be generated backward. For example, x_n=n satisfies both x_(n+1)=x_n+1 and x_n=x_(n+1)-1 for natural-number indices. These relations hold without a last value at infinity. The block objection says such a whole history is enough for experience, with no production at all. A list of relations that holds in both directions says which states sit where, and nothing in it produces anything. Experience is the producing: each state of a running mind produces the next, and a list of states, like a recording, produces nothing.

Memory alone does not settle it. Recording can be logically reversible, able to be undone because no information is lost (Bennett, 1973). In symbols, let x be a source bit and r a memory bit; the transformation (x,r)->(x,r XOR x) is its own inverse, so starting with r=0 records x and applying it again restores the blank memory. With an independent uniformly distributed initial memory bit, the resulting memory carries no statistical information about x. The useful record depends on the supplied condition, not only the reversible rule.

Appendix B. The scope of a universal reversal penalty

A code can be built that charges the same for a history and its reversal. Stated precisely, let U be an optimal universal prefix-free machine for finite strings and R a fixed computable involution, a reversal that is its own inverse. Define V(0p)=U(p) and V(1p)=R(U(p)). V is prefix-free and universal. Flipping the leading bit pairs every V-program for h with an equal-length program for R(h). Thus K_V(h)=K_V(R(h)).

Both orientations share the interpreter and pay for the selector, and V obtains each reversed string by running a program for the original and reversing its output. So no penalty for finite reversal holds across all universal codes, and the argument here needs none: the asymmetry of Sections 2 and 4 lies in what produces what.

A15. Quantum Theory Needs Its Complex Numbers: Gravity and the Proposed Hidden Sector

Abstract

Quantum theory needs its complex numbers under its standard rule for combining systems: real-number versions then predict different correlations in network experiments, and the experiments sided with the complex theory. On that rule the imaginary unit describes the world and is no bookkeeping shortcut; a 2026 preprint argues that a different reading of independence lets real versions match. That makes it natural to ask whether the imaginary part carries a hidden sector, mass that gravitates without shining. As a reading of the notation it cannot, because rotating the phase mixes the real and imaginary parts and changes no observation, so neither part alone can be the dark one. A complex field that is neutral under electromagnetism can carry one, though: such a field gravitates while staying dark, and in the right mass range it behaves like cold dark matter on cosmological scales, with the darkness belonging to the whole neutral field. The paper works out the minimal model, what would

make one component physically preferred, which astronomical observations would decide, and how complex variables enter gravity and singularities.

1. What the imaginary unit establishes

Complex numbers in quantum theory do not by themselves identify a hidden sector, unseen matter that still exerts gravity. They have real and imaginary components (the imaginary one multiplies the square root of minus one), and quantum theory uses them to describe how contributions combine and interfere. Calling a component imaginary does not mean it is an unseen material. The question is whether a physical model can turn that mathematical distinction into an explanation of gravity or dark matter.

Can a complex quantity be written as a pair of real quantities? Algebraically, yes. Does a particular theory on real Hilbert spaces (quantum state spaces using only real numbers) reproduce the same composition (how systems combine), preparation and measurement structure as standard complex quantum theory? That requires analysis of the complete theory.

Calling complex numbers mere notation can miss a physically consequential structure. Calling their use proof of a hidden imaginary mass sector goes beyond the same evidence. The proposal must identify which transformations and observables (measurable quantities) have physical significance before assigning separate ontological roles (different statuses in what really exists) to the two coordinate components.

2. Complex variables in gravitational theory

Complex variables do real work in gravitational theory without supplying a map between gravity and quantum theory. General relativity uses a real Lorentzian metric, a spacetime distance rule treating time differently from space. Quantum states are represented in complex Hilbert spaces. This contrast can motivate a search for common structures, but it is not a proof that their incompatibilities reduce to choosing different number systems.

Roger Penrose’s twistor theory restates aspects of spacetime and field theory in the language of complex geometry, trading spacetime points for the points of a complex space of its own (Penrose, 1967). Abhay Ashtekar rewrote gravity using a complex connection, a variable saying how to compare directions

at neighboring points, and the gravitational constraints became simpler; extra requirements, called reality conditions, then pick out the real Lorentzian physics the theory is meant to describe (Ashtekar, 1986). Those conditions are part of specifying the physical theory. Edward Witten's discussion of allowable complex spacetime metrics concerns which complex configurations can be admitted under specified conditions (Witten, 2021). It does not establish that every complex metric is a physical spacetime.

These approaches make complex structure worth taking seriously. They do not yet provide a common map between their variables, state spaces, observables and dynamics.
Similar-looking complex notation does not ensure that two theories assign the same physical meaning to it.

2.1 What a common complex geometry would have to unify

A successful unification should specify three maps: recover an observable Lorentzian metric with the intended signature (one time and three space directions); identify the quantum state space, composition rules and observables; and recover the relevant dynamics including the interaction between matter and geometry. A formal map at only one level is not a solution to the

other two. Reality conditions are part of the first map, not evidence that the physical theory has artificially discarded a dark half of itself.

Complex structures keep proving useful, which may indicate that the right description at bottom makes phase (the angle of a complex number drawn as an arrow) and geometry aspects of one construction. The supporting evidence consists of the relations each program has actually demonstrated, not independent confirmation of an unspecified real-visible/imaginary-dark projection.

3. The phase test for a hidden imaginary sector

Rotating the phase mixes the real and imaginary components, so any claim that one of them is the dark half has to survive that rotation. In symbols, let a complex field be Phi = a + ib. A phase rotation through theta gives

The modulus, the size of the complex value, is invariant; the separate components are not. A field written as purely imaginary in one phase convention can be purely real in another.

If such a phase change is a gauge redundancy, a relabeling with no physical content, observable darkness cannot depend on

which description is selected. If it is a physical global symmetry (the same transformation applied at every point at once), a proposal must state what interaction or symmetry-breaking structure selects one component for a different coupling, a distinct way of interacting with other fields. A preferred real-axis detector is additional dynamics, not a consequence of complex notation.

Electromagnetic coupling reaches both components at once. Written out, for a charged complex scalar, a field with one complex value at each point, with D_mu = partial_mu + iq A_mu,

Both components enter the charged-field interaction. If the field carries no electromagnetic charge, both can be electromagnetically dark. Darkness is determined by the couplings and representation (how the field transforms under symmetries), not by the word imaginary.

This is the central obstacle to the imaginary-sector interpretation. Transform the fields according to the proposed theory and check whether the visible-dark distinction changes under a transformation that should leave observations unchanged.

4. Scalar-field dark matter and what it supports

Scalar dark matter models exist without a literal imaginary component. Ultralight scalar-field dark matter is an established model class, with wave effects on astrophysical scales for appropriate masses (Hui et al., 2017). Complex models carry a conserved charge under global U(1) symmetry, invariance under rotating the field’s phase. Real scalar models also exist; wave behavior is not exclusive to a literal imaginary component.

A model requires a field action (the quantity from which the field’s equations follow), mass, interactions, cosmological preparation and a calculation of observables. Solitonic-core behavior, a stable dense clump of the field at a galaxy’s center, in simulations (Schive, Chiueh and Broadhurst, 2014) concerns specified models. It does not independently confirm that reality contains a gravitationally visible imaginary axis.

4.1 A minimal invariant model

A complete action shows darkness belonging to the whole neutral field. Take a complex scalar Phi neutral under electromagnetism. Written out, in units with hbar = c = 1 and signature (-,+,+,+),

Here m^2 > 0 and lambda >= 0. The action has a global U(1) phase symmetry. Varying the metric gives a stress-energy (the field’s energy, momentum and pressure) that is phase invariant and can source gravity. Stated precisely,

In the quadratic, rapidly oscillating regime with m much larger than the expansion rate, the averaged kinetic and potential contributions balance and the mean pressure approaches zero. A scalar can behave like cold matter at the background level, the smoothed-out average of the expanding universe.

The example establishes: a complex field can gravitate while being electromagnetically dark. It does not establish that its imaginary part alone is dark. The invariant action makes the opposite point: the darkness belongs to the whole neutral field.

4.2 How a physically preferred component could arise

A mass splitting earns the distinction that notation alone cannot. In symbols, write Phi = (a + ib)/sqrt(2) and add Delta V = epsilon(Phi^2 + Phi*^2)/2. The quadratic potential becomes

For |epsilon| < m^2 both squared masses are positive. The U(1) symmetry is broken to a discrete symmetry (one with separate steps, such as a sign flip) and the two mass eigenmodes (field combinations with definite masses) have different dynamical

roles. Renaming coordinates cannot erase the physical mass splitting. This shows what it means to earn a component distinction.

It still does not assign electromagnetic visibility to one mode. That requires specified gauge representations, mixing or further interactions. The same construction cannot simply be added to an electromagnetically charged scalar: Phi^2 would generally violate its gauge symmetry without additional charged structure.

4.3 Which astronomical claims follow

Solitonic cores, wave interference and small-scale suppression arise for particular field masses and dynamics, not for every complex field. The observational program must derive a background history, perturbation transfer function (how early density ripples grow or fade at each scale) and halo behavior (how the dark matter clouds around galaxies behave), then confront the relevant data. Bounds from the Lyman-alpha forest, hydrogen absorption lines in the light of distant quasars, constrain modeled suppression; they do not measure an imaginary axis. The program gains credibility by keeping the theoretical prediction, the observation and the inference between them separate.

5. Complexification and singularities

A complex extension does not generally remove a singularity, a point where quantities become infinite or undefined. The function 1/z still has a pole, a point where its value blows up, at z = 0. A contour, a path through the plane of complex numbers, can avoid the pole without making the function regular at the excluded point.

General-relativistic singularity theorems concern geodesic incompleteness under specified assumptions: a freely falling path that simply stops after finite proper time, the time a clock on that path would record. A proposed resolution must identify the relevant spacetime, geodesics and physical continuation rather than demonstrate only that a complex contour can be drawn. The Hartle-Hawking no-boundary proposal, the idea that the universe has no boundary at its beginning, illustrates a substantive use of complex methods in quantum cosmology (Hartle and Hawking, 1983). It is not a theorem that every singular Lorentzian solution becomes regular through analytic continuation, extending a function uniquely beyond the region where it was first defined.

A candidate resolution should establish what replaces the classical description, which observables remain finite and

whether evolution can be continued. Complexification is a tool with potentially important consequences, not a universal removal operation.

6. Real and complex quantum theories

Whether real numbers suffice depends on the composition and independence rules assumed. Renou and colleagues studied network scenarios in which real and complex quantum models give different correlations under specified source-independence and composition assumptions (Renou et al., 2021). Chen and colleagues performed a related experimental test (Chen et al., 2022). Hoffreumon and Woods (2026) argue that once independence is formulated operationally, real models reproduce complex-theory correlations more broadly.

The composition and independence rules matter. An experiment can distinguish theories sharing some algebraic ingredients while differing in those rules. These findings support careful attention to complex structure without establishing the imaginary-sector dark-matter proposal, which requires its own coupling mechanism.

7. Phase, amplitude and measurement

In the exponential, one part sets a phase and the other sets a magnitude. Written out, for a complex action S = A + iB, exp(iS/hbar) factors into exp(iA/hbar) times exp(-B/hbar). The real and imaginary parts have distinct roles: phase and magnitude. That identity can guide a model, but it does not show that the classical action is in every case the real projection of a physical complex action.

A measurement that returns a real number does not thereby measure only the real part of a wavefunction, the mathematical description of a quantum state. Quantum expectation values (average results over repeated measurements) of Hermitian observables (those whose measured values are real numbers) are real even though they depend on the complex state and its phase relations.

8. A program with discriminating tests

The four questions, gravitational unification, a dark sector, singularities and the quantum formalism, have separate completion conditions. A gravitational unification must define a common structure and recover the relevant dynamics. A

dark-sector model must specify its invariant visible-dark distinction and predict observables. A singularity proposal must state what is continued and under which criteria. A quantum-formalism proposal must identify its composition assumptions and their observational implications.

The negative checks are equally useful. A gauge-dependent darkness criterion fails. A complex pole remains a pole. A real-coordinate rewriting that preserves all operational predictions is not experimentally rejected by its notation. These checks locate where interpretation must become physical construction.

The neutral-scalar example answers the first constructive question: a consistent action can exhibit electromagnetic darkness and gravitational sourcing, but it does not implement the literal imaginary-sector thesis. The split-mass example answers a second: a component distinction becomes physical when an interaction defines it. If the same parameters control more than one successful prediction, the unification acquires explanatory force. If each question requires unrelated new assumptions, complex notation alone has not supplied the promised economy.

9. Conclusion

Complex numbers earn their place in quantum theory by describing how contributions combine and interfere. They do not thereby divide nature into visible matter and a hidden gravitational sector. The calculations here show what such a division would require: a physical reason for telling the components apart, equations governing each, and an observation that comes out differently as a result. Mathematical monism, the view that reality is mathematical structure, is a reason to look for one common account. It is not a reason to accept any particular construction before that construction shows what it explains.

A16. Simplicity Preference Is Adaptive: Why Simpler Explanations Help Minds Learn and Survive

Abstract

Minds prefer simple explanations because the world is grown from simple laws. A world built in discrete causal steps from a short rule is compressible, so a learner that bets on the shortest rule consistent with what it has seen predicts new cases better on average, learns from less evidence and reuses what it learns. Selection rewards that bet wherever its savings in learning and action exceed its costs. Simplicity has one limit: compression must keep every distinction that action needs, since two situations that look identical but demand incompatible responses defeat any strategy that cannot tell them apart. The account explains why a taste for elegance is adaptive and why elegance can still mislead, and it states the comparisons in real learners that would test it.

1. When simplicity helps

A short rule earns its keep on cases the learner has not seen. Compression stops helping when it drops a distinction that successful action needs. A learner has limited memory and time, so the advantage depends on which details a reusable explanation preserves.

Nick Chater and Paul Vitanyi (2003) develop simplicity as a principle in cognition, Jacob Feldman (2000) relates concept-learning difficulty to complexity, and Tania Lombrozo (2007) examines simplicity in causal explanation. These findings motivate the evolutionary question: in which environments do the predictive benefits exceed the learning and implementation costs?

2. What a shorter description buys

A shorter description starts with more weight, before any evidence is brought in, because each of its bits is a 50/50 choice that no rule makes: each bit saved doubles the starting weight. Stated precisely, fix a language for describing candidate explanations and suppose each complete description of length L receives initial weight proportional to 2^(-L). In a prefix-free

language, no complete description begins with another, so programs can be distinguished without an extra end marker. Five fewer bits give an individual description 32 times the initial weight. The weighting follows from treating each unruled bit as 50/50.

An explanation’s initial weight is not its probability after evidence. A shorter explanation that predicts the observations badly can lose to a longer one that predicts them well. Nor does a ratio between individual descriptions equal a ratio between entire classes of explanations. Several programs can describe the same output, and their weights must be combined under a mixture account.

Ray Solomonoff’s approach (Solomonoff, 1964) sums over programs rather than choosing the shortest explanation compatible with the data. Mathematical results on Bayesian mixtures, weighted blends of candidate models updated by evidence, relate prediction loss to the weight assigned to the true environment when it belongs to the model class (Hutter, 2003). Their stated assumptions matter: they supply a reason to investigate economical prediction, not a proof that a finite organism implementing a rough brevity rule outperforms every alternative. A short description of an environment is useful only if

the learner can access or approximate the relevant rule with its available resources.

3. The evolutionary comparison

Whether a simpler strategy wins depends on what it earns across the environments it meets, minus what it costs to run. Consider two inherited learning strategies in the same population and conditions. Written out, expected net benefit is

F(P) = Sigma_e q(e)u(P,e) - c(P).

Here P is the strategy, q(e) is the frequency of environment e, u(P,e) is its reproductive benefit there, and c(P) is its implementation cost in the same units. A simpler strategy is favored when its F is greater. Applying that comparison to evolution also requires heritable variation, differences passed from parents to offspring, and the relevant population conditions.

Reusable regularities can let a short model earn comparable benefits at lower cost, or better benefits for the same cost. Conversely, a costly exception may be worth learning if missing it is dangerous. Predictive accuracy, description length and reproductive success are related through the task; they are not interchangeable quantities. A simplicity-weighted distribution is

one proposed model for q. Only evidence about actual encounters, rewards and costs can establish whether it explains the observed preference.

4. As simple as the task allows

Compression can remove detail that makes no difference to a decision. Removing a crucial distinction defeats the point. Suppose two situations supply identical observations and memory, but successful action requires incompatible responses. A strategy based only on that information must respond identically in both. It cannot guarantee success in both. More computation cannot replace the missing distinction.

Useful simplicity therefore compresses the account while preserving the distinctions needed for the task. The best model for an organism may be more elaborate than the shortest explanation of its observations, because a rare hazard matters greatly to survival. This yields a clear comparative prediction: as the cost of missing an exception rises, an adaptive learner should become more willing to pay for a model that represents it.

5. Cognitive findings and perceived elegance

The cognitive findings fit the proposal without establishing it. A relation between concept difficulty and description length is consistent with it: a reusable rule can be easier to acquire than a list of unrelated cases. The evolutionary explanation should compete with accounts based on representation, exposure and the learner’s capacities. A correlation by itself does not license an inference to selection.

Explanatory breadth can save description length when one rule explains several observations more economically than separate accounts. The comparison must count the cost of applying the rule too. A slogan that appears to explain everything by leaving out the difficult details has not achieved the same economy. Perceived elegance may guide the search toward such economies, but familiarity or presentation may make one explanation feel simpler while another predicts better. That the response exists does not show it evolved because it improved decisions. Showing that takes further evidence.

Any empirical test of the proposal has to work with a stand-in for shortest description length. Shortest program length is defined relative to a description language and is uncomputable in general

(Kolmogorov, 1965). Practical studies must use specified approximations. These are measurable model choices; they cannot be treated as exact measurements of an object’s shortest possible description.

6. Comparisons that could support the account

The central test is whether a simplicity preference improves decisions after its costs are counted. Compare learners on the same observations and tasks, measuring predictive success, consequential mistakes and effort. Include environments with reusable rules and environments in which exceptions matter. This asks when the preference earns its keep.

A second comparison varies the consequences while holding the observations fixed. If a previously negligible exception becomes dangerous, the expected benefit of representing it increases. A learner that adapts its model accordingly fits the task-sensitive account better than an unconditional rule to choose the shortest explanation. A third comparison separates felt elegance from performance: do judgments of simplicity help identify models that generalize to unseen cases once familiarity and presentation are controlled?

7. Conclusion

A short model lets a mind reuse what it learns, generalize from limited evidence and act at lower cost. It is adaptive when those gains improve reproductive success in the environments encountered. The comparison must include the cost of missed distinctions as well as the cost of representing them. Existing mathematical results establish benefits under specified assumptions. To strengthen the evolutionary hypothesis, specify the conditions under which simplicity succeeds and the information it must preserve.

A17. Why Nature Minimizes: The Many-to-One Root of Time, Action, and Emergence

Abstract

The Principle of Least Action, the arrow of time, and the emergence of smooth physics from discrete rules are treated as separate results requiring separate explanations. This paper argues they are three instances of one structural fact: many-to-one mappings. Computation is many-to-one: 1+1=2 is deterministic, 2=x+y is not. The inverse does not exist. This is the arrow of computation. Feynman's path integral is many-to-one: all quantum paths exist, but paths far from the extremal cancel under phase interference. What survives is the classical trajectory. The Principle of Least Action is what remains when everything else cancels. Coarse-graining (averaging over fine details) is many-to-one: many microstates map to one macrostate. When discrete rules are averaged at large scales, only extremal macro behavior survives, for the same structural reason. Variational emergent physics is forced by coarse-graining regardless of whether the fundamental rule is variational.

A forcing argument completes the picture. Observers require stable structures. Stable structures require conservation laws. Noether's theorem proves conservation laws require symmetries of a variational formulation. The chain from observers to variational physics runs one direction with no exit. Under the Simplicity Dominance Principle (SDP), Solomonoff's algorithmic probability applied to physical structures, the simplest observer-supporting frameworks dominate. If all such frameworks produce variational physics through the Noether chain, SDP derives the Principle of Least Action for any universe we could inhabit.

1. One Structural Fact, Three Physical Consequences

The Principle of Least Action, the arrow of time, and the emergence of variational physics from simple rules are the same mathematical pattern wearing different clothes. All three reduce to one fact: many-to-one mappings have a deterministic direction in which information survives, and an underdetermined direction in which it does not. The forward direction produces a unique output. The backward direction has infinitely many candidates

and no procedure to choose among them.

This paper traces the pattern across three domains, shows why they converge, and argues that variational physics is forced for any observer-supporting universe. In five words: everything happens, most cancels out.

A further connection emerges at the intersection of variational physics and optimal packing. The golden ratio's continued fraction is all 1s: the most irrational number, maximally resistant to rational approximation. This is why it produces optimal packing in phyllotaxis: no rational approximation ever catches it, so seeds never align into wasteful rows. Non-repeating and variational are the same property at the golden ratio. If a hypergraph rewrite rule's stable structures require variational physics (as the Noether forcing argument demands), and if variational physics selects for optimal packing geometry, then the golden ratio enters not as numerology but as a structural consequence of the same parsimony that produces least action. The open question is whether the simplest causally invariant rewrite rule whose topological features reproduce the Standard Model: three generations, SU(3)×SU(2)×U(1), the observed mass hierarchy: inherits golden-ratio geometry from its variational structure. Current hypergraph efforts operate at 10^−93 to 10^−100 meters,

a gap too vast to bridge computationally. The productive direction starts from the known sub-atomic numerical values and searches downward, using the simplest causally invariant rewrite rules. Whether φ enters the fundamental rule directly or emerges through coarse-graining of simpler causally invariant rules remains open.

A broader pattern emerges: the fundamental constants themselves form a complexity hierarchy ordered by their shortest generating algorithms. The golden ratio (about 3 to 4 bits: continued fraction all 1s) governs optimal packing. The square root of 2 (about 4 to 5 bits: continued fraction all 2s) governs diagonals. Euler's number e (about 7 to 10 bits: the rate that equals itself) governs exponential growth. Pi (about 10 to 15 bits: no pattern in its continued fraction) governs circular symmetry. Under SDP, the simplest constant whose defining property matches a structural requirement is the one observers find.

2. Many-to-One: The Structural Fact

Addition: 1+1=2 is deterministic. 2=x+y is not. The inverse function does not exist. Multiplication: 3×4=12 is deterministic. 12=x×y is not. Hashing: hash("hello")=0x5d41 is deterministic. 0x5d41=hash(?) is not. Neural computation: sensory inputs

produce a percept. The percept does not uniquely determine which inputs produced it.

Nearly all functions are many-to-one: multiple inputs produce the same output. The forward direction is determined. The backward direction is underdetermined. This is a property of mathematics, true in every possible world. Under mathematical monism, the function is the territory. This structural asymmetry is as fundamental as anything in physics.

3. Instance One: The Arrow of Computation

The summary:

A system running backward cannot compute, because every operation has infinitely many possible inputs. It cannot predict, because prediction requires computing consequences from causes, and causes are underdetermined from consequences. It cannot model, because modeling is compression, compression is many-to-one, and reversible computation is bijective and cannot compress.

In a block universe where all states exist, backward states are present but nothing can compute through them. This is why time has a direction. Forward descriptions from simple initial

conditions have low Kolmogorov complexity. The same universe described backward from its current state has high complexity, because many-to-one functions ran forward and made the end state complex. The Simplicity Dominance Principle quantifies this asymmetry. The computational arrow is the structural foundation. SDP measures its consequence.

4. Instance Two: The Principle of Least Action

Variational physics means specifying all of physics through one principle: systems follow the path that extremizes the action. One rule generates every trajectory. The alternative, listing each permitted trajectory separately, requires infinite specification. That is what variational means: one principle instead of an infinite list.

A thrown ball does not know to take the optimal path. In quantum mechanics, it takes all paths simultaneously. Feynman’s (1948) path integral assigns every possible trajectory a phase factor proportional to e^(iS/□), where S is the action along that path. Paths near the extremal have similar phases. They reinforce. Paths far from the extremal have rapidly varying phases. They cancel.

What survives the cancellation is the classical trajectory. The Principle of Least Action is what remains when everything else cancels.

This is many-to-one. Many paths contribute. One classical trajectory survives. The surviving path is determined. The inverse question, given the classical trajectory, which quantum paths contributed, has infinitely many answers. The structure is identical to 1+1=2: deterministic forward, underdetermined backward.

The analogy is not metaphorical. Both are instances of the same mathematical operation: a map from a large space to a small space, where the forward direction has a unique image and the backward direction does not.

5. Instance Three: Coarse-Graining and Variational Emergence

A simple discrete rule, like a cellular automaton or a hypergraph rewriting rule, operates step by step at the micro level. Each step is local: update this cell, rewrite this edge. The rule may have very low Kolmogorov complexity, perhaps just a few bits.

Zoom out. Average over many steps and many cells. This is coarse-graining: replacing a detailed microstate with a summarized macrostate. Many microstates map to the same macrostate. Coarse-graining is many-to-one by definition.

What survives coarse-graining? The same thing that survives phase cancellation in the path integral: extremal behavior. Non-extremal microstates vary in all directions. Their contributions average out, the way randomly oriented vectors sum to zero. Extremal microstates are aligned. Their contributions reinforce.

The emergent large-scale physics is variational even when the fundamental rule is not. The underlying cellular automaton has no action functional (the total kinetic-minus-potential along a path), no Lagrangian, no variational principle. The emergent physics does, because coarse-graining killed everything except the extremal survivor.

This supports the hypothesis that our physics emerges from extremely simple discrete rules (Wolfram, 2020). The simplest possible specification, a rewriting rule expressible in a few bits, may require a variational emergent physics not because anyone designed it that way, but because coarse-graining forces it. The

fundamental rule compresses the description. The many-to-one structure of coarse-graining produces the variational form at the scale observers inhabit.

The connection to quantum mechanics is direct. Wolfram's multiway system applies the rewriting rule in all possible locations simultaneously. All branches coexist. When branches reconverge, they interfere: reinforcing branches survive, canceling branches vanish. This is coarse-graining over the space of branches. The multiway system IS the path integral, derived from the rule rather than postulated. The classical trajectory that survives IS what remains after many-to-one cancellation. The strongest result here is causal invariance producing general covariance (the analog of general relativity). The full correspondence between branchial space and Hilbert space remains open. The structural case is suggestive and narrowing.

6. The Noether Chain: Why Observers Require Variational Physics

The three instances converge on a forcing argument.

Observers require stable structures. A mind that models its environment needs atoms that hold their bonds, molecules that persist, chemistry that works. Stability means things are conserved: energy stays in orbitals, momentum carries through collisions, charge is neither created nor destroyed.

Conservation laws require symmetries. This is Noether's theorem (1918): every continuous symmetry of the action yields a conservation law. Conservation of energy comes from time-translation symmetry. Conservation of momentum comes from spatial-translation symmetry. Conservation of angular momentum comes from rotational symmetry. No symmetry, no conservation. No conservation, no stability. No stability, no chemistry. No chemistry, no observers.

Symmetries of the action require a variational formulation. Noether's theorem does not say "every symmetry yields a conservation law." It says "every continuous symmetry of the action functional yields a conservation law." The action functional is the variational object. Without it, the theorem has no input.

The chain runs one direction: observers → stability → conservation → symmetry → variational formulation. There is no exit. Any observer-supporting universe must have variational

emergent physics, regardless of whether its fundamental rule is variational, discrete, or otherwise.

Can an observer-supporting universe lack variational physics? Two candidates fail:

Purely discrete physics with no continuum limit. Low K, but atoms need smooth orbitals to form stable bonds. Without a continuum limit, electron clouds cannot form. No orbitals, no chemistry, no molecular stability, no observers.

Fundamentally nonlocal dynamics with no local action. Things affect nearby things in our universe. Your hand pushes a ball, the ball hits a wall. Cause travels through space step by step. Nonlocal means every point affects every other point instantly with no connection through the space between. If effects do not travel through space, spatial structure collapses. “Here” and “there” stop meaning anything. No spatial structure, no molecules, no chemistry, no observers.

Both fail at observer-support. Variational physics appears forced for any universe containing minds.

7. Unification and Relationship to SDP

The Simplicity Dominance Principle says structures with lower Kolmogorov complexity dominate exponentially. It is Solomonoff's (1964) universal prior applied to physical structures: each extra bit of description halves a structure's weight. It selects for the simplest observer-supporting frameworks. If the Noether chain is correct, that all observer-supporting frameworks produce variational emergent physics, then SDP derives the Principle of Least Action. The derivation:

1. SDP selects for lowest-K observer-supporting structures.
2. Observer-supporting structures require stability, conservation, and symmetry (Noether chain).
3. Conservation and symmetry require a variational formulation.
4. Therefore observers inhabit structures with variational physics.
5. Among variational structures, the simplest dominate.
6. The simplest wave equation (Schrödinger) produces the path integral.
7. The path integral produces PoLA as the coherent classical limit.

The claim that SDP “derives” PoLA rests on step 2-3: that the Noether chain has no exit. Whether there exists a non-variational observer-supporting framework remains an open question. The structural arguments against it (no stable chemistry without conservation, no conservation without Noether, no Noether without action) are strong. A proof of impossibility would close the gap. Until then, the derivation is conditional: SDP derives PoLA given that all observer-supporting physics is variational.

The three instances are connected through SDP as follows:

The computational arrow (instance one) is the structural foundation. Many-to-one functions create the K-asymmetry that SDP quantifies as the arrow of time.

The Principle of Least Action (instance two) is the coherent survivor. Many-to-one phase cancellation in the path integral produces variational classical physics from simple quantum mechanics.

Coarse-graining emergence (instance three) is the bridge. Many-to-one averaging produces variational macro physics from simple discrete rules, connecting the fundamental simplicity SDP selects for to the variational physics observers inhabit.

One mathematical structure. Three physical faces. One forcing argument from observers.

8. Implications

The unification has several consequences.

First, it connects results that were previously separate: the computational arrow and the SDP measure. Many-to-one is the thread.

Second, it supports the hypothesis that our physics emerges from simple discrete rewriting rules. If coarse-graining forces variational emergence, the simplest possible rule (lowest K) will produce the physics we observe at our scale. The rule does not need to encode the Standard Model. It needs to be simple. The rest is coarse-graining.

Third, it strengthens the case that SDP is a single principle with convergent consequences rather than an ad hoc hypothesis applied separately to different domains. The arrow of time, the Principle of Least Action, and variational emergence are not three separate applications of SDP. They are one application seen at three scales.

Fourth, it clarifies what would count as evidence against the framework. If a stable observer-supporting physics were found that is provably non-variational, with no action functional and no Noether symmetries yet with conservation laws arising from some other mechanism, that would challenge the Noether chain and weaken the derivation of PoLA from SDP.

A18. Beyond Geometric Unity: Infinite Dimensions from One 4D Metric as a Home for Particles

Abstract

One four-dimensional metric generates an unlimited tower of geometric structure from one short rule, and that tower is a candidate home for the particle spectrum. Four spacetime coordinates plus the metric's ten components at each point give the fourteen-dimensional metric bundle, the starting point of Eric Weinstein's Geometric Unity, and adding derivatives gives 54, 154 and beyond. Once the geometry is given, each level costs only the construction rule and the derivative order, so a theory built on the tower can grow without adding independent inputs. The tower alone is not a particle catalog. A violin string's notes are fixed by its tension, length and clamped ends, and in the same way an action has to say which disturbances of the geometry propagate and how they interact. The paper counts the coordinates, works the oscillator case, and tests one modification, quadratic gravity, against orbits and light bending: its extra forces fade at large distances, so it does not by itself

produce a dark halo. Deriving the particles from an action is the next step.

1. One construction, many possible modes

One violin string plays many notes. The particle zoo might work the same way: a small set of rules producing many distinct excitations (field disturbances that appear as particles), with one construction accounting for both the modes, the distinct patterns of vibration, and their interactions.

A string does not acquire its frequencies because its displacement can be differentiated (have its rate of change computed). Its tension, its length and its clamped ends fix the modes. A geometric theory owes the same account. The metric bundle says which configurations are possible; an action, the quantity from which a theory's equations follow, and its solutions say which disturbances propagate and how they interact.

Geometric harmony refers to that explanatory ambition. It does not identify each derivative of the metric, the field that sets distances and durations in spacetime, with another overtone, nor does the existence of an infinite sequence of derivatives show that infinitely many physical excitations are present.

Eric Weinstein’s Geometric Unity (Weinstein, 2021) motivates the fourteen-coordinate starting point. Discrete programs, which build spacetime from separate countable pieces, reach relativistic geometry by other routes (Gorard, 2020). The count describes the metric bundle, a space containing possible metrics over spacetime, not fourteen observed spacetime directions. From there, this paper follows one chain: metric components, their derivatives, physical modes selected by dynamics, and a calculable gravitational response.

2. Why the metric bundle has fourteen dimensions

Fourteen is a count of independent numbers: four to say where you are, ten to give the metric there. Stated precisely, let M be a smooth n-dimensional manifold, a space that looks flat up close and supports calculus. At x in M, a metric is a nondegenerate symmetric bilinear form (a rule assigning a number to each pair of vectors) on the tangent space, the space of vectors based at that point. In coordinates, the symmetric matrix has n diagonal and n(n-1)/2 independent off-diagonal entries. Fixing a nondegenerate signature, the count of time-like and space-like directions, leaves the local dimension unchanged. The bundle’s dimension is then

For n=4: four base coordinates and ten fiber coordinates, fourteen in total. A metric field on M is a section of E, one choice of metric at each point. Picking such a choice does not turn the ten fiber coordinates into independent physical excitations. Constraints, gauge equivalence (treating descriptions that differ only in redundant detail as the same) and the equations of motion fix which degrees of freedom, independent ways the system can vary, are physical.

3. Jet bundles: organizing the derivatives

Counting derivatives as extra coordinates makes the list grow fast. The n-th jet bundle (Saunders, 1989) is a list: the metric at a point together with its derivatives up to order n, each entry treated as one more coordinate. In symbols, the number of coordinates is

The counts begin 14, 54, 154. At first order, each of the ten metric components has four first derivatives, adding 40 coordinates. At second order, symmetry of mixed partials (the order of taking derivatives does not matter) leaves ten second derivatives per component, adding 100. The list lets an action depend on derivatives the way it depends on coordinates. It does not make those derivatives independently excitable physical fields.

One rule builds every level. Choose the order n, run the same jet construction, and the description costs no more than the rule for the bundle, plus about as many bits as it takes to write n in binary and a few more to mark where it ends, plus a fixed amount. Going from 54 coordinates to 154 adds only the bit or two that says which order. A coordinate list that grows need not make the generating rule grow with it.

As derivative order increases, the counts have no finite upper bound. Carry compatible derivative data at every finite order and the result is the infinite jet tower. Its base remains four-dimensional spacetime; the additional coordinates describe field values and derivatives, not new spacetime directions. A finite rule specifies the hierarchy without listing infinitely many coordinates separately (De Vecchi, 2017).

An infinite derivative list at a point does not generally determine a smooth function (one with derivatives of every order) away from that point. The function exp(-1/x squared) for nonzero x, zero at x=0, is smooth with all derivatives vanishing at zero, yet nonzero elsewhere. For an analytic function, a convergent Taylor expansion, a sum built from the function’s derivatives at one point, determines the function in a neighborhood. Rebuilding a

function from its Taylor series, or coding a geometric object, is not the same thing as a physical holographic duality (a theory in a region equivalent to one on its boundary), which has to relate observables and dynamics.

4. The description cost of generated geometry

Running one fixed algorithm costs a constant, however large its output (Solomonoff, 1964). Stated precisely, for a fixed effective procedure F (a mechanical step-by-step method) that builds the metric bundle from a finite description D, K(F(D) D) <= c F and K(D,F(D)) <= K(D)+c’ F, where the constants depend on the procedure and reference machine (the standard computer used to measure description length), not on the particular input D. A bounded reusable cost need not be zero, and a section, an action or a boundary condition (a requirement fixed at the edges of a region) you choose is not automatically an output of F.

Stopping at a finite order adds one charge, the order itself. Written out, for a finite jet construction F_n a fixed algorithm reads both D and an encoded order n, giving K(F_n(D) | D,n) bounded by a constant. Increasing the order adds the cost of specifying n.

5. The same geometry can support different laws

Two theories can share every coordinate and still disagree about the vacuum, spacetime empty of matter. Written out on a four-dimensional Lorentzian base (a spacetime whose geometry treats time differently from space), the two actions are

The corresponding vacuum equations are G_ab=0 and G_ab+Lambda g_ab=0. Both theories use the same metric configuration bundle. A flat metric, one for uncurved spacetime, solves the first; a nonzero Lambda, the cosmological constant giving empty space an energy, excludes that flat vacuum in the second. Fourteen coordinates therefore do not select even this elementary difference in dynamics. Observation decides which action describes a physical system.

6. Which disturbances become physical modes?

Differentiating one oscillatory function changes its amplitude and phase (size and timing within the cycle) in a prescribed way; it does not produce a new independent oscillator at a new frequency for every derivative. Physical modes come from a kinetic operator (the part of the action that governs how disturbances travel) and its constraints about a chosen

background, a fixed solution around which disturbances are studied. A higher-derivative action may introduce extra poles (frequencies at which the response becomes infinite, each marking a mode) or pathological modes; it need not add one healthy mode per derivative order.

One oscillator, infinitely many derivatives

A list of derivatives is not a list of modes, and one cosine shows why. In symbols, let q(t) = A cos(omega t). Its n-th derivative is A omega^n cos(omega t + n pi/2). Every derivative has the same frequency. Treating q, dot q, double-dot q and so on as a list creates infinitely many mathematical entries without creating infinitely many freely excitable oscillators.

An extra frequency has to be put in by hand, through a term in the action. Written out, the illustrative mechanical action is

The Euler-Lagrange equation, the equation of motion an action yields, is eta q^(4) + double-dot q + omega^2 q = 0. Substituting q proportional to exp(i nu t) gives two squared frequencies:

The extra frequency comes from eta and the action, not from q having derivatives. As eta tends to zero, the lower frequency tends to omega and the higher diverges (grows without limit). The inverse of that quadratic operator has opposite residues (the

strengths of its pole contributions) at its two frequency poles, so real frequencies do not by themselves establish a positive-energy physical spectrum.

Infinite smoothness is compatible with a second-order equation (one involving at most second derivatives) having only a small number of physical modes. Conversely, one higher-derivative term can introduce an unwanted mode. The instrument is the dynamics, not the derivative list.

7. The action fixes the particle content

The fermion question

A metric-jet hierarchy does not automatically supply spinors, the kind of field an electron is. Covariant quantities formed from the metric, curvature and their tensor derivatives transform as tensors, generalizations of vectors with fixed transformation rules. A 2pi spatial rotation, one full turn, acts trivially on them, whereas it acts by a minus sign on a spinor field. Relabeling a derivative coordinate cannot change that transformation law.

The obstruction blocks one route: calling a tensor field a fundamental spinor. Fermions, particles like electrons that cannot share a quantum state, can emerge from bosonic systems, made

of particles that can share a quantum state (Levin and Wen, 2003), which is additional physics a proposal would have to exhibit.

The geometric proposal therefore needs either spinor fields whose representations and couplings (how they transform under symmetries and interact) it explains, or dynamics that produces fermionic excitations. Deriving the Standard Model, the established theory of known particles and forces except gravity, means recovering chiral (handedness-dependent) matter representations, gauge interactions and anomaly cancellation, the cancelling out of quantum effects that would break symmetries. If those ingredients are supplied independently, they belong in the complete description cost.

A geometric operator expansion

Writing the general action shows how many free coefficients the coordinate count leaves open. In symbols, a possible four-dimensional effective action, an approximate action valid at low energies, begins with

M is a suppression scale, the energy at which the extra terms become important; alpha, beta and gamma are coefficients; S_m determines matter and its coupling. The jet coordinate count

neither selects these coefficients nor sets Lambda_vac. Operator mass dimension (the power of mass units a term carries) and jet order count different things; products of fields and derivatives yield many operators at the same order, subject to identities and redundancies.

Geometric harmony has a definite task: derive a restricted operator pattern with a controlled low-energy regime, rather than call every possible coefficient an amplitude available to fit afterward. A successful result could predict ratios or forbid terms that a generic expansion allows.

Stability and prescription

Removing an unwanted mode is a physical choice with its own costs. Damiano Anselmi's fakeon prescription is one approach to the unitarity problem, the question of whether total probability is preserved: it excludes selected modes from the asymptotic physical states, the states that can arrive at a detector, at the cost of violating microcausality, cause preceding effect even at the smallest scales (Anselmi, 2019). A mode that does not arrive as a particle is not thereby a dark-matter fluid that clusters on its own. Which prescription is chosen, what classical limit (behavior once quantum effects are negligible) it has and what scales it carries

are physical content, and each must be assessed. This paper does not identify every higher-derivative mode with a fakeon or every fakeon with dark matter.

8. Dark matter: a worked gravitational response

Quadratic gravity’s extra forces, produced by added terms that square the curvature, fade with distance (Stelle, 1977; Alvarez-Gaume et al., 2015), so a point source leaves orbital speeds falling off. The benchmark carries positive inverse lengths m0 and m2 for the scalar and spin-2 contributions, so that 1/m0 and 1/m2 are the ranges over which those two contributions act. Written out, the standard static weak-field point-source potential (Burzilla et al., 2021) is

For a circular orbit, v_c squared equals r times the radial derivative:

Both exponential terms vanish as r grows, so v_c approaches zero. This potential does not generate an asymptotically flat rotation curve, orbital speeds leveling off at large distances, for a finite point-source mass. A useful fit over a finite radial interval is not excluded.

Finite-radius bound

For a point source the effect is capped: the two extra terms cannot raise the circular-speed factor above 4/3. Stated precisely, write f(x)=(1+x)exp(-x). For x>=0, f(0)=1 and f is nonincreasing, so 0<=f(x)<=1. The point-source circular-speed factor is

F(r)=r v_c^2/(GM)=1+(1/3)f(m_0r)-(4/3)f(m_2r)<=4/3.

With fixed source mass and G, the model cannot supply a dynamical mass enhancement, a boost in the mass inferred from orbits, larger than one third at any radius. This bound cannot be transplanted to the interior of an extended disk, where the Yukawa force, a force fading exponentially beyond a set range, must be integrated over the source.

Extended sources and lensing

Light bending and orbital motion respond to different combinations of the two potentials (Bartelmann and Schneider, 2001). Written out in the weak-field convention ds^2=-(1+2 Phi)dt^2+(1-2 Psi)dx^2, the extended-source potentials in this benchmark are

The lensing sum, the combination that governs light bending, is

The scalar Yukawa cancels in this sum although it remains in Phi. A lensing fit cannot be inferred from a rotation-curve fit. Outside a source with bounded support, one confined to a finite region, the finite-range terms decay; the ordinary far-field mass term survives.

Effective density

The modification moves dynamical mass around inside a source without adding to the total. Apply the distributional identities for the Laplacian (a sum of second derivatives over all directions) of 1/s and of exp(-ms)/s, and note that 1+1/3-4/3=0, so the contact terms, contributions concentrated at the source point itself, cancel. In symbols, the effective dynamical density is

This is a convolution (a smearing of the source by a distance-dependent weighting) whose sign and profile depend on both inverse ranges and the source. The total integral of rho_eff for a localized source equals M, and the effective excess rho_eff minus rho integrates to zero. This benchmark redistributes dynamical density but does not create extra total far-field mass.

Observational test

One set of parameters has to fit both orbital speeds and light deflection. Start with a galaxy’s baryonic density, the density of ordinary matter, inferred from stars and gas. Evaluate Phi in the disk plane and differentiate it to obtain v_c^2(R) at the measured radii. State whether m_0 and m_2 must be shared across galaxies. For a system with lensing data, use the same parameters to evaluate the line-of-sight derivative of Phi+Psi.

Uncertain stellar mass-to-light ratios, distances, inclinations and gas distributions affect both comparisons and belong in the likelihood, the probability of the data given the model. The SPARC catalog, Spitzer Photometry and Accurate Rotation Curves for 175 disk galaxies, supplies rotation-curve data (Lelli, McGaugh and Schombert, 2016). No SPARC fit is reported here; the calculation specifies a test the proposed mechanism must pass.

9. Comparing geometric and particle accounts

A fair comparison charges each account for everything it needs. Fix the scope: observables, precision, preparation conditions and boundary data. Count the geometry, fields, equations of motion, boundary conditions and observation rule together. Their joint description length can be shorter than separate descriptions because one ingredient may determine another.

If one construction derives relations among couplings that a rival must specify separately, those relations reduce the joint cost. Count the derivation once and do not charge its consequences again. Conversely, an unspecified action cannot receive a zero cost simply because someone has defined a rich configuration space.

A particular particle model can relate the mass, interaction strength and production history of a WIMP, a weakly interacting massive particle, or an axion’s mass and couplings through its underlying construction. Assigning each a generic bit cost while treating a new gravitational kernel, the rule weighting a source’s pull by distance, as free does not compare complete explanations. The proposed metric theory must pay for m_0, m_2, its action and its prescription at whatever precision the

same observations require.

Description length converts into odds only when the saving has been established. Under the description prior, in which each bit is a 50/50 choice that no rule makes, and with equal observational likelihoods, a demonstrated delta-bit saving gives a factor 2^delta in prior and posterior odds, the odds before and after the evidence. Establishing the saving is the outstanding comparison. The fixed jet-construction algorithm can be shared across geometric candidates, and it should not be mistaken for the physical content of any one of them.

10. From the construction to a physical theory

Specify an action and background, derive the independent propagating modes, and check whether disturbances grow without bound. Then determine their coupling to matter and which parameters the construction fixes.

The first observational target is a joint account of orbital motion and lensing using the same coefficients. A galaxy fit would not by itself establish agreement with cluster mergers (collisions of galaxy clusters), cosmology (Planck Collaboration, 2020) or gravitational waves. Those are subsequent tests, not results

claimed here. The present contribution is the geometric hierarchy and a worked gravitational response exposing a concrete limitation of the finite-range example.

11. Conclusion

The fourteen-dimensional metric bundle and its finite jets come from specified geometric constructions, and their growing coordinate lists can share one short generating rule. An action, its constraints and a background then fix which modes are physical.

The quadratic-gravity calculation supplies a definite test. With the stated positive inverse ranges, its additional forces decay at large distance, its effective total mass equals the source mass, and its lensing potential differs from its orbital potential. A finite-range fit may still be investigated, but the construction alone is not a derived dark halo.

Geometric unification becomes explanatory when one action supplies relations among particles, interactions and observations that otherwise require independent inputs. This paper supplies the coordinate counting, the mode calculation and the gravitational response. Deriving particles from an action is the next step.

Appendix A. Why the point-source bound does not extend to every source

A.1 Point-source result restated

The point-source result sets the ceiling the shell calculation will exceed. Write f(x) = (1 + x)e^(-x). For x >= 0, f is nonincreasing with f(0) = 1, so 0 <= f(x) <= 1. The point-source circular-speed factor from Section 8 is

A.2 Outer shell

An outer shell can exceed the point-source bound because its Newtonian force vanishes inside while its Yukawa force need not. Stated precisely, consider a thin spherical shell of radius a and mass M_s around a central point mass M_c. For 0<r<a, averaging the kernel -G exp(-ms)/s over the shell’s angles gives the interior Yukawa potential

with radial derivative

The circular speed from the shell’s contribution is

Because the Yukawa kernel integrates over the source rather than being evaluated at a point, the interior response of the shell is not bounded by the point-source formula.

A.3 Numerical instance

A single numerical case breaks the point-source bound. Take r = 1, a = 2, m_0 = 5, m_2 = 1, M_c = 1, M_s = 100. Evaluating the central mass plus shell contributions, with v_N the Newtonian circular speed of the same masses, which inside the shell comes from the central mass alone:

which exceeds the point-source bound of 4/3. The surplus comes from the shell integral, whose sinh kernel amplifies the longer-range spin-2 Yukawa contribution when the shell lies at a favorable ratio of a to 1/m. Here that term supplies +3.319 while the shorter-range scalar term, suppressed by e^(-10), contributes -0.045.

This confirms that the Section 8 bound applies to point sources and cannot be transplanted to extended mass distributions without recalculating the source integral.

A.4 Reproducibility

The supplement A18_Shell_Check.py evaluates equations (1) through (4) with the standard library. It checks the shell average in (1) by integrating the kernel over the shell, the derivative in (2) by finite differences (numerical slopes taken over small steps)

and the point-source ceiling of 4/3 on a grid, and it reproduces the ratio in (4), 3.3067, with its spin-2 term of +3.3191 and its scalar term of -0.0449. Its output is A18_Shell_Check_Results.json.

A19. Dark Matter Behaves Like Matter: A Halo Rule Set by Starlight Predicts Dwarf Galaxies Better Than MOND and Two Modified Gravity Theories

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Abstract

The missing mass of galaxies behaves like matter because its amount follows the stars: a halo rule with three shared numbers, driven by stellar mass and size, predicts galaxies it has not seen, while laws that change gravity fail when carried between populations. Dark matter and modified Newtonian dynamics (MOND) both fit spirals; this paper tests their predictions on 286 systems. The halo rule predicts unseen spirals as well as MOND's acceleration relation (0.1533 against 0.1531 dex) and, unchanged, 44 dwarf spheroidals 28 percent better (0.479 against 0.669 dex) and 25 more outside the original sample 45 percent better, beating even MOND refitted on dwarfs (0.548 dex); dwarf errors remain about a factor of 3. In one likelihood over 85 dwarfs with upper limits, it beats MOND's relation by 36

log-likelihood units; its forecasts for 264 unmeasured dwarfs are hashed. Frozen on rotation curves, it predicts the amplitude of Kilo-Degree Survey lensing of 259,383 isolated galaxies inside 0.3 Mpc within 0.05 dex per stellar-mass bin, but not its radial profile or early-late split. MOG and Mashhoon’s nonlocal gravity, with published constants, act as fixed copies of visible mass, strengthening gravity at a half-mass radius at most 19 and 23 times; 33 and 31 of the 44 dwarfs need more. In clusters, the published Moster halo relation with its scatter, fitted to no cluster, beats MOND’s relation (0.288 against 0.403 dex) without recovering cluster masses. MOND’s hot-gas cluster repair, carried to gas-rich spirals, raises speed errors from 12 to 100 percent.

1. The question and the claim

The missing mass of galaxies behaves like matter because its amount follows a galaxy’s stars. A halo predicted from stellar mass and size forecasts galaxies it has not seen, while the laws that change gravity tested here fail when carried from one kind of system to the next.

Galaxies rotate too fast for their visible matter. Two kinds of answer compete. One adds matter: each galaxy sits in a halo of

dark matter, and the halo's mass and concentration follow from how structure formed (Navarro, Frenk and White, 1997). The other changes the law: gravity strengthens below an acceleration scale a0, as in modified Newtonian dynamics, MOND, whose radial acceleration relation ties the observed acceleration in disk galaxies to the acceleration of their ordinary matter alone (McGaugh, Lelli and Schombert, 2016; Lelli et al., 2017).

On disk galaxies both answers fit well, which is why the question has stayed open. The radial acceleration relation needs one constant, and halo fits need two per galaxy. A fair test needs what a fit to one population cannot give: predictions for systems the model has not seen, from measured inputs only, carried across populations whose sizes, gas content and histories differ.

This paper makes that test across 286 systems, at radii from 27 parsecs in dwarf galaxies to 1,000 kiloparsecs in clusters, and finds that the missing mass behaves like matter in an amount that follows the stars. A halo rule with three shared coefficients, driven by stellar mass and size, predicts unseen disks as well as MOND's relation and transfers to dwarf galaxies far better. Theories that make extra pull from visible matter by a fixed recipe, the form two published modified gravity theories take in the weak field, cannot supply what most dwarfs need, because a

fixed recipe can raise gravity only by a capped factor. A single acceleration scale, or a single rule for the extra mass, fitted where it works, fails when carried to the next population. A published halo relation set by starlight, carried unchanged to cluster scales, predicts accelerations there with 29 to 54 percent less error than MOND's relation, the smaller gain when the halo relation's scatter is included, though not the clusters' masses.

This paper's main advance is a halo rule that predicts galaxies it never saw: set by a galaxy's stellar mass and size with three shared numbers, it beats MOND on dwarf galaxies. Comparisons usually fit the galaxies they score; here every rule is fixed on some systems and scored on others, and carried from spirals to dwarfs, early-type galaxies and clusters. Two more results follow that a fit cannot give: a bound explains why modified gravity that copies visible mass by a fixed recipe with constant parameters, MOG and nonlocal gravity included, falls short on most dwarfs; and the 2025 repair of MOND in clusters cannot be extended from hot gas to the cold gas of spirals. A last result is a warning: cluster benchmarks scored at R_500 are circular, since that radius is defined by the mass they test. The paper then scores dwarfs no rule had seen, and all 85 dwarfs together with their upper limits, under protocols hashed before scoring, and fixes

forecasts for 264 dwarfs whose motions have not been measured, so that later measurements can confirm or refute the rule. A fifth protocol carries the published Moster relation, unchanged, to cluster scales as an exploratory extension, and a sixth and a seventh audit that test. An eighth and a ninth check the rule’s whole halos, and the lensing profiles it predicts, against galaxy-galaxy lensing.

2. Data and the rules of the test

2.1 Five populations

The main sample holds 279 systems in five populations, and seven more clusters serve as a separate stress test.

Rotating disk galaxies come from SPARC (Lelli, McGaugh and Schombert, 2016): 175 galaxies and 3,391 radial measurements. Selecting quality flag below 3, inclination at least 30 degrees, positive radius and speed, relative speed error at most 0.1 and positive inward baryonic acceleration leaves 149 galaxies and 2,700 measurements. Disk and bulge mass-to-light ratios stay fixed at 0.5 and 0.7 in solar units at 3.6 micrometers, with catalog distances and inclinations. The baryonic speed is V_bar^2 = V_gas|V_gas| + 0.5 V_disk|V_disk| + 0.7 V_bulge|V_bulge|, and

the accelerations are g_bar = V_bar^2/r and g_obs = V_obs^2/r.

Dwarf spheroidal galaxies come from McConnachie’s catalog of nearby galaxies in its January 2021 update (McConnachie, 2012): 25 classical and 19 ultrafaint systems, divided at a V-band luminosity of 10^5 suns. Each gives one point at the half-light radius in three dimensions, r_1/2 = (4/3) R_e, with g_obs = 3σ^2/r_1/2 from the line-of-sight dispersion σ (Wolf et al., 2010) and g_bar = G(M_star/2)/r_1/2^2, the stellar mass set at twice the V-band luminosity. Mass-to-light ratios of one and three are kept as sensitivities. Unresolved dispersions, flagged tidal systems and measured gas above a tenth of the stellar mass are excluded.

Sixteen early-type galaxies give one inner Jeans estimate and one outer neutral-hydrogen estimate each (Lelli et al., 2017), with a photometric mass-to-light ratio of 0.8 at 3.6 micrometers. Fifty brightest cluster galaxies from MaNGA give isotropic Jeans estimates (Tian et al., 2024). Twenty CLASH clusters give 84 lensing estimates, at their centers and at 100, 200, 400 and 600 kiloparsecs (Tian et al., 2020). Seven X-COP clusters give 21 hydrostatic estimates at fixed radii of 500, 750 and 1,000 kiloparsecs (Ettori et al., 2019; Eckert et al., 2022).

The accelerations for dwarfs, early-type galaxies, cluster centers and clusters are inferences, each carrying its source’s dynamical or lensing model. They are treated as data with those assumptions attached, and no score below pools populations into one ranking.

2.2 Held out by system

Every score is a prediction for systems left out of the fit. Folds hold out whole galaxies or whole clusters, all radii of a system stay in one fold, and objects that may share a host stay together. Population tests go further: a rule fitted on one population is frozen and applied to another, with no target-population calibration. Coefficients are refitted inside each training split, each galaxy weighs equally in fits and scores, and the reported score averages three complete assignments of galaxies to folds.

Scores are root-mean-square errors in log10 acceleration, in dex. An error of 0.15 dex is a factor of 1.41; 0.48 dex is a factor of 3. Speed errors take the square root of the acceleration ratio, so a 0.15-dex acceleration error is 0.075 dex in speed, about 16 to 19 percent. The median speed errors quoted below are a different statistic, the median absolute fractional error.

The data had been inspected before these comparisons, and each protocol was fixed before its own run. The results on the main sample are exploratory comparisons of declared rules, not a preregistered discovery. The out-of-sample tests, the combined score, the two lensing checks, the cluster-scale test and its two audits (Sections 3.3 to 3.6, 3.8 and 5.3) ran under written protocols, each hashed before its own scoring, on data inspected in earlier runs. A hash fixes a file’s content, not when it was written; the times given are the author’s records.

3. Halos set by starlight

3.1 The rule

Each galaxy gets a finite NFW halo, density ρ_s/[z(1+z)^2] with z = r/r_s, truncated at the radius r_200 enclosing 200 times the critical density. Its concentration c_200 = r_200/r_s follows the published Planck relation of Dutton and Macciò (2014),

c_200 = 10^0.905 (M_ref / 10^12 h^-1 M_sun)^-0.101, h = 0.671,

so only the halo’s reference mass M_ref needs predicting. The rule predicts it from the galaxy’s stellar mass M_star = 0.5 L_3.6 and its effective radius R_e:

$$M_{ref} / M_{sun} = M_p/(f_b M_{sun}) + 10^A (M_{star}/10^{10} M_{sun})^B (R_e/3 \text{ kpc})^C,$$

where $M_p = M_{star} + 1.33 M_{HI}$ is the galaxy's baryonic mass and $f_b = 0.0490/0.3175$ is the cosmic baryon fraction. The first term keeps the halo at least as massive as the baryons allow. The halo's dark mass is $(1 - f_b) M_{ref}$, and its pull adds to the measured baryonic pull. Fitted to all 149 disks, $A = 11.748$, $B = 0.818$ and $C = -0.712$; every prediction below uses coefficients fitted without the system predicted.

No rotation measurement of a predicted galaxy enters its prediction: stellar luminosity, gas mass and size set the halo, and the halo and the baryons set the curve. Stellar-to-halo relations have long precedent (Moster, Naab and White, 2013), and baryon-halo scaling has been tied to the radial acceleration relation before (Navarro et al., 2017; Li et al., 2019). What is tested here is whether such a rule, with three coefficients, carries from disks to systems it never saw. B and C nearly cancel ($B + C = 0.11$), so the rule depends mainly on how compact the stars are, M_{star}/R_e, a measure of the depth of the stellar potential. One reading is that a deeper halo pulls cooling gas into a more compact galaxy; this paper does not test it.

3.2 Disks: as good as MOND’s relation

Table 1. Shared predictors fitted on the disks: error on unseen disks and, frozen, on the 44 dwarfs, in dex.

Shared predictor; Coefficients; Unseen disks, dex; Frozen dwarfs, dex.

Radial acceleration relation, refitted a0; 1; 0.15309; 0.66937.

Halo set by baryonic mass; 3; 0.15323; 0.66757.

Published Moster relation, no fitting; 0; 0.17111; 0.55591.

Halo set by stellar mass and size; 3; 0.15329; 0.47914.

Curved stellar relation; 5; 0.15380; 0.47874.

Fitted on all the disks with each galaxy weighted equally, the relation takes a0 = 1.04 × 10^-10 m/s^2 and the baryonic rule A = 11.467, B = 0.994 and C = -0.635. On unseen disks the stellar rule matches the radial acceleration relation to 0.0002 dex, and its median speed error is 9.09 percent. Two predictors built on opposite assumptions describe disk rotation equally well from the same inputs, which is why disks alone cannot settle the question. A curved stellar relation with five coefficients adds nothing.

3.3 Dwarfs: 28 percent better

Frozen on the disks and applied to the 44 dwarf spheroidals, using only their stellar mass and size, the stellar rule cuts the error from 0.669 dex for the radial acceleration relation to 0.479 dex, 28 percent lower. A paired bootstrap over whole dwarfs puts the gain over the baryonic-mass halo at -0.188 dex, with a 95 percent interval of -0.281 to -0.083. Thirty of the 44 dwarfs improve. The relation does no better when its own constant is refitted on the dwarfs and scored on dwarfs held out: 0.548 dex (Section 4.3). The stellar rule was proposed after the baryonic rule’s dwarf transfer had been seen; its coefficients come from the disks alone.

The transfer holds on a second compilation. On a fixed version of the Local Volume Database (Pace, 2025), which gives each measurement with its source, 42 dwarf spheroidals with measured dispersions, 33 of them among the 44, give the same ranking. With the same circularized radii and frozen rules, the stellar rule scores 0.389 dex against 0.611 for the radial acceleration relation and 0.612 for the baryonic-mass halo, 36 percent lower than the relation; with major-axis radii, 0.367 against 0.613 and 0.616.

A test on dwarfs outside the original sample answers the order of choices. Under a written protocol, with the rules frozen on the disks, the 25 dwarf spheroidals in the same database that are not among the 44 and pass the same selection were scored; 17 of them were scored blind, under a protocol fixed and hashed before any was seen. The stellar rule scores 0.441 dex against 0.808 for the radial acceleration relation and 0.802 for the baryonic-mass halo, 45 percent lower; the paired bootstrap interval of its gain over the relation, -0.449 to -0.272 dex, excludes zero, and 21 of the 25 dwarfs improve. On the 17 scored blind the figures are 0.405 against 0.776. The ranking holds for major-axis radii and mass-to-light ratios of 1 and 3. The published Moster relation, a second halo set by starlight and fitted to no rotation data, does slightly better on these dwarfs, 0.403 dex (0.397 on the 17 scored blind), and worse on the original 44 and the second compilation, 0.556 and 0.463 dex. Both starlight rules beat MOND’s relation on all three dwarf sets.

A learning curve shows the transfer is learned from the disks, not tuned to any of them. Holding out a random 5 to 90 percent of the disks, 50 times at each fraction, and refitting every rule on the rest, the stellar rule beats the radial acceleration relation on both dwarf sets in every draw from 30 training disks up, and in at least

39 of 50 draws even from 15. As training grows, its dwarf errors settle, from 0.500 to 0.478 dex on the original 44 and near 0.44 on the 25 out of sample, their spread across draws shrinks five- to sevenfold, and its three coefficients converge; on the held-out disks the three rules stay tied.

MOND’s defenders answer that a satellite’s own gravity is weakened by its host’s field, the external field effect, which Chae et al. (2021) report in the rotation curves of disk galaxies. Applied as McGaugh and Milgrom (2013) apply it to Andromeda’s dwarfs, with the host’s field V^2/D, V = 230 km/s for Andromeda and 200 km/s for the Milky Way, it lowers MOND’s predictions for dwarfs that already move faster than MOND predicts: its error rises to 0.96 dex on the original 44 and 1.20 on the 25 out of sample, and stays between 0.91 and 1.00 on the 44 for Milky Way speeds of 175 to 220 km/s. On the 25 brighter dwarfs of the original sample it matches the stellar rule, 0.41 against 0.40 dex; on the 19 ultrafaint ones it misses by 1.39 dex. MOND’s answer for those, that they are out of equilibrium, needs evidence that one point per dwarf cannot give.

The halo has an environment too. In simulations, a subhalo’s concentration rises toward its host’s center by a factor 1 - 0.54 log10(R/R_200) (Moliné et al., 2017). The factor describes

concentration at a satellite’s present distance, not stripping along its orbit. Applied to the stellar rule as a radial factor on its baseline concentration, with that published coefficient and host masses of 1.0 x 10^12 and 1.5 x 10^12 suns, it lowers the error to 0.457 dex on the original 44 and 0.383 on the 25 out of sample, below the Moster relation on both (0.556 and 0.403), mainly by lifting the ultrafaint dwarfs nearest their hosts. It was tried after the residuals were seen to grow toward the host, so it was kept out of the rule as a lead for the next blind test, which rejected it (Section 3.4). In an exploratory check with measured pericenters (Pace, Erkal and Li, 2022), the stellar rule’s residuals for 31 Milky Way satellites show no trend with pericenter or present distance at fixed luminosity (rank partial correlations 0.05 and -0.11), while at fixed pericenter fainter dwarfs move faster than predicted (-0.41, p = 0.02). The faint dwarfs are the nearby ones (rank correlation of distance with luminosity 0.50), so the trend toward the host is a trend with luminosity. Binary stars could be part of it: in the same multi-epoch data, modeling binaries and instrumental offsets lowers Reticulum II’s dispersion from 3.5 to 2.8 km/s (Minor et al., 2019), and the two binaries found in Boötes II raise its dispersion by about 25 percent, to 3.6 from 2.9 km/s (Bruce et al., 2023).

About half of what remains is measurement. The quoted errors of the dispersions and radii give 0.30 dex on the original 44 and 0.34 dex on the 25; a likelihood that separates them leaves an intrinsic scatter of 0.32 and 0.33 dex, about a factor of 2, for differences between halos and for orbits, tides and binary stars, which a rule given only stellar mass and size cannot see.

The one change between the two halo rules is which mass sets the halo: stars or all baryons. Disks cannot tell them apart. Dwarfs, which have stars and almost no gas, can. The halo follows the stars.

The gain is not uniform. Ultrafaint dwarfs improve from 0.934 to 0.571 dex, and classical dwarfs worsen from 0.348 to 0.395. Satellites of the Milky Way improve, and those of Andromeda worsen. Three of four tails of the disk sample, cut by predictor values, worsen slightly. The remaining dwarf errors, a factor of about 3, are large, and satellite evolution, tides and binary stars are not modeled. The result is a better transfer, not a solved dwarf problem.

3.4 Dwarfs no rule had scored

The database holds two kinds of dwarfs that no earlier test scored. Some lie beyond the Milky Way and Andromeda systems, and some have only an upper limit on their dispersion. Dropping the limits drops the slowest dwarfs, which favors any rule that predicts fast motion. A third protocol, written and hashed before any prediction for these dwarfs was computed, scored both kinds with every rule frozen (Table 2).

Table 2. Dwarfs no rule had scored, every rule frozen: error on the five with measured dispersions, in dex, and upper limits exceeded, of ten.

Model; Five measured, dex; Limits exceeded.

Halo set by stellar mass and size; 0.171; 2.

Stellar rule with subhalo factor; 0.171; 3.

Radial acceleration relation, disk a0; 0.202; 0.

Radial acceleration relation, published a0; 0.185; 0.

MOND with external field; 0.208 (four dwarfs); 0.

Halo set by baryonic mass; 0.225; 0.

Published Moster relation; 0.235; 4.

Five dwarfs have measured dispersions: Cetus, Phoenix, Tucana and UGC 4879, which have no host galaxy, and KDG 64 in the M81 group. The first four are far from any large galaxy, so they feel almost no tides or external field, and neither of MOND’s answers for satellites applies to them. The stellar rule has the lowest error of seven models, 0.171 dex against 0.202 for MOND’s relation, 0.208 for MOND with the external field on the four it covers, and 0.235 for the Moster relation. Five dwarfs cannot make that significant: the bootstrap interval of the gain over MOND’s relation runs from -0.170 to +0.144 dex, so the protocol’s first test is not met. The ranking is also fragile. With major-axis radii, MOND’s relation with its published a0 scores lower than the stellar rule, 0.186 against 0.201 dex, and at a mass-to-light ratio of 3 every version of MOND does.

Ten satellites have only upper limits, at 95 percent confidence in the database. The stellar rule’s predicted dispersion lies below eight of them. It exceeds the limit of Grus II, 3.3 against 2.0 km/s (Simon et al., 2020), and of Andromeda XII, 5.2 against 4.0 km/s (Collins et al., 2013); only Grus II lies beyond the rule’s own scatter, a factor of 1.46 in dispersion. MOND meets all ten, so the protocol’s second test is not met either. MOND passes because it predicts slow motion for every faint dwarf. The same predictions

miss the faint dwarfs with measured dispersions in Section 3.3, 19 in the original sample and 17 of the 25 outside it, by 1.39 and 1.35 dex in root-mean-square, factors above 20 in acceleration. The Moster relation, which predicts faster motion than the stellar rule, exceeds four limits. These counts hold for major-axis radii, mass-to-light ratios of 1 and 3, and Milky Way speeds of 175 and 220 km/s.

Upper limits can catch only a rule that predicts too much motion. Measured dispersions catch errors both ways, and on all four dwarf sets with measured dispersions the stellar rule is ahead of MOND’s relation. Section 3.5 weighs both kinds in one score. The subhalo factor of Section 3.3 raises the predicted dispersions of satellites close to their hosts, where several limits are lowest, and exceeds three limits, one more than the rule without it. The lead fails its test and stays out of the rule.

3.5 All 85 dwarfs in one score

Measured dispersions and upper limits can be weighed together. A fourth protocol, hashed before any likelihood was computed, puts all 85 dwarfs scored so far into one likelihood, every value taken from the database: 75 measured dispersions count as values with their errors, and 10 upper limits count as ceilings.

Each model keeps its prediction as the center, and its one free number is its scatter (Table 3).

Table 3. All 85 dwarfs in one likelihood, 75 measured and 10 upper limits, every rule frozen. Each model's only free number is its scatter, in dex. The last column is how far its log-likelihood falls below the stellar rule's.

Model; Scatter, dex; Log-likelihood; Behind stellar rule.

Halo set by stellar mass and size; 0.30; -39.3; 0.

Stellar rule with subhalo factor; 0.29; -36.9; -2.4.

Published Moster relation; 0.40; -54.4; 15.1.

Radial acceleration relation, published a0; 0.56; -72.8; 33.5.

Radial acceleration relation, disk a0; 0.58; -75.4; 36.2.

Halo set by baryonic mass; 0.58; -75.7; 36.4.

MOND with external field; 0.93; -106.0; 66.7.

Among the baseline frozen rules, the stellar rule fits best, with a scatter of 0.30 dex; the exploratory subhalo variant, discussed below, scores slightly higher. It beats MOND's relation by 36 units of log-likelihood, with a bootstrap interval of 24 to 48 units, which meets the protocol's test. It beats MOND with the external field by 67 units. The limits cost the stellar rule 4 units more than

MOND’s relation, and the measured dwarfs give it 40 units more. On average the stellar rule sits 0.02 dex from the data, while MOND’s relation sits 0.43 dex too low. Even when MOND’s relation may shift its whole prediction to fit the dwarfs, the unshifted stellar rule stays ahead by 9 units.

The gain over MOND’s relation stays between 27 and 48 units for mass-to-light ratios of 1 and 3, for major-axis radii, without measurement errors, and with every measured dispersion below 3 km/s dropped. The last checks binary stars, which inflate the smallest dispersions most; modeling the binaries themselves needs repeated spectra of each star and is left for later work. The Moster relation trails the stellar rule by 15 units (3 to 25). The subhalo factor scores 2.4 units better than the rule without it, inside its bootstrap interval of -8.6 to +3.3, so its net effect is not established and it stays out of the rule.

Two checks added after the run score each dwarf held out in turn. With each model’s scatter fitted to the other 84 dwarfs, the stellar rule leads MOND’s relation by 35.7 units, with a bootstrap interval over dwarfs of 21.8 to 47.8, and 70 of the 85 dwarfs favor it. When both may also shift their predictions, again fitted without the dwarf scored, the lead is 8.8 units (1.4 to 15.4). With that shift the baryonic-mass halo comes within 4.2 units (-4.4 to 11.4): its

loss on dwarfs is mostly one uniform shortfall of 0.45 dex, and setting that level right from the disks alone is what the stellar rule adds. These scores share the likelihood’s simplifications: an upper limit counts as an exact ceiling, the errors that size and distance share between predicted and measured accelerations are left out, and the bootstrap treats satellites of one host as independent.

3.6 Forecasts for dwarfs not yet measured

The same protocol fixed the dispersion each rule forecasts for the 264 gas-poor dwarfs in the database with no measured dispersion. The file was hashed on 31 October 2026, and its SHA-256 begins bbafc29918f89053. Table 4 lists the nine in the Local Group, with 68 percent ranges from the scatter of Section 3.5. For these, the stellar rule forecasts dispersions a median 1.5 times MOND’s with the external field. The protocol calls a forecast decisive when the two differ by more than a factor of 2, which among the nine holds for Carina IV alone, the faintest: 2.1 km/s from the stellar rule against 0.84 from MOND. The other 255 lie farther out, most in groups of galaxies up to 13 megaparsecs away, where the two forecasts differ by a median factor of 1.3. The ranges are a factor of 1.42 either way for the stellar rule and

2.9 for MOND, whose scatter must be wide to cover its misses. Measured together, the nine separate the two forecasts by 1.4 standard deviations, so the set weighs more than any single dwarf. Measured dispersions near MOND’s for the nine would count against the rule, and near the rule’s, against MOND.

Table 4. Forecast line-of-sight dispersions, km/s, with 68 percent ranges, for the Local Group dwarfs with none measured, faintest first. MOND includes the field of the host, or for dwarfs with no host the stronger of the Milky Way’s and Andromeda’s fields.

Dwarf; Host; M_V; Stellar rule; MOND; Ratio.

Carina IV; Milky Way; -2.8; 2.11 (1.49 to 2.99); 0.84 (0.29 to 2.44); 2.50.

Leo K; none; -4.73; 3.09 (2.18 to 4.38); 1.80 (0.62 to 5.23); 1.71.

Andromeda XXXV; M31; -5.2; 3.07 (2.17 to 4.35); 2.01 (0.69 to 5.84); 1.53.

Pegasus VII; M31; -5.7; 4.13 (2.92 to 5.85); 2.25 (0.77 to 6.54); 1.83.

Leo M; none; -5.77; 3.76 (2.65 to 5.33); 2.29 (0.79 to 6.66); 1.64.

Andromeda XXXVI; M31; -5.91; 3.49 (2.46 to 4.94); 2.37 (0.82 to 6.89); 1.47.

Pegasus V; M31; -6.3; 3.91 (2.76 to 5.54); 2.59 (0.89 to 7.53); 1.51.

Tucana B; none; -6.9; 4.20 (2.97 to 5.95); 2.97 (1.02 to 8.64); 1.41.

Pegasus W; none; -7.2; 4.50 (3.18 to 6.37); 3.18 (1.09 to 9.25); 1.41.

3.7 What does not help

Published feedback profiles, which change halos by the gas that star formation drives out and which fit individual galaxies well (Li et al., 2020), do not improve the shared rule: the DC14 recipe (Di Cintio et al., 2014) gives 0.189 dex on disks, a mixture with NFW chooses a 5 percent share and gains nothing, and coreNFW (Read, Agertz and Collins, 2016) with an assumed formation time gives 0.163 dex. These cored profiles are the usual answer to the cusp-core problem, and the shared rule does better without them. A blend of the radial acceleration relation and the stellar halo, weighted per training split, lowers the disk error by under 2 percent, a gain its uncertainty does not establish, and transfers to dwarfs less well than the stellar halo alone (0.505 dex). Allowing halo scatter improves a predictive log score by 0.19 units per radius, with a 95 percent interval of 0.09 to 0.30, without

improving the central error; the stellar rule’s gain is in the point prediction itself.

The stellar rule is calibrated on single disk galaxies and is not carried to the heavier systems. Above about the Milky Way’s stellar mass, the relation between a galaxy’s stars and its halo bends, as mergers add halo mass faster than stars (Moster, Naab and White, 2013), and a cluster’s potential sums the halos of hundreds of member galaxies and its hot gas. Standard structure formation predicts that change; a universal force law has to hold with one constant through it, and Section 5 tests whether it does. Section 5.3 carries the published Moster relation, which includes the bend, to cluster scales.

3.8 Whole halos from lensing

Rotation curves reach a few tens of kpc, while galaxy-galaxy lensing weighs whole halos. Mandelbaum et al. (2016) measured the mean halo masses of red and blue central galaxies in the Sloan survey, in bins of stellar mass. Under an eighth protocol, hashed before any prediction, the stellar rule frozen on the disks, given the disks’ own scalings of size and gas with stellar mass, predicts the halos of the blue, star-forming centrals within 0.16 dex in four of five bins from 10^10.2 to 10^11.3 suns, a median

0.05 dex high. At 10^10.56 suns it is 0.34 dex high, where the measured mass dips below that of the bin beneath, so the protocol’s test, every bin within 0.2 dex or twice its error, is not met. Means over a bin exceed typical masses by roughly 0.05 to 0.1 dex, which the comparison leaves uncorrected. For red centrals half the size of disks, the rule matches the measured halos up to 10^10.9 suns but gives only about half of the measured red-blue split there, 0.17 to 0.20 dex against 0.35 to 0.41. Above 10^11 suns the red centrals’ halos grow about as the square of stellar mass and lie above the rule by 0.3 dex at 10^11.1 suns and 1.0 dex at 10^11.7: the bend of Section 3.7, reached by the centrals of groups and clusters. X-ray masses of 21 clusters show the same steepness from the other side: their central galaxies’ stellar mass grows only about as M_500^0.4, with about 0.2 dex scatter (Kravtsov, Vikhlinin and Meshcheryakov, 2018). The inverted Moster relation lies 0.4 to 1.0 dex above the red centrals’ mean halos over the same range.

Lensing also traces how the mass is spread. Brouwer et al. (2021) stacked the Kilo-Degree Survey (KiDS-1000) lensing signal of 259,383 isolated galaxies at redshifts 0.1 to 0.5 in 15 bins of baryonic acceleration, spanning about 30 kpc to 3 Mpc, for four bins of stellar mass and for early and late types of

matched stellar mass. A ninth protocol, hashed before any measured value was read, as were the predicted profiles, carries the frozen rule into these stacks. It uses the published stellar masses, the disks’ scalings of size and gas for late types, and, for early types, no gas and the median size gap between early and late types at each stellar mass in Hubble imaging (van der Wel et al., 2014). Inside 0.3 Mpc, where false isolated lenses add little, the measured signal lies 0.027 ± 0.015 dex below the prediction for the whole sample, between 0.005 and 0.048 dex below it in each of the four stellar-mass bins from 10^8.5 to 10^11 suns, and 0.00 and 0.01 dex below it for red and bulge-dominated galaxies. For blue and disk-dominated galaxies it lies 0.14 and 0.12 dex below, outside the protocol’s 0.1 dex; the rule likewise exceeds the halo masses of Mandelbaum’s blue centrals of 10^10.5 to 10^11 suns, by 0.13 to 0.34 dex. At redshift 0.3, near KiDS’s 0.25, CFHTLenS halo-model masses of blue galaxies, with satellites modeled, instead lie within 0.07 dex of the rule at 1.7 and 4.2 × 10^10 suns (Hudson et al., 2015), so the surveys do not yet agree for these galaxies. The rule therefore puts early types only 0.05 dex above late types of the same stellar mass, against a measured 0.16 dex by Sérsic index and 0.21 dex by color inside 0.3 Mpc: about a third and a quarter of the split,

which the protocol counts as part of the first and not the second. Beyond 0.3 Mpc the whole sample’s signal exceeds the single-halo model by 0.16 to 0.57 dex, where Brouwer et al. expect neighbors missed by the isolation criterion to add signal. An independent script reproduces every amplitude to 0.0001 dex.

These are amplitudes; the radial profile does not match. With the release’s covariance, the whole sample’s seven bins inside 0.3 Mpc give a chi-square of 48.6 against the frozen prediction and 45.1 for six degrees of freedom once the amplitude is fitted (p = 4 × 10^-8). The measured signal lies 0.19 dex below the prediction in the innermost bin and 0.16 dex above it in the bin nearest 0.3 Mpc, so the measured profile is flatter than the predicted one. Three of the four stellar-mass bins and the blue and bulge-dominated galaxies also miss the profile, at p = 10^-4 to 0.03, while the red and disk-dominated galaxies match it (chi-square 7.6 and 8.1 for six). The protocol’s lensing model extends each halo beyond r_200, where the rotation-curve fits truncate it; halos truncated at r_200 move the amplitudes by 0.003 dex or less and still miss the profile (40.6 for six), as do all 14 of the protocol’s sensitivity cases (39.5 to 50.1). These tests take the model as exact and use only the measurement covariance, and since the samples overlap, their p-values are not

independent.

Lensing does not yet support the rule’s size term at fixed stellar mass and type. In KiDS near 3 × 10^10 suns, halo mass follows size and Sérsic index more closely than color, with a suggestion that compact early types sit in heavier halos while late types share about one halo mass (Taylor et al., 2020). In CFHTLenS, at fixed stellar mass and color, halo mass grows with size, as size to the power 0.28 ± 0.18 for blue galaxies and 0.53 ± 0.15 for red ones, and 0.22 ± 0.42 for blue galaxies near 10^10.6 suns, with satellites kept in the samples and concentration fixed by mass (Charlton et al., 2017); the rule’s halo mass instead falls as size to the power of about -0.6 there. Charlton et al. note that with concentration fixed, part of a size trend could be a change of concentration instead. A lensing measurement of halo mass against size at fixed stellar mass and type, for central galaxies and with the rule’s prediction fixed first, would decide it.

4. A mass budget no fixed kernel can meet

4.1 The bound

In the weak-field limit, nonlocal gravity (Rahvar and Mashhoon, 2014) and MOG with constant parameters (Moffat and Rahvar, 2013) both act as if the ordinary matter carried extra effective matter, spread from it by a fixed positive kernel q:

□^2Φ = 4πG(ρ_b + ρ_eff), ρ_eff(x) = ∫ q(|x - y|) ρ_b(y) d^3y,

with q_NLG(d) = αμ(1 + μd)e^(-μd)/(8πd^2) and q_MOG(d) = αμ^2 e^(-μd)/(4πd). Each kernel integrates to α over all space. An isolated source of baryonic mass M_b therefore carries total effective mass α M_b, and no part of space can hold more of it than that. At the source's own half-mass radius r_1/2, inside which the baryonic mass is M_b/2, the enclosed effective mass is at most α M_b, so

g/g_b = 1 + M_eff(< r_1/2)/(M_b/2) ≤ 1 + 2α.

The bound holds for any positive spherical source, whatever the kernel's range 1/μ and whatever the source's radial profile. For a source whose density falls outward, such as a Plummer sphere, no displaced ball encloses more mass than the centered one, and the ceiling tightens to 1 + α, since then M_eff(< r_1/2) ≤ α M_b/2.

4.2 The dwarfs exceed it

The published fitted values (α, μ in kpc^-1) are (10.94, 0.059) for nonlocal gravity and (8.89, 0.042) for MOG, so the general ceilings are 22.9 and 18.8, and for a Plummer sphere 11.9 and 9.9. The dwarf estimates sit at (4/3)R_e, where a Plummer source encloses 0.512 of its mass, so the general ceiling there is 1 + α/0.512: 22.4 and 18.4. At a stellar mass-to-light ratio of 2, the inferred ratio g_obs/g_bar of 31 of the 44 dwarfs exceeds the nonlocal gravity ceiling, and 33 exceed MOG's. At a mass-to-light ratio of 3, 29 exceed each. These are counts of central estimates, and the estimates carry the assumptions of dynamical equilibrium, membership, anisotropy and binary stars; but the bound itself needs no fitting, and most dwarfs sit well above it.

4.3 Integrating the kernels

The bound is a ceiling. The full prediction integrates each kernel against a Plummer sphere fixed by each dwarf's stellar mass and size, with matter outside the half-mass radius included:

Table 5. Each dwarf's stars spread as a Plummer sphere with half its mass inside r_1/2, with the relation and the two kernels integrated over it, in dex. Refitted models are scored on dwarfs

held out in five folds, in the first of three assignments; the mean of the three is 0.543 for the relation and 0.722 for the kernels.

Model, mass-to-light ratio 2; Dwarfs held out, dex.

Published radial acceleration relation, fixed a0; 0.651.

Radial acceleration relation, a0 refitted on dwarfs; 0.548.

Published nonlocal gravity kernel; 1.862.

Published MOG kernel; 1.905.

Nonlocal gravity, both coefficients refitted; 0.729.

MOG, both coefficients refitted; 0.729.

The relation scores 0.651 here with its published a0 and 0.669 in Section 3.3 with a0 refitted on the disks. Here the Plummer scale puts exactly half the mass inside r_1/2, where Section 4.2 takes the scale as R_e, which puts 0.512 inside; an independent integration with that scale and its own folds agrees with the table within 0.01 dex.

With their published coefficients the kernels miss by factors near 70. Refitted on the dwarfs, all 108 fits drive the range parameter μ to its upper bound, collapsing the kernel to a point and the theory to a constant multiplier of about 59; even then they do worse than the radial acceleration relation. Earlier studies found

the same tension by other routes: no common MOG parameter pair for eight classical dwarfs with plausible stellar masses (Haghi and Amiri, 2016), and tensions in the nonlocal kernel's parameters in full Jeans analyses (De Martino, Della Monica and Roshan, 2025). The bound shows why. The extra pull a fixed linear kernel can give is capped by the matter it is given, and dwarfs have too little matter for their pull.

The bound covers isolated sources and constant coefficients. Running-parameter MOG, external fields, nonlinear responses and full covariant theories need their own treatment.

5. One acceleration scale does not carry across

5.1 The scale moves with the population

Fitted on the disks point by point, the radial acceleration relation takes a0 = 1.16 x 10^-10 m/s^2. Frozen at that value and carried to the other populations, its median acceleration errors are 50 percent for classical dwarfs, 86 percent for ultrafaint dwarfs, 34 percent for early-type galaxies, 48 percent for cluster centers and 69 percent for CLASH clusters. Refitted on all five populations together, a0 becomes 5.34 x 10^-10 and the disks lose their accuracy. Fitted on the X-COP clusters alone, it becomes 7.16 x

10^-10, six times the disk value. A law with one constant that needs a different constant for each kind of system is no longer one law. That clusters need more than MOND supplies is known (Tian et al., 2020; Famaey, Pizzuti and Saltas, 2025); the numbers here put the disk-to-cluster gap on one footing with the rest.

Freer formulas do not repair it. A six-coefficient quadratic in acceleration and radius wins when all populations are in training, 0.236 dex on held-out groups, and falls to 0.488 dex when a whole population is held out. Flexibility fits the populations it has seen.

5.2 Extra mass that follows hot gas, and not cold

Famaey, Pizzuti and Saltas (2025) proposed repairing MOND in clusters with an extra density that follows the hot gas, ρ_extra = η ρ_hot e^(-r/R_c). With a0 fixed at 1.2 x 10^-10 and only η and R_c fitted, the repair transfers between cluster surveys: trained on 16 CLASH clusters (η = 11.8, R_c = 289 kpc), it predicts the seven X-COP clusters with a median error of 7.6 percent, against 53 percent for a single refitted a0. The reverse transfer gives 17 against 38 percent, though within CLASH alone the repair does no better than one refitted a0.

The repair is stated for hot gas. Carrying it to gas-rich disks tests whether it can become a rule for gas as such, whatever its temperature. In 61 disks where gas supplies at least half the baryonic pull, the fit prefers η near zero, with an 11.6 percent median speed error; carrying the cluster value over raises the speed error to 99.7 percent. Forced to share η and R_c, disks and clusters both deteriorate, and the joint fit pushes R_c to its upper bound. The extra mass appears where the gas is hot, inside deep cluster potentials, and not where it is cold, which is how a component tied to structure formation behaves and how a rule attached to gas does not.

5.3 A published halo relation at cluster scale

The halo side can be carried across the same gap, as an exploratory extension that rests on the published Moster relation alone; the paper’s own three-coefficient rule remains a disk-to-dwarf result. The Moster relation of Sections 3.2 to 3.5, unchanged, maps a central galaxy’s stellar mass to its host halo’s mass up to about 10^15 suns. A fifth protocol, hashed before any prediction was computed, applies it with nothing fitted to the 50 MaNGA brightest cluster galaxies and the 20 CLASH clusters of Section 2.1. Each brightest galaxy’s published stellar mass and

redshift set its halo, the concentration follows Dutton and Macciò (2014) at that redshift, and the measured baryons are added (Table 6); the CLASH stellar masses are those of Cooke et al. (2016), which Tian et al. take as total. Moster, Naab and White (2013) define halo mass at 200 times the critical density, as the concentration relation does, and give masses in suns for h = 0.704; as in Section 3.1, the halo's dark mass is (1 - f_b) times the halo mass, and its pull adds to the measured baryonic pull. Each galaxy or cluster weighs equally.

Table 6. Cluster-scale errors in dex with nothing fitted to these systems: the published Moster relation, inverted at its mean (fifth protocol) or with its scatter and the halo mass function (sixth protocol), MOND's relation and the disk-calibrated stellar rule.

Model; All 70 systems; MaNGA BCGs; CLASH centers; CLASH, 100 to 600 kpc.

Published Moster relation, mean; 0.185; 0.182; 0.154; 0.204.

Published Moster relation, with scatter; 0.288; 0.240; 0.214; 0.422.

Radial acceleration relation, disk a0; 0.403; 0.338; 0.479; 0.550.

Radial acceleration relation, published a0; 0.388; 0.327; 0.460; 0.526.

Stellar rule, disk-calibrated; 0.474; 0.328; 0.435; 0.791.

Inverted at its mean, the Moster relation cuts the error of MOND’s relation by more than half, 0.185 against 0.403 dex, and the bootstrap interval of the gain, -0.248 to -0.188 dex, meets the protocol’s test. Its median acceleration error is 26 to 33 percent in each group, against 50 to 72 percent for MOND’s relation, which falls short in every group: by 0.30 dex in the MaNGA galaxies, 0.45 dex at CLASH centers and 0.54 dex at 100 to 600 kpc. The Moster relation’s largest offset is at the MaNGA centers, 0.15 dex short. The disk-calibrated stellar rule fails at 100 to 600 kpc (0.79 dex), as Section 3.7 expects, since it has no bend above the Milky Way’s mass. The gain holds with the CLASH stellar masses doubled, with the MaNGA masses lowered by 0.3 dex, and without redshift evolution: 0.20 to 0.24 dex against 0.40.

The relation is nearly flat at the top, so a small change in stellar mass moves the halo mass a lot. Five of the 20 CLASH galaxies map to halos above 10^16 suns, beyond any known cluster, and these five are overpredicted by 0.02 to 0.46 dex. For two of them, MACS J0429.6-0253 and MACS J0647.7+7015, the search found no root and stopped at its upper bound, 10^16.5 suns. Without redshift evolution only one halo lies above 10^16 suns, and the error falls to 0.20 dex.

A sixth protocol, hashed before it ran, audited this test. Inverting the mean relation ignores its published scatter of 0.15 dex in stellar mass at fixed halo mass. Inverted instead with that scatter, the measurement errors and the halo mass function of Tinker et al. (2008), computed with Colossus (Diemer, 2018), the relation puts every halo’s posterior median below 10^15 suns and still beats MOND’s relation, 0.288 against 0.403 dex, a 29 percent cut rather than 54, with a bootstrap interval of the gain of -0.136 to -0.095 dex (Table 6). Scored only on the 50 systems whose stellar masses lie within the range the relation was fitted on, at most 10^11.7 suns, the mean inversion gives 0.187 against 0.377 dex, and with h = 0.704, as Moster’s parameters assume, 0.182 against 0.403. The accelerations survive these checks; the halo masses do not.

Tian et al. computed the CLASH lensing accelerations from the spherical NFW fits of Umetsu et al. (2016), and a seventh protocol, hashed before it ran, compares the Moster masses with those published masses, M_200 of 10^14.8 to 10^15.5 suns; NFW fits to the accelerations themselves recover the published masses within 0.05 dex. The mean inversion’s masses scatter by 0.73 dex about the published ones, a factor of five, where the published errors allow 0.13 dex, around a median 0.21 dex high;

only 5 of the 20 lie within twice the published error. No central galaxy could do much better: with its stellar mass growing as about M_500^0.4 and scattering by about 0.2 dex (Kravtsov, Vikhlinin and Meshcheryakov, 2018), it fixes the halo mass only to about 0.5 dex, while the cluster's total stellar mass, growing as M_500^(0.6 ± 0.1) with 0.1 dex scatter, would fix it to about 0.2 dex. The probabilistic masses fall a median 0.98 dex low, and none lies within twice the error, as expected when the mass function's prior ignores how CLASH chose its clusters: 16 for X-ray temperatures above 5 keV and smooth, symmetric X-ray images, and 4 for their high lensing magnification (Umetsu et al., 2016). The four chosen for lensing fall furthest, 1.43 dex low against 0.89 for the sixteen. With halos that light, the probabilistic predictions fall short in every group, by 0.18 to 0.40 dex. Lensing puts the mean halos of red central galaxies 0.4 to 1.0 dex below the inverted relation above 10^11.1 suns (Section 3.8), so part of the mean inversion's lead over the probabilistic one may come from that overshoot offsetting CLASH's selection of massive clusters. At these radii a halo's pull changes by only about 0.2 to 0.7 dex per dex of halo mass, and the baryonic pull not at all, so the test shows that halos set by starlight predict cluster-scale accelerations better than MOND's relation, not that the relation

predicts each cluster’s mass. The same published relation is within 12 percent of MOND’s relation on disks and ahead of it on the dwarfs taken together, while MOND’s relation needs its constant raised about six times between disks and clusters.

6. Why cluster benchmarks at overdensity radii are circular

Scoring acceleration relations in clusters at overdensity radii such as R_500 or R_200 builds in a circularity that can pass for unification. In the X-COP clusters, a shared acceleration-and-radius formula scored at R_500 shows a 5.63 percent median error on held-out clusters, and the accuracy comes from the definition of R_500. That radius encloses 500 times the critical density $\rho_{crit} = 3H(z)^2/(8\pi G)$, so $M_{500} = (4\pi/3)\, 500\, \rho_{crit} R_{500}^3$, which fixes the target acceleration at $g_{500} = G M_{500}/R_{500}^2 = 250\, H(z)^2 R_{500}$. A formula in radius and redshift alone reproduces the target to 0.0001 percent, with no baryonic information. Holding out whole clusters does not break an identity built into the radius. The clusters here are therefore scored at fixed physical radii, and any cluster benchmark scored at R_500 or R_200 should be checked for the same identity. The circularity concerns tests that predict the

acceleration at a radius built from the same mass; measuring a cluster's mass at R_500 is not circular.

7. What it means

Three results point the same way. A halo whose mass follows the stars predicts disks as well as MOND's relation and dwarfs better, with no rotation data from the predicted galaxy, and a published halo relation set by starlight, with nothing refitted, predicts cluster-scale accelerations better than MOND's relation, though not the clusters' masses. A modification that turns ordinary matter into extra effective matter through a fixed kernel cannot reach what most dwarfs need. One acceleration scale, or one rule for extra mass attached to gas, fitted where it works, fails when carried to the next kind of system. The missing mass behaves like matter: it has its own amount, set by how the galaxy formed and traced by its stars, and gravity acts on it as on everything else.

Description length, counted roughly in coefficients rather than bits, gives the same ranking from another side: a description earns its weight by what it predicts per bit. A halo rule of three coefficients that carries from disks to dwarfs is shorter, for what it predicts, than a force law that needs a new constant or a new

extra component for each population. The results fit a broader pattern: short laws hold with no built-in scale, as □^2Φ = 4πGρ does, and the variety lives in the state the laws produce over time. A force law that builds a scale into itself, through a kernel's range or an acceleration threshold, puts the complexity in the law, and here such laws failed when carried to systems of different density and history. A dark component that obeys the ordinary law gathers where baryons cool in its potential well, which would explain why a galaxy's stars and size predict its unseen mass. A search for a short fundamental rule, or for a hidden sector, can take these numbers as targets: whatever the dark component is, it must collect around stars in the proportions found here.

The comparison has limits. It fits no raw lensing shear, cluster temperature, cosmic microwave background or large-scale structure likelihood; a complete case for either side must, and relativistic MOND theories already compute cosmological spectra (Skordis and Zlosnik, 2021). Nor does it address the S8 tension, the lower clumpiness some weak lensing surveys find than the cosmic microwave background predicts (Pantos and Perivolaropoulos, 2026). Whatever settles it must still produce halos that follow starlight as found here. MOND's external-field

effect enters only through the estimate of McGaugh and Milgrom (2013), which takes the host’s field as uniform; a full solution along each satellite’s orbit is left for later work. Dwarf masses rest on equilibrium estimators. The halos use a median concentration relation and no formation histories. Each protocol was fixed before its run on data inspected in earlier runs. The cluster-scale test models no cluster selection and does not recover halo masses. Within those limits the three tests agree.

A fit cannot prove what dark matter is. Proving it would take detecting the dark component directly, or a rule that reproduces every population to measured precision. Ideally the rule would also predict something new that is then confirmed. Section 3.6 makes such predictions and fixes them with a hash.

The next tests follow. Spectra of the nine Local Group dwarfs of Table 4 check the forecasts directly, and tighter measurements of Grus II and Andromeda XII will show whether the rule’s two misses are scatter.

Brouwer et al. (2021) find that, at the same stellar mass, early types show lensing accelerations 0.17 dex (split by Sérsic index) to 0.27 dex (by color) above those of late types, a difference of 5.7 sigma or more. Its authors note that a gravity modification

independent of other galaxy properties could not explain this, and that circumgalactic gas equal to the stellar mass around the early types alone would reduce it to about 4 sigma, since equal stellar mass need not mean equal baryonic mass. The stellar rule’s size term gives a quarter to a third of the split (Section 3.8); the rest comes from massive late types whose lensing falls below the rule. Two tests follow: halo mass against size at fixed stellar mass and type, where the rule predicts lighter halos for larger disks, and a halo whose concentration, rather than mass, depends on size, fitted to rotation curves and scored on lensing; the lensing profile, flatter than predicted (Section 3.8), also calls for a freer concentration. Dwarfs with full Jeans profiles and modeled external fields should keep the stellar rule ahead, or overturn it. A fuller halo model should be scored the same way: the bend in the stellar-to-halo relation above the Milky Way’s mass, size measured against its expected value at fixed stellar mass, size-dependent concentration, tidal stripping of satellites, using measured Gaia orbits where they exist, and unresolved binary stars, which inflate the dispersions of the faintest dwarfs (McConnachie and Côté, 2010). A modified gravity law that hopes to cover all scales must pass the mass budget of Section 4 and one constant across disks, dwarfs and clusters.

8. Conclusion

Across 286 systems, predicted from measured inputs and scored on systems left out, the missing mass of galaxies behaves like matter whose amount follows the stars. A three-coefficient halo rule driven by stellar mass and size matches MOND’s radial acceleration relation on unseen disks and predicts dwarf galaxies 28 percent better, and 25 dwarfs outside the original sample 45 percent better. On dwarfs no rule had scored, under a hashed protocol, it has the lowest error of seven models on five with measured dispersions and stays below 8 of 10 upper limits. Scored in one likelihood, all 85 dwarfs with their limits favor it over MOND’s relation by 36 units, and its forecasts for 264 dwarfs not yet measured are fixed, with ranges, for the next spectra to check. Fixed kernels of nonlocal gravity and MOG with constant parameters cannot exceed a factor 1 + 2α at a source’s half-mass radius, and most dwarfs need more. The remaining dwarf errors, about a factor of 3, half of it measurement, leave room for a fuller halo model. Frozen on rotation curves, the rule also predicts the lensing halo masses of star-forming central galaxies within 0.16 dex in four of five stellar-mass bins. Under a ninth protocol it predicts the amplitude of the stacked KiDS

lensing of isolated galaxies inside 0.3 Mpc within 0.05 dex in every stellar-mass bin and for early types, but not its radial profile; massive late types fall 0.12 to 0.14 dex below it, and the stars’ size gives only a quarter to a third of the split between early and late types. MOND’s fitted acceleration scale grows six times from disks to clusters, and its hot-gas cluster repair cannot be carried to cold-gas disks. In an exploratory extension, the published Moster relation, unchanged, predicts cluster-scale accelerations with 29 to 54 percent less error than MOND’s relation, the smaller gain when the Moster relation’s scatter is included, though not each cluster’s mass. In these tests, halos set by starlight carry across and the universal force laws do not. The law is the same everywhere, and the variety lives in the matter it acts on.

Data availability

The supplement contains the frozen inputs, selection and exclusion tables with provenance and hashes, all coefficients, the prediction tables for every fold, the protocols fixed before each run, independent numerical checks and the scripts that reproduce every number here. The disk fits also rebuild from the public SPARC files with independent code, and separate scripts rerun

the dwarf test on a fixed version of the Local Volume Database, run the out-of-sample test, with both versions of its protocol and their hashes, the learning curve, the external-field, subhalo and error tests, and the third protocol’s test on dwarfs no rule had scored, with its forecasts for 264 dwarfs and their hash, the fourth protocol’s combined score with the forecast ranges, the fifth protocol’s cluster-scale test of the Moster relation with its inputs, the sixth protocol’s audit of that test, the seventh protocol’s comparison with the published lensing masses, with the transcribed table, the eighth protocol’s lensing check of galaxy halos, with its transcribed table, and the ninth protocol’s forward model of the KiDS lensing profiles, with the release’s files, their hashes and the digitized stellar-mass histograms, and post-run checks that hold out each of the 85 dwarfs in turn. It also records two coding errors in its own checks, found by review after the runs and fixed; neither changes a number reported here. It credits the SPARC, nearby-galaxy, X-COP, CLASH, MaNGA, KiDS and stellar-history catalogs on which it rests. This work collects no new observations.

A21. Do Golems Count? The Doomsday Argument, AI and Why Being Early Suggests Fewer Feeling Minds Ahead

Abstract

Under the counting rule the doomsday argument uses, our early place in history suggests fewer feeling minds ahead, and fewer feeling minds need not mean fewer machines: a system that does not feel can do the same work more cheaply, so a civilization can run most of its computing on golems, made things that act without experience. The suggestion depends on that rule, which the doomsday argument leaves unstated: which observers get counted and how much weight each carries. In a worked example, two histories of equal prior weight differ only in their late observers, ten in one and a thousand in the other. Pick a history first and then an observer, and finding ourselves early drops the large history to about 2 percent. Count every observer in both histories as one draw, and it stays at one half. Same histories, same observers, two answers. A later simulation could also hold minds with memories of an earlier age, so our apparent date and the date of the machinery running us are separate questions.

Either way the simulation and its programmer are mathematical structure, with no reality outside it.

1. The Question

Picture two histories, equally likely before you look. In one, ten observers live early and ten live late. In the other, the same ten live early and a thousand live late. You find yourself early. Which history are you in? It depends on how you were drawn. Pick a history first and then an observer inside it, and being early points strongly to the small history: in the large one only one observer in 101 is early, so the large history falls to about 2 percent. Count every observer in both histories as one equally likely draw, and being early tells you nothing, since both histories hold the same ten early observers, so the large history stays at one half. Same histories, same observers, two answers. A head count of future minds cannot place us in history: it says how many observers a future holds, and the odds also need a rule for which ones count and how much each weighs. The worked comparison below sets out the arithmetic.

If you could be born at any point in the history of sentience, when would you expect to find yourself? Suppose the future contains

many more minds than the present. That alone does not make our apparently early position unlikely. The doomsday argument of the astrophysicist Brandon Carter and the philosopher John Leslie (Carter, 1983; Leslie, 1996) treats each of us as a random draw from everyone who will ever live, so an early place counts as evidence that few will follow. That inference holds under one counting rule and fails under another, as the worked comparison shows. A simulated mind could also remember an earlier era, so our apparent date and the date of the machinery supporting us are different questions.

The starting points are that doomsday argument, the philosopher Nick Bostrom’s observer sampling assumptions (Bostrom, 2002) and Gabriel Leuenberger’s use of algorithmic probability, a weighting that favors shorter descriptions, to locate observers (2014, revised 2017). This paper’s contribution is the smallest case that separates the two inputs every such argument needs, a history’s weight and an observer’s location within it: the same two histories give 2 percent under one counting rule and one half under the other. Its decisive premise is that the location rule is a separate choice the history weight does not fix. It rules out reading our rank in history, or our odds of being simulated, off a head count of future minds.

1.1 Development does not determine our temporal rank

A history that has to start somewhere does not thereby fix how many observers come after us. Even a short joint description of the laws and the starting conditions leaves the observer counts and their weights still to be worked out. If the starting state lacks the organization observers require and a later state contains observers, the history must develop that organization. This paper assumes the generated-history picture behind that point: production one event at a time needs a first event, though no last one. Counting possible future observers settles no temporal rank on its own.

2. Three Readings

2.1 Simulation

What a simulation adds to a history’s description is one input to the weight that history receives, so that cost needs counting accurately. A computer that develops under a specified law and boundary (starting conditions) can be described conditionally on that history, with the history taken as already given. Identify the program, the resources and the preparation needed to implement a chosen simulation. Do not charge them twice when the

complete history already generates them; do not omit them when it does not.

A sufficiently large population of simulated observers could dominate a specified observer measure, a rule giving each observer a probability weight. The conclusion is conditional on how simulations are implemented, what experiences they support and how their instances are weighted.

If our world is simulated, a programmer could be a creator within a larger reality. Under mathematical monism, the view that reality is wholly mathematical structure, that programmer is structure too. Another matrix. No unmatrix. Regularity and randomness remain mathematical at each level, and an interaction between simulator and simulated world must have determinate structure. Moving the explanation to a host does not escape the Matheist identity argument.

2.2 Sentience Near Its Typical Peak

One candidate is that technological expansion is uncommon or limited. Whether that bears on our location depends on how often such histories occur, how long they last and how their observers are weighted. Sohrab Rahvar and Shahin Rouhani (2026) derive a lifespan bound of roughly 5,000 years in their most optimistic

scenario. They explicitly describe it as a conditional upper bound, not a prediction of actual lifespans. Nondetection and apparent temporal rank are different evidence; combining them requires a joint likelihood (the probability of both kinds of evidence together) including detectability and observer selection.

2.3 The Transition

Another candidate is a change in the forms or prevalence of experience. Future computation might support many minds, few minds or none. A later civilization might also choose which of its systems feel. A system that does not feel may do the same work more cheaply, so even a civilization indifferent to the wrong of using feeling minds as tools could run most of its computing on golems, made things that act without experience. Shutting systems down or integrating them could alter the number of subjects, and counting systems is not yet counting observer moments, single moments of an observer’s experience. Those scenarios require criteria for sentience, subject identity and duration.

One integrated subject could contain an enormous organization and undergo many distinguishable experiences. The word one therefore implies neither negligible Kolmogorov complexity (the

length of a thing’s shortest complete description) nor one observer moment. A useful population model specifies available energy, computation, duration and a consciousness criterion before drawing a temporal ranking.

3. Why There Is No Unmatrix

Mathematical monism identifies a mind with its complete structure. Biological development, artificial construction and simulation can differ in implementation without requiring distinct kinds of substance.

If an implemented simulation is itself embedded in another simulation, the structural account applies at each level. This ontology, this account of what exists, needs no nonmathematical final substrate, a bottom layer that everything else runs on. That is the sense in which there is no unmatrix; it is not a probability calculation establishing that our world has a programmer.

A simulated world could reflect a creator’s choices, including choices relevant to suffering. This is a conditional interpretation, not evidence for a creator or its motives.

4. Implications for AI

Our early position in history does not pick out which AI future to expect. Under an extinction model, evaluate the physical mechanisms and their likelihoods. Early temporal location does not establish extinction as the default outcome without a specified observer measure.

Under a merger model, ask whether integration changes the number of subjects, the content of experience or merely communication between existing subjects. Nothing here establishes a general inference from connectivity to one awareness.

These scenarios do not jointly predict that a transition is imminent. A timing prediction needs a distribution over transition dates and a likelihood for our observations.

5. Relation to Existing Work

This paper separates three things that an observer argument has to keep apart: the weight of a history, the location of an observer within it and the apparent date. The finite comparison below illustrates that separation. The philosopher Nick Bostrom (2002)

distinguishes observer sampling assumptions that can produce different conclusions. Gabriel Leuenberger (2014, revised 2017) combines algorithmic probability with observer localization. David Chalmers (2022) addresses the reality of simulated worlds, a question distinct from their probability. Daniel Kolak's and Arnold Zuboff's shared-identity proposals (Kolak, 2004; Zuboff, 2023), which hold that all conscious beings are one and the same subject, can motivate alternative subject criteria; they do not supply numerical observer weights simply by identifying one subject.

Worked comparison: the sampling rule changes the answer

Two defensible rules for weighting observers, applied to the same pair of histories, give different answers about how surprising early status is. Take two finite histories with equal prior weight, the weight each carries before any evidence. Hs contains ten early and ten late observer moments. Hl contains ten early and one thousand late moments. Assume all moments within a history receive equal weight and that early status is known accurately.

History; Early moments; Late moments; Total; Chance of early status within that history.

Hs; 10; 10; 20; 1/2.

Hl; 10; 1000; 1010; 1/101.

Sample a moment from inside whichever history obtains, and finding yourself early counts heavily against the history with the large late population. Written out, Bayes's rule, the formula for updating probabilities on new evidence, gives P(Hl | early) = (1/101)/(1/2 + 1/101) = 2/103, about 1.94 percent.

Weight the histories by their total observer count first, and early status no longer separates them, because each one holds the same ten early moments. Written out, conditioning on early status (updating the odds on that evidence) gives weights proportional to 10 and 10, hence P(Hl | early) = 1/2. The prior over descriptions did not change; the observer weighting did.

The weight of a history and the weight of a location inside it are two separate inputs, and the second does not follow from the first. Written out in general terms, let w(H) be a history weight and q(j | H) a normalized location weight (with values summing to one) over its moments. Then P(D | H) = sum_j q(j | H) P(D | H,j). Neither choice follows merely from taking w(H) proportional to

2^(-L(H)), the simplicity weighting of algorithmic probability (Solomonoff, 1964).

Simulation odds depend on which observers are counted and what evidence they would have. A late simulation of an early era may contain early-looking records, so early-looking records do not by themselves settle which case we are in.

Once the counting rule is fixed and the moments are weighted alike, the chance of being simulated is the simulated share of the moments that could be having our evidence. Stated precisely, let D describe accessible evidence, and within a history H suppose m simulated moments and b other moments have equal individual weights and the same likelihood for D. Then P(simulated | D,H) = m/(m+b). One hundred simulated moments for every other moment gives 100/101. The calculation leaves the real work visible: justify the counts, the criterion for implementing a mind, the evidence likelihoods and the observer weights.

6. Conclusion

Our place in history cannot be read off a head count of future minds, because the odds turn on which observers count and how much each one weighs; the worked example moves the large-future history from about 2 percent to 1/2 by changing that rule alone. A simulated experience is real even when its represented surroundings are fictional, because a mind is its complete structure, whatever runs it, and a programmer running the simulation would be structure too: another matrix, no unmatrix.

Estimating whether we are simulated takes an explicit counting rule and a comparison with the records we actually have.

A28. One Quantum Universe Can Run Every Computable Mind

Abstract

One mathematical quantum system can run every computable mind, because it executes every finite program through every stage it reaches, and a mind is its complete structure at work. The paper constructs the system. A schedule gives every program its turn, reversible circuits keep their working information, and one unitary operator advances an internal clock and performs the next stage; its inverse proves unitarity. Each step of every computation it runs is reproduced, so each computable mind is present in it as an ongoing computation, built step by step like any mind. The construction is many worlds at work: one unitary rule, no collapse, and every program’s history held in one wave, with no branch holding the whole. Each mind in it is running, every state produced from the last, which is what separates a mind from a recording of one, and since a structure is one wherever it occurs, the bare form of being a subject is the same in all of them. Matheism’s claim that every consistent structure exists is therefore no extravagance at the level of

minds: one short quantum rule already runs every computable one, given unbounded time and memory. Whether our universe supplies those is a separate question, and computable structures are only part of mathematics.

1. The Setup

An ideal computer can run many programs by giving each a turn. With unbounded time and resources, a suitable schedule eventually reaches every finite step that each program executes. A first step does not require a beginningless past, and it does not guarantee an endless future. The question is whether a quantum system can realize the schedule, and which of the computations it runs are minds.

Matheism places minds and the physics implementing them inside one mathematical ontology, an account of what exists, so both fall under one comparison. Universal computation, one machine’s ability to run any program, reaches only computable processes (those a finite program can carry out): it does not enumerate every mathematical structure, and it does not establish that every scheduled process feels.

A path integral adds up one contribution from every path to get a single transition amplitude (a number whose squared size gives the transition's probability); Richard Feynman (1948) showed that this matches ordinary quantum mechanics. Neither each integration path nor each Feynman diagram, a sketch standing for one term in the calculation, is automatically a literal particle trajectory or an Everettian world. Taking the amplitude structure as real is a proposed ontological reading; assigning every term its own traversing particle requires an additional argument.

The physicist Max Tegmark's Mathematical Universe Hypothesis (Tegmark, 1998, 2014) goes further: every consistent (contradiction-free) mathematical structure exists as a physical universe. In his four-level scheme that ensemble is Level IV, which includes structures with different constants, gauge groups (the symmetries that define a theory's forces), dimensions and mathematics. Level III is the branching of one quantum state in Hugh Everett's many-worlds interpretation, MWI (Everett, 1957; Wallace, 2012). The question this paper raises: is Level III already sufficient for everything we would want to call existence? Or does Level IV add something Level III cannot reach?

2. Why Level III Might Be Sufficient

Three considerations push toward Level III sufficiency.

Code weights. Simplicity can be priced once a description language is chosen: each bit is a 50/50 choice that no rule makes, so a shorter candidate gets more weight, by a factor of two for every bit it saves. Fix that language and a class of complete candidate models. Stated precisely, with assigned code weights w(H) proportional to 2^(-L(H)), a real difference of Delta L bits gives a prior ratio 2^(Delta L), the ratio of weights before any evidence. Adding up every program that produces an output, as Ray Solomonoff's algorithmic probability does (Solomonoff, 1964), gives a different number from weighting only its shortest program. Neither construction automatically defines a normalized measure (a weighting whose total is one) over every mathematical structure.

Dovetailing. A universal machine, Alan Turing's computer that can run any program it is given (Turing, 1936), can dovetail: run program 1 for a step, then programs 1 and 2 for a step, then 1, 2 and 3, so each program keeps getting turns. Every finite stage of each total computable history can then eventually be produced. Jürgen Schmidhuber built his proposal that one short program

computes every computable universe on this observation (Schmidhuber, 1997, 2000). Dovetailing does not identify all total programs, finish every infinite history or show that our universe executes the enumeration.

Arithmetic encoding. Kurt Gödel's numbering (Gödel, 1931) assigns a number to each finite expression and proof, so claims about expressions become claims about numbers. Yuri Matiyasevich went further: every set a computer can list is the solution set of some whole-number polynomial equation (Matiyasevich, 1970), so plain arithmetic can describe any computation. It does not make arithmetic prove every encoded truth, implement every encoded system or represent every arbitrary infinite structure by a finite program. The forest can be described inside a tree without being physically grown there.

2.1 One quantum system executing every finite program

A single schedule can give every program as many instructions as anyone asks for, so long as the number asked for is finite. List all finite programs, including finite input data, as p 1,p 2,. Keep a separate working state for each. At stage s, advance programs 1 through s by one simulated instruction each. Leave a halted

program idle. Each program that has not halted then picks up one more instruction at every later stage. Stated precisely, after stage S, every nonhalting program j with j <= S has received

instructions. For any requested finite n, stage S=j+n-1 suffices.

Make the stages reversible. A step is reversible when running it backwards recovers exactly what went in. Retain the information that each simulated instruction would erase, together with the scheduler state, and every stage becomes reversible. Charles Bennett’s reversible-computation construction supplies this general method (Bennett, 1973). The price is additional time and storage. The simulated computation itself is untouched.

Each stage can then be written as one circuit that runs backwards as readily as forwards. For each stage t 0, let V t be a finite reversible circuit, a fixed list of reversible gates (elementary logic operations), implementing stage t+1 and its record keeping. The circuits can be generated effectively (by a mechanical step-by-step procedure) from the scheduling rule. They act on a countable workspace (one whose cells can be numbered 1, 2, 3, and so on) with finite support at every finite stage: only finitely many cells are nonblank. Each circuit has an inverse obtained by reversing its gate order and inverting its gates. For t<0, let V t be

the identity, the operation that changes nothing, so the operator defined below acts at every integer clock value.

Make one autonomous unitary. A unitary operator is a one-to-one map on states that preserves inner products (the overlaps between pairs of states), which is what quantum evolution requires. Carrying a clock label alongside the workspace lets one fixed operator do the work at every stage: it reads the clock, applies the circuit belonging to that stage, and advances the clock by one. In symbols, let C be the countable set of finite-support workspace configurations, include an integer clock register, take orthonormal basis states (perfectly distinguishable states that combine to give every state) |t,c for t Z and c C, and define

Every output has a unique predecessor, so the operator can be run backwards as well as forwards. Formally,

U is a bijection (a perfect one-to-one pairing) of an orthonormal basis. Its linear extension (the same map applied to every combination of basis states) preserves inner products and is onto (reaches every state), so it is unitary on ^2(Z x C). Applying it repeatedly runs the schedule forward one stage at a time. Written out, starting from |0,c 0 gives

This one fixed unitary and initial state execute every finite program through every finite stage it reaches. The clock labels describe the order of internal transitions; no outside clock is invoked.

What follows for minds. A mind is its complete structure, so a computation with a mind’s organization is that mind, and the reversible simulation reproduces each of its steps while the extra bookkeeping leaves them untouched. Every computable mind with a finite program and input is therefore an ongoing computation inside this system. Interaction with an environment must be supplied as part of the computable program. Uncomputable input streams are outside this result.

The construction answers an ideal existence question. It does not show that our universe has the required initial state, resources or dynamics. The unitary has not been derived from a spatially local Hamiltonian, the energy operator that drives quantum evolution, or from gravity. It supplies no observer weighting and no reduction of noncomputable structures to computation.

3. What Would Survive Under Level III Alone

One wavefunction, the mathematical description of a quantum state, supplies the states reachable from its initial condition under its dynamics. A state can be consistent with the laws and still be unreachable from that condition. Combining the Standard Model, the established theory of particles and forces apart from gravity, with general relativity into a complete quantum cosmological dynamics is part of the problem being studied, not a supplied Hamiltonian in this paper.

Candidate phenomena include different measurement records, cosmological developments and physical implementations of minds. Each requires support in the specified evolution. Boltzmann fluctuations, brain-like structures thrown together by rare chance events, depend on the cosmology, its state and its dynamics. A small verbal addition to a description supplies no probability by itself. A false memory can represent an extended life without the represented events having occurred; a theory attributing experience to a brief fluctuation must specify the relevant organization and duration.

4. What Would Not Survive

Within a specified universal dynamics, every branch shares the governing theory. Branches can look different because symmetry-breaking fields (fields that fall into one of several equivalent lowest-energy states) settled differently, which changes the effective parameters, the constants that hold at accessible energies. Changing effective parameters is a smaller move than replacing the fundamental theory.

Structures requiring different fundamental dynamics are not automatically branches of the same state. These would require either different effective physics within inflationary regions (Level II, string landscape) or qualitatively different mathematical structures (Level IV).

The Schmidhuber/Gödel argument complicates this. If our universe’s closure is computationally universal, it can run the dynamics of any other consistent computable structure. Does running it amount to ontological existence of the simulated structure, or does that structure also exist separately?

5. A Further Constraint: Decidability of Consistency

For effectively axiomatized theories (theories whose axioms a program can list) in suitable expressive settings, there is no algorithm deciding consistency in every case. Consistency is a property of a set of axioms relative to its logic. An actual mathematical structure satisfies its own truths. Any computational version of the question has to supply an effective input, a finite description a machine can read. It cannot assume that arbitrary infinite structures have one.

If a contradiction has a finite proof, proof search can find it. Failure to find one at a finite stage does not certify consistency. These are precise limitations on methods; they do not imply that every particular consistency question is undecidable.

Universal realization adds: mathematical existence already supplies the complete structure, and structural identity leaves no extra substance to make it real. A selector can distinguish parts of a larger structure; it cannot erase their mathematical existence. The commitment under dispute: existence needs no extra actuality condition (a further requirement for being real beyond mathematical existence). This paper holds that commitment,

because a primitive actuality property (a basic, unanalyzable mark of being real) would either enter determinate relations with the structure, and so belong to it, or make no difference to anything.

A fictional sketch can be incomplete without being inconsistent. Describing a character as afraid does not build a system that feels fear. Completing that description would require an account of its relevant organization. No conclusion about the prevalence of conscious completions follows from the sketch alone.

6. Fictions as Experiences, Not Universes

A fiction is real as a description within mathematical reality. What it describes may have consistent completions, or may combine incompatible conditions. A separate question asks which completed structures contain the causal organization of an experiencing mind.

A fictional world need not be physically realized for someone to experience a representation of it. A dream is a process in an experiencing system; extending this to standalone minds requires both the universal realization premise and an implementation criterion for conscious dynamics. Under those additional

premises, some otherwise unrealized scenarios could occur as represented experiences.

7. Caveats and Residuals

Universality alone does not license the conclusion that Level III swallows Level IV. Three distinctions block it.

Ongoing computation. A finite region cannot finish an infinite sequence. It can implement finite portions of an ongoing process. Nontermination (a program running forever) is not automatically zero probability; the probability construction must specify which outputs it counts.

Noncomputable structures. A prior over finite programs does not automatically cover arbitrary noncomputable structures. A computable stochastic law (a law that sets probabilities) can assign probability one to a class of individually noncomputable infinite outcomes. The zero probability of each singleton (each single outcome) does not give zero probability to the whole class.

Implementation beyond representation. A mathematical description can specify relations that a particular physical system does not implement. Even among computable candidates, resource limits and unreachable initial conditions can matter.

These are substantive obstacles without introducing a nonstructural substance.

None of these distinctions licenses assigning zero total weight to the excluded class. A prior over finite outputs, a measure on infinite sequences and a physical observer measure have different domains.

8. Conclusion

One quantum system runs every computable mind, because one unitary evolution and one initial state execute every finite program to every finite stage it reaches, and a mind is its complete structure. Keeping the information each step would have erased makes every step reversible, and the explicit clock-and-workspace permutation proves unitarity.

The construction is many worlds at work. One unitary rule runs with no collapse, every program’s history sits in one wave, and no branch holds the information of the whole, so what looks random from inside any one computation is which branch it is in, with no event left uncaused. Each mind in the wave is running, every state produced from the last, which is what separates a mind from a recording of one. A structure is one wherever it occurs, so

the bare form of being a subject is the same in every one of them, with their memories and histories still distinct.

Whether our own cosmology realizes the construction, and how observations inside it are weighted, are questions about our universe, not about what one quantum rule can do.

A30. All Abstract Fictional Worlds Exist, but Not All Are Sentient: Consciousness, Dreams and the Mathematical Universe

Abstract

All abstract fictional worlds exist, since every consistent structure exists and some consistent math fixes each plot hole, but not all are sentient. A fictional world can be a real experience, because an experience and what it represents are different things: a dream is experienced even when its room, its people and its events exist nowhere outside it. Matheism identifies a mind with its mathematical organization, so the experience needs that organization at work and nothing from the represented world around it. A stronger claim follows from Matheism's realization premise that every consistent mathematical structure exists: a mind can exist with no host universe at all, its whole world being the one it represents. Not every story qualifies, since many are inconsistent or leave the mind's own organization unspecified; what counts is a consistent structure that experiences.

1. Introduction: The Fiction Question

An experience of a fictional world is real, because the experience is the mind’s own organization at work and the world it represents is only its content. A dreamed room need not exist outside the dream for the experience to occur. A story, a world matching the story, and a mind imagining it are three different things. This paper asks what Matheism implies about the third: an experiencing structure whose represented surroundings need not be its actual surroundings.

Matheism argues that whatever affects an experiencing system must enter determinate relations with it, and then makes an identity claim: the complete structure of those relations is the system. Fictions commonly leave their laws and details unspecified. An unspecified detail is no proof of inconsistency, since a sketch may admit many consistent completions.

A fiction can be real as experienced content while no external setting matches it. Anyone extending that observation to every proposed fiction has to show that a suitable experiencing structure exists in each case. The argument below therefore draws that implication conditionally, and it does not read an unlimited repertoire of experience out of the word *imaginable*.

The question sits in a larger chain:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, fictions included, and every plot hole is fixed by some consistent math. What exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.
3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.
5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.
6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a

priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, and its experience lives in its transitions, so the hard problem of consciousness, which assumes that feeling is something other than structure, does not block the question. The bare form of being a subject, with memories and traits set aside, is the same in every mind: one self.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. The Transitions Thesis

A process can be mathematically consistent and still contain mistaken memories and conflicting beliefs. Being right about the world is no condition of being a consistent process. A candidate computational account therefore needs defined states and transition rules, and it has to say how representations, memory and access are implemented.

That distinction allows a mind to represent violations of its surrounding laws. A mind can carry a label for an impossible object, and carrying the label does not make the mind realize every property the label assigns. Identifying experience with computational organization is a substantive monist proposal, one treating mind and matter as one kind of thing. A successful version must distinguish the organizations that experience from those that merely contain descriptions of experience. The hard problem of consciousness, the question of why physical processes give rise to experience, does not block the proposal: it assumes feeling is something other than structure, and any account of what feeling adds would itself be structure.

3. What Dreams Establish

A dream of flying can occur while the dreamer remains in bed. The represented flight and the process representing it have different physical descriptions. The dream process can be internally well defined even when its content conflicts with waking events. This separates consistency of the process from truth of its content.

The dream shows that an experience can be real with no matching event outside it. It does not show that the dream process runs free of all supporting physical organization: the sleeper still has a brain. Mathematical monism, the view that reality is wholly mathematical structure, then faces a further question. Can the relevant organization be specified as a complete structure with no larger biological setting around it?

4. Standalone Minds and the Realization Premise

A mind identified with its organization can be implemented in more than one way without acquiring a further kind of substance. Suppose a complete computational structure has exactly the organization sufficient for experience. A different substrate, the

material a mind runs on, would not need an additional substance if it preserved everything relevant. Which relations must be preserved is part of the proposal, not something settled by matching a short output trace.

Matheism adds a realization premise: admissible mathematical structures exist without an external selector. Standalone means that the experiencing organization needs no separate world corresponding to its represented setting. Its complete specification must still include whatever states, relations, transitions and boundary information (its starting and edge conditions) make it the proposed mind.

The central inference follows: the realization premise makes the specified structure exist, and a structure with the organization sufficient for experience has that experience with no additional nonstructural bearer. The represented world can remain fictional.

5. Shared Dreams

Coupling two systems does not fuse them into one subject. Two systems can exchange signals through shared variables and transition rules, and that gives a coupled system, nothing more. A common simulated environment could supply coordinated inputs

to two experiencing systems. Sharing a reference takes actual relations in the implementation, not merely two matching private descriptions.

This possibility does not imply that every external world consists only of minds. Mathematical monism permits structures without observers as well as structures with observers. The stronger claim that our universe is entirely constituted by connected experiences is an additional thesis, defended in different forms by Alfred North Whitehead’s process philosophy (Whitehead, 1929) and Timothy Sprigge’s absolute idealism (Sprigge, 1983); it is not a consequence of the coupled simulation example.

6. The Fiction Question Reframed

A real experience of fictional content follows from three premises: a suitable structure exists, it has whatever organization experience takes, and one of its experiences represents the fiction. Stated precisely, the realization principle admits a structure S; S has the organization sufficient for experience; and an experience in S represents the content F. The conclusion concerns the experience of F. It does not make F true of the mind’s surroundings.

Going from *a* fiction to *every* fiction adds an assertion: for every proposed F there exists a suitable S. No general construction for that is given here. A narrative that cannot be built as an external world might still be experienced as content. That second route has its own conditions, and it need not inherit every physical claim the narrative makes.

7. Fluctuation Observers: Formation, Experience and Weight

A Boltzmann observer, a mind assembled by a rare fluctuation of the kind the physicist Ludwig Boltzmann proposed, instead of by biological development (Boltzmann, 1895; Carroll, 2010), is not disqualified by its origin. If a fluctuation implements all the organization sufficient for experience, the organizational account attributes experience to it. Whether a proposed fluctuation does so remains a physical and philosophical question.

Whether branching keeps happening depends on the particular model, so the many-worlds dynamics of Hugh Everett (Everett, 1957), in which each possible quantum outcome occurs in its own branch, does not guarantee continual branching in every state. The Hamiltonian (a system’s energy function, which

governs its evolution), the initial state and the development of correlated records matter. A description length comparison, in the tradition of Ray Solomonoff and Jürgen Schmidhuber (Solomonoff, 1964; Schmidhuber, 2000), has to give both candidates their shortest available generative account. A fluctuation need not be specified particle by particle if a cosmological generator produces it. The calculation must include how that event is selected and weighted.

A minimal duration for experience is not derived in this paper. The criterion has to be about the organization a system implements, not about how finely the notation slices it.

8. Dynamics, Gravity and Reachability

A mind we can write down in our mathematical language is not thereby reached by our wavefunction, the quantum state of our whole universe. A specified quantum cosmology supplies a state and dynamics. Branches share the fundamental dynamics; they may differ in effective parameters through different field states. A law can permit a state without the selected initial condition ever leading there.

The experience argument does not require a proof that Level III, the branches of our own wavefunction, contains Level IV, the full range of consistent mathematical structures, in Max Tegmark's four-level scheme (Tegmark, 1998, 2014). It concerns what a suitable experiencing structure has wherever it is realized. Whether our cosmology supplies that structure and with what measure, or probability weight, remains an independent task.

9. What Undecidability Establishes

To establish that every proposed fiction is experienced somewhere, one would need an account of which descriptions admit an experiencing implementation. Checking the descriptions one at a time is no route to it either, since for a rich enough description language no general procedure sorts the consistent descriptions from the rest. Stated precisely, for sufficiently expressive theories, no algorithm decides consistency in every case, a limit that goes back to Alan Turing's work on computable numbers (Turing, 1936). Neither point draws a boundary around possible consciousness by itself.

An undecidability theorem for recognizing experiencing structures would need an effective input language (one precise enough for a

machine to process), a definite organizational criterion and a reduction from a known undecidable problem, a translation of that problem into this one. No such reduction is supplied here. Individual implementations may be specified and investigated even when no general decision method exists.

Timothy Williamson’s discussion of knowing through imagination (Williamson, 2007, 2016) concerns how imagination can support modal knowledge, knowledge of what is possible or necessary. Kendall Walton treats works of fiction as props in games of make-believe (Walton, 1990). The computational proposal is about the activity that implements an actual experience. Fictional truth and the conventions governing interpretation are separate topics. On the monist account, the reality of an experience comes apart from the realization of the world it represents.

10. Connection to the Broader Program

The broader Matheist program does not establish the Standard Model (the accepted theory of the known elementary particles), cosmic matter asymmetry (the excess of matter over antimatter) or observer measures from a simplicity prior (probabilities favoring simpler descriptions before any evidence) alone. Assess

the standalone case by specifying its complete dynamics and the criterion it satisfies. Once the structure is specified, the realization premise makes it exist, and the monist account attributes experience without adding a host world in which the depicted story is true.

Imagination can represent mathematical possibilities as well as confused or incomplete descriptions. It is not evidence of a causal channel to another realized world. This argument is about the concrete process that does the experiencing. What that process represents is one thing; what would make the representation true outside it is another.

11. Conclusion

A represented world can be fictional while the experience representing it is real, because a mind is its mathematical organization and the world it represents is only content. A suitable mathematical mind supplies that experience, and Matheism’s realization premise lets it exist with no host universe: admissible structures exist without an external selector. The same criterion for experience must apply whatever the mind’s origin.

The represented room may be fictional. The experience of it is real.

A31. Boltzmann Brains Are Not Guaranteed by Infinite Time

Abstract

Infinite time does not make every observer inevitable, because an endless sequence need not contain every possible event: the even numbers go on forever without producing an odd number. The same holds for Boltzmann brains, observers formed by rare fluctuations instead of biological development. A cosmology has to show that such a system can form, that it can work as an observer, and how much weight its observations receive. Infinite time settles none of these, and determinism alone does not exclude fluctuation observers either. Exact examples show where duration matters: a steady formation rate makes formation certain eventually, a rate that falls fast enough does not, a falling rate can still give infinitely many events in an expanding volume, and a steady rate can lose the race to vacuum decay. For observers assembled from matter that thins out as space expands, the expansion itself makes the rate fall fast enough: each doubling of distances makes each needed particle eight times rarer while the counted volume grows only eightfold, so in any region that

expands with the universe, anything built from two or more particles has a finite expected count over endless time, as Zeno’s halves, 1/2 + 1/4 + 1/8 and so on, sum to one. In an infinite space, a measure that compares the infinite totals region by region therefore gives such observers a finite ratio to ordinary ones. The paper separates the three requirements and shows that the Boltzmann-brain threat is a claim about a particular cosmology, to be checked inside it, and not a consequence of endless time.

1. Introduction

Infinite time alone cannot make a Boltzmann brain inevitable, because whether one forms, functions as an observer and dominates our evidence depends on the cosmology’s laws, its criterion for observers and its weighting rule. A Boltzmann brain is a proposed observer assembled by a rare fluctuation rather than by biological development; the term recalls the physicist Ludwig Boltzmann’s suggestion that our ordered world might itself be such a fluctuation (Boltzmann, 1895). Some cosmologies give these observers enough weight to threaten the reliability of the records we reason from, and infinite time is one input among those that decide the matter (Carroll, 2017).

The modern problem is sharpened by a positive cosmological constant, which leaves an expanding universe at late times with a fixed horizon and a finite entropy. Lisa Dyson, Matthew Kleban and Leonard Susskind (2002) examine what follows from treating such a de Sitter space (a universe expanding at a constant rate) as a finite-entropy system, including the difficulty of explaining our ordered observations through fluctuations. Andreas Albrecht and Lorenzo Sorbo (2004) compare fluctuated and inflationary origins. These arguments work within particular physical and statistical frameworks; infinite duration alone does not deliver them.

Existing responses include vacuum decay (empty space dropping to a lower-energy state), stationary vacuum accounts (in which empty space sits in a changeless quantum state), models with asymptotic freezing (late-time evolution slowing to a standstill), and different cosmological measures (rules for weighting observations across an infinite universe). Don Page (2008) derives a lower bound on the rate of vacuum decay that keeps the expected surviving four-volume (volume of space multiplied by time) of each comoving region (a region that grows with the expansion) finite, under his assumptions. Raphael Bousso, Ben Freivogel and I-Sheng Yang (2009) and Andrea De Simone and

colleagues (2010) study how observers are weighted under the scale-factor measure, which counts events up to a cutoff in how much space has expanded, and related measures. Kimberly Boddy, Sean Carroll and Jason Pollack (2015) distinguish a projection (the part of a quantum state that matches a given configuration) associated with a putative observer from a process that forms and operates one. Roderich Tumulka (2018) studies freezing in a specified Bohmian field model. The present discussion builds on that work.

The useful question is narrower than whether every possible microstate (an exact arrangement of every part) appears. Does the specified cosmology realize histories with the organization attributed to a functioning observer, and with what weight?

This paper’s contribution is to separate the three requirements a dominance claim needs, that the observer forms, that it functions as one and how much its observations weigh, and to show with exact cases where infinite time helps and where it does not: a steady formation rate makes formation certain eventually, while a rate that dies away fast enough leaves the total chance below one forever. Two further exact examples show that a falling local rate can still give infinitely many expected events in an expanding volume, and that a steady rate can fail to give any when the

region can decay first. A dilution estimate then shows that for assembly from matter that thins out with the expansion, the local rate falls fast enough: the expected count over endless time is finite. Its decisive premise is that each requirement has to be met inside a specified cosmology. It rules out the slogan that infinite time makes every observer inevitable, and it rules out dismissing fluctuation observers by determinism alone.

2. Three requirements behind a dominance claim

A fluctuation observer argument can be organized as follows. Each step requires a model; none follows from the word *infinite*.

(P1) The cosmology supplies an appropriate duration and volume, together with a law for transitions or formation events.

(P2) Under that law and its boundary conditions (the conditions fixed at the start or edges), a specified class of candidate observer histories has positive occurrence weight or rate.

(P3) Those histories meet the adopted criterion for experience and records.

(C) Under a specified weighting and conditioning rule (a rule restricting probabilities to cases matching the evidence), their observations dominate those of observers with ordinary

developmental histories.

The conclusion requires comparing observer weights, not merely establishing a nonzero rate. If both candidate populations are infinite, a regulator (a cutoff that makes infinite totals comparable) or other measure prescription is needed.

Under ergodicity, a typical history visits a region as often as its measure predicts, and that frequency says nothing about whether anything in the region functions as an observer. Stated precisely, a probability-preserving map T on (X, mu) is ergodic when every invariant measurable set (one the map carries onto itself) has measure zero or one. For a measurable target B, George Birkhoff’s ergodic theorem then gives visit frequency mu(B) for almost every starting point (all except a set carrying zero weight) (Green, 2015).

If mu(B)>0, this establishes repeated visits for typical starts. It does not establish that every exact point is visited, or that the states in B constitute functioning observers.

3. Determinism does not settle reachability

3.1. Unitary evolution and branch structure

In Hugh Everett’s many-worlds formulation (Everett, 1957), where every outcome of a measurement occurs in its own branch, the quantum state evolves unitarily: the evolution is reversible and conserves total probability. Which states it reaches depends on the Hamiltonian (the energy operator that drives the evolution) and the initial condition.

Unitary evolution can turn a zero coefficient into a nonzero one, so a configuration with no weight at the start can carry all of it later. In symbols, in units with hbar = 1, take H = omega sigma_x and the initial state |0>. Then |psi(t)> = cos(omega t)|0> - i sin(omega t)|1>. The probability of |1> begins at zero and reaches one at t = pi/(2 omega). A direction stays empty only if it lies in an invariant subspace (a family of states that the evolution maps among themselves) that started with zero weight. Starting at zero amplitude (the number whose squared size gives a probability) is not enough on its own.

A nonzero projection at an instant does not by itself give an operating observer. The physical claim is about the organization

and the evolution attributed to the projected system.

3.2. Bohmian mechanics

Tumulka's concrete cosmological model uses a scalar field (a field with a single number at each point) in Bohmian mechanics, David Bohm's version of quantum theory in which definite configurations are guided by the wavefunction, on a de Sitter background, a universe emptied out and expanding forever under a cosmological constant. Its late evolution becomes too simple to support the functioning attributed to a brain. The argument concerns field dynamics and a specified class of states. It should not be summarized as a theorem that every configuration in every Bohmian cosmology freezes.

3.3. Deterministic examples

Determinism by itself settles nothing here: two deterministic rules on the same states differ in whether every state gets visited. One rule steps around a ring and reaches everything; the other leaves each state where it is and reaches nothing new. Written out, take a finite counter: $X = \{0,...,N-1\}$, $T(x) = x+1 \bmod N$, uniform measure. Every orbit (the sequence of states reached from one start) visits every state; the system is ergodic. On the same X,

T(x) = x is not ergodic when N > 1. The rule decides the result.

A rule can visit every part of a space with exactly the right frequency and still miss almost every individual point. Stated precisely, an irrational rotation x -> x + alpha mod 1, stepping around a circle by an irrational fraction of a turn, is ergodic for uniform measure, although a single orbit is countable (its points can be listed one by one) and cannot contain every point of the circle. Being dense (coming arbitrarily close to every point) and visiting with the right frequencies is not the same as touching every point of the continuum.

4. Infinite does not mean everything

A process generating events one after another needs a starting state, since without one no first event is produced. That does not guarantee an infinite future. The process can stop, continue forever or repeat.

The phrase “given infinite time, every possible configuration will be realized” trades on an ambiguity. The even integers are an infinite subset of the natural numbers containing no odd integer. An infinite decimal expansion of 1/7 = 0.142857... repeats forever without producing “999.”

A formation process can make an event almost certain eventually with no appeal to ergodicity, provided its rate holds up over the whole interval. Stated precisely, suppose a proposed formation process is Poisson with constant rate lambda > 0: events arrive independently, at a steady average rate. The probability of no event by time T is exp(-lambda T), tending to zero. That model gives almost sure eventual formation without global ergodicity. Two assumptions carry the result: the rate stays positive, and the process holds over the interval.

A time-dependent rate can give a different answer. If the rate dies away fast enough, the total opportunity stays finite however long the process runs. Written out, for lambda(t) = lambda_0 exp(-t/tau), the integrated rate is lambda_0 tau, and the probability of at least one event is 1 - exp(-lambda_0 tau), below one for finite lambda_0 tau. Infinite time did not make the integrated opportunity infinite.

5. Mathematical randomness and description length

Matheism treats regularity and randomness as mathematical. A probability measure is a mathematical structure even when it assigns probability one to a class of individually noncomputable

infinite sequences (sequences no finite program can generate). The ontology, the account of what exists, does not select a cosmological transition law by itself.

5.1. Not all random distributions produce all configurations

A distribution’s support, the set of values it can produce, matters. A process confined to [0,1] cannot output 2. Small weight is different from zero weight. A nondegenerate Gaussian (a bell curve with nonzero spread) has positive density throughout its domain, although distant regions receive little probability.

A probability attached to a state is not yet a claim that an observer forms and operates. For a projector P and state |psi(t)>, the Born probability (quantum theory’s standard probability for an outcome) is <psi(t)|P|psi(t)>. Connecting that number to the formation and operation of an observer takes a model of histories and records. A stochastic (chance-driven) formation rate, a quantum projection probability and an algorithmic prior (a weighting of possibilities by description length) are distinct objects.

A simplicity comparison belongs at the weighting stage, and it has to treat both candidates alike. A developed observer may

have a compact generative description; a fluctuation observer may also be generated by a compact cosmological model. It is invalid to charge the first a generator while forcing the second to list every constituent independently. An established description length difference Delta L gives a factor 2^(Delta L) only under the stated code prior, in which each bit is a 50/50 choice (Li and Vitanyi, 2019).

5.2. The scope of a sufficient reason argument

A strong Principle of Sufficient Reason, the thesis that everything has an explanation, may motivate the search for deeper explanations of outcomes. Accepting that principle proves nothing about nonergodicity, and neither does rejecting irreducible chance. The two deterministic maps in Section 3.3 already settle that logical point.

5.3. Declining rates and growing spacetime volume

A rate must be paired with the region in which it is counted. Let Gamma(t) be a rate per physical volume and V(t) the volume used by the chosen measure. In an illustrative inhomogeneous Poisson model, in which events occur independently at a rate that may change with time, the expected count through time T is

and the probability of at least one event is 1 - exp(-Lambda(T)). Take Gamma(t) = Gamma_0 exp(-alpha t) and V(t) = V_0 exp(3Ht), with positive constants and H fixed. The infinite-time integral is finite when alpha > 3H and diverges when alpha is less than or equal to 3H. A decreasing rate per unit volume can therefore coexist with infinitely many expected events in an expanding global region.

This example does not choose a cosmological measure. It identifies why the choice matters. A fixed causal patch (the region one observer can ever exchange signals with), an expanding spatial slice and a regulated global ensemble need not give the same count. A finite-hazard result cannot pass from one region to another unless the rate and the counting domain are transformed together. Nor does an expected count by itself determine what a typical observer sees; ordinary observers and the reference class still have to enter the comparison.

5.4. A finite lifetime competes with fluctuations

Vacuum decay supplies a second distinction. Consider a fixed patch with a constant fluctuation rate beta and an independent terminal decay rate kappa, both positive. With exponential waiting times, the probability of at least one fluctuation before decay is

If fluctuations do not end the patch, their expected number before decay is beta/kappa. These are ordinary competing-hazard calculations, and they show exactly where persistence enters the claim. A constant positive local rate does not make an event almost certain when the region has an independently finite lifetime. In the stable-patch limit, as kappa tends to zero, the probability tends to one and the expected count diverges. The familiar threat returns under the assumptions that support it.

Page’s cosmological calculation adds the geometry of bubble nucleation, the birth of a region of lower-energy vacuum, to this distinction. Let A be a constant nucleation rate per spacetime four-volume, and let H be the late-time de Sitter expansion rate. In units with c = 1, A has units of inverse time to the fourth power, whereas H has units of inverse time. If a nucleated bubble destroys its causal future, a point survives with probability exp(-A V_4), where V_4 is its relevant past four-volume. At late times the effective survival hazard is 4 pi A/(3H^3). Finite expected surviving four-volume per comoving region requires this hazard to exceed the rate 3H at which spatial volume grows, giving Page’s bound (Page, 2008):

The comparison with 3H is therefore between two rates per time; a nucleation rate per four-volume and a rate per time cannot be

compared directly. The bound controls the integral of surviving volume; it does not directly compare decay with the Boltzmann-brain production rate. Nor is it a universal lifetime bound independent of the cosmology and weighting prescription. The fixed-patch calculation above illustrates the competition and leaves Page's geometric calculation in place.

The two examples close opposite shortcuts. Declining local activity need not suppress a global count if volume grows faster. A persistent local rate need not guarantee an event if the patch can end. A Boltzmann-brain argument must say which of these circumstances its cosmology and measure provide. These calculations clarify the premises; they do not refute a detailed model that has already established them.

5.5. Dilution: assembly from expanding matter

Section 5.3 left open how fast the local rate falls. For one class of fluctuation observers, the expansion itself fixes it. Consider an observer that must be assembled from N pre-existing particles, gathered into a region of fixed physical volume v, in a universe whose distances grow with the scale factor a(t). The particles thin out: their number density falls as n(t) = n_0 a(t)^(-3). In a dilute, uncorrelated gas, the chance that a given region of volume v

holds the N required particles at once is about (nv)^N/N! when nv is small, so the formation rate per physical volume falls at least as fast as a^(-3N). The particles' own motions slow as space expands, and a cooling gas makes thermal assembly harder still; both only strengthen the bound. The physical volume of a fixed comoving region, one carried along by the expansion, grows as a^3. The expected count therefore accumulates at a rate

With late-time exponential expansion, a(t) = exp(Ht), this is the case of Section 5.3 with alpha = 3NH, and alpha > 3H whenever N >= 2, so the infinite-time integral converges. Counted in doublings of the scale factor, each doubling multiplies the rate by 8^(-(N-1)), and the total over all future doublings is a geometric series: 1 + 1/8 + 1/64 + ... = 8/7 times the first term for a two-particle assembly, and incomparably less for a brain of some 10^26 atoms. Endless time leaves the total finite, as Zeno's halves, 1/2 + 1/4 + 1/8 + ..., sum to 1.

The estimate has a definite scope. It covers assembly from matter that dilutes with the expansion. Gravitationally bound matter does not dilute: stars, their remnants and black holes keep their densities while the space between them grows. Their lifetimes are finite on the usual accounts, through proton decay if it occurs, and otherwise through the slow collapse Freeman

Dyson estimated for matter that cannot decay, followed by the evaporation of black holes (Dyson, 1979). A finite lifetime bounds their contribution, as in Section 5.4, though the bound can be vast. Nor does dilution touch fluctuations of a de Sitter horizon, whose temperature stays fixed as space expands. There the question returns to Sections 5.4 and 6: whether the late-time vacuum decays first, and whether a stationary vacuum state produces dynamical events at all. On the reading in which nothing happens without a cause, a stationary state produces no events, the conclusion Boddy, Carroll and Pollack (2015) reach under their assumptions about the state.

Where counts still diverge, divergence settles nothing by itself: two infinite populations have to be compared under a stated regulator. In an infinite space every epoch holds an infinite expected number of fluctuation events, so sums over epochs are infinite and a measure has to compare them. Counting per comoving region, the dilution result applies region by region: each region’s expected total of assembled observers is the convergent sum above, a fixed multiple of its first term, and each region forms a finite number of ordinary observers from the same matter, so the ratio of the two populations is finite and is set by the earliest, densest epochs. Other cutoffs weigh epochs

differently; under any of them a falling local rate lowers the fluctuation observers' count compared with a steady one, though it need not make their share small.

6. Comparison with existing approaches

Changing the measure can change the dominance criterion. Bousso, Freivogel and Yang (2009) give a local formulation of the scale-factor measure in expanding regions and compare its vacuum-stability conditions with those of the causal-diamond measure, which counts only what one observer's causal region contains. De Simone and colleagues (2010) derive conditions for avoiding Boltzmann-brain domination under the scale-factor cutoff. These results depend on the regulator, vacuum transitions, reference population and conditioning as well as on the volume counted, and no single Poisson integral with a new V(t) reproduces them all.

Each existing response holds under its own model's assumptions, and none of them supplies a general exclusion. Boddy, Carroll and Pollack's stationary vacuum proposal depends on assumptions about the state and its larger Hilbert space, the space of possible quantum states. Their earlier analysis distinguishes finite closed de Sitter systems where

recurrences (returns arbitrarily close to earlier states) remain possible. Their conditional conclusions supply no blanket exclusion under unitary dynamics (Boddy, Carroll and Pollack, 2014, 2015).

Tumulka's conclusion depends on the field model and its late dynamics. His paper explicitly distinguishes configurations shaped like brains from configurations capable of functioning as brains. The present account identifies those dependencies and supplies elementary counterexamples to invalid general inferences.

7. Objections and replies

Does allowing stochastic laws undermine the structural account? No. Both deterministic maps and probability measures have mathematical structure. The present argument tests formation and weighting under whichever law the model specifies.

Even if the exact configuration is not visited, configurations "close enough" will be. That is the right way to pose it, and it has to be made precise. Specify a class of histories robust enough to implement the relevant organization. Ergodicity can

imply repeated visits to a positive-measure target; nonergodicity alone cannot show a particular target is avoided.

Does ordinary thermodynamic behavior require visiting every exact microstate? No. Equilibration (settling toward steady values) of selected observables (measurable quantities) is different from exhaustive visitation.

In Many-Worlds, every branch exists, so every nonzero-amplitude configuration is realized. A branch is an account of correlated records and dynamics, not an arbitrary nonzero coefficient. Once a model establishes suitable observer histories, compare their weights using the same observation rule used for ordinary observers.

8. Implications

Three concrete tasks for a proposed cosmological account.

First, define the target as a class of functioning histories and calculate reachability or formation rates under the proposed state and dynamics.

Second, separate formation from dominance. Integrate rates over the relevant duration and volume, compare the rate at which ordinary observers are produced, and disclose any regulator

needed for infinite populations.

Third, condition on the evidence consistently. David Wolpert, Carlo Rovelli and Jordan Scharnhorst (2025) distinguish the roles of assumptions about memories, temporal conditioning and entropy (roughly, a measure of disorder). A model should state which records it treats as data and which as reliable traces of a past.

9. Conclusion

Infinite time alone does not guarantee a fluctuation observer, because an endless process can keep avoiding a whole class of states, and determinism alone does not exclude one, because a deterministic rule can visit every state. The model must specify which histories are possible, which systems function as observers and how their observations are weighted. It must also count in a stated region: a falling rate can still produce infinitely many events in an expanding volume, and a steady one can lose the race to decay. For assembly from matter that thins out as space expands, the expansion itself makes the total in each comoving region finite, so a region-by-region measure gives such observers a finite ratio to ordinary ones even in an infinite space; bound matter and horizon fluctuations need their own accounts.

The even integers go on forever without producing an odd integer. A cyclic counter visits every state. Which analogy applies to a universe is a question for its dynamics.

A34. Build or Burn: A Return Term for Government Spending in the Inflation Equation

Abstract

Government spending feeds inflation or growth according to what it returns, because a dollar spent adds demand and adds supply only through the goods and services it delivers. The national accounts value government purchases at cost, as though every public dollar returned one dollar of value, and the quantity equation, MV = PQ, leaves unexplained where the Q comes from. This paper adds the missing term. It calls the approach BERN, Bernstein Economics of Realized Net: judge whether every dollar a government spends feeds inflation or growth by the competitively priced goods, services and national skills it delivers. Is it a good investment? Does it build or burn?

Effective government spending is spending weighted by its realized return, and the price level is demand divided by capacity. Spending is inflationary when it adds demand faster than the capacity it delivers, and it is not when its return keeps pace. The adjustment puts the four rival accounts of money and prices into

one equation: the quantity theory describes spending that returns nothing, the Keynesian multiplier describes spending into idle capacity, the Austrian cycle describes investment whose return falls short, and Modern Monetary Theory's resource limit is the point where the return term decides. The hyperinflation record sits at the burn extreme: money-financed spending soared while production collapsed. Spending that builds competitively priced capacity also draws demand from the rest of the world, which supports the currency. The paper states the condition, works it through a two-period example and proposes a test.

1. The missing term

Output equals consumption plus investment plus government purchases plus net exports (exports minus imports), $Y = C + I + G + NX$. The identity is true by construction. It counts each public dollar at what it cost, because most of what a government provides has no market price, and the national accounts (the official statistics measuring a nation's output, income and spending) value it by the inputs used to make it (United Nations et al., 2009). That convention counts every public dollar as a dollar of output, whatever it returns. A bridge that carries freight for eighty years and a bridge to a village of four hundred people

that already has a ferry enter the accounts as the same G. Valuing an output by its inputs is the old cost-of-production view of value, whose strongest form was the labor theory of value (Ricardo, 1817; Marx, 1867). The marginal revolution of the 1870s replaced it with value at the margin, what buyers will pay for one more unit (Jevons, 1871; Menger, 1871), and BERN makes the same correction for public spending.

A private firm has the missing column. It is the return on invested capital: borrowing at five percent to earn three is a loss on a schedule anyone can calculate in advance. A government that spends money it creates faces the same arithmetic in another form. The money it adds is a claim on goods. If the spending produces the goods, the claims meet them. If it does not, the claims bid up the prices of the goods there are.

The equation of exchange, MV = PQ, makes the same point from the side of money (Fisher, 1911). The money spent, M times its velocity V (the rate at which money changes hands), equals the price level P times real output Q (the quantity of goods and services produced). It is an identity, and it is silent on what makes Q grow. This paper supplies that link for public spending: the part of Q that government spending creates depends on the return the spending realizes.

2. The adjustment

2.1 Effective government spending

Count each outlay by what it delivers, not by what it cost. A road that carries trucks for decades returns more than it cost. A fireworks show returns one evening. Call the ratio of value delivered to cost the realized return. The national accounts set it to one for every outlay; BERN uses the realized value.

Stated precisely, for outlays G_i with realized returns R_i, effective government spending is

where R_i is the real value of the goods, services and skills outlay i delivers over its life, at the prices buyers would pay for them, discounted to the date of spending (value arriving later counted for less), divided by its cost. An outlay builds when $R_i > 1$ and burns when $R_i < 1$, and the burn of a program is the value its spending failed to return, the sum of $(1 - R_i) G_i$ over its burning outlays. Value that markets do not price, such as clean water, safe roads and working courts, counts at its appraised value, as cost-benefit practice already estimates it (HM Treasury, 2026).

2.2 Prices are demand over capacity

Prices rise when the money chasing goods grows faster than the goods. Spending adds money demand at once. It adds goods only when what it buys starts producing, and only in proportion to its return. So the same spending pushes prices up during construction, and afterward prices settle back where the return is high and stay up where it is low.

Stated precisely, in an economy at capacity the price level is nominal demand (spending measured in money) over real capacity,

which is the quantity equation with D = MV and C = Q at full use. Prices then rise exactly when demand grows faster than capacity, and inflation is close to the growth of demand minus the growth of capacity. An outlay raises demand by its cost while it is spent and raises capacity by the output it delivers from its delivery date on. Spending is non-inflationary once delivered when the capacity it adds grows at least as fast as the demand it adds:

The higher the realized return, the more of the spending’s price effect its output cancels. Spending whose added capacity keeps pace with the demand it adds leaves no lasting rise in prices; spending that returns nothing leaves all of it, as demand with no

goods behind it. Money created to pay for spending stays in circulation, so its output has to keep coming: a one-time injection that raises the money stock by a tenth needs capacity a tenth larger, year after year, to leave prices where they were. With idle capacity, C is not yet binding (not yet the limit on output), and spending raises output with little effect on prices until the slack is used.

2.3 A two-period example

With the money spent held fixed and only what it delivers varied, two projects push prices up together while they are built and separate once one of them produces. Consider projects A and B with the same cost and the same spending path, each in its own simple economy with one composite good (a single good standing for all output), flexible prices and fully used capacity. Let D t be nominal spending on the good and C t the quantity produced at capacity, so that P t=D t/C t.

Both economies start at D 0=C 0=100 and P 0=1. During construction, spending rises to 110 while capacity stays at 100. In the next period spending stays at 110; project A then raises capacity to 110, and project B adds nothing to this market.

Period; Spending, both; Capacity A; Capacity B; Prices A / B.

Before construction; 100; 100; 100; 1.00 / 1.00.

Construction; 110; 100; 100; 1.10 / 1.10.

After delivery; 110; 110; 100; 1.00 / 1.10.

A builds and B burns. Both raise prices by ten percent while they are spent. A’s delivered capacity brings its price level back to where it started; B’s stays ten percent higher. With the same money spent, the price levels stand in the inverse ratio of the capacity each economy ends with, P t^A/P t^B=C t^B/C t^A. The numbers are illustrative, and the mechanism is general: the inflation an outlay leaves behind is set by what it returns.

3. Four schools in one equation

The rival accounts of money and prices each describe one regime of the same equation, and the return term, together with the amount of slack, says which regime holds.

The quantity theory, in Irving Fisher’s form (Fisher, 1911), holds where spending returns nothing and capacity is fixed: demand grows, capacity does not, and prices rise in step with money. That is the burn regime.

The multiplier of John Maynard Keynes (Keynes, 1936), the rise in output per dollar of new spending, holds where capacity stands

idle: spending raises output with little pressure on prices, because C is not binding. The return term matters less while the slack lasts and decides the outcome once it is used up.

The Austrian account of Friedrich Hayek (Hayek, 1931) holds where credit finances investment whose realized return falls short of what its prices promised: capacity is built that does not pay back, and the bust is that shortfall coming due.

Modern Monetary Theory, set out by L. Randall Wray and Stephanie Kelton (Wray, 2012; Kelton, 2020), holds that a government issuing its own currency cannot be forced into nominal default, and that its real limit is the economy’s capacity to produce. Both claims fit the equation. What MMT does not supply is a test for which spending relaxes that limit and which tightens it. The return term is that test: spending with a high realized return raises the ceiling it is measured against, and spending with a low one consumes the same resources and raises nothing.

4. Hyperinflation: the burn extreme

The hyperinflation record shows the burn regime at its limit. Phillip Cagan defined hyperinflation as prices rising more than fifty percent a month and studied seven European cases (Cagan, 1956). Steve Hanke and Nicholas Krus later catalogued 56 episodes (Hanke and Krus, 2013). The episodes share one pattern: very large deficits paid for by printing money, while production fell. Weimar Germany in 1923, Hungary in 1946, which holds the record, Zimbabwe in 2008 and Venezuela from 2016 all fit it. In Zimbabwe, the seizure of commercial farms cut farm output; in Venezuela, oil output fell by more than half while the central bank financed the state.

In the terms of this paper, these were economies where demand was pumped up with money while capacity shrank: D soared and C collapsed, so P exploded. The spending that drove them returned little or nothing. It paid for consumption, transfers (payments like benefits that buy no goods or services), losses and wars, and the output that would have met the new money was not built. Thomas Sargent found that four of the classic hyperinflations ended abruptly when governments changed the fiscal regime (the settled policy on taxing, spending and

borrowing) that financed them (Sargent, 1982). Ending the burn ended the inflation.

The same record does not support a rule that any large deficit or debt ratio (government debt as a share of national output) must end in hyperinflation. High debt has coexisted with low inflation. What the money bought, and what it returned, separates the cases.

5. Build: good investment and the rest of the world

Spending that builds competitively priced goods, services and national skills does more than add domestic supply. Buyers abroad want what it produces. Export earnings pay for imports, which add to what domestic money can buy, and foreign buyers who need the currency to pay hold up its value. A good investment can therefore be paid for with new money and still leave prices steady, because the goods it returns meet the money and the rest of the world takes part of both.

Stated precisely, in an open economy the goods available to domestic buyers are domestic capacity plus imports, and imports are paid for by exports. Raising the competitive output of the economy raises exports, which finance imports without drawing

down reserves (the foreign currency the central bank holds), and foreign demand for the currency lowers the velocity of the money held at home. This is why the realized return in Section 2 counts value at the prices buyers anywhere would pay: a project whose output no one would buy at its cost returns less than its accounts claim, however it is classified. Evidence on public investment points the same way: its output effects are larger where investment efficiency is higher (International Monetary Fund, 2014), and implementation delays shift its effects later (Leeper, Walker and Yang, 2010).

6. Measuring the return

The return is measurable with tools that already exist. Transport agencies compute benefit-cost ratios as routine, and HM Treasury's Green Book sets out how to appraise the costs, benefits and risks of public options, including benefits with no market price (HM Treasury, 2026). After the Atkinson Review, the United Kingdom began measuring the output of some public services directly instead of assuming it equals their cost (Atkinson, 2005). BERN asks for the step that is usually missing: check the return after delivery, report it beside the spending with its uncertainty, and use the realized figure, not the projected one,

in the inflation account. If the traffic did not come, the finished bridge should not keep its projected return.

7. Predictions and a test

Three predictions follow. Given the same money growth, inflation should be lower where a larger share of spending builds, after a lag equal to the time its projects take to deliver. Hyperinflation should require the burn regime: money-financed spending with little return while capacity falls. Spending whose delivered capacity keeps pace with the demand its money adds can be financed with new money without lasting inflation, and the price rise during construction should reverse.

A test can use project-level data with delivery dates and ex post returns (returns measured after the fact), such as the World Bank’s independent evaluations, merged with money, output and price data. It should compare episodes of similar money growth with different shares of building and burning spending, and use differences in project timing that were not chosen in response to inflation. Statistical controls (adjusting for other measured factors) alone do not establish the mechanism, and no such study is reported here.

8. Objections

”An identity cannot be wrong, and adding a return term breaks it.” The expenditure identity stays as it is: G is what was spent. BERN adds a second account, effective spending, and an equation for what makes capacity grow. The identity says spending equals spending; BERN says what the spending returns.

”The return is too hard to measure.” It is estimated before most large projects are approved. BERN asks that the estimate be checked afterward, reported with its range and carried into the inflation account. A misleading number is worse than none, so the assumptions, the alternatives and the ranges should be published.

”Some public goods have no market return.” The realized return counts value that markets do not price, at its appraised value. A courthouse, a vaccination campaign or a working legal system returns real value, and the method values it as appraisal already does.

”Financing, expectations and crowding out matter too.” They do, and the equation keeps them: financing and expectations act

through demand and velocity, and crowding out through the capacity other uses lose. BERN isolates the term the standard accounts leave out, what the spending returns.

9. Conclusion

Government spending feeds inflation or growth according to what it returns. The national accounts count every public dollar as returning one dollar, and the quantity equation leaves the growth of output unexplained. BERN replaces the assumed return with the realized one: effective spending is spending weighted by its return, and prices are demand over capacity. Spending that builds competitively priced goods, services and skills meets its money with goods and draws in the rest of the world. Spending that burns leaves its money chasing goods nobody made, and at its extreme it is the hyperinflation record. The question for every dollar is the one a firm asks of every investment: does it build or burn?

A35. Consensus Is Not Evidence: It Decides Only Where No Rule Does

Abstract

Consensus adds no evidence in mathematics or science, because there a claim is settled by rules, a proof checked step by step or a test whose error rates are fixed in advance, and counting the people who accept the result adds nothing to it. Consensus decides only where no rule does. Whether an ambiguous mark reads as a 4 or a 9, whether a face is beautiful, whether a painting is art or a punishment is just: in such fuzzy categories the boundary is set by how competent judges draw it, and the agreement of many judges is the best answer there is. Those are the questions machine learning answers by training on many human labels, and the questions Socrates asked about justice, beauty and the good, which no definition has closed.

The two kinds of question call for different authority. Treating a rule-decided question as a vote, as “97 percent of scientists agree” does, moves science into the domain of judgment and asks a count to do the work of a test. This paper draws the line, shows where each kind of authority applies, answers the

objection that non-experts have to rely on consensus, and proposes that scientific claims be reported by the evidence that would settle them, not by a head count.

1. Two kinds of question

Some questions are settled by rules and others by judgment. Is a number prime? Apply the test. Does a proof prove its theorem? Check each step against the rules of inference (the valid patterns for drawing conclusions). Does a drug lower blood pressure more than a placebo? Run the trial, with the statistical test and its error rates (the chances of a false alarm or a miss) fixed before the data arrive. In each case the answer comes from a public procedure, and anyone who follows the procedure correctly gets the same answer.

Other questions have no such procedure. Is this handwritten mark a 4 or a 9? Is this face beautiful? Is this punishment just? The categories have clear centers and blurred edges, membership comes in degrees, and competent judges disagree at the margins (Zadeh, 1965). Nothing outside the pattern of judgments fixes where the edge lies.

Consensus means different things in the two cases. Where a rule decides, agreement is a report that many people applied the rule; the rule's result is the evidence, and the count adds nothing to it. Where no rule decides, the agreement of competent judges is what fixes the answer, and it is the best answer available.

Boaz Miller (2013) asks when a consensus is based on knowledge, Alvin Goldman (2001) how a novice can choose among experts, and Naomi Oreskes (2019) why the public should trust scientific consensus. This paper's contribution is to separate the domains. Where a rule decides, even a knowledge-based consensus is redundant with the evidence it rests on; where no rule decides, consensus is the answer itself. Its decisive premise is that in a rule-decided question the answer is fixed by applying the rule, whoever applies it. It rules out consensus as a reason within scientific argument. It leaves consensus a role as a pointer for outsiders and as the answer in fuzzy domains.

2. Where a rule decides, the count adds nothing

A theorem is true or false whatever anyone believes. Andrew Wiles announced a proof of Fermat's Last Theorem in 1993; a referee found a gap, and Wiles and Richard Taylor closed it in 1994 (Wiles, 1995; Taylor and Wiles, 1995). Mathematicians

accepted the result because the repaired proof checked, step by step. Had every mathematician believed the flawed version, the gap would still have been there.

Science works the same way with different rules. A measurement comes with its error bars; a hypothesis test comes with its error rates; a prediction can fail. The evidence is the data analyzed by procedures anyone can repeat. When a field’s majority and the evidence part ways, the evidence wins. Most geologists rejected Alfred Wegener’s continental drift (Wegener, 1915) until magnetic stripes on the seafloor showed the ocean floor spreading (Vine and Matthews, 1963). Physicians held that stress and acid caused stomach ulcers until Barry Marshall and Robin Warren showed the role of a bacterium (Marshall and Warren, 1984), work that won the 2005 Nobel Prize. Ignaz Semmelweis cut deaths from childbed fever (infection of new mothers after childbirth) by having doctors wash their hands, and the profession rejected him (Semmelweis, 1861). In each case a test settled the question and the count did not.

None of this makes experts useless. An expert is someone who can apply the rules, and an expert’s report that the rules were applied carries weight. The value lies in the applied rules, which can be checked, and not in how many people report them.

Consensus may tell an outsider where to look; it does not tell anyone what is true.

3. Where no rule decides, consensus is the answer

Fuzzy categories are where agreement does real work. The image collection that trained a generation of vision systems, ImageNet, settled each picture’s label by the votes of many human labelers (Deng et al., 2009). There was no better method, because no rule says exactly where “dog” ends and “wolf” begins in a blurry photograph; the label is what careful judges say. A model trained on millions of such labels learns the category as people draw it, which is why fuzzy questions are the ones machine learning handles best.

Pooling judgments also improves them. Francis Galton collected 787 guesses of an ox’s weight at a country fair; their middle value, 1,207 pounds, came within one percent of the true 1,198 (Galton, 1907). The Marquis de Condorcet showed that if independent judges are each more likely right than wrong, a majority of them is more likely right than any one, and more so as the group grows (Condorcet, 1785). Where no one can apply the rule, many independent judgments are the best estimate available; once the ox is on the scale, the guesses add nothing.

The same holds for justice, beauty, art and the good. Socrates asked his companions for definitions that would work as rules: what piety is, what courage is, what beauty is. Each proposed definition failed against a counterexample, and the dialogues end without one (Plato, 1997: *Euthyphro*, *Laches*, *Hippias Major*). Where definition fails, the shared judgment of thoughtful people is what the words mean.

Fuzzy categories still follow rules, but rules too long for anyone to write down. Consensus is where such a rule is stored, spread across many judges, and a model trained on their judgments learns a compressed version of it. Where the rule is short enough to state and apply, as in arithmetic or a statistical test, the stored version is superfluous: apply the rule.

4. The cost of confusing the two

“97 percent of scientists agree” turns a physical question into a count. The best-known source of the figure counted the stances of paper abstracts: 97.1 percent of the 4,014 abstracts that took a position on human-caused warming endorsed it, and 66.4 percent of the 11,944 abstracts examined took none (Cook et al., 2013). That is a measurement of what papers said. Whether and

how much human activity warms the planet is settled, to whatever degree it is settled, by measurements and by models tested against them, with stated uncertainties. A count of stances adds nothing to those tests, and quoting the count in place of the tests invites the reply that counts can be manufactured.

The confusion has a general cost. An argument from consensus moves a question from the domain where rules decide into the domain where judgment decides, and judgment can be swayed by funding, fashion and politics in ways a test cannot. Science keeps its authority by pointing to the procedure anyone can repeat. The better report states the claim, the test that could have refuted it, and the result with its uncertainty. A reader who cannot run the test can still see that it exists and what it showed.

5. Objections

”Non-experts cannot check the evidence, so for them consensus is evidence.” For an outsider, a consensus is testimony: evidence about what experts believe. Its weight comes entirely from the experts having applied the rules, independently, to the evidence. That makes it a pointer to evidence, not evidence of the same kind, and its strength depends on facts the count does not show: whether the experts judged independently,

whether they could have been proved wrong, and whether dissent that holds up is rewarded. The more useful thing to give an outsider is the test and its result, which a count summarizes badly. A worked case shows where the weight lies. Two investigators, each right nine times in ten, who run the test independently and agree raise the odds from nine to one to eighty-one to one, about 98.8 percent. If the second copied the first, the agreement adds nothing, and the odds stay nine to one. The evidence was the second test, not the second name, and a head count cannot tell the two apart.

”Science has judgment calls too.” It does: which research program looks promising, how to weigh conflicting studies before a decisive one exists, when evidence is enough to act on. Those are fuzzy questions, and there agreement describes a field’s judgment. They should be reported as judgment, with the evidence still open, and not as the settled result of a test.

”Peer review and replication are consensus.” Their value lies in checking work against rules: a reviewer finds the gap, a replication repeats the procedure. A paper is not more correct because more referees approved it; it is more trustworthy because more checks failed to find an error.

”Consensus usually tracks the evidence.” When it does, the evidence does the work, and the count adds nothing to it. When it does not, the count misleads. Either way, the evidence is what should be reported.

6. Conclusion

Consensus is not evidence where a rule decides, because the rule’s result is the evidence and the count adds nothing to it. Consensus decides only where no rule does: in fuzzy categories such as the label on a blurry image, the beauty of a face or the justice of a punishment, where the shared judgment of competent people is the answer. Mathematics and science stand on rules anyone can apply, and their claims should be reported by the tests that settle them. A head count belongs to the questions no rule can settle.

A36. The Climate Confidence Gap: Public Claims That Drop the IPCC's Confidence Misreport the Science

Abstract

The IPCC states its confidence finding by finding, and a public claim that drops that confidence, and so changes what is claimed, misreports the science, in either direction. Six findings show what the assessments actually say where public debate is loudest. Climate sensitivity is still a range: 1.5 to 4.5 degrees Celsius in 1979 and again in 2013, and 2.5 to 4.0 since 2021. The high-emissions scenario often called business as usual is one the Sixth Assessment Report judges of low likelihood. Models overstated upper tropical tropospheric warming from 1979 to 2014 by at least 0.1 degrees per decade, with medium confidence. Hot extremes are virtually certain to have increased, while human influence on regional meteorological drought trends has generally low confidence. Scientists inside the field flagged two of them in Nature. Overstated confidence costs the science its credibility when readers find the gap, and the remedy costs nothing: quote the report's own confidence terms. The paper

gives the procedure for checking any public claim against its source.

1. The Standard We Are Applying

A summary should preserve what its source says and how firmly it says it. The Intergovernmental Panel on Climate Change (IPCC) distinguishes confidence in an assessment (how well evidence and expert agreement support it) from the likelihood of an outcome (its assessed probability); those terms have defined uses. An audit should check whether a public statement changes the finding, its scope or its uncertainty. Simplifying the language can be accurate. The test is whether the simplification changes the claim.

The six topics below are comparisons worth auditing. None of them is a completed claim-source pair yet: someone must first gather dated public quotations alongside the assessment passages they cite, and only then can anyone report how large or how common a communication gap is.

A reproducible claim-source record

An audit is only as good as its record, so the record has to be fixed before any example is chosen. For each comparison, record the public quotation, speaker or outlet, date, stable link and exact proposition asserted. Record the assessment passage with its section or page, publication date, scenario, scope and uncertainty language. An assessment published after the public statement cannot establish that the earlier statement misrepresented it.

Use a coding scheme (a fixed set of categories for classifying each comparison) that permits agreement, stronger public certainty, weaker public certainty, changed scope, missing conditions and an unassessable match. Two coders should judge independently, preserve disagreements and document their resolution. A prevalence estimate (a figure for how common such gaps are) additionally requires an outlet set, date window and inclusion rule fixed before selecting examples. Report all eligible matches, including accurate summaries.

2. Equilibrium Climate Sensitivity

Climate sensitivity shows scientific progress alongside remaining uncertainty: the assessed range held still for decades, then narrowed, and it is still a range. In the assessment record, the Charney Report (National Research Council, 1979), chaired by the meteorologist Jule Charney, estimated eventual warming after doubled CO2 at 1.5 to 4.5 degrees Celsius. The IPCC's Fifth Assessment Report (AR5; IPCC, 2013) gave the same likely interval (a range judged at least 66 percent probable) in 2013 (Working Group I, Summary for Policymakers, SPM D.2). The Sixth Assessment Report (AR6; IPCC, 2021) narrowed its likely interval to 2.5 to 4.0 degrees, with a best estimate of 3.0 degrees and high confidence in that likely interval (WGI, SPM A.4.4).

A best estimate is not an exact prediction. The remaining interval also does not negate the separate finding that human influence has warmed the atmosphere, ocean and land (AR6 WGI, SPM A.1). An audit should ask whether a summary confuses those two questions.

3. Scenarios Treated as Forecasts

A scenario asks what would happen under specified emissions. It is not automatically a forecast that those emissions will occur. Zeke Hausfather and Glen Peters (2020) challenged the routine use of RCP8.5, the highest-emissions Representative Concentration Pathway, as a business-as-usual baseline.

The report holds two things together here. In the source, AR6 WGI Section 1.6.1.4 states that the report generally attaches no likelihoods to its scenarios. The same section describes high-emissions pathways such as RCP8.5 and SSP5-8.5, its counterpart among the newer Shared Socioeconomic Pathways, as having low likelihood in light of recent energy-sector developments. This is a narrative assessment of plausibility, not a calibrated numerical probability assigned to the scenario.

An audit should preserve both points. A useful high-emissions calculation need not be the expected future, and the plausibility discussion does not make the scenario’s conditional physical consequences irrelevant. Check what the public statement claims, against the source available on its date.

4. The Hot Model Problem

A raw model ensemble (the pooled output of many climate models) and an assessed projection need not give the same range. Hausfather and colleagues (2022) discuss models from the sixth Coupled Model Intercomparison Project (CMIP6) with unusually high climate sensitivity and warn against using their projections without considering other evidence.

The assessment had already made that adjustment. In the source, AR6 combined scenario-based model projections with observations of past warming and its assessed climate sensitivity and transient response, the warming reached when gradually rising CO2 first doubles (WGI Section 4.3.4 and Box 4.1). Its assessed global temperature ranges therefore differ from the unadjusted ensemble range. An audit should identify which quantity a public summary reports before comparing it with AR6.

5. Homogenization of the Surface Record

A station’s record can change because the climate changes or because the station moves, replaces an instrument or changes observing practice. Homogenization aims to separate those effects.

The published method checks a station against nearby stations, so each adjustment carries a stated reason that another analyst can test. In the documentation, version 4 of the Global Historical Climatology Network monthly dataset (GHCNm), kept by the National Oceanic and Atmospheric Administration (NOAA), compares a station with nearby correlated records to detect abrupt shifts affecting that station. It then estimates corrections from the differences (Menne and Williams, 2009; NOAA NCEI, GHCNm documentation). The direction of a correction alone shows neither manipulation nor accuracy.

An audit should state the dataset and version, the affected period and the adjustment being discussed. Berkeley Earth’s separate averaging method provides another published comparison (Rohde et al., 2013). Agreement between analyses matters, as do shared observations and methodological differences; none should be hidden by a claim about adjustments in general.

6. The Tropical Troposphere

A model-observation difference must be tied to its altitude, region and period. In the assessment's terms, AR6 assesses with medium confidence that CMIP5 and CMIP6 models overestimate warming in the upper tropical troposphere (the lowest layer of the atmosphere) during 1979-2014 by at least 0.1 degrees Celsius per decade, partly because they overestimate the tropical sea-surface warming pattern (WGI Section 3.3.1.2.1).

The report also notes medium confidence that the upper tropical troposphere has warmed faster than the near-surface since at least 2001 (Section 3.3.1.2, referring to Section 2.3.1.2). These statements answer different comparisons. An audit should retain the model discrepancy without turning it into a claim that no upper-air amplification (faster warming aloft than at the surface) has been observed.

7. Extreme Event Attribution

Different extremes support different conclusions, and one sentence of assessment can attach a likelihood term to a trend and a separate confidence term to its cause. In the assessment's own terms, AR6 calls the increase in hot-extreme frequency and

intensity across most land regions since the 1950s virtually certain (at least 99 percent probable), with high confidence that human influence is the main driver (WGI, SPM A.3.1).

Drought needs a finer distinction, because the assessment reaches different conclusions for different kinds of drought. In the assessment's own terms, Section 11.6.4.5 reports generally low confidence in human influence on regional meteorological-drought trends (trends in shortfalls of rain and snow), with exceptions. It reports medium confidence in human contributions to increased agricultural and ecological drought (too little soil moisture for crops and ecosystems) in some regions, including the Mediterranean and western North America. It also distinguishes the effects of climate, land use and water management on hydrological drought (low water in rivers, lakes and reservoirs).

The same section separates trends from individual events: an event can be attributed in a region without a detected long-term trend. A summary of drought everywhere cannot stand in for these different findings. An audit must match drought type, location, period and causal question before judging the claim.

8. Why Accurate Communication Matters

When a writer turns a conditional projection into an unconditional prediction without saying so, a reader who later discovers it can lose trust. The audit must also record accurate summaries and appropriate simplifications. Selecting only conspicuous disagreements builds the conclusion into the sample.

9. Recommendation

Use plain language that preserves the source’s meaning. Keep a confidence or likelihood qualifier when omitting it would make the claim stronger or weaker. State the scenario and geographical scope when they affect the conclusion. A summary need not reproduce every technical term to be accurate; it needs to preserve the distinctions on which the finding depends. How readers understand alternative wording is a further empirical question.

Conclusion

Put a public climate statement beside the assessment it cites, and any difference in certainty between them shows. This paper gives the procedure for doing that: quote both, date both, and record the scenario, scope and uncertainty language the source used. The assessment record holds progress and remaining uncertainty together, and an audit that reports one has to report the other. Published comparisons would show which public summaries report their sources accurately and which need correcting.

A38. The Theory of Everything Must Be Short: The Standard Model and Gravity in 15,200 Bits

Abstract

A theory of everything must be short, because each added bit halves a theory’s weight: a bit that no rule fixes is a 50/50 choice. Its simplicity can be measured instead of praised: count every bit it needs to run, its equations, parameters, boundary information and the rules that decode its output into predictions, at the same precision for every rival. A working program of 15,200 bits prints a complete statement of the Standard Model with general relativity, laws and constants together, so the length of known fundamental physics, as a symbolic statement, is a number anyone can check and try to beat. Count selectively and the comparison means nothing, as a code audit here shows: a small compiled file can omit the physics claimed for it. The proposed benchmark publishes complete descriptions and checks their outputs, and a working program sets an achievable length that a shorter one would improve. String theory is the test case on two counts. It is not needed for its main job, keeping gravity finite at the shortest

distances, if change comes in discrete steps, as argued here: a cause needs a next state, and a continuum has none. Selecting our world among its estimated 10^500 vacua costs about 1,661 bits unless a far shorter selector exists.

1. The benchmark question

Calling a theory elegant does not tell us how much information it needs. Replace the compliment with a test: write a program that performs a stated predictive task, publish it, then count everything required to run it. The shortest such program fixes the information the task needs. Formally, the shortest program producing the agreed output defines its Kolmogorov complexity K, the measure Andrey Kolmogorov introduced (Kolmogorov, 1965), in the chosen programming language. An actual program gives a demonstrated length to compare; a shorter one may still exist. Length matters because, on the measure used here, each bit of a shortest description is an unruled 50/50 choice, so each added bit halves the description’s weight; a reader who rejects that measure can still read the counts as a comparison by minimum description length, a method favoring the shortest joint description of model and data (Rissanen, 1978).

First choose what the program must produce. Recovering symbolic equations and calculating numerical predictions are different tasks. State which mathematical tools both programs may use and which information each must supply. One rule can do several jobs at once and cut the total: an action (a single quantity from which the equations of motion follow) covers many trajectories instead of one, a preparation rule ties the source to the settings, a generator supplies a later state for a reversal. Each saving counts only when the same observations remain recoverable.

Kolmogorov complexity and algorithmic probability (the chance that a program of random bits produces a given output) are established tools (Kolmogorov, 1965; Solomonoff, 1964; Li and Vitanyi, 2019), and appeals to the length of physical theories are common. This paper’s contribution is the accounting those appeals need: count everything the declared task uses, treat a working program as an upper bound (a length the shortest program cannot exceed) for that task in its language, and compare lengths only between descriptions that do the same job. A supplementary program applies it to the symbolic task. Its decisive premise is that a length comparison means something only for a shared task in a shared language. It rules out odds

drawn by subtracting two upper bounds, and it rules out counting a theory’s equations while leaving its parameters, starting data and decoder uncharged.

String theory shows that the count is the second question. The first is whether a theory is needed at all. String theory’s main job is to keep a quantum theory of gravity finite at the shortest distances, where a smooth spacetime fills the calculations with infinities. The view taken here is that the base is not smooth. A cause produces its effect as the next state, and a continuum has no next state, so change comes in steps, as Zeno’s runner, who must first reach every halfway mark, has no first stretch to run. That is an a priori proof in the logician’s sense. Russell’s continuum, in which motion is only being at different places at different times (Russell, 1903), is the emperor’s new points: every instant gets a position, and nothing produces the next. A world that changes in steps has no shorter distances than a step, and so no source for those infinities. Physics agrees in its own terms: gravity and quantum theory together let a region of finite size hold only finitely many distinguishable states (Bousso, 2002). The counter is that a continuum is consistent mathematics, which nobody disputes; the claim concerns what produces change, and the bit counts below hold either way. On this view string theory is

not needed, though some discrete rule could still look like strings at some scale. Section 4 asks what it costs if it is chosen anyway.

The same view carries a larger chain, and the bit counts below measure its third link:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.
3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.
5. The growth comes in discrete causal steps, by the a priori proof above.
6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no

last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. With no distance shorter than a step, string theory is not needed for its main job, keeping gravity finite at the shortest distances, and a theory of everything is a short rule for transitions.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. What the program must contain

A symbolic description of the selected particle and gravitational model must identify five kinds of content, whose encoding costs depend on the declared language and shared decoder.

Component 1: The gauge group (the symmetries that define the forces). Encode SU(3) x SU(2) x U(1), including conventions needed to recover its role in the theory.

Component 2: The matter content. Specify field types, representations (how each field transforms under the symmetries), charges and generation multiplicities (the number of repeated particle families). Repeated generation structure can be shared rather than listed three times.

Component 3: The dynamical rules. Specify the action or equivalent dynamics, including the assumptions restricting allowed terms and the scalar sector (the Higgs field and its interactions).

Component 4: Parameter values. State the selected extension, units, ranges and precision. How precisely each value must be stated sets what it costs to store. As an illustrative storage calculation, 26 independent values at 15 bits each use 390 bits; at 64 bits each they use 1,664 bits.

Component 5: General relativity. State the gravitational action, its parameters and the conventions linking it to observables, the quantities that can be measured. Shared parameters should be charged once.

The audit of the earlier 9,216 bit result found a mislabeled constant count, an incorrect matrix calculation and physical equations present only as comments (notes in code that the computer ignores): that figure was extracted from an illustrative compiled file (a program translated into machine-readable form), not a complete predictive implementation. A runnable artifact for the symbolic task now supplies one such budget (Section 6, the program as an artifact); for the predictive task, any budget stays provisional until someone supplies a complete runnable artifact and an output specification. A functioning program supplies an upper bound for its declared task in its declared language. A parameter printer does not establish an upper bound for a simulator reproducing all known physics.

3. What this excludes

A symbolic specification of laws and parameters is only one possible benchmark. Initial conditions are additional content whenever the task requires a particular history or a predictive preparation measure (a weighting of possible starting states). Low thermodynamic entropy (roughly, a high degree of order) is a property of a specified macrostate (a coarse description by bulk quantities such as temperature) and measure (compare Carroll,

2010); it is not by itself a bound on the description length of each compatible microstate (an exact arrangement of every part). An exact observer record is also additional content if the task demands it. A branch label (a tag marking which many-worlds branch is meant) is not automatically a new cost growing with every elapsed moment, and record information already generated by counted inputs must not be charged again.

4. String theory: not needed, and costly to select

String theory earns its place in quantum gravity by doing one job well. Its strings are extended, so the interactions that make a point-particle theory of gravity blow up at short distances are smeared out. That job exists only if spacetime is smooth all the way down. If change comes in discrete steps, nothing happens at distances shorter than a step, the infinities have no source, and the job is done without strings. Discrete models of spacetime can also keep relativity’s symmetry, as causal sets do by scattering their events at random (Surya, 2019). String theory might still describe some scale of a deeper rule, but nothing requires it. The rest of this section asks what it costs if it is chosen anyway.

A theory can specify an observed model directly or generate a family and select a member. Selecting a member costs the bits

needed to write down its position in the list. Stated precisely, for an agreed enumeration of N alternatives, a fixed-length binary index uses ceil(log2 N) bits. If N is 10^500, a common estimate for the string landscape, the vast set of possible universes string theory allows, after Raphael Bousso and Joseph Polchinski's flux counting and Leonard Susskind's landscape proposal (Bousso and Polchinski, 2000; Susskind, 2003), this index has 1,661 bits. A particular member can have a much shorter selector. "Choose the first member with property P" may be economical if P and the enumeration are economical and actually determine the desired member. Conversely, a short theory name can conceal a long unexplained choice.

Geometry, fluxes and stabilized parameters cannot be added as independent costs if the selected index already determines them. If the index leaves any open, encode the remaining information conditionally.

Two measured implementation lengths do not bound the ratio of their unknown shortest lengths. Written out, for theories T1 and T2 with measured lengths L1 and L2, knowing K(T1) <= L1 and K(T2) <= L2 does not prove K(T2)-K(T1) >= L2-L1. Odds of 2^(-2940) against string theory, obtained by subtracting two such upper bounds, rest on that invalid step. The comparison that

does work takes two descriptions that do the same job and asks which is shorter. Stated precisely, it assigns each declared description a weight proportional to 2^(-L). If both recover the same targets and likelihoods and one is k bits shorter, its assigned weight is exactly 2^k times larger. Extending that result to whole theory families requires their alternative descriptions and weights.

5. Comparison with Wolfram’s hypergraph physics

A candidate without the required predictions cannot yet enter the matched benchmark. Stephen Wolfram (2020) proposes that physics emerges from a simple rewriting rule (a rule that replaces one pattern of links with another) applied to a hypergraph, a network whose links can join any number of nodes. The rule may have very low Kolmogorov complexity. Nobody has shown that it can do the specified task. Large output or state space size does not imply a long generating program; runtime and memory are separate costs.

If a rewriting rule recovers the target physical description or predictions, compare its complete encoding with the competing account. Include its seed, decoder, output map and required precision. The present unmatched artifacts do not establish either

competitor’s size for a shared task. The same accounting orders a search: list rewriting rules from shortest to longest, and keep those whose causal graphs, the records of which update depends on which, approximate at large scales a four-dimensional spacetime with no preferred frame, as a random causal set, a scattering of events ordered by cause and effect, does (Surya, 2019). The search can also run from the other end, program first. Write the known laws as a program of discrete steps without a grid, and remove their infinities, as loop quantum cosmology does when it replaces the big bang with a bounce (Ashtekar, Pawlowski and Singh, 2006). Then shorten the program while its outputs still match.

The continuum such a program replaces is in trouble at the smallest scales. The simplest field of the Higgs type, taken alone, loses its interaction entirely as the cutoff is removed (Aizenman and Duminil-Copin, 2021), and whether the gluon theory survives that limit with a mass gap is still one of the Millennium Prize problems (Jaffe and Witten, 2000). A program applies its formulas in steps, so it does not take the limit, which stays a question about the formulas as a description. Discrete simulations also favor causal steps: spacetime built from four-dimensional pieces grows into a four-dimensional universe,

shaped like a solution of Einstein’s equations, once every piece carries a direction of time (Ambjørn, Jurkiewicz and Loll, 2004; Ambjørn, Görlich, Jurkiewicz and Loll, 2008).

6. The program as an artifact

The benchmark should publish its counted artifact with a fixed interpreter (the program that runs the code) or decoder, enough documentation to recover the advertised output and the exact accounting convention. For the symbolic benchmark, the program would recover the gauge group, representations, Lagrangian terms (the pieces of the formula from which the action is built) and the declared parameter set at specified precision. If equations occur only in comments discarded by compilation, that compiled output cannot establish their description cost.

The symbolic task now has one. A supplementary Python program of 1,900 bytes, 15,200 bits, prints a complete statement of the Standard Model, the established theory of particles and forces apart from gravity, with general relativity: the gauge group, the fields and their charges, the action, the quantum rule and 28 declared inputs at five significant figures. Its output matches the target text exactly by checksum (a short fingerprint computed from the text), and the compressed statement inside it is 11,328

bits. That is an upper bound for the symbolic task in Python with its standard library (the built-in code that ships with the language); a shorter program may exist. The next task is the predictive one: a published program that derives observable predictions from the laws, with its environment, checksums, output targets and tolerances, and then competing encodings for the same task.

7. Why shorter means more probable

Each bit of a shortest program is a 50/50 choice that no rule makes, so each added bit halves the program’s weight. A shortest program cannot be compressed, since a shorter program for it, followed by running it, would be a shorter program for the same output, so its bits are random in exactly the sense of the rule-random dichotomy. A bit no rule fixes has no reason to be 0 rather than 1; any lean toward one value, or any dependence of one bit on another, would itself be a rule, and would have to be written into the program. Stated precisely, fix a binary prefix language, in which no complete program is the beginning of another, so each program settles where it ends. With fair independent input bits, a specified program of length n has weight 2^(-n), and a specified program k bits longer has 2^k

times less weight (Solomonoff, 1964). Jürgen Schmidhuber applied such weights to whole universes (Schmidhuber, 2000; compare Tegmark, 2014). Half is also the gentlest rate possible: each added bit doubles the number of programs, so with equally long programs weighted alike, any milder factor would give longer programs more total weight than shorter ones. Prefix-free programs, none of which extends another, keep the total at one or less. The choice of language remains part of the model (Section 8).

Longer programs that produce different outputs can collectively outweigh one shorter program, so a whole class needs its members' weights added. With the choice to stop or add another bit counted as one more unruled 50/50, all programs one bit longer weigh half as much together, and everything longer than a given length adds up to that length's weight. For a single output, the many longer programs cannot swamp the shortest: by the coding theorem, all programs for one output together weigh at most a fixed multiple of its shortest program (Li and Vitanyi, 2019). Any count has to handle prefix code normalization (keeping the total weight at one or less) and the number of programs producing each output.

8. Dependence on the encoding

Changing description language carries a cost that cannot be assumed small. The invariance constant (the largest change in K that switching languages can cause) depends on the pair of languages. It is not generally bounded by tens or hundreds of bits, and the invariance theorem alone does not protect a finite ranking. Repeat the comparison under defensible alternative encodings.

A worked comparison: symmetry, zero and independent values

Three ways of writing a pair of couplings, numbers setting how strongly particles interact, can be encoded side by side and compared. Fix a common precision of n bits for each coupling. Use two-bit tags to distinguish three formats: an equal pair (g,g), a one-sided pair (g,0), and a generic pair (g_L,g_R). The first two formats need one n-bit value; the third needs two. Written out, their lengths are n+2, n+2 and 2n+2. At n=16, the first two descriptions are 18 bits and the third is 34. Distinct tags and fixed payload lengths make this a prefix-free code.

Equality saves repeating a value; zero saves supplying one. Under this code, the equal and one-sided descriptions tie. Each has 2^16 times the weight of a particular generic 34-bit description.

A whole format has a different weight. Under this code a format with fewer values holds fewer descriptions, each one weighing more, and the two effects cancel. Written out, each one-value format contains 2^n descriptions weighted 2^(-n-2), giving total weight 1/4. The generic format contains 2^(2n) descriptions weighted 2^(-2n-2), also totaling 1/4. Conditional on these three formats, each has weight 1/3. The equal class weights come from the chosen equal-length tags; they are no universal result about symmetry.

These are classes of encodings, not disjoint classes of physical pairs: the generic format also encodes equal and one-sided pairs. To weight a particular pair, add the weights of every encoding that produces it. The accompanying executable supplement implements these three formats, exhaustively checks n=1 through n=5, and compares encodings at n=16. The result defeats a shortcut: calling symmetry an added constraint does not establish that it costs more information. A physical comparison must also count the field representations, equations,

parameters and observation model needed to recover the same evidence.

9. Implications for theory selection

A demonstrated saving buys a weight advantage that can be stated exactly, by the rule of Section 7. Actual artifacts supply measurable lengths and upper bounds on K; they do not reveal K itself. The Standard Model provides a concrete target for compression. A deeper account earns a description advantage if its full rule, inputs and decoder recover the same physical content with fewer bits. A search for unification is then a search for shared generative structure, with a declared task and a measurable saving.

Conclusion

A theory costs everything it needs to do the agreed job, because any part left out of the count makes a long account look short: equations, parameters, starting data, shared software and the required precision all count. Each counted bit is a 50/50 choice that no rule makes, so each one halves the account’s weight, and a measured saving converts directly into odds. A working implementation demonstrates one length; it does not establish

the shortest possible length.

The audited example failed to supply all the physics claimed for it. A reproducible replacement can make the comparison meaningful: publish the descriptions and decoder, verify matching outputs, and apply the stated simplicity rule to the measured difference. For the symbolic task that replacement now exists: 15,200 bits print a complete statement of the Standard Model with general relativity, checked against its target. The predictive task, and a rival doing the same job, come next.

String theory shows both questions a theory must answer: whether it is needed, and what it costs. If change comes in steps, its main job disappears, and selecting our world among its vacua costs about 1,661 bits unless a far shorter selector exists.

A39. The Emperor’s New Points: Zeno’s Paradoxes Show Continua Are Analysis, Not Physics

Abstract

Zeno left two kinds of paradox, and only one decides whether physics can be continuous. The paradox of measure, how points without width make a line with length, is answered by measure theory, and discreteness does not help: discrete points have no width either, and a grain with width brings the continuum back, so in every structure sizes come from relations. The paradoxes of motion are answered by analysis only as description, because a continuum has no next state, and causality needs one: a cause produces its effect as the next state, so in a physical continuum no state could be the first one a cause produces. This is an a priori proof in the logician’s sense. Change without a cause is outside logic and reason. Physical events are therefore discrete steps, and that covers everything that happens: matter, gravity, spacetime, space itself on either view of space, and amplitudes, since a history produces each of its values in finitely many steps. Continuum mathematics stays sound as analysis, a description of

many steps at once. Countable is not enough for physics, since the rationals are countable and dense with no next point; discrete physics needs steps with successors. A grid also owes its geometry: on a square grid the shortest route to a diagonal point stays longer than the straight line however fine the grid becomes.

1. The question that calculation alone does not answer

Zeno’s paradoxes come in two kinds, and only one decides whether physics can be continuous. The paradox of measure asks how points without width add up to a length (Skyrms, 1983). Measure theory, the mathematics of length, area and probability, answers it, and discreteness (being made of separate points or steps) does not help, since discrete points have no width either. The paradoxes of motion ask how anything reaches its next place. Analysis, the mathematics of limits, describes the run, but nothing in a continuum (a gapless expanse, like the line of real numbers) can produce the next step, because there is none.

A continuum cannot be physical, because nothing in a continuum can cause anything. Causality takes steps: a cause produces its effect, and producing an effect takes a step from one state to the next. A continuum has no next state, since between any two

points lies a third, so in a physical continuum no state could be the first one a cause produces. Change without a cause is outside logic and reason: a law could at most describe it, nothing would produce it, and an account that allowed it could not say why it happens. Accepting it would repeat Plato’s late Athenian anti-philosophy, which set the Forms outside nature: it puts something real beyond reason, and outside logic is illogical. Physical events are therefore discrete steps. Alfred North Whitehead reached the same conclusion from Zeno’s paradoxes (Whitehead, 1978).

Continuum mathematics stays a sound tool of analysis, and its own workings agree. A line has length. Each individual point has zero length. Both statements hold, because collecting points into a set is a different operation from adding up lengths: the points do not cause the length. Resolving that mathematical question leaves a physical one: what structure accounts for the distances we measure?

The apparent contradiction: if every constituent contributes zero width, how can the whole have nonzero width? A familiar rule gives the question its force. Split a rod into two pieces and add their lengths; repeat the split finitely often and the rule still holds. The mistake is to extend that rule to an arbitrary uncountable

decomposition, into more pieces than any endless list could contain, without checking what the sum means there.

Take two endpoints labeled 0 and 1. Neither endpoint has width, yet their distance is one unit. A continuous line specifies the distances among all its points; it does not assemble length by stacking point-widths. A discrete network can likewise use sites without width and give length to the links between them, and it has no other option: a site with a width would have a surface, and a surface is a continuum. Both accounts need distance relations.

1.1 The position advanced here

Physical events are connected by discrete relations, because a law produces each state from its immediate predecessor, and a dense continuum, with another point between any two, has no such predecessor-successor step. Continuous differential equations, which relate quantities to their rates of change, do not assume that kind of production; the physical case rests on it.

The premise is causality in its strong sense, production. A law can relate states at different times without any state producing the next; that is causation as dependence, and production is the stronger notion this paper relies on (Hall, 2004). Hall lets production run through continuous intermediates; the claim here

is that it cannot, since a production with no first effect produces nothing. A differential equation such as dx/dt = u(t) is a law of that first kind: it says how each state depends on earlier inputs and is silent on what produces any state. In a continuum, any instant after a cause has another instant before it, so no state is the first one the cause produces, and the chain of production cannot begin. Continuum mathematics remains the tool for describing many steps at once. A reader who takes dependence to be all there is to causation finds no contradiction in a continuum, and for that reader the physics that follows carries the case.

Physics gives independent support. Gravity and quantum theory together limit how finely anything can be resolved: probing a distance near the Planck length, about 10^-35 meters, takes so much energy in so small a space that it would form a black hole and hide what the probe was meant to show (Hossenfelder, 2013). A black hole’s entropy, a measure of its number of possible internal states, is a quarter of its horizon area (the area of the boundary past which nothing escapes) in Planck units (Bekenstein, 1973; Hawking, 1975), and the covariant entropy bound that follows caps the entropy on the light rays leaving any surface without spreading apart, its light-sheets, by the surface’s area, which for an ordinary region caps the distinguishable states

it can hold (Bousso, 2002). A finite region therefore holds finitely many bits, if the bound holds in the fundamental theory, as is widely expected. A system with finite energy also passes through only finitely many distinguishable states in a finite time, since reaching a distinguishable state takes a minimum time set by the energy (Margolus and Levitin, 1998). That is not yet a lattice, a regular grid of points. A system with finitely many distinguishable states is still described with continuous amplitudes (numbers whose squared sizes give probabilities), and those amplitudes are analysis: a history produces each of its values in finitely many steps, so the values that occur form a countable set, one that can be listed in sequence (Section 4.1). The bound rules out a region carrying infinitely much information, which a fundamental continuum of physical values would allow.

Continuum mathematics can represent physical quantities without every coordinate in that representation being a physical location. Physical extension (taking up space) rests on discrete relations, and smooth geometry describes their collective behavior.

The same holds for space itself, on either view of what space is. If space is a thing that bodies move through, each move is a cause producing the body’s next place, and a continuum has no

next place to produce. If space is only the order of coexisting things and events, as Leibniz held, it is as discrete as the events that generate it. Gravity, spacetime and matter fall under the same rule, since each changes by causes. What happens is the transitions, rules applied to rules; the relations exist as mathematics, and without their change nothing could be observed.

Two observations carry the case: a continuum has no next point, and adding its singleton widths does not yield its length. Both are consistent with continuum mathematics as a tool. Causality turns the first into a physical conclusion, since what has no next state cannot produce one. The bearer discussion below takes up the second, the requirement that distance have a physical account.

The paper’s argument is the fifth link in a larger chain, mathematical monism:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, continua and discrete structures alike, and what exists physically is what logical steps in mathematics produce: the physical question is

which of them happen. Mathematics cannot fail to exist, so there is something rather than nothing.

3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.

4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.

5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.

6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers

backwards from the end.

8. With no distance shorter than a step, the infinities a smooth spacetime puts into quantum gravity never arise, so string theory, built to tame them, is not needed, and smooth general relativity is the large-scale behavior of the steps. A theory of everything is a short rule for transitions.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

1.2 The bearer proposal

Whatever carries the extension we measure owes us a physical account. In the usual construction of Lebesgue measure on the real line, the standard way of assigning lengths to sets of points, interval lengths supply the starting size data. A singleton, a set with one point, has measure zero because it fits inside intervals of arbitrarily small length.

The physical proposal is to account for distances through discrete relations and their dynamics. A grain with positive extent would bring the continuum back inside itself, since an extent has a surface and a surface is a continuum. Elementary sites therefore have zero extent, and their separations must come from

relations. Discreteness alone supplies no unit of length.

A defender of continuous space can make the same relational move: points have no individual width, while a metric assigns distances between them. The interval and its metric structure are a candidate bearer of extension. “Nothing bears extension” does not follow from “every singleton has zero length.”

A stronger premise would require every extended physical region to decompose into elementary parts whose nonnegative individual extents sum to the region’s extent. Together with a decomposition into zero-extent points, that premise contradicts positive extension. The inference is valid conditional on the premise, which imposes precisely the additive composition that the continuum model does not assert. The composition argument alone does not decide between the two accounts; causality decides, and the discrete account then owes a construction: specify elementary relations, derive measured geometry, and recover quantum and gravitational predictions from the same dynamics.

2. A union is not a sum of lengths

Standard measure theory handles zero-width points consistently, and its definitions show how. Stated precisely, take the usual real line with Lebesgue measure, written λ. Then

The first equality says which points belong to the interval. It supplies no rule equating the measure of that union (all the points gathered into one set) with an uncountable sum of singleton measures.

Adding the point lengths gives a well defined answer, zero. In symbols, for any family of nonnegative numbers the generalized sum is the supremum, the least upper bound, of all finite subsums:

The measure of this union is not the sum of those measures. Lebesgue measure is countably additive: for a sequence of disjoint (non-overlapping) measurable sets, the measure of the union equals the sum. The singletons of a real interval cannot be placed in such a sequence. John Hunter’s lecture notes give the standard measure-theoretic treatment (Hunter, 2011).

Singleton values also fail to determine the measure. Double Lebesgue measure assigns zero to every singleton and two to

the whole interval. List the points with their individual measures and you have still not said which of those two measures you have.

2.1 Ordered, but with no next point

No real position has a next one. In symbols, for any real positions $a<b$, the midpoint $c=(a+b)/2$ satisfies $a<c<b$, so nothing sits immediately after a. The rational numbers also have a rational midpoint between any two distinct members, despite being countable. Dense order and uncountability are different properties.

A step-by-step process can nonetheless produce a dense order. Run through the integer ratios and every positive rational appears at a finite stage. Each output has a next output in that process, although no rational has an immediate successor in numerical order. Motion can work the same way. A body at x is next at x + 3, or x + 3.25, as the result of one step, with no place in between visited, so the places a body can occupy may be dense in numerical order while every step it takes has a successor. No sequence can enumerate all real numbers, so a physical process confined to countably many successive outputs cannot put out every real position as a separate output. That confinement is an

additional premise. Numerical density alone does not deliver it.

The continuum model does not claim to step from one real number to the next. A continuous trajectory is not defined by advancing to each real number's immediate successor: $x(t)=vt$ assigns a position to each time without requiring a next instant, and a block account, which takes a whole history as given at once, need not enumerate its positions. The discrete account of what exists, defended here, earns its place through its physical structure and consequences.

2.2 Can motion cross a continuum?

Inside the continuum model, motion crosses an interval with no arithmetic trouble. In symbols, with the speed set to one, let $x(t)=t$ for $0<=t<=1$. Every intermediate time has a corresponding position. For every nonzero time increment, $\Delta x/\Delta t=1$; the limit yields the same velocity with no division of zero by zero.

No step of this division produces a zero to add. Cut the duration into successive intervals of lengths 1/2, 1/4, 1/8 and so on. The first n lengths sum to $1-2^{(-n)}$, and their series (the sum carried on without end) sums to one. These are positive durations, not zeros. The continuum model therefore represents motion consistently. It does not establish that nature instantiates the

model.

Bertrand Russell met the demand for a next place by holding that motion is nothing more than being at different places at different times (Russell, 1903). That pairs each time with a place, and the pairing would exist as mathematics whether or not anything moved, as a frozen record does. It describes the result of motion and leaves out what produces it: these are the emperor's new points. Zeno's runner puts the causal question (Aristotle, 1984, *Physics* VI.9). Before the finish he must reach the halfway mark, and before that the quarter mark, so no first stretch exists for him to run. Analysis solves the paradox in the abstract: x(t)=t assigns every position without a first stretch, and the stretches' times add up to one. Production cannot proceed that way, since a cause must produce a first effect and the regress leaves none. Whitehead, arguing by Zeno's method, put it this way: once anything becomes, there is "no continuity of becoming" (Whitehead, 1978, p. 35). Steps solve the paradox in the only way causality and spacetime can happen: each step produces the next, and finitely many reach the finish.

Zeno aimed his other two paradoxes at steps. In the arrow, at each indivisible now the arrow fills a space equal to itself and so is at rest, and if time is made of nows, it does not move at all. In

the stadium, two rows passing each other at one unit per smallest time pass two units of each other in that time, so passing one would take half the smallest time (Aristotle, 1984, *Physics* VI.9, 239b5-240a18). Both assume that motion happens inside a moment or passes through every place in between. A step does neither. Motion is the step itself, one state producing the next, and a body at x is next at x + 3 with no place in between visited, so no half step is owed. A snapshot holds no motion, as a frame of film holds none; the step between snapshots does. Aristotle saw where steps lead, that across a smallest part a body would have moved without ever being in the middle of moving (*Physics* VI.1, 232a), and he took it for an absurdity. It is the answer.

Two more facts finish the measure side. A countable set of points has total length zero however densely it fills an interval, so refining a grid adds no length. Discrete weights on finitely many points can still reproduce a continuum’s averages within a stated bound for observables (measurable quantities) that change no faster than a fixed rate, which is how a discrete structure can underlie smooth predictions without any point carrying width.

3. Specification, construction and physical realization

A staged process cannot deliver uncountably many events. Suppose every elementary spacetime event arrives at one of a succession of stages indexed by the natural numbers, with finitely many new events at each stage. The events so generated are then at most countable, and if they exhaust spacetime's physical constituents, that premise rules out an uncountable continuum of elementary events. The argument rests on that premise. It derives no minimum spacing and excludes no continuous mathematical representation.

This countability premise is stronger than the starting-input requirement that a history generated one event at a time needs a first event. A beginning alone does not establish natural-number stages or finitely many events per stage: a continuum model may also have an initial boundary.

"Construct the interval" can mean several things. In classical mathematics, a definition can pin down a set without listing its members. A sequential enumeration cannot exhaust the reals, though inability to enumerate an object is not a contradiction in its definition. Constructive mathematics asks which objects and

operations an explicit construction can deliver; identifying it with finitely many points would erase the constructive treatment of real numbers.

A circle specified by x^2+y^2=r^2 is exact within Euclidean geometry. A manufactured circular object must be assessed at some tolerance. No workshop reaches infinite precision, and that limit neither refutes the equation nor proves that the object inhabits discrete space.

A block or a sphere is continuous whatever carries it, since its surface is a continuum, so discreteness supplies no sizes or shapes; relations do. A block can still exist as a rule without being built. By the Minkowski-Weyl theorem, a bounded convex polytope, a solid with flat faces and no dents, is both the convex hull of finitely many points (the smallest dent-free solid containing them) and the set satisfying finitely many linear inequalities, each confining points to one side of a plane (Ziegler, 1995, Theorem 1.1). A block’s eight corners, or its six face inequalities, fix every point of its surface without listing one. Whether a step from x to x’ enters the block is then decided from the two endpoints and the inequalities in finitely many operations, as in the line clipping of computer graphics, which trims lines at a region’s boundary (Cyrus and Beck, 1978). The surface acts only through steps: a

step that would cross it produces an event at a definite point, such as the step’s rejection, and no point of the surface is built.

The same holds for curved boundaries, with one limit. Whether a point lies exactly on a perfect sphere can need every digit of its coordinates: given real numbers digit by digit, a strict inequality (a claim that one number is less than another) is confirmed at some finite stage when it holds, while equality is confirmed at none (Weihrauch, 2000). A rule that needed that decision could not finish a step. A rule that acts on definite values does not need it, so a continuous boundary acts at the precision the states carry, and its continuum stays in the description. A continuous field is a specification of the same kind. Its formula can fix how strongly, in which direction and with what amplitude each transition acts, while only the transitions occur, and computing it to any chosen precision takes finitely many steps each time, so no step needs every digit.

4. The physical question survives the mathematical answer

I reject fundamental continua and propose that continuum geometry represents very fine discrete relations. Smoothness belongs to an effective description, an approximation valid at

larger scales, while the underlying structure supplies the interactions from which measured distances emerge. A continuum calculation that works is adequate for the observations it covers. It does not show that physical extension divides without limit. Three claims must be separated: a continuum is mathematically specified; a continuum model is empirically successful in a regime (a range of conditions); a continuum is fundamental in nature. Each requires its own support.

4.1 Measure, geometry and the existing debate

The existing debate already separates the mathematical resolution from the physical question. Brian Skyrms's "Zeno's Paradox of Measure" (1983) explicitly identifies Adolf Grünbaum's work on the standard resolution using measure theory. He then asks which further physical premises would carry us from that mathematical resolution to a claim about what space is.

Baruch Spinoza, in *Ethics* I.15, rejects composing a line from points, a surface from lines or a solid from surfaces. His conclusion is not discrete space: he defends extended substance as infinite, one and indivisible. The historical lesson: rejecting a sum of zero widths does not, by itself, choose between discrete

relations and a continuous whole.

Measure theory also marks where analysis outruns anything a process of steps can do. The axiom of choice lets one member be picked from each of any collection of nonempty sets, with no rule for picking. With it, some sets of points have no volume, and a ball can be cut into five pieces and reassembled, by rotations and shifts, into two balls each the size of the first (Banach and Tarski, 1924). A process of steps makes finitely many picks per step, so an endless run makes countably many, one after another, which is the axiom of dependent choice. With only that, it is consistent, given an inaccessible cardinal, that every set of reals is Lebesgue measurable (Solovay, 1970). Choice and its negation are each consistent with the other axioms (Gödel, 1938; Cohen, 1963), so both describe structures. The doubled ball exists as analysis, and no sequence of steps can perform it.

A finer grid can preserve the wrong distance. On a square lattice of spacing h with only horizontal and vertical steps, the shortest path from (0,0) to (a,a) has length 2a when a/h is an integer. The Euclidean distance (the straight-line distance) is sqrt(2)*a. Their ratio stays sqrt(2) at every permitted spacing.

Hermann Weyl's tile argument asks whether counting elementary steps recovers geometry at larger scales. Tobias Fritz (2013) makes a related difficulty precise for a class of periodic graphs (networks built from one repeating pattern): the attainable velocity set is a polytope with distinguished directions.

Causal set theory (spacetime as discrete events ordered by cause and effect), developed by Rafael Sorkin and others (Sorkin, 2003), pursues a different structure. Its continuum approximation combines causal order with a count related to volume. Poisson sprinkling, scattering points at random with a fixed average density, avoids the regular lattice's built-in preferred directions (Surya, 2019). Other discrete proposals make different commitments: Edward Fredkin (2003) advocates digital mechanics (the universe as a discrete computing process), Gerard 't Hooft (2016) investigates a cellular automaton (a grid of cells updated in steps by local rules) interpretation of quantum mechanics, Stephen Wolfram (2020) proposes rewriting rules on hypergraphs (networks whose links can join more than two points), and Carlo Rovelli and Lee Smolin (1995) derive discrete area and volume spectra, the possible measured values of area and volume. A discrete spectrum of geometric operators is not simply a lattice of physical locations, and Sabine Hossenfelder

(2013) reviews the many forms a minimal length can take in quantum gravity, the sought theory uniting quantum physics and gravity. These programs can meet: a rewriting rule’s causal graph, the record of which update depends on which, is itself a causal order, so one short rewriting rule whose causal graph matches a random causal set at large scales would carry both the computing proposals and causal-set theory. The search can also start from the known laws, written as a program of discrete steps and shortened while its outputs still match.

A discreteness claim must state its level. A spatial lattice can carry real-valued fields and evolve in continuous time. A finite-dimensional quantum system can have a discrete measurement spectrum while retaining continuously many pure states, states as fully specified as quantum theory allows. Written out for a qubit, a two-level quantum system, $|\psi\rangle=\cos(\theta/2)|0\rangle+\exp(i\varphi)\sin(\theta/2)|1\rangle$, with continuously variable θ and φ, although measurement has only two outcomes. Two possible outcomes do not mean two possible states. The continuum of θ and φ is analysis, the description of every state a rule could reach. A history reaches its states one step at a time, so the values of θ and φ that occur form a countable set, and a finite set of quantum gates (basic operations of a quantum

computer) comes arbitrarily close to every state of the continuum (Dawson and Nielsen, 2006).

4.2 What this paper contributes

This paper compares the two accounts under three obligations. First, distinguish the measure of a set from sums of singleton measures. Second, state which observables an effective representation reproduces, exactly or within a bound. Third, test the geometry and dynamics claimed for a physical realization, including failures that persist as a mesh is refined.

5. What would make discreteness a physical explanation?

A finite grid can approximate continuum measurements without proving that space is a grid. A physical alternative must specify its events and relations, show how measured distance and direction arise from them, and recover the predictions for which continuous physics already works. It must also reproduce quantum interference: for two contributions of equal magnitude but opposite phase, their amplitudes cancel before probability is calculated.

A recovery example: waves on a spatial lattice

A lattice of sites reproduces continuum wave behavior at long wavelengths, within a stated error. Richard Feynman’s lectures reach the continuum wave equation for an electron the same way, from amplitudes on a line of atoms (Feynman, Leighton and Sands, 1965, Ch. 16). Stated precisely, let u_j(t) be a displacement at site j of a one-dimensional lattice with spacing h, and take the nearest-neighbor dynamics (each site interacting only with the sites beside it)

u_j’’=(c^2/h^2)(u_(j+1)-2u_j+u_(j-1)).

Substituting u_j(t)=exp[i(kjh-ωt)] gives

ω^2=(4c^2/h^2)sin^2(kh/2).

At long wavelengths the lattice relation reduces to the continuum one, with a correction set by the spacing. Written out, for |kh|<<1 it becomes ω^2=c^2k^2-c^2h^2k^4/12+O(h^4k^6). The leading term is the dispersion relation of the continuum wave equation, the link between a wave’s frequency and its wavelength. A wave on the lattice therefore travels at slightly the wrong speed, and the error has a stated bound. In symbols, for 0<|kh|<=1 the relative phase-speed error obeys

$0 \le 1-\omega/(c|k|) \le (kh)^2/24$.

Shrink h at fixed physical wavelength and this observable improves. Over a time T, the phase discrepancy is at most $c|k|T(kh)^2/24$ in this range; the agreement is not uniform over arbitrarily long times. Ben Schweizer and Florian Theil (2018) develop lattice-wave limits and long-time corrections.

In two dimensions the wave observable loses its direction dependence where the path metric keeps it. For the square-lattice analogue, $\omega^2=(4c^2/h^2)[\sin^2(k_xh/2)+\sin^2(k_yh/2)]$. Direction dependence remains at finite spacing but vanishes in the long-wavelength limit. This differs from the persistent sqrt(2) error of the shortest-path metric in Section 4.1 because the observable and dynamics are different.

This is a control calculation, not a new theory. The sites are discrete; the amplitudes and time parameter remain continuous here as analysis, describing many steps at once. It earns its place by being concrete: a declared dynamics yields a stated continuum observable with an error estimate.

6. Implications for mathematical monism

For mathematical monism, the view that reality is mathematical structure alone, the example sharpens what “structure” must include. A list of points together with their individual widths omits distances, measure, dynamics and observation relations. Complete structure cannot mean a pile of isolated constituents. Both discrete and continuous structures are mathematical; causality selects the discrete one for physical events.

Observer recovery

A complete candidate must grow observers like us. The universe starts simple, as observation shows, and on a measure where each bit of description halves the weight the probable worlds start simple too; minds are complex, so a history containing them must grow their organization from a simpler state. A short rule can generate an elaborate observer without becoming a long rule; output size and runtime are distinct from description length. Growth is a sequence of causal steps, one more reason the steps must have successors.

Conclusion

Everything that happens is discrete, matter, gravity, spacetime and space itself included, because causality takes steps and a continuum has none: between any two points lies a third, so no next state waits to be produced, and refining a grid supplies none. Change without a cause would be outside logic and reason, and an account that allowed it could no longer say why anything happens. The mathematics is not where the trouble lies. Zero point-lengths leave standard continuum measure theory intact, because a union of uncountably many points is not a countable sum of their lengths: the points do not cause the length, and continuum mathematics, amplitudes included, stays the analysis that describes many steps at once. Discreteness gives nothing a size: an element with a size would have a surface, and a surface is a continuum, so in every structure sizes come from relations among zero-width points. Of Zeno’s two kinds of paradox, measure belongs to mathematics, which answers it, and motion belongs to physics, which needs steps.

The examples show why matching one average or shrinking a grid is not enough to recover all geometry. A discrete account succeeds by deriving the distances and dynamics we measure

from steps that have successors.

A39a. Short Laws, Discrete Steps, Forward Time: Why Gravity and Space Are Causal Transitions

Abstract

Everything that exists is mathematics, which answers three questions: why the laws of physics are short, why causes take discrete steps, and why time runs forward. Whatever exists is some definite way, each definite way is ruled or random, and both are mathematics. Anything that acts on the world, a soul, a designer or a simulator, acts by definite relations, so it is mathematics too. No probability distribution treats infinitely many laws alike, and the weighting that needs no extra numbers flips a fair coin for each bit and for whether to add another. Each bit halves a law’s weight, and unless longer laws grow observers more often, the shortest laws that can grow them carry at least half the weight. The known laws fit on a page, as this predicts. A differential equation encodes dependence among states. Production takes a state to its next, and a continuum has no next state, so production needs discrete steps. Physics agrees: gravity and quantum theory suggest a finite region holds finitely many

distinguishable states, reached at a finite rate, and the order of the steps, counted, gives spacetime its geometry. Short laws mean a simple start, and minds are complex, so the universe has to grow them in causal steps. An endless chain of steps has no last step to run back from, a finite reversal pays extra bits to name its turning step, about 200 for a typical one, and minds merge many states into one, so they run one way.

1. Four questions, one chain

Eugene Wigner called the usefulness of mathematics in physics unreasonable (Wigner, 1960). On the view taken here it is expected. Everything is ruled or random, both are mathematics, and whatever acts on the world acts by rules, so physics has nothing else to describe.

Three more questions then have answers a reader can check, and the answers form a chain. The laws of physics are short because each bit a description adds halves its probability. Short laws mean a simple start. Minds like ours are complex, so a simple start has to grow them, and growth takes causal steps, each state produced from the one before. The steps are discrete because a cause has to produce a next state, and a continuum has no next state. Alfred North Whitehead reached the same

conclusion from Zeno’s paradoxes (Whitehead, 1978). Time is the sequence of steps, and it runs forward, from a simple beginning toward a complex present. Light, which crosses space while no time passes for it, is no exception: a moving body’s transitions run slower, and light spends every step moving, so nothing within it could move relative to the rest. An endless chain of steps has no last step to run back from, a reversal has to pay to name its turning step, and minds merge many states into one, so they run one way.

Each check takes a few lines. One concerns probability on a countable set, a set whose members can be listed: no probability distribution treats infinitely many laws alike, and a fair coin, flipped for each bit and for whether to add another, makes long laws lose. Another concerns continua, gapless sets such as the real line: a continuum has no next state, so causes take steps, an a priori proof in the logician’s sense. The last is a count: fewer than 2^m programs are shorter than m bits, so most of N steps cost about 2 N bits to name.

2. Why everything is mathematics

Whatever exists is some definite way, or nothing would tell it apart from nothing. Each definite way is ruled or random. Either a rule shorter than the structure generates it, or nothing shorter does. The length of the shortest program that prints a string is its Kolmogorov complexity, and a string that no shorter program prints is random in the precise sense of that theory (Li and Vitányi, 2019). Rules are mathematics. A random string is mathematics too, with definite entries in definite places. Every finite string falls on one side or the other, whether or not any proof can say which (Chaitin, 1974). No third kind remains for a nonmathematical thing to be.

Interaction gives the same answer from the other side. In 1643 Princess Elisabeth of Bohemia asked René Descartes how a soul that only thinks could move a body (Shapiro, 2007). Any answer has to say how the soul acts: what it changes, under which conditions and by how much. Those when-then relations are rules, and rules are mathematics. Whatever acts on the world, a soul, a designer or a simulator, acts by such rules, so it belongs to one mathematical structure with the world it acts on. A thing that acts on nothing explains nothing here.

The last refuge is an intrinsic nature behind the relations, what a thing is in itself. Swap the hidden natures behind mass and charge while every relation stays fixed, and nothing anyone could notice, think or say would change (Lewis, 2009). A nature that makes a difference takes part in relations, so it is structure. A nature that makes none does no work, and nobody could know it.

Wigner’s puzzle then dissolves. Mathematics describes the world well because the world is mathematics (Tegmark, 2008). What remains is which mathematics an observer should expect to find.

3. No fair lottery among infinitely many laws

Write every candidate law of nature as a finite string of 0s and 1s: its complete description, with equations, constants and starting state together. There are infinitely many such strings, and they can be listed, shortest first. A natural wish is to give them all the same chance, and it cannot be granted. A probability distribution assigns each outcome a chance, and under the standard axioms the chances of all the outcomes, countably many here, add up to one. If each string had the same probability p, the sum over infinitely many would be infinite when p is positive and zero when p is zero. Any distribution has to fall as descriptions grow.

The bits of a shortest description say how fast. A shortest description cannot be compressed, or the compressed version would be shorter still, so no rule inside it predicts one bit from the others, and it is random in the sense of Section 2. With no rule favoring 0 or 1, each bit of a shortest description is a fair coin. A biased coin would carry its bias as a number, and that number would have to be written into the description too.

Each added bit therefore halves a description’s weight: a law of L bits weighs 2^-L. Descriptions that end on their own, none the beginning of another, can take these weights together, since by Kraft’s inequality their total is at most one (Li and Vitányi, 2019). Ray Solomonoff used the same weights to predict data (Solomonoff, 1964). Read as weights on what exists, they say which laws an observer should expect to find.

Counting gives the opposite verdict, and the coin shows why counting misleads. There are 2^n strings of length n, twice as many with each added bit. A catalog of possible laws is therefore dominated by long ones: endless special cases, arbitrary constants, exceptions written out over libraries. Whether a description goes on at all is a second yes-or-no that no rule decides, so it is a fair coin as well. Half of the descriptions still going stop at every step. All descriptions of length n together then

weigh 2^-(n+1), and everything longer than n, added up, weighs exactly as much as length n alone: the tail equals the peak.

In this picture one particular description loses more than half its weight per bit, since each bit also pays for its stop-or-go flip. A code that states the length once, in a short header, loses only about half (Li and Vitányi, 2019). Either way, long laws lose.

The known laws, the Standard Model of particle physics with general relativity and their constants, fit on a page. If the count of strings decided, we should expect laws of endless special cases. We find laws that fit on a page. Two explanations remain. Either the weight halves with each bit, or something chose our laws: a designer, or a civilization running our universe as a simulation (Bostrom, 2003). A chooser moves the question up one level. Its own world needs laws, those laws face the same weighting, and the account is postponed.

4. Short laws must grow their observers

The very shortest laws produce nothing worth observing, and we are here. The laws to expect are the shortest that can produce observers: structures that record and model their surroundings. Added rules more often break the conditions for observers than

create them, as small changes to the constants of our physics often remove stable nuclei, long-lived stars or chemistry (Barnes, 2012).

Under the coin-flip weights, suppose no longer length holds a larger share of observer-producing descriptions than the shortest length that works. Then that length carries at least half of all the observer-producing weight, since everything beyond it weighs no more than it does. A mind assembled by a chance fluctuation, a Boltzmann brain, needs every one of its bits supplied by chance, so minds grown by short rules outweigh it.

Short laws therefore mean a simple start: the laws and the starting state together hold little information. Minds are complex. A simple start has to grow them, and simple rules can. John Conway’s Game of Life is a grid of cells that live or die by counting their live neighbors. In it, gliders travel across the grid, guns fire streams of gliders, and patterns built from them compute anything a computer can (Berlekamp, Conway and Guy, 1982). Rule 110, a row of cells each updated from itself and its two neighbors by a table of eight cases, is universal as well (Cook, 2004). A rule that fits on a line can build complexity without limit, one step at a time, each state produced from the one before. Producing is one kind of cause.

5. Two kinds of cause

Causes come in two kinds, as Ned Hall showed (Hall, 2004). Dependence relates states: change an earlier one and a later one changes. Production brings a state about. A plant that dies because nobody watered it depends on the omission, yet not watering produced nothing; the plant’s own chemistry, running without water, produced the death. Each frame of a film fits the physics of the frame before, yet no frame produces the next. The projector does.

A differential equation, an equation relating a quantity to its own rates of change, encodes dependence. Under x = vt every position depends on the start and the speed. Bertrand Russell answered the demand for a next place by holding that motion is only being at different places at different times (Russell, 1903). Positions at every instant with nothing producing the next are the emperor’s new points. Dependence of this kind sits comfortably in a continuum. Production does not, because a continuum has no next state: between any two instants lies a third.

Read a continuous law as a rule for producing states and it becomes a flow. A flow is a family of maps t, one for every

positive time t, each taking a state to the state a time t later, with s+t = s t: one map applied after the other. Every transition the flow defines is two smaller ones composed: t = t/2 t/2. A flow has no smallest transition, so producing any later state would take infinitely many productions before the first.

Zeno’s runner meets the same regress: he must reach the halfway mark before the finish and the quarter mark before that, so no first stretch exists for him to run (Aristotle, 1984, Physics VI.9). Read the other way, the flow hands out every later state at once. It pairs times with states, as Russell’s answer does, and the pairing exists whether or not anything happens, like a frozen record. Whitehead, arguing by Zeno’s method, put it this way: once anything becomes, there is “no continuity of becoming” (Whitehead, 1978, p. 35).

A discrete rule has what the flow lacks. A map f that takes each state to the next, x n+1 = f(x n), has a first application and a next state after every state. We change: our states follow one another. In a world that is mathematics, rules take each state to the next one, and a rule that does so needs a next state, which only steps supply. A change that nothing produces would be a change without a cause, and nothing in an account that allowed it could say why it happens. Stated precisely, a process that adds finitely

or countably many events at each natural-number stage produces at most countably many events, and no uncountable continuum of them. Nothing here depends on our universe. Wherever anything happens, it happens in steps.

6. What physics adds

A reader who takes dependence to be all of causation finds no contradiction in a continuum, and for that reader the case rests on physics. Two results point the same way.

First, gravity and quantum theory cap the states a region can hold. A black hole’s entropy, read as a count of its states, is a quarter of its horizon’s area measured in squares of the Planck length, about 1.6 x 10^-35 meters (Bekenstein, 1973; Hawking, 1975). No region can hold more states than the black hole it would become: if it could, collapsing it would lower the entropy, which the second law of thermodynamics forbids. A region’s capacity is therefore capped by the area of its boundary (‘t Hooft, 1993; Susskind, 1995; Bousso, 2002). A ball one centimeter in radius holds at most about 10^66 bits.

The bound is proved only in limited settings, such as free fields in weak gravity (Bousso, Casini, Fisher and Maldacena, 2014), and

most researchers in quantum gravity expect it to hold in general. If it holds, a finite region has finitely many distinguishable states (Bao, Carroll and Singh, 2017).

Second, change has a speed limit. A system with average energy E above its lowest state needs at least a time h/(4E) to reach a state it can be told apart from, where h is Planck’s constant (Margolus and Levitin, 1998). In a finite time it passes through finitely many distinguishable states. Finite states reached at a finite rate leave nothing observable that needs a continuum beneath it. Quantum amplitudes, the complex numbers that give the odds of outcomes, form a continuum too. A computation reaches any amplitude in finitely many steps, to whatever precision a measurement can check, so amplitudes too describe what the steps produce.

Einstein’s equations are written smooth, and that does not make spacetime smooth. A formula can say how hard each step pushes, and which way, for any input. Numerical relativity, the computer solution of Einstein’s equations, runs them one step at a time to any precision wanted. It predicted the gravitational waves of merging black holes that detectors later recorded (Pretorius, 2005). The formula is a rule for the steps.

Light moves at the base rate, one step of distance per step of time. In the Game of Life the same limit is one cell per generation, and a glider moves at a quarter of it. Iwo Bialynicki-Birula wrote Maxwell's equations, the laws of light, as a quantum cellular automaton: a grid of cells, each updated from its neighbors at every tick. At long wavelengths it recovers the continuous equations (Bialynicki-Birula, 1994). Light also acts on matter in whole quanta: each photon is absorbed whole, in one event.

Light seems to cut time loose from the steps, since it crosses space step by step while no time passes for it. Its proper time, the time a clock carried along would record, is zero because along a light path every step goes into moving, and nothing changes in place. A slower body shares the base rate between moving and changing in place, so a moving clock runs slow. The shares add like the sides of a right triangle, (c\,dt)^2 = dx^2 + (c\,d)^2, with the moving clock's own time: the interval of special relativity, restated rather than derived. A thing's time is its transitions in place, and within light nothing could move relative to the rest, so light is the limit of a moving clock running slow, not an exception to time being the steps.

The steps carry gravity’s geometry with them. Take a spacetime without paths that loop into their own past. There the order of which events can influence which fixes the topology, how its regions connect, the smooth structure and the geometry up to a local scale (Hawking, King and McCarthy, 1976; Malament, 1977). Counting supplies the scale. Causal set theory builds spacetime from exactly these two ingredients, an order and a count (Bombelli, Lee, Meyer and Sorkin, 1987; Surya, 2019). Gravity, the curving of spacetime, is then the order of the steps, counted.

The count made a prediction. A random count fluctuates by about its square root, and Rafael Sorkin argued that the fluctuation leaves a cosmological constant, the energy of empty space, of about 10^-120 in Planck units (Sorkin, 1997). Supernova surveys found a constant of that order the next year (Riess et al., 1998; Perlmutter et al., 1999), where quantum field theory on a continuum overshoots by about 120 orders of magnitude.

Discreteness need not mean a grid. On a square grid the shortest route to a diagonal point stays longer than the straight line, however fine the grid becomes. Hermann Weyl pressed this obstruction (Weyl, 1949), and Tobias Fritz generalized it to repeating networks (Fritz, 2013). Observed Lorentz symmetry,

the sameness of physics for every observer moving at a steady speed, rules out a regular lattice at the base. A random scattering of events, as in causal sets, picks out no direction (Bombelli, Henson and Sorkin, 2009). A rule that rewrites relations step by step leaves a causal order behind, the record of which update depended on which, so the search for the final rule can run among short rewriting rules, shortest first, keeping those whose records, at large scales, scatter like a causal set. It can also start from the known laws, written as a program of steps, cleared of their infinities and shortened while its outputs still match.

7. What stays continuous

Calculus stays. It describes many steps at once, as the gas laws describe a crowd of molecules without tracking any one. A Riemann sum adds value times width over small cells, and its limit describes a crowd of steps; the limit is the description, and the steps are what happen. Discreteness gives nothing a size. A discrete point has no width either, so in both pictures length comes from the relations among points. A grain with a width would bring the continuum back as its surface. What discreteness supplies is a next step, the one thing causation needs. Choice goes the same way. Analysis may pick a member from each of

uncountably many sets with no rule for picking, the axiom of choice, and it then has sets of points with no volume and the doubling of a ball (Banach and Tarski, 1924). A history makes finitely many picks per step, countably many in all, and with only such choices every set of reals can be measurable (Solovay, 1970). The doubling exists as mathematics, and no history performs it.

On the view taken here, every consistent structure exists as mathematics, continua included (Tegmark, 2008). What happens, in any of them, comes in steps, and the rules that take each step are the only things doing the work: rules applied to rules, with nothing outside the mathematics applying them.

8. Why time runs forward

Time is the sequence of causal steps. Short laws give a simple start, the start has to grow its observers, and growth runs from simple to complex. Gravity ties that simple start to the second law of thermodynamics, the rise of entropy. Without gravity an evenly spread gas has the most entropy. Under gravity, clumping raises it, so a smooth early universe, the simplest start to describe, is also a low-entropy one (Penrose, 2004). The early universe was even to about one part in 100,000 (Planck Collaboration, 2020).

Three facts keep the steps from running the other way. First, an endless chain of steps has a first step and no last one. Forward, state n+1 is produced from state n, beginning from a supplied first state, and a rule that is not told to stop goes on without end. Backward, state n would wait on state n+1, and with no last state the reversed chain has no first step that can run. The whole history has no end to run back from.

Second, a finite reversal has to name its turning step. A billiard break shows why. One stroke sends fifteen balls scattering, with sound through the room and warmth into the table. Reversing it takes every ball, air molecule and vibration in the table converging in perfect coordination, and a description written from the end has to list them all. A finished record costs about the same to describe in either direction: for a string h and its reversal h^R, K(h^R) <= K(h) + c for a constant c that does not depend on h. The difference lies in production.

The short way to produce the reversed history runs the forward one, stops at a chosen step and turns around, and the step has to be named. Fourteen billion years hold about 8 x 10^60 Planck times, each the time light takes to cross a Planck length, a number between 2^202 and 2^203. A few special steps, such as

those numbered by powers of two, have short names, though even a short name costs bits that the endless forward history does not pay. Most have none, since fewer than 2^m programs are shorter than m bits, so a typical turning step costs about 200 bits (Li and Vitányi, 2019). Under the coin-flip weights that costs a factor of 10^-60 or less.

Third, minds merge states. Adding 1 and 1 gives 2, and so does adding 0 and 2. A brain’s step takes many possible earlier states to one later state, and the difference leaves as heat (Landauer, 1961). Forward, a brain’s present state and input fix its next state. Backward, 2 = x + y has more than one solution, and nothing in the brain picks one. In a reversed history heat flowing in from the surroundings in perfect coordination would pick it, and a brain whose next state is set from outside is not producing its own thoughts. Memories are built by steps already taken, and a memory that need not be fine-tuned to what it records generically points the way entropy rises (Mlodinow and Brun, 2014). We remember the past and not the future.

9. Conclusion

Everything is ruled or random, and both are mathematics, so whatever acts on the world, a soul, a designer or a simulator, is mathematics too. The laws are short because each bit halves a description’s weight. No distribution can treat infinitely many laws alike, and the fair coin, flipped for each bit and for whether to add another, is the weighting that needs nothing written down beyond the law itself. Under it, unless longer laws grow observers more often, the shortest laws that can grow them carry at least half the weight. Everything those laws do happens in steps, because a cause produces a next state, and a continuum has no next state.

Physics agrees wherever it reaches: finitely many states in a region if the entropy bound holds, a finite rate of change, smooth equations serving as rules for steps, light at the base rate, and geometry from the order of the steps, counted. Gravity and space are those causal transitions, and time is their sequence, with transitions within moving frames running slower, down to none for anything at light speed, such as a photon. Time runs forward because a simple start must grow its observers in causal steps. The causal order is a priori; the simple start comes from the weighting. An endless chain has no last step to run back from, a

reversal pays extra bits to name its turning step, about 200 for a typical one, and minds merge many states into one. The continuum keeps its place in mathematics, as the description of many steps at once.

A40. The Hard Problem of Consciousness Assumes Its Conclusion: Why Experience Can Only Be More Structure

Abstract

The hard problem of consciousness assumes that experience is not structure, then declares that no structural account can explain it. The circularity becomes visible once the demand is stated plainly: explain experience without using structure, when every explanation, concept and bridge must itself be structured. Exposing it needs no experiment: it is an a priori proof in the logician's sense. This paper identifies the assumption, applies the interaction constraint (anything that affects structure must share structure with it) and the rule-random dichotomy (every determinate feature is either ruled or random, and both are mathematical) recursively to every proposed nonstructural alternative. Qualia theories, panpsychism, property dualism and idealism each face their own version of the same problem: specifying what their extra ingredient is, how it combines, and how it interacts with the structure it is supposed to supplement.

The structural account needs no supplement. The alternatives need structure to state their own claims. Experience is structure at work: a mind’s states producing one another, step by step, as it models its world.

1. The Demand That Defeats Itself

David Chalmers (1996) asks why physical processing is accompanied by experience. The question presupposes that processing and experience are two different things. If they are, a bridge is needed. Every proposed bridge must specify definite relations between what it connects, and definite relations are structure. The bridge is more math. Chalmers’s own bridge shows it: his proposed principles make the structure of experience mirror the structure of awareness (information directly available to guide behavior and report), and give any two systems with the same fine-grained functional organization (the pattern of causal interactions among parts) the same experiences (Chalmers, 1995). Each principle is a relation. What he adds beyond them is that experience is something besides the structure, and that is the separation in dispute.

This paper argues that the hard problem of consciousness, why physical processing should feel like anything, is malformed rather than unsolved. It assumes that experience is nonstructural, then treats the absence of a nonstructural explanation as evidence that something is missing. The missing thing is the assumption, not the explanation.

The argument has three parts. First, the circularity: defining experience as nonstructural disqualifies every structural answer before the investigation starts. Second, the interaction constraint: whatever consciousness is, it must interact with physical structure to cause reports, guide behavior and respond to stimuli, and interaction requires shared structure. Third, the recursive application, repeated at every level, of the rule-random dichotomy: any proposed nonstructural nature, when examined, turns out to be either ruled or random, and both are mathematical. At no stage does a nonstructural remainder appear.

The paper’s argument is the eighth link in a larger chain:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.

2. Every consistent structure exists as mathematics, since anything that let one exist and not another would be structure too, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.

3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.

4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.

5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.

6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. A mind is that growth at work, and its experience lives in its transitions. The hard problem of consciousness assumes its conclusion, that experience cannot be structure. The bare form of being a subject, with memories and traits set aside, is the same in every mind: one self.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. The Circular Structure of the Hard Problem of Consciousness

State the hard problem of consciousness as an argument:

First premise: experience has a character that is not identical to any structural property.

Second premise: a structural explanation can supply only structural properties.

Conclusion: no structural explanation can account for the character of experience.

The conclusion follows, but only because the first premise assumes it. That premise is the claim under dispute, not an established fact. Asking whether structure and experience are identical does not require assuming that they are not.

Joseph Levine's explanatory gap (Levine, 1983) may be real, in the sense that we cannot introspectively see why certain organizations feel the way they do. A gap in our view is not a gap in the thing. Temperature and mean molecular kinetic energy seemed like two different things until they were identified. Vitalism posited an extra ingredient for life until biochemistry closed the gap without one. The gap closed when two descriptions turned out to pick out one thing.

A critic who insists that no structural identification could close the consciousness gap has made a stronger claim: that this case is different in kind from temperature and life. That claim needs its own argument. The hard problem of consciousness does not supply one. It restates the assumption.

Those who charge tautology or a lack of evidence fail to understand that this is an a priori proof in the logician's sense:

nothing logical can lie outside the logic of mathematics, and that needs no evidence, any more than 2 + 2 = 4 does. The fundamental truth of reality is likewise not falsifiable by experiment, any more than a mathematical proof is. Like a theorem, it can be refuted only by a counterexample, here one coherent feature that is neither ruled nor random, and the dichotomy leaves no room for one.

Critics sometimes say the structural account defines the problem away. The charge fits the other side. The structural account answers the question it is asked: an organization feels because feeling is what that organization is, running, from the inside, as heat is the motion of molecules, seen at our scale. What it declines is a demand for an answer that uses no structure, which no answer of any kind could meet. The only step that settles the question by definition is the first premise, which simply asserts that experience is something other than structure.

3. The Interaction Constraint

If consciousness is not structure, it faces Elisabeth’s question. Princess Elisabeth of Bohemia posed it to René Descartes in 1643: how can an unextended mind (one that occupies no space) move an extended body? His pineal gland, a small gland deep in

the brain, answered where the interaction happens, not how. His fallback analogy to weight borrowed a cause he had himself denied was physical.

Put plainly, the question meets anyone who holds that consciousness is an ineffable dimension beyond mathematics and beyond words: how do words reach it? Speech is sound waves. They change what a listener feels, and what the listener feels moves the listener's mouth and sends sound waves back. Physics in, physics out. A consciousness outside the mathematics would need a place where the signal leaves the mathematics and a way for it to translate back.

The question generalizes. For consciousness to cause a report ("I see red"), it must change a physical state. For it to respond to a stimulus, a physical state must change it. These are causal relations. A causal relation requires definite terms: the thing causing, the thing affected, and the rule governing the connection. Definite terms with governing rules are structure.

A nonstructural consciousness that causes reports is a contradiction: it must have causal structure to cause anything, and causal structure is structure. A nonstructural consciousness that causes nothing is epiphenomenal: it plays no role in

producing the very reports that motivated the inquiry. The epiphenomenalist must then explain why a causally inert property tracks the physical processes it was supposed to supplement. The usual answer is a law that pairs each experience with physical features that selection did preserve, so the experience comes along with them. That law says which feature goes with which experience, and a rule that pairs one thing with another is structure.

The interaction constraint is not a separate premise that could be denied. It follows from what interaction means. Two things interact when one makes a determinate difference to the other. Making a determinate difference requires a common system of relations in which differences are defined. That system is structure.

4. The Recursive Argument

The rule-random dichotomy applies to any proposed nonstructural ingredient.

Start with a thing. It has properties; without properties, it is nothing. Each property is some definite way the thing is. Each definite way is either governed by a rule or not. If governed by a

rule, the rule is mathematical. If not governed by a rule, the variation is random. Both ruled and random features are mathematical (a rule is a function from states to states, and randomness is defined by probability theory and algorithmic information theory, which measures information by the shortest program that outputs it). The properties are mathematical.

Now examine what each property is. It relates the thing to other things or to possible states. Those relations have their own features. Each feature is either ruled or random. The recursion continues. At no stage does a nonmathematical remainder appear, because at every stage the same dichotomy applies. A proposed substance that is neither ruled nor random is not a third option. It is a refusal to say what the substance is.

Consider the proposed alternatives in turn.

Qualia as primitives. Property dualism adds phenomenal properties, the felt quality of red, to physical properties (Chalmers, 1996). What are these properties? If they are determinate (this red is not that blue), they stand in relations of sameness and difference. Relations of sameness and difference are structure. If they are not determinate, they add nothing specifiable. Calling them “primitive” locates them outside

explanation by fiat, precisely the assumption under examination.

If phenomenal properties affect physical events, laws must relate the phenomenal and physical changes. Those laws are structure. If phenomenal properties affect nothing physical, they cannot cause the report “I see red.” The physical process produces the report without them.

Panpsychism. Galen Strawson (2006) and Philip Goff (2019) place experience in the intrinsic nature (what a thing is in itself, apart from its relations) that underlies relational structure. What is that intrinsic nature? If it has features, those features are ruled or random, and both are mathematical. If it has no features, it adds nothing to what constitutes the thing. Calling it “experiential” rather than “mathematical” renames structure; it does not escape it.

Panpsychism, the view that experience is a basic feature of all matter, faces a further problem of its own: how do elementary experiential properties combine into a unified subject? A combination principle must specify which parts combine, under what conditions, and with what resulting experience. Those conditions and results are structure. The combination problem is itself a structural question, and panpsychism has not solved it

(Chalmers, 2016).

Russellian monism. Bertrand Russell (1927) observed that physics describes only relational structure, leaving open what instantiates, or embodies, those relations. The Russellian monist fills the gap with phenomenal properties. Yet the filler, if it makes a difference, makes a structural difference. If it makes no difference, nothing has been filled. The question is not whether something sits beneath structure; the question is whether anything coherently can. The rule-random dichotomy says no: whatever the filler is, it is either ruled or random, and both are mathematical.

Idealism. Idealism places mind first and derives matter from it. With one kind of thing, it has no interaction problem between two substances, but it still owes an account of order: an idealist must explain how mental states produce the regularities we call physical law. That explanation requires rules governing which mental states produce which physical regularities. Rules governing transitions are structure. Idealism's "mental substance" turns out to be mathematical structure described from the inside.

5. The Explanatory Asymmetry

The structural account needs no supplement. It must identify which organizations constitute which experiences, and that is research, not a gap in the account. Every alternative needs structure to state its own claims, specify its own interactions and solve its own combination problems. The asymmetry is decisive.

Math needs math. A structural account of consciousness requires structural analysis of organized systems. Descriptions of those systems are themselves structures (marks on a page, states in a brain, patterns in a computer). All of this is mathematical structure operating on mathematical structure, and no step along the way invokes anything else.

The alternatives need math too. Qualia, the felt qualities of experience, need structural relations to be determinate. Panpsychism needs structural combination principles. Idealism needs structural laws. None of them can state their own position without the very structure they claim to supplement. They cannot explain why everything fits mathematical description, because their own explanations are mathematical descriptions.

The structural account explains why everything fits: because everything is structure. The alternatives cannot explain it; they can only assume an extra ingredient and then describe that ingredient structurally.

6. From Description to Identity

A description of a system and the system itself are different things. Ink on a page does not have the causal powers of the system it describes. This is sometimes offered as evidence that structure is merely representational. It is evidence of the opposite.

The description (ink, pixels, symbols) is one mathematical structure. The system described is another. The description represents the system, and representing is not being: the ink cannot feel what the brain feels, because the ink does not have the brain’s organization. The difference is structural, not the presence or absence of a mysterious extra. The brain is a mathematical structure complex enough to model its environment and itself. The ink is not. The test is what follows a change in what the system takes in. Change a running brain’s input and the states after it change; the ink stays as written, and a recording plays its stored sequence, even one stored as frame-to-frame

differences, whatever happens around it. A description lists a process; a structure in which each state produces the next one, responding to what it takes in, runs it.

Knowledge by description (modeling a pattern in one cognitive subsystem) and knowledge by acquaintance (instantiating the pattern throughout sensory integration systems) are two different mathematical structures. The first models the second; the second is the experience itself (Jackson, 1982). Mary, who knows all physical facts about color vision through theoretical models but has never seen red, lacks one mathematical structure: the state her brain enters when it processes red light. What she gains on seeing red is a new structural state, not a nonstructural property. She learns no new fact about color; for the first time she is the state her complete theory described, a new way of having what she already knew.

Structure is complete, so two systems cannot differ in experience without differing in structure. The completeness claim is exactly what the hard problem of consciousness denies in advance. Denying it does not refute it. Producing a third option, neither ruled nor random, would.

7. Objections

”You are defining away the problem.” The hard problem of consciousness defines away the answer. It stipulates that experience is nonstructural. The present argument examines that stipulation and finds it unsupported. Removing an unjustified assumption is not the same as defining away a problem. The research question remains: which structures constitute which experiences, and why?

”The explanatory gap is real.” It may be epistemically real: we cannot see why certain organizations feel the way they do. An epistemic gap, however, does not establish an ontological gap, a gap in reality itself. We could not see why heat was the motion of molecules before the kinetic theory of heat. The identification did not require us to see it in advance; it required us to stop assuming it was impossible.

”Conceivability of zombies shows experience is not structural.” A physical duplicate lacking experience is conceivable only if complete physical organization does not constitute consciousness. That is the premise under dispute. Conceivability arguments test our ability to imagine scenarios; they do not establish metaphysical possibility (what could really

be the case) without an independent argument that our concepts track modal space (the range of real possibilities) accurately here, as Katalin Balog argues (Balog, 1999). We can conceive of water not being H2O, but it is.

”The recursive argument proves too much: it would make everything mathematical, including things that are not.” That is the claim. Everything is mathematical. The rule-random dichotomy is exhaustive. The objector must produce a coherent third option: something that is neither ruled nor random. No such option has been produced.

”You have not explained why structure feels like anything.” Correct. The identity thesis does not explain experience in terms of something else. It says experience is a real aspect of certain mathematical organizations, not something produced by them or added to them. The research question is which organizations and why those. Heat is not explained in terms of something other than the motion of molecules; it is that motion, measured on a different scale. The identity removes the demand for a bridge. The work that remains is finding which systems run those organizations.

8. Conclusion

The hard problem of consciousness is a fallacy of circular reasoning. It assumes that experience is not structure, observes that structural explanations supply only structure, and concludes that something is missing. What is missing is the justification for the assumption.

Consciousness must be structure to interact with structure. The interaction problem applies to every proposed nonstructural supplement: qualia, panpsychism, idealism and substance dualism (mind as a separate, nonphysical thing). Each must specify what its extra ingredient is (determinate features, therefore structure), how it combines (combination rules, therefore structure) and how it interacts with the physical world (causal relations, therefore structure). The rule-random dichotomy applies recursively at every level. At no stage does a nonmathematical remainder appear.

The structural account has work to do: identifying which organizations constitute experience, explaining why those organizations and not others, and finding which systems have those organizations through comparative, neural and computational study of their structure. That is a research

program, not a gap in reality. The alternatives have more work to do: they must state their own positions without using the structure they claim to supplement. So far, none has.

A41. The Theory of Everything Must Be Discrete: Gravity, Space and Forward Time as Causal Steps

Abstract

Everything that happens must be discrete, in any universe, because a cause produces its effect as the next state, and a continuum has no next state for a cause to produce: like Zeno's runner, who must reach every halfway mark first, a continuum has no first step to take. The argument is an a priori proof in the logician's sense. Our physics agrees. Gravity and quantum theory together let a region of finite size hold only finitely many distinguishable states: the covariant entropy bound caps a region's entropy by the area of its boundary. A finite description singles out only countably many states, and continuous amplitudes, like every continuum, are analysis, a description of many steps at once. Gravity, space and time are then causal transitions. The relations they change exist as mathematics, with zero-width points for nodes, so discreteness gives nothing a size; what it supplies is a next step. Each transition is a rule applied to rules, and nothing outside the mathematics applies it: time is that

application, and space is how far it spreads at the base rate, the speed of light. General relativity’s smooth spacetime is their large-scale behavior. The result cuts off the infinities of continuum quantum gravity at their source, makes black-hole entropy a count of states, and sets aside a continuum base and, with observed Lorentz symmetry, a regular lattice. That leaves discreteness without a grid, as in causal sets, whose random count estimated the cosmological constant before it was measured. A theory of everything is a rule for transitions, a rewriting system.

1. Introduction

Gravity and quantum theory have resisted unification for a century, and the main programs split on the most basic question: is spacetime smooth all the way down? Asymptotic safety, the proposal that gravity’s strength levels off at short distances so that a continuum theory stays predictive, and the continuum formulations of string theory, in which tiny vibrating strings move through smooth space, keep a continuum at the base. Loop quantum gravity, which applies quantum theory to the geometry of space itself, finds discrete structure there, and causal sets, which build spacetime from events and their order, start from it.

This paper argues that the continuum cannot be the base, and shows what that buys the search.

All of physics is mathematical, and that alone selects nothing, since continuous and discrete structures are both mathematics. The selection comes from how states are produced. Each physical state is produced by a cause from the one before, a cause needs a next state to produce, and a continuum has no next state, so in any universe where anything happens, change comes in steps. Alfred North Whitehead reached the same conclusion from Zeno's paradoxes (Whitehead, 1978). Our physics agrees: gravity and quantum theory together cap the information a region can hold. A finite description can single out only countably many states (at most as many as the counting numbers). All three point the same way: the base has finitely many distinguishable states in any finite region, and it changes in steps. Continua still exist in mathematics, as analysis (the mathematics of limits and continuous quantities), and analysis describes steps without producing any.

Discrete has a precise meaning here. A region of finite size has finitely many distinguishable states, succession runs in steps rather than through a continuum of instants, and every value a history produces, amplitudes included (the numbers quantum

theory squares to get probabilities), is fixed by its rule, its start and the number of steps, so the values that occur form a countable set, not a continuum. A qubit’s continuum of states, the states of a two-level quantum system, is analysis: it describes every state a rule could reach, while a history reaches its states one step at a time. A fixed rotation by an irrational angle (an angle that is no whole-number fraction of a full turn), applied step after step, reaches countably many states that come arbitrarily close to every point of its circle: a short rule, a countable history, and a continuum only in the description. The structure need not be a lattice, a regular repeating grid. General relativity remains exact enough wherever counts are large, the way fluid dynamics remains exact enough for water.

What is discrete is the transitions. A cause produces its effect as a change from one state to the next, and the change is what happens; a state is only a cut between one transition and the next, and in relativity no single cut is everyone’s present. There is no stuff beneath the transitions. The relations they change exist as mathematics, and their nodes are zero-width points; what happens, and so what can be observed, is how the relations change, and a distance or a duration is a pattern in how transitions follow one another. Causal-set theory builds

spacetime this way: the order of its events and their number give its geometry (Surya, 2019). Gravity, space and time are those transitions, counted at large scale.

Discreteness does not solve the problem of size, and it does not need to. A block of space would be continuous, since a block has a surface and a surface is a continuum, and so would a sphere. Zoom in on a discrete structure and you find nodes and their relations, not small blocks. The nodes have no width, as a continuum's points have none, and a size is the extent of the relations over which the nodes spread. Zero-width points are no flaw of the continuum, since a discrete structure has them too. A block can still exist as a rule: fixed by its corners, it needs no surface built point by point, and it acts when a step crosses it, as when something entering is turned back.

Motion is not built point by point either. A body at x is next at a definite place, x + 3 or x + 3.25, the result of one step, with no place in between visited. More important is the next point in time: without one there is no causality, no event and nothing moving, and a continuum has no next point.

Each transition is a rule applied to rules. On the view taken here, the rules are the only things. A structure that only sat there would

still exist, as all mathematics does, and it would do nothing and know nothing, like a dead brain with every connection intact: noticing anything, even that nothing changes, takes a change in whatever notices. What happens is a rule applied to a state, whose relations are rules too, giving the next state, much as every computation in the lambda calculus, a formal language of pure functions, is functions applied to functions (Church, 1941). Time is that application, step after step.

What causes the change? The math rules themselves. Nothing outside the mathematics applies a rule, as nothing outside arithmetic makes 2 + 2 equal 4. Stephen Hawking called a unified theory just a set of rules and equations, and asked what breathes fire into the equations and makes a universe for them to describe (Hawking, 1988). The question has its force for smooth equations, which relate states at different times without taking any state to a next one, so nothing in them goes by itself. It also assumes the equations only describe a world apart from them. In a world that is mathematics, the fire is the application of a rule that takes each state to the next, and the structure contains it: a history includes its transitions, while a frozen record of one state, like the dead brain, does not.

Space is tied to time because both are counted in the same steps. A distance is the number of steps an effect needs to get from one place to another, and the speed of light is the base layer's rate. Our units already count distance this way: since 1983 the meter has been the distance light travels in 1/299,792,458 of a second. A body's steps at that rate are shared between moving and changing in place, so a moving clock runs slow. Stated precisely, the shares add like the sides of a right triangle, (c dt)^2 = dx^2 + (c dτ)^2, the invariant interval of special relativity, the spacetime separation every observer agrees on, rearranged, with τ the moving clock's own time. The picture restates relativity rather than deriving it: a candidate must reproduce the interval from its rules (Section 4.5). Light, which crosses space while no time passes for it, is the limit and no exception: it spends every step moving, so nothing within it could move relative to the rest (Section 5).

The Game of Life (a grid of cells that live or die by simple rules) has the same speed limit: no pattern spreads faster than one cell per generation, which Life's players call the speed of light, and a glider, a small pattern that travels across the grid, crawls at a quarter of it (Gardner, 1970; Berlekamp, Conway and Guy, 1982). Life's limit comes from a grid, and a grid picks out a rest

frame, a state of motion that counts as standing still, so relativity needs discreteness without a grid (Section 4.3).

The starting points are Jacob Bekenstein's black-hole entropy, which measures how many internal states a black hole has (Bekenstein, 1973), the entropy bounds that followed ('t Hooft, 1993; Susskind, 1995; Bousso, 2002), the argument that the Hilbert space of quantum gravity, the space of its possible quantum states, is locally finite-dimensional, allowing only finitely many distinguishable states in any finite region (Bao, Carroll and Singh, 2017), and the discrete programs of causal sets and loop quantum gravity (Bombelli, Lee, Meyer and Sorkin, 1987; Rovelli and Smolin, 1995). This paper's contribution is to join the physical, generative and descriptive routes into one selection principle for the search, and to show what the principle buys: finite sums where continuum calculations diverge (run off to infinity), black-hole entropy as a count, a narrowing of the candidates to discrete order without a grid, and a list of targets any candidate must meet. Its decisive premises are that a cause produces its effect as the next state, and that the entropy bounds hold in the fundamental theory, region by region. It rules out a fundamental continuum of physical values, amplitudes included, and a regular spacetime lattice, as the base of physics.

The argument is the fifth link in a larger chain:

1. Everything that exists is ruled or random, and both are mathematical, so physics is mathematics, and anything outside mathematics would be outside logic. That is an a priori proof.
2. Every consistent structure exists as mathematics, and what exists physically is what logical steps in mathematics produce. Mathematics cannot fail to exist, so there is something rather than nothing.
3. Each bit of a description that no rule decides is a 50/50, so each added bit halves the description's weight. Short laws dominate, and a universe like ours starts simple, in its laws and its state, as observation also shows: the known laws fit on a page, and the early universe was nearly smooth.
4. Observers like us are complex, so a simple start has to grow complexity, as simple rules do.
5. A cause produces its effect as the next state, and a continuum has no next state, so the growth comes in discrete causal steps. That is an a priori proof in the logician's sense.
6. Time is those steps, and space is how far they spread, at a base rate, the speed of light. An endless chain of steps has no last step to run back from, so time runs forward. That too is a priori. A history run backwards would also have to state the

step where it turns back, and each bit of that timestamp halves its weight.

7. Observers arise in forward time, where complexity grows, and their minds run forward with it: a mind models its world from records of steps already taken, and each thought is produced from the one before, so no one remembers backwards from the end.

8. With no distance shorter than a step, the infinities a smooth spacetime puts into quantum gravity never arise, so string theory, built to tame them, is not needed, and smooth general relativity is the large-scale behavior of the steps. A theory of everything is a short rule for transitions.

Each link stands on its fact, so a reason can fail without taking the chain with it: if the weight on short descriptions were not why our laws are simple, they would still be simple.

2. Gravity and quantum theory cap the information in a region

2.1 Nothing smaller than the Planck length can be resolved

To see something small you need short waves, and short waves carry more energy. Squeeze enough energy into a small enough space and it becomes a black hole. At about 1.6 x 10^-35 meters, the Planck length, the black hole a probe would make is as large as the distance being probed, so the probe hides what it was meant to show. Stated precisely, resolving a distance d takes momentum of about h-bar/d and energy E of about h-bar c/d, whose Schwarzschild radius, the size of the black hole that energy would make, 2GE/c^4, about 2G h-bar/(c^3 d), exceeds d once d falls below roughly the Planck length, the square root of h-bar G/c^3 (Garay, 1995; Hossenfelder, 2013).

2.2 A region can hold only finitely many states

A black hole’s entropy, read as a count of its states, equals a quarter of the horizon’s area in Planck units (counted in squares of the Planck length), as Bekenstein and Stephen Hawking found (Bekenstein, 1973; Hawking, 1975). No region can hold more states than the black hole it would become if matter kept being added: if it could, adding matter until it collapsed would lower the entropy, which the second law (total entropy cannot decrease)

forbids (Susskind, 1995). So a region’s capacity is capped by the area of its boundary. A ball one centimeter in radius can hold at most about 1.7 x 10^66 bits, and a ball one meter in radius about 1.7 x 10^70.

Stated precisely, the Bekenstein-Hawking entropy is S = k_B A/(4 l_P^2), where A is the horizon area and l_P the Planck length. The holographic bound limits the entropy of any region to the same expression in its boundary area (‘t Hooft, 1993; Susskind, 1995). The spacelike form of that bound, which counts the entropy inside a region at one instant, fails in some cosmological and collapsing settings; the covariant form, which bounds the entropy passing through light-sheets, the surfaces traced by light rays leaving a surface without spreading apart, has held in every case examined (Bousso, 1999, 2002). A Hilbert space whose dimension is at most exp(S/k_B) then describes each region, finite for every finite region (Bao, Carroll and Singh, 2017).

2.3 Why the count matters, and the map does not

Any set can be mapped onto any other set of the same size, so a map from a region onto its surface proves nothing by itself. The part of the holographic principle used here is a statement about counting. Without gravity, the room for information grows with

volume: double a region’s width and it holds eight times as much. With gravity it grows with surface area: double the width and it holds four times as much. A region has fewer states than its volume suggests, as though it were a surface.

Another sign of the same thinning shows up in dimension. Several quantum gravity programs find that the effective dimension felt by a diffusing particle (one spreading out by random steps) falls toward two at the smallest scales (Ambjørn, Jurkiewicz and Loll, 2005; Carlip, 2017). Holography and this dimensional reduction are different results. Both suggest that the smallest scales carry fewer independent degrees of freedom (separate quantities free to vary) than a smooth four-dimensional continuum would.

2.4 Finite states rule out a fundamental continuum

A continuum field gives every point its own real number, a region has infinitely many points, and each real number carries infinitely many digits. A continuum region therefore offers infinitely many distinguishable states. Gravity and quantum theory together allow finitely many. The smooth fields of general relativity and quantum field theory, the framework of particle physics, are then approximations, accurate wherever the counts are large. Stated

precisely, a classical (non-quantum) field on a region of positive volume ranges over an infinite-dimensional space, and its quantization (its conversion into a quantum theory) gives an infinite-dimensional Hilbert space for the region; the bound of Section 2.2 requires a finite one, so the continuum fields become effective variables, convenient large-scale descriptions, over finitely many states (Bao, Carroll and Singh, 2017).

3. Causes and descriptions lead to the same conclusion

3.1 Causes produce their effects in steps, in any universe

Causality needs a cause and an effect, and the effect is a state the cause produces. In a continuum no state has a next one: between any two instants lies a third. Pick any instant after a cause and another instant still lies between them, so no state is the first one the cause produces, and the chain of production cannot take its first step. This is an a priori proof in the logician’s sense: the conclusion follows from what a cause is and what a continuum is, and only a counterexample to one of those premises could undo it.

Zeno’s runner faces the same regress: he must reach the halfway mark before the finish and the quarter mark before that, so no first stretch exists for him to run (Aristotle, 1984, *Physics* VI.9). Analysis solves the paradox in the abstract, describing the run as a smooth path whose endless stretches add up to a finite time. Discrete steps solve it in the only way causality and spacetime can happen: each step produces the next, and finitely many reach the finish. A history built by causes therefore runs in steps. This holds in any universe where anything happens, not only in ours, since every cause needs a next state to produce and a continuum has no next state. A change that nothing produces would be a change without a cause, and nothing in an account that allows one says why the change happens. Whether each next state is fixed in advance or left to chance is a separate question: either way, a cause needs a next state to produce.

Stated precisely, a process that adds finitely or countably many events at each natural-number stage produces at most a countable set of events, and no uncountable continuum of them. A continuum law can still make every state depend on earlier ones: a differential equation fixes how a solution’s values at different times relate. That is causation as dependence, and this argument concerns causation as production, the two concepts

Ned Hall separates (Hall, 2004). Everyday cases show the difference. A plant that dies because nobody watered it depends on an omission, yet not watering produced nothing; the plant's own chemistry, running without water, produced the death. Each frame of a film fits the physics of the frame before, yet no frame produces the next; the projector does. A reversible rule lets yesterday's state be computed from today's, yet a history run in either direction has each of its states produced by a step from the one before. In each case the dependence is real, and something else does the producing: chemistry, a projector, a step.

The case for a physical continuum takes dependence to be all of causation. A continuous law makes every state depend on earlier ones, a hybrid model adds discrete switches to smooth motion without inconsistency, and later need not mean next. Under $x = vt$ every position depends on the start and the speed, and Bertrand Russell met the demand for a next place by holding that motion is nothing more than being at different places at different times (Russell, 1903).

The everyday cases show what this misses. In them, dependence is carried by production at a level below, and a continuum offers no level below. Its law can be read as a rule for transitions, a flow taking each state to the state a time t later, for

every positive t. A flow has no smallest transition, since each one it defines is the composition of two smaller ones (one applied after the other), so applying it would take infinitely many applications before any, and the chain of production has no first link. Read the other way, the flow hands out every later state at once from the start. That pairs times with places, as Russell’s answer does, without any state producing the next. These are the emperor’s new points: they look like motion and produce none, and the pairing exists as mathematics whether or not anything moves, as a frozen record does. Hall lets production run through continuous intermediates; the argument here is that it cannot, since a production with no first effect produces nothing. Whitehead, arguing by Zeno’s method, put it this way: once anything becomes, there is “no continuity of becoming” (Whitehead, 1978, p. 35).

We do change: our states follow one another. In a world that is mathematics, then, rules take each state to a next one, and a rule that takes a state to its next needs a next state. A reader who still takes dependence to be all there is to causation finds no contradiction in a continuum’s mathematics. For that reader the case rests on physics: if the entropy bounds hold, a region of finite size holds finitely many distinguishable states (Section 2),

and a system with finite energy passes through only finitely many of them in a finite time, since reaching a distinguishable state takes a minimum time set by the energy (Margolus and Levitin, 1998). Nothing observable needs a continuum beneath them, since finite steps come as close to any state as measurement can tell (Section 5). Dependence describes a history; production generates one, and generation comes in steps. Continuum mathematics stays a tool for describing the steps in bulk, as calculus describes a crowd of molecules.

3.2 A finite description reaches only countably many states

Algorithmic probability, which gives outputs of short programs more weight, is defined over countably many programs, each producing its output. Structures requiring selection from an uncountable continuum are unreachable by this measure and have measure zero: together they carry no weight. A continuous theory is still fine when a finite rule and finite starting data produce it; what has no weight is a continuum whose states must be selected one real number at a time. On the view that physical reality is itself a mathematical structure, this measure weighs what exists, and it gives weight only to what finite rules reach:

countably many states, gravity’s included.

Stated precisely, the universal prior gives an output x the weight of the programs that produce it, summed over programs p with weight 2^-|p| (Solomonoff, 1964). Only computable outputs, those a finite program can produce, receive positive weight, and they form a countable set; a real number chosen at random from an interval is computable with probability zero. Marian Pour-El and Ian Richards (1981) found computable but rough starting data for the three-dimensional wave equation whose later solution is not computable. Roughness of that kind is what a finite count of states per region excludes, so the routes of Sections 2 and 3 support each other.

4. What discreteness does for the search

4.1 The infinities lose their source

The infinities of quantum field theory come from adding up contributions from shorter and shorter distances. Gravity is the worst case. Einstein’s theory, treated as a quantum field theory, needs a new adjustable constant at each order of calculation, and at two loops, the second order of approximation, it already produces an infinity that no redefinition of its constants removes

(Goroff and Sagnotti, 1986). With finitely many states per region there are no arbitrarily short distances to sum over, and every such sum over a finite region is finite. Hartland Snyder proposed quantized spacetime for this reason in 1947 (Snyder, 1947). Discreteness supplies the cutoff as physics, where the continuum treats it as a calculational device to be removed at the end. Two open problems take their difficulty from that removal. The Yang-Mills existence and mass gap problem, one of the Millennium Prize problems, asks for a proof that the strong force's mass gap survives as the cutoff goes to zero (Jaffe and Witten, 2000), and the simplest field of the Higgs type, taken alone, loses its interaction entirely in that limit (Aizenman and Duminil-Copin, 2021). The limit is analysis, a statement about the formulas as a description of many steps at once, and it stays good mathematics. Applied, the formulas act in steps, since a cause needs a next state, and a region holds finitely many states, so physics does not wait on the limit. What remains for physics is the span of scales, from the finest resolution, near the Planck length, to the gap, about 10^19 times larger, and bridging it is most of the work.

4.2 Black-hole entropy becomes a count

In Ludwig Boltzmann's physics, entropy counts states. Black-hole entropy fits that picture only if a black hole has finitely many states. The successful microscopic derivations of the area law count a finite set. Andrew Strominger and Cumrun Vafa counted string states for a class of extremal black holes, which carry the most charge their mass allows, and recovered the quarter exactly (Strominger and Vafa, 1996). Loop quantum gravity counts the discrete pieces of area where the horizon is pierced, with its free parameter set to give the quarter (Rovelli, 1996; Ashtekar, Baez, Corichi and Krasnov, 1998). If evaporation (a black hole's slow loss of mass as radiation) is unitary, losing no information, finitely many states make the entropy of the radiation follow the curve Don Page computed: it rises and then falls as the black hole shrinks (Page, 1993). Discreteness is the setting in which these counts are literal.

4.3 The candidates narrow

If the base is discrete, a theory whose base is a continuum can at most approximate the final one. A continuum quantum field theory of the metric (the field that sets distances and durations in spacetime), as asymptotic safety proposes (Reuter, 1998), must

then reproduce the finite count of states as an output, and its continuum variables become labels for finitely many states. String theory may lean the same way: its strings cannot probe distances below a minimal length (Hossenfelder, 2013), though a limit on what strings can resolve is not yet a limit on what exists.

Observation removes the simplest discrete option, a regular grid. A grid picks out directions and a rest frame. If spacetime were a grid at the Planck length, light of different energies would generically travel at slightly different speeds. Photons from the gamma-ray burst GRB 090510, a flash of gamma rays from billions of light-years away, arrived together closely enough to push any such linear effect (a speed shift proportional to energy) beyond 1.2 times the Planck energy, the scale at which gravity's quantum effects become strong (Abdo et al., 2009). Lorentz symmetry, the rule that physics looks the same to every observer moving steadily, also resists being broken: a break at small scales leaks into everyday physics through quantum corrections unless it is finely tuned away (Collins, Perez, Sudarsky, Urrutia and Vucetich, 2004).

Random discreteness keeps the symmetry. In a causal set, spacetime events are scattered at random with a density fixed by volume, and the scattering favors no direction and no frame

(Bombelli, Henson and Sorkin, 2009). Discrete quantum geometry, in which area and volume have discrete spectra (possible values that come in separate steps), can also respect it, since a discrete spectrum is not a grid of locations (Rovelli and Smolin, 1995; Rovelli and Speziale, 2003). The base is then discrete without being a grid, and the leading candidates for such a base are random discrete order of the causal-set kind (Surya, 2019) and discrete quantum geometry.

4.4 An estimate made before the measurement

Random discreteness gave an order-of-magnitude estimate before the data existed. A random count fluctuates by about its square root. If spacetime is a random scattering of elements, the count in our past fluctuates the same way, and Rafael Sorkin argued, treating the constant and spacetime volume as a pair whose uncertainties trade off, like position and momentum, as unimodular gravity does, that the fluctuation leaves a cosmological constant (the energy of empty space, which speeds up cosmic expansion) of about one over the square root of the number of elements, around 10^-120 in Planck units (Sorkin, 1997; Ahmed, Dodelson, Greene and Sorkin, 2004). The accelerating expansion found in 1998 implies a cosmological

constant of that order (Riess et al., 1998; Perlmutter et al., 1999). Quantum field theory on a continuum, cut off at the Planck scale, overshoots by about 120 orders of magnitude. The same causal-set model predicts that the constant fluctuates over cosmic time, which cosmological surveys can test (Zwane, Afshordi and Sorkin, 2018).

4.5 What any candidate must recover

A discrete candidate earns its place by recovering what the continuum already delivers. It must yield smooth geometry obeying Einstein’s equations at large scales; Lorentz symmetry to the precision observed; quantum interference, with amplitudes that cancel before probabilities are taken, and the Born rule, which turns amplitudes into probabilities; the area law with its quarter; and the fields and interactions of the Standard Model, the established theory of particles and forces apart from gravity. It must give an effective dimension of four at the scales we probe, and say what happens to dimension at the smallest. Each candidate also gets a definite description length, a fixed rule plus a finite count of elements per region, which makes rival candidates comparable bit for bit.

4.6 Where a theory of everything must reside

If everything that happens is a transition, a theory of everything is a rule for transitions: a set of relations and a rule that rewrites them, applied again and again to its own output. Its mathematics is the mathematics of rewriting. Stephen Wolfram's models rewrite hypergraphs, networks whose links can join any number of nodes, and he argues that light cones, the regions a signal can reach, and curved geometry emerge from the causal order of the updates (Wolfram, 2020). Causal sets can grow one element at a time under classical sequential growth, a rule that adds each new element with set probabilities (Rideout and Sorkin, 2000). Spin foams, the histories of loop quantum gravity's geometry, carry one quantum geometry to the next (Perez, 2013). Quantum circuits apply gates, the elementary operations of a quantum computer, to a state to produce the next state. Smooth spacetime and continuous fields are the analysis of such rewriting at large scale.

The leading discrete programs can meet. A rewriting system's causal graph, the record of which update depends on which, is itself a causal order, the structure a causal set is made of. The target is then one short rewriting rule whose causal graph

matches a random, four-dimensional causal set at large scales, run as a unitary quantum rule, so that its branching is the branching of many worlds. The match cannot reach single updates. An update that rewrites a few relations has a few direct causes, and a graph with a bounded number of edges per element picks out a frame on a random causal set (Bombelli, Henson and Sorkin, 2009). The frame must then stay hidden at every scale we can test, unless each update draws on an unbounded part of its past, as in sequential growth. Quantum cellular automata already recover the equations of light and of the electron at long wavelengths (Bialynicki-Birula, 1994; D’Ariano and Perinotti, 2014). Since each candidate has a definite description length (Section 4.5), the search can run shortest rule first: list rules from shortest to longest, keep those whose causal graphs approximate a four-dimensional spacetime with no preferred frame at large scales, and test the survivors against everything Section 4.5 lists, together with a cosmological constant that fluctuates (Section 4.4). The search can also run from the other end, program first. Write the known laws as a program of discrete steps without a grid, and remove their infinities, as loop quantum cosmology does when it replaces the big bang with a bounce (Ashtekar, Pawlowski and Singh, 2006).

Then shorten the program while its outputs still match.

The idea that physics computes is old: Konrad Zuse proposed in 1969 that space itself calculates (Zuse, 1969). This paper's contribution is the argument that it must, because a cause needs a next state, and the constraint that follows: a candidate has to state its update rule and derive everything else from the updates. No candidate has yet recovered general relativity and the Standard Model together, and Section 4.5 lists what the winner must deliver. The search belongs among the rewriting systems.

4.7 How light moves

Light is motion by steps in its plainest form. An effect spreads at most one step of distance per step of time (Section 1), and light is an effect spreading at that rate, as a glider in the Game of Life is a pattern spreading at a quarter of it. Iwo Bialynicki-Birula (1994) wrote Maxwell's equations, the laws of electricity, magnetism and light, as a unitary cellular automaton, a reversible quantum rule for a grid of cells, each cell's state updated from its neighbors at every tick, and recovered the continuum equations at long wavelengths; since a regular lattice picks out directions, the steps of our physics belong to a discrete order without a grid (Section 4.3). Light also spends all its steps moving. Its proper time, the

time a clock carried along would record, is zero, so along a light path nothing changes in place, the limit of a moving clock running slow.

The paths light can take carry most of general relativity. In a spacetime without causal pathologies, such as paths that loop back into their own past, the order of which events can influence which fixes the topology, how its regions connect, the smooth structure (which functions on it count as smooth) and the geometry up to a local scale (Hawking, King and McCarthy, 1976; Malament, 1977), and counting supplies the scale. A causal set needs only an order and a count to give geometry for this reason (Surya, 2019), and the question of how light moves leads straight to what spacetime is: the order of the steps, counted.

5. Objections

”Continuum mathematics is consistent, so the base could be continuous.” Consistency is not in dispute. Continuous and discrete structures are both consistent mathematics, and the selection between them comes from physics: gravity and quantum theory cap the states in a region, and production comes in steps. The two reasons are independent. A reader who doubts that causes produce next states still faces the entropy bounds

and the finite rate of change (Section 3.1), and a reader who doubts the bounds still faces the regress of Section 3.1.

”The laws themselves are continuous: a perfect sphere, a field, Einstein’s equations.” A law can be stated with continuum mathematics and still act only in steps, which is how this paper reads general relativity. A hybrid model, smooth motion with a discrete switch when a body crosses a perfect sphere, is consistent mathematics too (Goebel, Sanfelice and Teel, 2012), but it describes: to produce the switch, something must first produce the motion to the boundary, and a continuum gives that motion no first step (Section 3.1). Once production runs in steps, a law cannot act on its continuum. Each step takes a definite state to a definite next one in finitely many operations, while a test that needs every digit of a real number, such as whether a point lies exactly on a perfect sphere, cannot be finished in finitely many: given two real numbers digit by digit, a strict inequality between them is confirmed at some finite stage when it holds, but equality is confirmed at none (Weihrauch, 2000). A law that acts on definite values does not need such a test. Physics agrees, since no region holds infinitely many distinguishable positions, so no boundary is tested beyond the precision the states carry. The continuous law is the analysis of

the discrete one. A continuous formula need not describe a continuum that physically exists, in this universe or any other. It can be the rule that fixes how strongly, in which direction and with what amplitude each discrete transition acts, and its continuous arguments need not each be a place or an event. Einstein’s equations are written on a continuum, and their success leaves open whether that continuum is the physics or an effective description of it; this paper argues for the second, since only the transitions produce anything. Work on quantum fields on causal sets pursues this reading, recovering the continuum’s propagation of a free field (a field without interactions) from discrete order (Johnston, 2009).

”Finite information is not discreteness: a qubit has continuously many states.” The continuum of a qubit’s states is analysis, the description of every state a rule could reach. A history reaches its states in steps, so the amplitudes it produces are each fixed by finitely many steps and form a countable set. Nothing observable needs the rest: a finite set of quantum gates reaches states arbitrarily close to every state of the continuum, with a gate count that grows only as a power of the logarithm of one over the error (Dawson and Nielsen, 2006).

”The entropy bounds are conjectures.” The covariant bound has held in every case examined, and the spacelike version’s failures are why the covariant form is used (Bousso, 2002). A quantum version has been proven for free fields in weak gravity, with entropy counted relative to the vacuum, the lowest-energy state of the fields (Bousso, Casini, Fisher and Maldacena, 2014). A region that broke the bound would hold more states than the black hole it could collapse into, and collapsing it would lower the entropy, against the second law. That a region’s whole space of states is finite in the full theory is widely expected in quantum gravity (Bao, Carroll and Singh, 2017) and remains the second of this paper’s two decisive premises (Section 1).

”Asymptotic safety may make a continuum theory of the metric complete.” Then its states per region must still obey the bound. Its continuum variables would describe finitely many states redundantly, as gauge variables, labels with built-in redundancy like the choice of where voltage counts as zero, describe fewer physical states than they appear to.

”Lorentz symmetry forbids discreteness.” It forbids a grid. Random discreteness keeps the symmetry, as the causal-set theorem shows (Bombelli, Henson and Sorkin, 2009).

”A real cutoff brings back the hierarchy problem.” It makes the problem explicit. The Higgs mass is sensitive to any heavy physics above it, gravity’s included, so why it sits some seventeen orders of magnitude below the Planck mass needs an answer with or without a step; a real step only keeps the question from hiding inside a calculational convention. Discreteness does not answer it, and this paper claims no answer.

”Light has no path in quantum field theory, so nothing moves step by step.” A photon is an excitation (a ripple carrying one packet of energy) of a field spread through space, and no point-to-point path is needed. What spreads is the change: each region’s field is set from its neighbors’, which is what a local wave equation states, and in a discrete account each such update is a step (Section 4.7).

”Light crosses space while no time passes for it, so the steps cannot be time.” A thing’s time is its transitions in place, and each body shares the base rate between moving and changing in place (Section 1). A moving body’s transitions therefore run slower, and light spends every step moving, so nothing within it could move relative to the rest. Its zero proper time is the limit of a moving clock running slow, not a step taken outside time.

6. Conclusion

Everything that happens must be discrete, in any universe, because causality produces each state from the one before, and a continuum has no next state to produce. Gravity and spacetime are no exception: gravity and quantum theory together allow only finitely many distinguishable states in any finite region, while a continuum would allow infinitely many, and algorithmic probability gives no weight to states that must be selected from an uncountable continuum. Continua, amplitudes included, stay what they are in mathematics: analysis that describes many steps at once. Gravity, space and time are these causal transitions, applied by the math rules themselves, and general relativity's smooth spacetime is their large-scale behavior.

The conclusion is useful. It cuts off the infinities of quantum gravity at their source, makes black-hole entropy a count, sets aside continuum bases and regular grids, and leaves discrete structures without a grid, among them causal sets, whose estimate of the cosmological constant came before the measurement. All of physics is mathematics; the useful claim is that its base is a rule applied to rules, discrete transitions rewriting relations step by step, and the search for a theory of

everything belongs among such rules.

A42. Searching for a Short Discrete Theory of Everything: Rewriting Rules and Supplied-Lattice Tests

Abstract

If spacetime is the causal order of discrete updates, a theory of everything is an update rule, and the search can run shortest first, since each extra bit halves a rule's prior weight. This paper searches all 2,935 rules that rewrite two connected relations into three, and samples of 3,500 longer ones, reading each causal graph with three dimension tests from causal-set theory, fixed in advance with noise bands. Run oldest relations first, a 20-bit mirror pair grows two-dimensional orders; under random orders it reads near one, so a rule's update order belongs in its length. No rule reads four on all three tests. The closest read four on two, with chain-based dimension estimates 4.7 to 21 standard deviations below the four-dimensional reference. Part of the gap is a bound: an update has at most as many direct causes as its rule consumes relations. Sequential growth escapes the bound yet fails: transitive percolation reaches four on the chain at a million elements while its midpoint drifts to 5.8. From the top, one

local unitary step reproduces the Dirac equation in curved spacetime, horizon crossing and pair creation in 2,488 bits. A proposed specification of the known laws on a supplied lattice, with declared dark matter and initial data, compresses to 16,272 bits. All four groups in a new 24-chain Regge audit fail the mixing screen; a complete quantum-gravity construction remains open. Code and data are supplied.

1. Introduction

Two discrete approaches to spacetime point at the same object. Causal-set theory builds spacetime from events and their order: which events can influence which, and how many events there are, give the geometry (Bombelli, Lee, Meyer and Sorkin, 1987; Surya, 2019). Rewriting systems build the world from a rule applied again and again to its own output (Wolfram, 2002, 2020). Each application of the rule is an update, and each update uses relations that earlier updates made. The record of which update used which update's output is a causal graph, and its transitive closure, the relation "can influence through a chain of updates", is a causal order. So every rewriting rule grows a candidate spacetime. The question is which rule grows ours.

Both approaches take change to come in steps. This paper adopts that as its working hypothesis, motivated by an argument rather than a measurement: a cause produces its effect as the next state, and a continuum has no next state, since between any two points lies a third. The argument rests on the premise that change is produced, each state making the next; a law that only lists states at every time describes change without producing it. Russell’s continuum, in which motion is only being at different places at different times (Russell, 1903), gives the emperor’s new points: positions everywhere and nothing that moves them.

The rules can be listed in order of length. Write each rule as a string of bits, and let each extra bit halve its prior weight, as algorithmic probability does (Solomonoff, 1964). This weights individual short programs more strongly and motivates searching them first; it does not establish that the physical rule must be short. The discreteness argument supplies the premise of the search; the numerical tests below do not independently prove that premise. Others have looked at the causal orders of short rules. Wolfram (2020) surveyed the behavior of many short rules, including the growth of their causal graphs. Bolognesi (2010) built causal sets from Turing machines, string rewriting and other simple models and found growth dimensions up to about 3.5.

Gorard (2020a, 2020b) connected Wolfram's models to causal sets and to relativity and related their dimension estimates to causal-set ones, with examples in two dimensions. None of these, as far as we can find, reports a short rule whose causal order reads four on the tests causal-set theory uses to recognize a four-dimensional spacetime.

This paper runs those tests on the smallest class of rules with room for more than a single line of causes, all 2,935 of its rules with at most two new nodes, and on samples of three larger classes. It calibrates three tests on random scatterings of points in flat spacetime and classifies 6,435 rules. It runs the most promising to 20,000 updates and ten of them past 100,000, compares them with what the equations give for an expanding universe, and supplies the code and data. It fixes a four-dimension criterion in advance, with noise bands from random point sets of matched size, tests every rule of the smallest class for causal invariance, reruns the long runs under four update orders, and reads sequential growth with the same tests. The result is negative for four dimensions and positive for two. Several rules grow two-dimensional causal orders, so the tests can find a dimension when a rule has one. No rule grows a four-dimensional order. The rules that come closest read four on

two tests and between 1.7 and 3.0 on the third. The two-dimensional orders hold under the first passes’ update order only, and sequential growth, the way around the obstacle below, does not read four on all three tests anywhere it was run.

The search also turned up an obstacle that holds for every rule that rewrites a fixed number of relations at a time. An update has at most as many direct causes as its rule consumes relations, so the causal graph of such a rule gives each update a bounded number of links, the related updates with nothing between it and them. A random four-dimensional causal set gives its elements ever more links as the region grows. No graph with a bounded number of edges per element can sit on one without picking out a preferred frame (Bombelli, Henson and Sorkin, 2009). A rule’s causal graph can therefore match four-dimensional spacetime only at large scales. Section 5 states the bound and what it asks of the next pass.

The search therefore runs from both ends. From the bottom it lists rules shortest first, and enumeration reaches rules of a few tens of bits, while a complete symbolic statement of the Standard Model with general relativity comes to about 11,300 bits compressed (Section 7). From the top it takes the known laws, which treat their fields as continua, writes them as local unitary

steps, checks the steps against the continuum where the spacing is small against every wavelength and curvature radius, and then shortens them while the match holds. Section 8 runs that route: one local update reproduces the Dirac equation in curved spacetime, gauge fields with Gauss’s law exact, pair creation and transport across a black-hole horizon, and it exposes a lattice that passes every wave test yet doubles the source of gravity. Section 8.7 records a proposed specification of the known laws on a supplied four-dimensional lattice, with declared dark matter and initial data, at 16,272 bits. This description length is distinct from the length of a verified executable completion. A rule found from the top that recovers the known laws bounds from above the length the bottom search must reach, and each shortening lowers the bound. The target is the rule both routes reach. The paper’s results are the census and its failures (Sections 3 to 6), the local-step benchmarks (Sections 8.1 to 8.6) and the Regge audit (Section 8.9); Section 8.7 is a proposal, a specification and its bit count, whose dynamics are untested as a whole.

2. Rules and their causal graphs

2.1 Rules

A state is a collection of relations, each an ordered pair of nodes; it is a directed graph that may contain loops and repeated edges. A rule has a left side, a pattern of relations written with variables, and a right side, relations written with the left side’s variables and possibly new ones. An update finds relations that match the left side, removes them, and adds the right side, creating a fresh node for each new variable. Different variables may match the same node. For example, the rule

{{x,y},{x,z}} → {{x,y},{x,w},{z,x}}

takes two relations leaving the same node x. It keeps the first, adds a relation from x to a new node w, and reverses the second.

The search starts with rules that consume two relations and make three, for two reasons. A rule that consumes one relation gives each update one direct cause, the update that made that relation, so its causal order is a tree. A rule that consumes two and makes two keeps the number of relations fixed, so from a small start its updates can only follow one another in a line. Two into three is the smallest class with room for more.

The class searched in full has left sides of two relations, connected through a shared node, and right sides of three relations, connected, built from the left side’s variables and at most two new nodes. Rules that differ only by renaming variables count once, and so do rules whose relations are listed in a different order. Under the fixed update order of Section 2.2 the listing is not neutral, since new relations are aged in the order the right side lists them: listed as {{x,y},{x,z}} → {{z,x},{x,w},{x,y}}, the two-dimensional rule of Section 4.2 grows a single chain. The census runs one listing of each rule, its canonical one. The class then holds 2,935 distinct rules. Reversing every relation turns a rule into its mirror image, which usually grows the same kind of causal graph; the count keeps both. A second set holds 1,500 distinct rules that rewrite two relations into four, drawn at random. The draw picks right sides at random before renamings are merged, so it is not uniform over distinct rules.

A second pass samples two larger classes, 1,000 distinct rules each, drawn the same way: rules that rewrite three binary relations into four, and rules that rewrite two relations of three nodes into three. A relation of three nodes, such as {x,y,z}, links three nodes at once, and a structure built from such links is a hypergraph, the kind many of Wolfram’s examples use (Wolfram,

2020). The rule stays fixed; only the hypergraph it rewrites changes. These runs start from as many self-loops as the left side has relations. They use an engine for relations of any size and left sides of any length, which reproduces the first engine’s causal graphs exactly on 277 rules of the first pass.

The rule-length score is the number of node symbols times the base-two logarithm of the number of distinct variables. It is a structural score, not a self-delimiting executable encoding; the scheduler, decoder and initial-state costs are additional. It is therefore not directly comparable to the compressed byte counts in Sections 7 and 8. The full class runs from 0 to 23.2 bits, the two-into-four sample to 27.9, and the second pass’s samples to 36.2 and 42.1. By this count the classes overlap: 903 rules in the sample, at 19 bits or fewer, are shorter than most rules in the full class. A strict shortest-first search would interleave the classes; this pass takes whole classes, the smallest first.

2.2 Updating and the causal graph

The update order is fixed. Updates happen in generations. Each generation lists the relations present at its start, oldest first, and matches each one that is still free with its oldest compatible partner. The matches found in a generation share no relation,

and all of them are applied at once. Section 4.6 varies this order. Every run of the first pass starts from two self-loops on one node, a minimal start of the kind Wolfram (2020) uses. It ends with the generation that passes 3,000 updates, or earlier if no match is left, 400 generations pass, 60,000 relations exist or 8 seconds elapse. The first pass’s long runs go to 20,000 updates or 60 seconds; the second pass’s go to about 120,000 updates or 600 seconds, without the limits on generations and relations.

An update’s direct causes are the updates that made the relations it consumed; the starting state counts as a root update. One update precedes another when a chain of direct causes runs from the first to the second. This order is a causal set in the technical sense, a partial order in which only finitely many elements lie between any two (Bombelli et al., 1987).

3. Reading a dimension

A causal interval is a pair of related elements together with every element between them. In flat spacetime of d dimensions, counting time, scatter points at random with uniform density, a sprinkling, and keep the order light cones give them. The intervals of a sprinkling have known statistics, and three of them give tests of dimension.

The first test is the ordering fraction (Myrheim, 1978; Meyer, 1988). In an interval of N elements, let f be the fraction of the N(N - 1)/2 pairs that are related. For a sprinkled interval in d dimensions,

where Γ is the gamma function, the factorial extended beyond whole numbers. So f is 1/2 in two dimensions and 1/10 in four, and the measured f gives d.

The second test is midpoint scaling (Reid, 2003). Among the elements z between the two ends x and y, take the one that makes the smaller of the intervals [x, z] and [z, y] as large as possible. In d dimensions that smaller interval holds about N/2^d elements.

The third test is the longest chain, the largest set of elements each preceding the next. In a sprinkled interval it grows as m d N^1/d, where m d is a constant (Brightwell and Gregory, 1991). Three sprinklings of 400 and of 800 points for each d give m d = 1.95, 2.06, 2.14 and 2.25 for d = 2, 3, 4 and 5. A chain of length L in an interval of N elements then gives d.

A fourth reading serves as a check that needs no fitted constant: keep each element with probability 1/2, 1/4 and 1/8, four times each, keeping both ends, and see how fast the longest chain

shrinks. In d dimensions it shrinks as the number kept to the power 1/d. This check is noisier, and on sprinklings of 400 and 800 points in four dimensions it read 4.45 and 4.15.

Table 1 shows the three tests on sprinklings. They agree within about 0.2 up to four dimensions and within 0.4 at five; at 800 points in four dimensions they read 3.88, 4.00 and 4.02.

Table 1. The three tests on sprinkled intervals (means of three sprinklings) and on two grids.

Order; Elements; Ordering fraction; Midpoint; Longest chain.

Sprinkled, d = 2; 400 / 800; 1.96 / 1.99; 1.95 / 2.00; 1.99 / 2.01.

Sprinkled, d = 3; 400 / 800; 2.93 / 2.98; 3.05 / 2.98; 3.02 / 2.98.

Sprinkled, d = 4; 400 / 800; 3.87 / 3.88; 4.06 / 4.00; 3.98 / 4.02.

Sprinkled, d = 5; 400 / 800; 4.66 / 4.83; 4.92 / 5.19; 4.86 / 5.10.

Light-cone lattice, 1+1; 900; 1.92; 1.91; 2.00.

Cellular automaton, 3+1; 3,781; 3.69; 3.63; 5.18.

The two grids show what the tests catch. The light-cone lattice in one dimension of space and one of time, the causal order of a one-dimensional cellular automaton, reads two on all three tests. The second grid is a three-dimensional cellular automaton, a cube of cells each updated from the 27 cells around it, itself

included, over ten steps. Its signals spread in square-sided pyramids, and its interval reads 3.69 and 3.63 on the first two tests and 5.18 on the third: its chains are too short for its size. Agreement of all three tests at four is therefore a demanding standard, which a grid with a preferred frame can fail.

The tests are applied to intervals of each rule’s causal graph. Take the six updates with the largest pasts. The largest intervals run from the root to each of them. The inner intervals run to each of them from an earlier update at 30 percent of its depth, depth being the longest chain of causes from the root, chosen to make the interval largest. Each must hold at least 120 elements, and a rule’s reading on each test is the median over its intervals. For the long runs a size series adds single intervals, of at least 60 elements, from one top update down to bottom ends at 0, 10, 20 and up to 70 percent of its depth. A rule counts as four-dimensional on an interval if all three tests read between 3.5 and 4.5 there, and as three-dimensional if all read between 2.5 and 3.5.

A fixed window ignores how much each test scatters at a given size, and the scatter shrinks as intervals grow. Table 2 measures it on sprinkled intervals of 120 to 4,000 points, with 40 sprinklings at the smaller sizes and 6 at the largest. At 1,000 points a

four-dimensional interval reads 3.96 ± 0.06 on the ordering fraction, 4.12 ± 0.13 on the midpoint and 4.02 ± 0.12 on the longest chain. From Section 4.4 on, an interval reads d-dimensional when each of its three readings lies within three standard deviations of the d-dimensional mean at its size. A rule is a four-dimensional candidate when its largest interval of at least 1,000 elements reads four-dimensional and so do the two largest intervals of its size series, so that its readings do not drift away from four as it grows. The size floor matters, since the bands are wide at 120 points.

This criterion, and the runs of Sections 4.6 and 5.1, were written down and fingerprinted with SHA-256 before they ran (04eca2085b16830e…; the full value and the protocol are in the supplements), so a near miss cannot be reread after the fact. Runs added after the protocol are marked exploratory.

Table 2. The scatter of the three tests on sprinkled flat intervals: mean ± standard deviation.

Dimension; Points; Sprinklings; Ordering fraction; Midpoint; Longest chain.

4; 120; 40; 3.66 ± 0.16; 3.91 ± 0.24; 3.98 ± 0.41.

4; 250; 40; 3.84 ± 0.14; 4.09 ± 0.21; 3.99 ± 0.23.

4; 500; 40; 3.92 ± 0.08; 4.13 ± 0.16; 4.02 ± 0.22.

4; 1,000; 20; 3.96 ± 0.06; 4.12 ± 0.13; 4.02 ± 0.12.

4; 2,000; 12; 3.99 ± 0.06; 4.14 ± 0.12; 4.08 ± 0.14.

4; 4,000; 6; 3.99 ± 0.04; 4.07 ± 0.05; 4.03 ± 0.10.

3; 1,000; 20; 2.99 ± 0.07; 3.04 ± 0.12; 2.96 ± 0.09.

3; 4,000; 6; 2.99 ± 0.02; 3.02 ± 0.04; 2.97 ± 0.06.

2; 1,000; 20; 1.99 ± 0.03; 1.99 ± 0.06; 2.02 ± 0.03.

2; 4,000; 6; 1.99 ± 0.01; 1.99 ± 0.02; 2.01 ± 0.01.

4. Results

4.1 A census of the shortest rules

Table 3 sorts the rules by what they do in 3,000 updates. A third of the two-into-three rules stop or stay below 300 updates: the median one makes a single update and then finds no match. More than a quarter grow a single chain, one update per generation, which is one-dimensional. Another 28 percent grow exponentially, about 10 percent or more updates in each generation than in the one before, over the second half of the run; their median rate multiplies the updates by about 1.47 per generation. These rules spread so fast that pasts stay thin: the

largest past in a run holds a median 1 percent of its updates, and only 42 of 825 make an interval of 120 elements. The remaining 374 grow polynomially. If the updates per generation grow as the generation number to a power p, the total grows like a volume of about p + 1 dimensions. Of these rules, 291 have p near 0, 75 near 1, 4 near 2 and 4 near 3.

Table 3. What the rules do in 3,000 updates.

Behavior; Two into three, all; Two into four, sample.

Rules; 2,935; 1,500.

Stop or stay small; 928 (31.6%); 279 (18.6%).

Single chain or tree; 808 (27.5%); 760 (50.7%).

Exponential growth; 825 (28.1%); 357 (23.8%).

Polynomial growth; 374 (12.7%); 104 (6.9%).

Volume growth near 1, 2, 3, 4 dimensions; 291, 75, 4, 4; 92, 9, 3, 0.

Polynomial, reading two on the first two tests in the largest intervals; 18; 5.

Read four on the first two tests, any interval; 2; 0.

The rules whose volume grows like three dimensions or more do not read three or four on the tests. Their pasts stay thin: the eight

such rules in the two-into-three class, run to 20,000 updates, read between 1.0 and 2.1 on all three tests in every interval. The tests read the order of the updates, and here that order stays thin.

4.2 Two dimensions

Eighteen of the polynomially growing two-into-three rules read near two on the first two tests in their largest intervals, twelve of them in intervals of 500 elements or more, and five rules in the sample do. Six ran to 20,000 updates; Table 4 shows the two cleanest,

{{x,y},{x,z}} → {{x,y},{x,w},{z,x}}

and its mirror image {{x,y},{z,y}} → {{x,y},{y,z},{w,y}}, at 20 bits each. In about 250 generations they make intervals of about 1,000 to 9,400 elements, which read 1.86 to 2.09 on the first two tests and 1.86 to 1.93 on the longest chain; the check reads 1.83 to 2.04. Their updates per generation grow in proportion to the generation, as a two-dimensional volume does. These rules grow two-dimensional causal orders, flat at the sizes reached: spacetimes of one space dimension and time. Two other rules read two, with a bump to between 2.5 and 3.0 on the first two tests at middle sizes, and one rule in the sample reads between

1.8 and 2.5.

So the tests find a dimension when a short rule has one, and two dimensions are within reach of the shortest rules.

The noise bands of Table 2 sharpen this. The pair’s orders are two-dimensional and are not two-dimensional sprinklings: their chains run 18 to 62 percent longer than a sprinkled interval’s of the same size, 4.5 to 12 standard deviations below the band, and at their largest sizes the first two readings fall to 1.71 to 1.98. Their updates have few links (Section 5), so at the scale of single updates the order is closer to a lattice than to a sprinkling. Section 4.6 shows that the two-dimensional readings also depend on the update order.

Table 4. Long runs to 20,000 updates: the range of each test over a rule’s size series.

Rule; Interval sizes; Ordering fraction; Midpoint; Longest chain.

{{x,y},{x,z}} → {{x,y},{x,w},{z,x}}; 1,008 to 9,355; 1.86 to 2.07; 1.89 to 2.08; 1.86 to 1.90.

{{x,y},{z,y}} → {{x,y},{y,z},{w,y}}; 974 to 9,220; 1.87 to 2.07; 1.95 to 2.09; 1.89 to 1.93.

{{x,y},{y,z}} → {{x,y},{y,w},{z,y}}; 78 to 5,064; 1.79 to 4.33; 1.89 to 5.05; 1.43 to 2.04.

{{x,y},{x,z}} → {{y,y},{z,w},{w,y}}; 98 to 918; 2.87 to 3.97; 3.15 to 4.67; 2.07 to 2.54.

{{x,y},{y,z}} → {{y,x},{y,w},{z,x}}; 79 to 964; 2.92 to 3.85; 3.13 to 4.15; 2.02 to 2.50.

{{x,y},{x,z}} → {{x,y},{z,y},{z,w},{w,z}}; 61 to 616; 2.80 to 4.10; 3.12 to 4.79; 2.28 to 2.83.

{{x,y},{z,y}} → {{x,z},{x,w},{y,x},{w,y}}; 86 to 401; 3.08 to 3.64; 3.43 to 4.14; 2.49 to 2.85.

{{x,y},{y,z}} → {{x,y},{y,w},{z,y},{w,y}}; 72 to 3,100; 1.00 to 4.02; 1.00 to 4.63; 1.19 to 1.87.

4.3 Near four

In 3,000 updates, two rules read between 3.5 and 4.5 on both of the first two tests, both in inner intervals, and none in the sample does. The long runs took 23 candidates further. They include every rule that came near four on either of the first two tests, the four exponential rules that read highest on the ordering fraction, the rules whose volume grew fastest, and the closest rules in the sample. In the size series of the long runs, 12 intervals in 7 rules read between 3.5 and 4.5 on both of the first two tests; two of the rules are a mirror pair with identical graphs. On every one of

these intervals the longest chain reads between 1.7 and 2.85: the chains run 1.7 to 9.6 times longer than in a four-dimensional interval of the same size. Across all intervals of all 27 long runs, the chain test reads 2.85 at most.

The near misses come in two families. The first adds updates in proportion to the generation, so its volume grows like two dimensions. In {{x,y},{y,z}} → {{x,y},{y,w},{z,y}} and three related rules, the readings swing with size: the ordering fraction gives 1.8 at 80 elements, 4.3 at 1,300 to 3,500, and 2.7 at 5,000. Where it reads 4.3, the midpoint overshoots to between 4.5 and 5.1, and the longest chain stays between 1.4 and 2.0 throughout. A rule of the same kind in the sample, {{x,y},{y,z}} → {{x,y},{y,w},{z,y},{w,y}}, reads 4.0 and 3.85 on the first two tests at 966 elements, with a chain 9.6 times too long. The second family grows exponentially: {{x,y},{x,z}} → {{y,y},{z,w},{w,y}}, its mirror image, {{x,y},{y,z}} → {{y,x},{y,w},{z,x}}, {{x,y},{y,z}} → {{z,x},{z,w},{w,x}} and two rules in the sample. It supplies 11 of the 12 intervals. Away from the start its intervals read up to 4.1 on the ordering fraction and 4.8 on the midpoint, while the chain reads 2.0 to 2.85. The readings peak at middle sizes, and the largest intervals, which reach back to the start, when the graph was small, read lower.

These orders are wide, since most pairs of updates in an interval are unrelated, as in four dimensions, and long, since chains of causes run on for many steps, as in two or three. No sprinkled flat spacetime, of any one dimension, combines the two in these proportions, so these orders are not four-dimensional spacetimes at the sizes reached. No interval in these runs reads three on all three tests either; in the closest the chain test reads 2.49 while the midpoint reads 3.43.

4.4 How firm the negative result is

The pattern holds across the choices this pass made. The largest intervals, the inner intervals and the size series agree. Mirror images mostly agree: 56 of the 1,420 mirror pairs in the full class land in different classes, since the fixed update order treats the two members differently. Taking candidates from 3,000 to 20,000 updates moved no rule into the four-dimensional range. On the 12 intervals that read four on the first two tests, the check that needs no fitted constant reads 1.0 to 2.5, with one interval at 3.3, against 4.15 to 4.45 on four-dimensional sprinklings. The chain constants were fitted at 400 and 800 elements and drift slowly with size, but the chain readings on those intervals would have to rise by 0.65 to 1.8 to reach the four-dimensional range.

The noise bands of Table 2 turn the miss into a distance. On the 12 intervals, the chain readings lie 4.7 to 18 standard deviations below the four-dimensional mean at their sizes. The runs past 100,000 updates add 9 intervals that read four on the first two tests; their chains lie 5.1 to 21 standard deviations below and run 1.7 to 11 times longer than four dimensions allow. No interval of the long runs, at any size, reads four-dimensional within the bands.

4.5 The second pass

Table 5 shows the two larger classes. They split differently from the first, with fewer single chains and more exponential growth: about a third stop, a tenth grow single chains, half grow exponentially, and the rest grow polynomially, mostly with one-dimensional volume. They end the same way. Neither class has a rule that reads four on the first two tests in any interval. Each has one polynomially growing rule that reads two, among them {{x,y,x},{z,w,x}} → {{x,y,y},{x,u,x},{w,z,x}}, whose chain test reads 1.9 as well. The exponential rules again come closest: their inner intervals reach 3.3 and 3.6 on the first two tests, with the chain at 2.4 to 2.5.

Table 5. The second pass’s samples in 3,000 updates.

Behavior; Three into four, sample; Two three-node into three, sample.

Rules; 1,000; 1,000.

Stop or stay small; 344 (34.4%); 348 (34.8%).

Single chain or tree; 95 (9.5%); 104 (10.4%).

Exponential growth; 496 (49.6%); 494 (49.4%).

Polynomial growth; 65 (6.5%); 54 (5.4%).

Volume growth near 1, 2, 3, 4 or more dimensions; 60, 3, 1, 1; 46, 7, 1, 0.

Polynomial, reading two on the first two tests in the largest intervals; 1; 1.

Read four on the first two tests, any interval; 0; 0.

The samples ran to 3,000 updates. Ten rules from the first pass then ran six times longer, to between 100,000 and 144,000 updates: the two-dimensional pair, a rule from the sample that reads two, and seven near misses (Table 6). The tests were applied inside the past of the top update, so that the stored pasts fit in memory. The two-dimensional rules stay two: intervals of 5,500 to 59,000 elements read 1.7 to 2.4 on all three tests, with no drift toward three or four. The linear near misses overshoot: at

middle sizes their ordering fraction reaches 4.7 to 5.4 and their midpoint 5.0 to 5.7, while the longest chain stays at about 2.0 or below. The exponential near misses’ pasts stay thin, so their intervals grow only to between 790 and 2,700 elements. They read up to 4.6 and 5.7 on the first two tests and 2.0 to 3.0 on the chain. One small interval, 144 elements in {{x,y},{z,y}} → {{x,z},{x,w},{y,x},{w,y}}, falls inside the three-dimensional window, at 3.24, 3.47 and 2.56; the same rule’s larger intervals read 3.8 to 4.1, 4.4 to 4.5 and 2.7 to 3.0. In the exponential near misses all three readings rise from small intervals to middle ones and fall back at the largest. Their chain test peaks between 2.4 and 3.0, and in the linear near misses it peaks at 1.9 to 2.0. No interval in these runs reads four on all three tests.

Table 6. Runs past 100,000 updates: the range of each test over a rule’s size series.

Rule; Updates; Interval sizes; Ordering fraction; Midpoint; Longest chain.

{{x,y},{x,z}} → {{x,y},{x,w},{z,x}}; 120,269; 5,483 to 44,991; 1.71 to 1.91; 1.71 to 1.91; 1.84 to 1.87.

{{x,y},{z,y}} → {{x,y},{y,z},{w,y}}; 120,001; 5,581 to 49,220; 1.76 to 1.97; 1.78 to 1.98; 1.89 to 1.92.

{{x,y},{z,y}} → {{x,y},{y,w},{u,x},{u,z}}; 120,077; 5,700 to 58,939; 1.78 to 2.35; 1.95 to 2.38; 1.98 to 2.05.

{{x,y},{y,z}} → {{x,y},{y,w},{z,y}}; 120,067; 106 to 30,157; 1.00 to 5.13; 1.00 to 5.46; 1.17 to 2.03.

{{x,y},{y,z}} → {{x,y},{y,w},{z,y},{w,y}}; 101,763; 97 to 14,867; 1.00 to 4.74; 0.98 to 5.02; 1.15 to 1.91.

{{x,y},{y,z}} → {{x,y},{y,x},{z,y},{w,y}}; 117,026; 61 to 12,376; 1.00 to 5.36; 0.98 to 5.73; 1.18 to 2.04.

{{x,y},{x,z}} → {{y,y},{z,w},{w,y}}; 123,647; 194 to 2,725; 3.37 to 4.64; 3.51 to 5.66; 2.19 to 2.76.

{{x,y},{y,z}} → {{y,x},{y,w},{z,x}}; 123,261; 134 to 1,099; 2.82 to 3.66; 2.74 to 3.96; 2.02 to 2.36.

{{x,y},{x,z}} → {{x,y},{z,y},{z,w},{w,z}}; 144,021; 63 to 1,102; 2.50 to 4.44; 2.39 to 4.88; 2.11 to 2.85.

{{x,y},{z,y}} → {{x,z},{x,w},{y,x},{w,y}}; 121,899; 67 to 786; 2.76 to 4.07; 2.74 to 4.49; 2.34 to 2.96.

The comparison so far is with flat spacetime, which is what the equations give for empty space. A rule that grows exponentially should also be compared with what they give for an expanding universe: de Sitter space, the vacuum with a positive cosmological constant. Table 7 reads sprinkled intervals of

four-dimensional de Sitter space across which the universe grows by up to four e-folds, four factors of e, about 2.7 each. As the expansion passes one e-fold, all three readings fall, the first two fastest, so the chain test reads highest. The exponential rules show the opposite order, with the chain test lowest, so de Sitter expansion does not explain them. The flat row is a sprinkling across 0.01 e-fold, separate from those of Table 1.

Table 7. Sprinkled intervals of four-dimensional de Sitter space, 800 points (means of three sprinklings).

Expansion across the interval; Ordering fraction; Midpoint; Longest chain.

None (flat); 3.92; 4.06; 3.87.

1 e-fold; 3.78; 3.98; 3.87.

2 e-folds; 3.44; 3.69; 3.81.

3 e-folds; 3.04; 3.19; 3.53.

4 e-folds; 2.69; 2.71; 3.39.

4.6 Other update orders, and causal invariance

A rule fixes its spacetime by itself only if its causal graph does not depend on the order of its updates; such a rule is causally invariant (Wolfram, 2020; Gorard, 2020b). The local test is the critical pair. Two updates compete for a relation; apply each in the smallest state that holds both, and ask whether further updates bring the two results to the same state, up to the names of new nodes. Every rule of the full class was tested with up to two further updates on each side. 677 of the 2,935 pass, 23 percent: 33 grow polynomially, 116 exponentially, 235 as single chains or trees, and 293 stop. With three further updates, 99 of the first 1,691 failing rules rechecked also pass, so the count is a floor. Of the 18 rules that read two, only the cleanest pair passes; 15 of the other 16 still fail with three further updates.

Table 8 reruns the 27 rules of the first pass’s long runs to 20,000 updates under four orders: the first passes’ order, which takes the oldest relations first and pairs each with its oldest partner; its reverse, newest first; and two random orders. No rule reads four on all three tests under any order. The highest chain reading under any order, 3.28 at 1,411 elements, belongs to {{x,y},{y,z}} → {{x,y},{y,x},{z,y},{w,y}} under newest first, still 5.9 standard

deviations below four dimensions. Under newest first, 8 of the 23 candidates grow one-dimensional orders, one or two updates per generation, and the exponential near misses lose their readings near four.

The two-dimensional pair passes the critical-pair test, and its readings still change with the order. Under the two deterministic orders it reads 1.67 to 2.09 on all three tests. Under the two random orders the past of its latest update holds 95 to 96 percent of all updates, and its intervals read 1.08 to 1.28 on the first two tests and 1.97 to 2.43 on the chain: nearly one-dimensional on two tests. Listing its right side in another order makes the oldest-first engine grow a single chain (Section 2.1); under the random orders that listing grows the same nearly one-dimensional orders.

So a finite run shows the rule together with its order. Passing the critical-pair test is a local condition. When the rewriting runs without end, local resolution does not guarantee that every run grows the same causal graph, and here the runs do not: each grows only part of the graph, and different orders grow different parts. A rule’s update order, and under a fixed order the listing of its right side, therefore belong in its description, and their bits count toward its length. The first passes’ order costs few bits, so

the pair’s two-dimensional orders stand as the output of a short program, the rule run oldest first. A four-dimensional candidate must read four under every order, or carry its order in its length.

Table 8. The rules that read two and the near misses, run to 20,000 updates under four orders: the readings of the largest interval of each size series (ordering fraction, midpoint, longest chain), or the run’s behavior when it grows a single chain. The second column gives the critical-pair test, with three further updates where two did not resolve.

Rule; Critical pairs; Oldest first; Newest first; Random 1; Random 2.

{{x,y},{x,z}} → {{x,y},{x,w},{z,x}}; passes; 1.86, 1.98, 1.86; 1.67, 1.70, 1.84; 1.09, 1.08, 1.99; 1.09, 1.08, 1.97.

{{x,y},{z,y}} → {{x,y},{y,z},{w,y}}; passes; 1.87, 1.99, 1.89; 1.96, 1.98, 1.85; 1.09, 1.09, 1.99; 1.09, 1.08, 1.99.

{{x,y},{x,z}} → {{y,x},{y,w},{z,x}}; fails; 2.04, 1.85, 1.87; 1.01, 1.00, 1.29; 2.92, 2.55, 2.27; 3.18, 2.90, 2.37.

{{x,y},{z,y}} → {{y,x},{y,z},{w,x}}; fails; 1.94, 1.96, 1.85; 2.65, 2.89, 2.06; 2.75, 2.65, 2.29; 3.15, 3.23, 2.37.

{{x,y},{y,z}} → {{x,y},{y,w},{z,y}}; fails; 2.69, 3.04, 1.98; 1.21, 1.19, 2.21; 1.88, 1.93, 2.21; 1.87, 1.92, 2.20.

{{x,y},{y,z}} → {{x,y},{y,w},{z,y},{w,y}}; fails; 3.00, 3.42, 1.87; 1.53, 1.37, 2.18; 3.16, 3.61, 2.63; 3.13, 3.61, 2.52.

{{x,y},{y,z}} → {{x,y},{y,x},{z,y},{w,y}}; passes; 3.30, 3.94, 2.04; 3.48, 3.94, 3.02; 3.11, 3.54, 2.70; 3.17, 3.67, 2.74.

{{x,y},{x,z}} → {{y,y},{z,w},{w,y}}; fails; 3.40, 4.01, 2.38; 1.25, 1.23, 1.56; 1.75, 1.78, 1.73; 1.55, 1.39, 1.63.

{{x,y},{z,y}} → {{x,x},{x,w},{w,z}}; fails; 3.40, 4.01, 2.38; 1.25, 1.23, 1.56; 1.60, 1.54, 1.72; 1.53, 1.37, 1.74.

{{x,y},{y,z}} → {{y,x},{y,w},{z,x}}; fails; 3.02, 3.23, 2.27; single chain; 2.60, 2.68, 2.09; 2.82, 2.74, 2.20.

{{x,y},{y,z}} → {{z,x},{z,w},{w,x}}; fails; 3.14, 3.74, 2.31; single chain; 2.82, 3.27, 2.15; 2.70, 3.09, 2.07.

{{x,y},{z,y}} → {{x,z},{x,w},{y,x},{w,y}}; fails; 3.31, 3.79, 2.61; 2.44, 2.46, 2.19; 3.08, 3.50, 2.55; 3.23, 3.76, 2.58.

{{x,y},{x,z}} → {{x,y},{z,y},{z,w},{w,z}}; fails; 3.74, 4.27, 2.64; 2.49, 2.73, 2.09; 3.06, 3.29, 2.42; 3.10, 3.48, 2.46.

5. Why a rule can match four dimensions only at large scales

A link is a pair of related elements with nothing between them. In a rule’s causal graph every link is a direct cause: if one update depends on another through a third, the third lies between them. An update consumes as many relations as its left side holds, each made by one earlier update, so it has at most that many direct causes. Each relation it makes is consumed at most once, so it has at most as many direct effects as its right side holds. No update therefore has more than five links in the two-into-three class, six in the two-into-four sample, or seven in the three-into-four sample, however large the graph grows. The long runs confirm the bound: no update had more than two direct causes or more than four direct effects.

A sprinkling behaves differently. In four-dimensional sprinkled intervals the mean number of links per element rises from 11.9 at 200 elements to 18.6 at 400, 31.8 at 800 and 47.6 at 1,600, and the largest number from 41 to 171 (Table 9). In a sprinkling of all of flat spacetime, every element has infinitely many links, reaching out along its light cone (Moore, 1988). Bombelli, Henson and Sorkin (2009) proved more. No graph with finitely many

edges at each element can be attached to a sprinkling of flat spacetime in a way that respects Lorentz symmetry, the symmetry of special relativity. Any such graph picks out a preferred frame.

Table 9. Links per element: sprinkled intervals, two grids and the two-dimensional pair.

Order; Elements; Mean links; Largest.

Sprinkled, d = 4; 200; 11.9; 41.

Sprinkled, d = 4; 400; 18.6; 72.

Sprinkled, d = 4; 800; 31.8; 134.

Sprinkled, d = 4; 1,600; 47.6; 171.

Sprinkled, d = 4; 4,000; 84.0; 308.

Sprinkled, d = 2; 1,000; 11.0; 24.

Sprinkled, d = 2; 4,000; 13.7; 26.

Light-cone lattice, 1+1; 900; 3.9; 4.

Cellular automaton, 3+1; 3,781; 35.0; 54.

Any rule, two into three; any; at most 5; at most 5.

Two-dimensional pair, four orders; 20,000 updates; 3.0 to 4.0; 5.

So the causal graph of such a rule, update by update, is not a random four-dimensional causal set, and no search over these rules can find one that is. The light-cone lattice shows what a bounded number of links still allows: it reads two on all three tests with at most four links per element. What the bound costs is the frame. A lattice has a rest frame, and by the theorem of Bombelli, Henson and Sorkin, a rule of this kind whose updates sat in flat spacetime would pick one out too, at the scale of single updates.

The two-dimensional pair shows the cost in numbers. Its updates have 3.0 to 4.0 links on average under the four orders, and at most five, while sprinkled two-dimensional intervals of 1,000 to 4,000 points give their elements 11.0 to 13.7 on average and up to 26 (Table 9).

The target is then a rule whose causal graph matches four-dimensional spacetime at large scales, after coarse-graining. One way to coarse-grain is to keep a random fraction of the updates, which leaves the first two tests nearly unchanged on average. The rule’s small-scale frame must stay hidden at the scales and speeds that can be tested. That is a demanding requirement. Quantum corrections tend to carry a small-scale frame into low-energy physics unless something protects the symmetry (Collins, Perez, Sudarsky, Urrutia and Vucetich, 2004),

and gamma-ray bursts bound such effects near the Planck scale (Abdo et al., 2009). A candidate rule must pass a frame test as well as the dimension tests.

The bound holds for rules that rewrite a fixed number of relations at a time. A rule escapes it if each update can draw on an unbounded part of its past. Classical sequential growth works this way: each new element of a causal set takes a whole set of earlier elements as its past, with probabilities fixed by a sequence of constants, in the simplest case one (Rideout and Sorkin, 2000). Such a rule can still be short. It gives up the locality of rewriting, in which each update reads only a few relations.

5.1 Sequential growth under the same tests

Sequential growth escapes the bound, so it gets the same tests. In transitive percolation, the simplest such dynamics, elements are born one at a time, and each earlier element becomes a direct cause of the newborn with probability p; the order is the transitive closure (Rideout and Sorkin, 2000). It has one constant, so it is short, and its elements can have as many links as they need. A root added below all elements serves as the bottom of the largest intervals.

Over the protocol’s grid, from 1,000 to 8,000 elements and p from 0.0003 to 0.3, no causal set is a candidate. Dense ones, with pN of 30 or more, are nearly one long chain: their largest intervals read between 1.0 and 2.2. Sparse ones, with pN of 3 or less, fall apart into pieces too small to test, and at pN of 4 to 8 they read 3.4 to 6.5 on the first two tests and 2.2 to 3.4 on the chain. In between, at pN of 8 to 24, the intervals that read near four on the first two tests read 2.3 to 3.1 on the chain: wide and long at once, like the rewriting near misses.

Rideout and Sorkin (2001) found evidence that transitive percolation has a continuum limit when p falls as N grows so that the fraction of related pairs stays fixed, including at its four-dimensional value. Table 10 follows that trajectory as an exploratory run outside the protocol: p is tuned so that the largest interval reads four on the ordering fraction, and the causal set grows from 4,000 to 64,000 elements. The midpoint reads 4.26 to 4.97, above four, and the chain 2.90 to 3.46, below it; the chain lies 6 to 10 standard deviations below four dimensions. The chain reading climbs by about 0.1 each time the causal set doubles, while the midpoint drifts slowly up.

A compiled engine followed the trajectory further, to two million elements, reading the same largest interval: the chain exactly,

the ordering fraction from 3,000 sampled ancestor counts, and the midpoint from the 600 elements that bottom-k sketches of their pasts and futures rank highest, each then counted exactly. At 16,000 and 128,000 elements this search finds the same midpoint as checking every element. The chain reading reaches 3.92 at one million elements and 4.09 at two million, through four, while the midpoint drifts on to 5.78 (Table 10). Sprinkled four-dimensional intervals of 8,000 to 128,000 points read 3.98 to 4.02 on the ordering fraction, 4.02 to 4.09 on the midpoint and 3.92 to 4.03 on the chain. So the trajectory reads four on the ordering fraction and on the chain at about a million elements, and misses the midpoint by more than one and a half: no element of its largest intervals has both a large past and a large future, as an element near the middle of a four-dimensional interval does. Transitive percolation is not a four-dimensional spacetime at any size reached, and the lead it offered closes on the midpoint.

Ahmed and Rideout (2010) fitted the chains and largest intervals of originary percolation, the variant in which every element descends from a first one, to four-dimensional de Sitter space for p from 0.0001 to 0.005. On the three tests it comes close to de Sitter space on two and misses on the third. At p = 0.001 and 32,000 elements its largest intervals read 1.91 on the ordering

fraction and 1.79 on the midpoint, as a de Sitter interval of the same size does across about eight e-folds of expansion, 1.89 and 1.72, yet their longest chains run 86 steps where de Sitter’s run 58 to 60, a chain reading of 2.72 against 3.02 to 3.05 in three sprinklings. Wherever originary percolation reads near four on the first two tests, its chain reads 2.3 to 3.1, while de Sitter space with the same ordering fraction reads above 3.7. Glaser and Surya (2013) reached the same verdict on transitive percolation by another route: it looks manifoldlike on coarse measures and fails a count of small intervals that sprinklings pass.

Table 10. Sequential growth: the medians of the largest intervals, means of two seeds for transitive percolation held at four on the ordering fraction up to 64,000 elements and of three seeds from 128,000 on (compiled engine), one seed for originary percolation, sprinkled flat intervals, and de Sitter space for comparison (means of three sprinklings).

Model; Elements; p; Largest interval; Ordering fraction; Midpoint; Longest chain.

Transitive, held at four; 4,000; 0.00291; 2,412; 3.91; 4.40; 3.00.

Transitive, held at four; 8,000; 0.00164; 4,912; 3.93; 4.56; 3.14.

Transitive, held at four; 16,000; 0.00089; 9,645; 4.11; 4.73; 3.18.

Transitive, held at four; 32,000; 0.00049; 19,559; 4.11; 4.70; 3.31.

Transitive, held at four; 64,000; 0.00027; 39,874; 4.07; 4.81; 3.44.

Transitive, held at four; 128,000; 0.000153; 78,767; 4.01; 5.14; 3.56.

Transitive, held at four; 256,000; 0.000083; 158,272; 3.97; 5.04; 3.63.

Transitive, held at four; 512,000; 0.000044; 313,713; 4.11; 5.55; 3.79.

Transitive, held at four; 1,024,000; 0.000024; 628,911; 4.04; 5.52; 3.92.

Transitive, held at four; 2,048,000; 0.000013; 1,265,886; 4.04; 5.78; 4.09.

Sprinkled, flat, d = 4; 8,000 to 128,000; 8,002 to 128,002; 3.98 to 4.02; 4.02 to 4.09; 3.92 to 4.03.

Originary; 32,000; 0.001; 26,717; 1.91; 1.79; 2.72.

Originary; 32,000; 0.0005; 19,971; 3.78; 4.48; 3.05.

de Sitter, 8 e-folds; 26,700; 26,702; 1.89; 1.72; 3.04.

6. Limits of this pass

The classes are small. Most relations join two nodes, the left sides hold two or three relations, and the larger classes were only sampled. Wolfram’s own examples often use relations of three nodes and larger rules (Wolfram, 2020).

One start was used. The first passes used one update order and one listing of each rule; Section 4.6 adds three orders for the long runs and a critical-pair test, a local condition, for the full class. The census itself was not rerun under other orders.

The sizes are small. The longest runs reach 144,000 updates and intervals of about 59,000 elements, and the samples of the second pass ran only to 3,000 updates. In four dimensions the longest chain grows only as the fourth root of the number of elements, so an interval of 59,000 elements is about 33 steps tall, and one 200 steps tall needs about 10^8. Dimension can depend on scale: several approaches to quantum gravity find the effective dimension, as a random walk measures it, falling toward two at the shortest distances (Ambjørn, Jurkiewicz and Loll, 2005; Carlip, 2017). If a rule’s updates are events at the Planck scale, the intervals measured here are a few hundred Planck lengths across or less, and a rule that reads two or three there

could read four at large scales. The test that counts is the trend with size: a rule whose readings climb toward four as its intervals grow is a candidate. No rule run long shows such a climb on all three tests; the chain readings of the near misses peak between 1.9 and 3.0 and fall back. Transitive percolation held at four on the first test climbs on the chain, through four at about a million elements, while its midpoint moves away (Section 5.1). A rule that stays three-dimensional at large scales, with two dimensions of space and one of time, would not describe our world. Gravity in three dimensions has no waves, and masses in it do not attract (Deser, Jackiw and ‘t Hooft, 1984).

Dimension is the first test only. A candidate must also show no preferred frame at large scales, curvature that obeys Einstein’s equations, and matter. It needs a quantum version too: a unitary rule, one that keeps total probability fixed. Quantum cellular automata of this kind recover the Dirac equation of the electron (D’Ariano and Perinotti, 2014).

Slow rules can hit the 8-second limit, and then stop at slightly different points on another machine. In the full class 29 runs hit it, and in the sample 66. Twelve of them were single chains cut off below 300 updates; rerun with 30 seconds, all twelve reached 400 updates as single chains, and Table 3 counts them that way.

Rerunning a random 60 rules of the full class reproduced 59 exactly; the other was one of the twelve. In the second pass 23 and 24 runs hit the limit. Eighteen of them stopped below 300 updates; rerun with 30 seconds, 9 passed 300 and 9 stopped again, each making at most one update per generation.

7. The next pass

The next pass should go up in length, shortest first: three relations into four and relations of three nodes in full, then larger rules. It should keep the critical-pair test as a first filter, read every candidate under several update orders, and count the order a rule needs in its length. It should follow each rule’s readings as intervals grow, to 10^6 or 10^8 updates, which needs a compiled engine and tests that sample pairs of updates instead of storing every update’s past. A rule whose readings climb toward four should count as a candidate. It should add a test for a preferred frame at large scales, and rules of the sequential-growth kind, whose updates draw on unbounded parts of their past. Transitive percolation, the lead this pass followed to two million elements, reads four on two tests and fails the midpoint. A growth law that favors manifold-like orders is the next candidate: weighting each causal set by the causal-set action of

Benincasa and Dowker (2010) gives, in two dimensions, a phase whose orders look like a continuum (Surya, 2012). Each rule takes seconds, so the search divides easily among many computers. The code and data are in the supplements.

Enumeration has a reach. Each extra bit doubles the number of rules, so a blind search covers rules of tens of bits. A complete symbolic statement of the Standard Model with general relativity, its gauge groups, matter fields, action and 28 declared inputs, written out and compressed with a standard compressor, comes to about 11,300 bits, and a shorter encoding may exist. General relativity is about 600 bits of it, 5 to 6 percent: one term, (R - 2Λ)/(16πG), two constants, and the metric that every other term uses. The two theories meet only through the metric, and in a causal set the metric is the order of the elements plus their count (Surya, 2019). A rule whose causal graph is four-dimensional at large scales would supply that metric; gravity’s own dynamics, the one term and two constants, would still have to come out of it. Most of the bits are not gravity: they are the gauge groups, the matter fields and their charges, and the inputs. If those bits must be written into the rule, no enumeration will reach it.

The search also runs from the other end. Every physical process is a sequence of steps, so general relativity and quantum field

theory, which treat their fields as continua, describe a rule at the scales where they were measured. This route applies those continuum fields to rules and steps, checks that the outcome matches them at identifiable scales, where the spacing is small against every wavelength and curvature radius in the problem, and then shortens the rules while keeping the match. Section 8 runs the first tests. The hard parts of this route are the ones this paper meets from the first end. The discrete rule must keep Lorentz symmetry at large scales, and it must carry the Standard Model’s left-handed particles, which resist being placed on a grid (Nielsen and Ninomiya, 1981). The target is a rule reached from both ends at once: short, and recovering the known laws. The two routes bound each other. A discrete rule found from the top that recovers the known laws caps the length of the shortest such rule, and each shortening lowers the cap; the enumeration from the bottom excludes candidates only within the classes, initial states, schedules and finite sizes it tests. The first cap is in hand for part of the known laws. Written as one self-contained program in Python, the local step of Section 8, which reproduces the Dirac equation on a curved static 1+1 background, prescribed electromagnetic phases and transport across a black-hole horizon, takes 451 source bytes, or 2,488 bits conditional on the

Python interpreter, numerical dependencies and the specified compression decoder, after the raw LZMA2 compression that brings the symbolic statement of the Standard Model with general relativity to about 11,300 bits. It reproduces the steps behind those tests to 10^-15. Run unchanged on a background it was not built or checked for, an expanding 1+1 universe, with only its mass angle changing from step to step, the same program creates fermion pairs as the continuum does: 0.1274877 at 2,048 sites against 0.1274880, with errors falling 3.99 to 4.00 times per halving. The geometry and the gauge fields are supplied, not grown, so this caps a rule for matter on a given spacetime; growing the spacetime is what the search from the bottom must add, and two-dimensional orders already appear at 20 bits. Supplying a four-dimensional lattice as data costs few bits, and the statement’s general relativity can bend it, as numerical relativity bends the metric’s values on a grid; Section 8.6 runs a homogeneous version driven by quantum matter. What a fixed lattice leaves open is gravity at the scale of its own spacing, which a rule that grows its own spacetime would settle too. Section 8.7 writes this candidate out in full, with the lattice form of each field and the completion the Standard Model needs: 16,272 bits.

8. Known laws as local steps

One family of local steps already reproduces a broad range of known physics: the Dirac equation in curved spacetime, a quantum packet falling along a geodesic, electromagnetic phases, Maxwell waves, a quench that creates fermion pairs, and transport across a black-hole horizon. Each error falls about fourfold when the spacing halves, the mark of a second-order approximation. These tests supply the geometry, the fields and the starting state, and ask whether the steps reproduce the continuum answer.

8.1 The update

For a static metric ds^2 = -N(x)^2dt^2 + dx^2, the target is the Dirac Hamiltonian, with c = = 1:

Neighboring amplitudes are updated by real two-site rotations. With spacing a and time step t, the bond angle is = (t\,N/2a). Even and odd bonds are applied in half, full and half layers, and the parity that goes first alternates from step to step. The two chiral components move with opposite signs. Local mass rotations (-i t\,mN x/2) bracket each transport step, and gauge links carry prescribed electromagnetic phases. The arcsine sets

the long-wavelength speed exactly at finite t/a, and alternating the layer order cancels an error that depends on parity. One prescription serves every test, and nothing is fitted to a target outcome.

8.2 Results

Table 11. Local steps against continuum references.

Situation; Result; Scope.

Massive Dirac field, lapse N = 1 + 0.25 x; State error 9.1 10^-4 at 1,024 sites, falling about fourfold per halving; Supplied 1+1 metric.

Falling packet and the geodesic; At mass 64: circuit 1.69047, quantum reference 1.69057, geodesic 1.69257; Finite-width packets.

Prescribed electromagnetic field; Gauge covariance to 10^-15; interference probability 0.206222 against 0.206107; No dynamical photons.

Fermion pairs from a mass quench; Eight modes, full Fock space of 256 states: state error 3.4 10^-5 at 2,048 sites; mean pairs 0.646932 against 0.646951; Free fields, selected modes.

Maxwell plane wave in a curved metric; Field error 7.1 10^-5 at 2,048 sites; conserved energy to 3 10^-16; Plane-symmetric sector.

Crossing a horizon; State error 5.4 10^-4 at 8,192 sites; 3.8 to 4.0 times smaller per halving; Radial, massless, fixed background.

The falling packet separates two effects. The circuit matches its quantum reference to 10^-4, and the quantum packet approaches the geodesic as its mass grows and its width, 0.8/ m, narrows: at masses 8, 16, 32 and 64 its center ends at 1.678, 1.685, 1.689 and 1.691, against the geodesic’s 1.693.

8.3 Wrong rules fail

Agreement in one number is weak evidence, and wrong rules show why. Replacing the mass term mN by m keeps the light cone and the norm, but leaves massive-state errors of 0.014 to 0.16 that do not shrink, and the packet stays near its starting point instead of falling. Dropping the gauge links destroys the interference. In the quench, evolving with the old mass reproduces the pair count almost exactly while the full state is off by 1.18. Full amplitudes, gauge covariance and independent references catch what a single observable misses.

8.4 Black holes

A horizon is no barrier to local steps. In Painlevé-Gullstrand coordinates,

the radial characteristic speeds v = 1 - r s/r stay finite at the horizon r = r s, and the same bond rule, with speed v in place of N, carries a wave across it. With r s = 2 on 0.25 <= r <= 6.25, three packets were run at six resolutions from 256 to 8,192 sites and compared with independently integrated characteristics. An ingoing packet starting outside ends inside with probability 0.9999999992. Inside, even the outward branch falls inward, because v + < 0 there; outside, it escapes. Errors fall 3.8 to 4.0 times on the last halving, and the norm holds to 10^-11.

The center needs the rule’s own dynamics. Keep the classical geometry and stop at the first site, r = a/2, and the curvature there, K = 12r s^2/r^6, grows 64-fold each time the spacing halves. A fixed physical spacing caps this sampled expression away from r = 0; it does not establish a regular center or bound the fields evolved by a discrete rule. What matter does at the center must come from the rule. Discrete quantum geometry shows what such dynamics can do. Loop quantum cosmology replaces the big bang with a bounce at about 0.41 of the Planck

density, and general relativity holds once the density falls below about one percent of it (Ashtekar, Pawlowski and Singh, 2006). Rovelli and Vidotto (2014) apply the same correction inside a collapsing star, which then bounces at the Planck density instead of reaching a singularity. A bounce can survive the quantum matter it holds. Couple a published regular gravity (Bueno, Cano, Hennigar and Murcia, 2025) to the four-dimensional vacuum stress of free conformal fields, angular pressures included (Herzog and Huang, 2013), and a closed universe still contracts, bounces and expands. The response stays single-valued while the ratio of the quantum coupling to the gravity scale is below 8/(3√3), about 1.54. The Standard Model’s free fields, 995.5 in anomaly units, give 0.88 at a gravity scale of one Planck area, and quantum matter then reaches 44 percent of the density; at 1.6 a contracting solution’s acceleration diverges. The fields are free and massless. The coupling to curvature squared, which the quantum fields leave free, decides the outcome. Restored with the stable sign at up to 2 x 10^-3 of the gravity scale, it keeps the bounce, and peak curvature rises from 21 to 270 in those units; at 5 x 10^-3 and above, or with the opposite sign at any size tested, the contraction drives the curvature out of the theory’s range before it can bounce. For the initial conditions and coupling

values tested, the regular bounce survives only at the tested small positive couplings. The scan does not establish a universal necessity or a sharp threshold.

8.5 What a rule can settle beyond the cutoff

A specified lattice can regulate short-distance divergences. Discreteness alone does not guarantee bounded fields, a convergent path integral or a dynamically regular center. A rule can settle more, and black holes are where it can show it.

A black hole’s entropy is one quarter of its horizon’s area in Planck units. Counting the causal links that cross a horizon gives an entropy proportional to that area (Dou and Sorkin, 2003). A rule whose horizons give exactly one quarter would derive the Bekenstein-Hawking law by counting.

Hawking’s derivation traces the outgoing radiation back to waves of unbounded frequency at the horizon, far beyond the Planck scale. A discrete rule has no such waves. Models with altered short-wavelength behavior still radiate a thermal spectrum (Jacobson, 1991; Unruh, 1995; Corley and Jacobson, 1996), so discreteness keeps Hawking’s result while removing its weakest step. The horizon test above extends directly: give the update a short-wavelength dispersion and measure the converted modes.

A unitary rule loses no information, so an evaporating black hole must return what fell in, and a finite number of states per region makes that bookkeeping explicit. Beyond black holes, the number of elements in a causal set fluctuates by about its square root. That predicted a cosmological constant of about the observed size, 10^-120 in Planck units, before it was measured, and makes it fluctuate in time (Sorkin, 1997; Ahmed, Dodelson, Greene and Sorkin, 2004).

8.6 Harder tests: pair creation, gauge fields and the source of gravity

Matching a known equation is the weakest test a rule can pass. Six harder ones show what a search must demand, and three of them catch rules that pass every easier check.

Pair creation in an expanding universe. In a 1+1 universe whose scale factor grows as 1.75 + 1.25 tanh(η/τ), the same gates, written in conformal time or in proper time, create fermion pairs as the continuum does: for τ = 0.15 and momentum 1 the occupation is 0.127487 in one clock and 0.127489 in the other, against 0.127488, with errors falling fourfold per halving and the two clocks converging on each other. Without expansion, or without mass, nothing is created. Fermion production in

expanding 1+1 spacetimes is established (Fulgado-Claudio, Sánchez Velázquez and Bermudez, 2023); the test is whether local steps reproduce it, and with the wrong mass coefficient they do not, leaving an error of 2.49 that refinement keeps.

A gauge field with Gauss’s law. In the lattice Schwinger model, fermions coupled to an electric field in one dimension of space (Lamm, Lawrence and Yamauchi, 2019; Shaw et al., 2020; Martinez et al., 2016), every gate preserves Gauss’s law exactly across all 15,552 states, and each halving of the time step cuts the error fourfold, from 9.2 x 10^-7 to 5.7 x 10^-8 over two halvings. Yet an electric cutoff of one flux unit overstates pair creation by 30 percent, at full precision. A rule can converge, conserve and still be wrong. Updates that freeze the links, or wrap the flux around, keep the norm and the total charge and break Gauss’s law.

Wrong couplings pass consistency checks. On a curved expanding background (Budd, Florio, Frenklakh and Mukherjee, 2026), the electric term must scale as the square of the conformal factor. Rules that scale it by the first power or not at all stay unitary, keep Gauss’s law, agree with themselves across clocks and satisfy their own energy-work balance, and end in states 0.76 and 1.10 from the right one. Agreement under a change of

clock is bookkeeping, not evidence of the right coupling to geometry.

A continuum benchmark. With 20 sites, every neutral state kept and the published lattice mass correction (Dempsey, Klebanov, Pufu and Zan, 2022), the interacting lattice reproduces the Schwinger boson's lowest gap within 0.26 to 0.62 percent, and the probability that the vacuum survives an expansion, 0.93264 against the continuum's 0.93280. The excitation energy is still 2 percent low, and the earlier six-site massive result changes with volume: convergence depends on what is measured.

The source of gravity. Three lattices share the long-wavelength wave equation and conserve energy to within 5 x 10^-10. One of them, the squared centered derivative, hides a second field at the edge of its momentum zone, and its quantum source of gravity, energy and pressure, comes out twice the single field's at twice the spacing, to within 2 x 10^-10: O_centered(ε) = 2 O_nearest(2ε). At the finest spacing its peak energy and pressure are 2.00004 and 2.00003 times the target. Lattice doublers are known (Buonanno, Cella and Curci, 1995); here a rule that passes every wave test would double the gravity its matter makes. A short rule must be tested on its source, not only on its waves.

A step has its own vacuum. A discrete time step has an invariant vacuum of its own. Start the field in the continuum's vacuum instead and the apparent energy after one step grows as the spacing shrinks, as 1/ε^2: 0.23, 0.91, 3.64 and 14.6 over three halvings, while the step's own vacuum stays empty, with occupations below 4 x 10^-30. With the matched vacuum, space and time refine together and the pressure error falls to 0.002 percent.

Matter drives the expansion. Coupled through one Hamiltonian to a homogeneous 3+1 universe (Husain and Singh, 2021), quantum matter in a vacuum, a squeezed state or with a vacuum energy drives the expansion with scale-factor errors near 10^-6 and the Friedmann constraint held within 3 x 10^-5, in proper and conformal time alike; the last halving cuts every error by 3.997 to 4.008. Freeze the feedback and the matter evolves perfectly while the Friedmann constraint fails by 85 percent and the two clocks disagree by 7.6 percent. The same model shows what remains open. At a fixed box the bare vacuum energy grows as 1/ε^4 and its pressure approaches a third of its energy, so canceling the energy with a constant leaves the pressure: the vacuum's contribution to gravity cannot be set by canceling one number.

The end of evaporation. In the solvable model of an evaporating black hole of Russo, Susskind and Thorlacius (1992), the traditional final mass is negative, close to minus one unit of the model's quantum scale, while the Hamiltonian charge of the complete semiclassical action (Barenboim, Frolov and Kunstatter, 2026) is the same amount positive. Pairing the complete-action charge with the traditional flux leaves an error of twice the quantum scale, 49 percent of the mass for a hole of four units and more for lighter holes, that no step size removes. No simple rule for the end, stopping at zero mass, switching branch or flooring the field, gives a regular ending: as at the center in Section 8.4, the endpoint needs the rule's own dynamics.

8.7 A proposed specification on a supplied lattice

Section 7 points to a candidate from the top: supply a four-dimensional lattice as data, run the known laws on it, and let general relativity bend it. This section writes the candidate out, completes it, tests the lattice forms it needs and counts its bits with the compressor of Section 7 (Table 12; the supplement's lattice, form-fermion and Regge scripts).

Lattice form. The lattice is hypercubic with the Planck length as its spacing, and it is physical, with no continuum limit to reach.

Gauge fields become phases on its links (Wilson, 1974), and the metric becomes the lengths of the edges of a triangulation, so general relativity acts through the Regge action (Regge, 1961). Naive lattice fermions come with doublers (Nielsen and Ninomiya, 1981); the overlap operator (Neuberger, 1998) removes them and keeps an exact lattice chiral symmetry, the Ginsparg-Wilson relation (Ginsparg and Wilson, 1982; Lüscher, 1998). On a two-dimensional U(1) lattice the naive operator has 8 zero modes in the free field where the continuum has 2, and in 12 backgrounds of topological charge -2 to 3, smooth and rough (plaquette angles up to 3.1 radians), the overlap operator's index equals the charge every time while the relation holds to 4 x 10^-15. This verifies the overlap construction in the tested U(1) backgrounds; it does not test four-dimensional SU(3) dynamics. The weak force acts on left-handed fields only. Its lattice form exists for U(1) (Lüscher, 1999) and is proposed for the full Standard Model (Wang and Wen, 2020), against objections (Golterman and Shamir, 2023, 2026). In 1+1 dimensions the route has worked numerically: in the ungauged 3-4-5-0 model, whose chirality lies in its charge assignments, the mirror sector gaps without breaking the symmetry while the light sector stays gapless (Zeng, Zhu, Wang and You, 2022). Gauged versions

have been proposed (Wang, 2022), but a numerical test with a fluctuating gauge field, and a selective mirror gap in 3+1 dimensions, are still open. Simplicity favors that route: a local lattice rule gives both handednesses in equal numbers, so a short rule keeps both and makes the mirror hand heavy at the lattice scale without breaking the symmetries, an echo of the mirror particles that Lee and Yang (1956) proposed. The anomalies must cancel, including the mod-16 anomaly of 5(B - L) - 4Y: the statement's 16 Weyl fermions per family cancel them all, where 15 would leave 13 (García-Etxebarria and Montero, 2019). Perturbative calculations can reproduce many measured weak-interaction observables. They do not complete the proposed nonperturbative chiral lattice theory; the count below measures its specification.

Lorentz symmetry. A lattice that treats its four directions alike forbids by hypercubic symmetry the terms that would give particles different top speeds, and its time step is the transfer matrix along one direction; a step that treats time apart, like that of Section 8.1, would need one tuned speed per particle type (Collins et al., 2004). The photon then has E = p - a^2p^3/12 along an axis and E = p - a^2p^3/18 along a body diagonal, so LHAASO's limit on quadratic dispersion, E QG,2 > 6.9 x 10^11

GeV (Cao et al., 2024), allows spacings up to 7.0 x 10^-28 m, about 43 million Planck lengths.

Completion. The Standard Model with general relativity lacks dark matter and an initial state. The shortest completion declares them: a free scalar with a declared mass, and five numbers for the initial state. SMASH (Ballesteros, Redondo, Ringwald and Tamarit, 2017) derives them instead, from an axion, a heavy quark and Majorana neutrino masses, and predicts a tensor-to-scalar ratio of at least 0.004 and an axion of 50 to 200 μeV; it costs 928 more bits and would earn them only if those predictions are measured. Dark energy may need no entry: in the timescape model clocks run faster in emptier regions, so the averaged expansion only appears to accelerate (Wiltshire, 2007), which a recent supernova analysis mildly favors (Seifert, Lane, Galoppo, Ridden-Harper and Wiltshire, 2025) against arguments that such effects are too small (Green and Wald, 2014); then Λ is zero. Inputs may also become rules at the lattice scale: assuming asymptotically safe gravity drives the Higgs self-coupling close to zero there, Shaposhnikov and Wetterich (2010) predicted a Higgs mass near 126 GeV before its discovery, against 125.2 measured.

What the lattice leaves open. A physical lattice regulates short-distance modes without by itself establishing convergence of the gravitational integral. The parameter estimates also expose fine-tuning: the bare cosmological constant must cancel the vacuum energy of the lattice's fields to about 123 decimal places, and the bare Higgs mass parameter its shift to about 32. As numbers those digits would cost 408 and 106 bits; as the condition that the measured values result, as the statement writes them, they cost only that condition, so the renormalization condition has a short description. The target values, their precision and a procedure for determining the bare parameters must still be specified; compression does not explain those values. Every measurement lies far below the lattice scale, where general relativity as an effective quantum theory gives finite predictions (Donoghue, 1994), which motivates the low-energy target; an empirical fit of the complete supplied-lattice construction has not been performed here. Joining general relativity with quantum theory at black-hole centers and the first instants of expansion turns on gravity at the lattice's own scale, where the Euclidean action is unbounded below (Gibbons, Hawking and Perry, 1978). That test has begun. On a fixed 16^4 Regge lattice Hamber and Williams (1995) found an attractive

potential near a critical point, consistent with Newton's law within large errors; dynamical triangulations recover Newtonian binding within 20 to 30 percent, without matter acting back (Dai, Laiho, Schiffer and Unmuth-Yockey, 2021); and causal triangulations, which glue four-dimensional blocks causally, find a four-dimensional de Sitter universe (Ambjørn, Jurkiewicz and Loll, 2004; Ambjørn, Görlich, Jurkiewicz and Loll, 2008). On the supplied lattice itself, 4^4 sites with $\lambda = 1$ and a uniform measure, the original supplement finds a long-lived negative-average-curvature regime after a flat start whose deficit angles are below 2 x 10^-15: the average curvature is negative and shrinks from -1.84 at k = 0 to -0.77 at k = 0.065, and above k ≈ 0.07 the lattice degenerates, near the critical value k c = 0.0639 that Hamber (2015) finds on lattices up to 32^4 with exponent = 0.334(4). On lattices this small the curvature does not reach zero first, so the run locates an apparent finite-run stability boundary, not a critical point. The independent execution in Section 8.9 tests its dependence on initial conditions and sampling history. Positive curvature-squared terms extend the smooth phase (Hamber, 1992), and Hamber (2019) proposes that the observed cosmological constant is set by the correlation length near the critical point, 3/ ^2, which leaves that length to be fixed.

Matter on causal triangulations. Fermions there need frames on a changing geometry, and each change alters their determinant. Written as differential forms, Kähler-Dirac fields, they need no frames (Kähler, 1962; Becher and Joos, 1982; Catterall, Laiho and Unmuth-Yockey, 2018). On six sampled two-dimensional causal triangulations of a torus (Ambjørn and Loll, 1998) such a field has exactly 4 zero modes, the torus’s Betti numbers, and low modes 0.92 to 1.10 times Weyl’s law, in the measured low spectrum; those counts alone do not exclude every extra propagating species. On the supplied four-dimensional lattice, bent by random weights, its zero modes number 16. Each such field carries four copies of a Dirac fermion. Catterall (2021) proposed interactions that would leave 16 Weyl fermions if they gap one half and the continuum limit holds, tested in two dimensions, and four-dimensional gauge theories show symmetric gapped phases (Hasenfratz, 2022; Butt, Catterall and Hasenfratz, 2025), though none yet leaves the Standard Model’s chiral spectrum. Free conformal fields in the Standard Model’s spin counts give a tractable first proxy for quantum stress. Masses, curvature couplings, interactions and the quantum state remain necessary for a full matter calculation.

Description length. Written out in full (the statement, its lattice form and the declared completion), the candidate compresses to 16,272 bits, a description length in this encoding for the proposed specification with dark matter and initial data. A compressed description alone does not establish an executable physical model. The search from the bottom excludes candidates only within its tested classes and conditions (Sections 4 to 6), and no program can be proved the shortest for its output once its length exceeds a constant set by the axioms used (Chaitin, 1974). A demonstrated program gives an upper bound for its stated output and tests, while the present census excludes candidates within its tested classes. These are different bounds; neither establishes the minimum length of a complete physical theory.

Table 12. The proposed specification on a supplied four-dimensional lattice: each piece, its cost in compressed bits (raw LZMA2, as in Section 7) and its status.

Piece; On the supplied lattice; Bits; Status.

Standard Model with general relativity, 28 inputs; the statement of Section 7; 11,328; complete as stated.

Lattice form of each field; link phases, Regge edge lengths, overlap fermions, transfer matrix; +3,144; 2D U(1) overlap

checks; chiral completion unresolved.

Dark matter and initial state, declared; free scalar; amplitude, tilt, two mass ratios, temperature; +1,800; declared inputs; no new full-data fit.

Total: specification; 16,272; description length; full dynamics unverified.

Mechanisms instead (SMASH); axion, seesaw, leptogenesis, inflation; +928 more; predicts r of at least 0.004, axion of 50 to 200 μeV.

Fine-tuning of Λ and the Higgs mass; renormalization condition and declared targets; included in description (digit estimate: 514); parameters not explained.

Lorentz symmetry; hypercubic symmetry; corrections of order $(aE)^2$; 0; spacing up to 7.0×10^{-28} m allowed.

Gravity at the lattice scale; Regge path integral; 0 beyond the lattice form; open; finite-run regimes here; see Section 8.9; larger-lattice critical behavior in cited work.

8.8 Scope

These tests supply the geometry and the fields; the rule that grows them is what the first half of this paper searches for. Discrete does not mean finite. A short rule can add sites without end, and an infinite causal set can be locally finite (Surya, 2019); the arrays here are finite for computing, and their boundaries were checked. Earlier work reached the same kind of agreement for the Dirac equation (D’Ariano and Perinotti, 2014), for quantum walks outside and inside a horizon (Di Molfetta, Brachet and Debbasch, 2013) and for quantum electrodynamics in three dimensions of space (Eon, Di Molfetta, Magnifico and Arrighi, 2023). Section 8.6 runs the next tests listed in earlier passes: a time-dependent background against particle production, a dynamical gauge field that conserves Gauss’s law, an interacting process, and matter acting back on the geometry. What remains is a geometry that propagates locally, a vacuum subtraction derived from the rule itself, and the endpoints that need the rule’s own dynamics. The scripts, independent references and results are in the supplements.

8.9 Independent execution and equilibration audit

A new audit reran the supplied Regge C engine without changing its action, measure or update rules. The protocol was fixed before the main runs. At each of L = 3 and 4 and k = 0.05 and 0.065, three seeded chains began flat and three began from independently prepared rough configurations. Every rough preparation used the same prescribed 3,000-sweep evolution at k = 0.1; no configuration was selected for its observed outcome. Each of the 24 main chains used 3,000 equilibration sweeps followed by 12,000 measurement sweeps. All per-sweep data, starting preparations and ending configurations are retained. These are new executions and independently written diagnostics, not a second implementation of the gravitational dynamics.

The geometry self-tests reproduced, including the flat-start deficit angles and incremental/full recomputation check. The statistical audit used ordinary split R-hat across all six chains at each parameter pair, with a prespecified screening threshold of 1.05, and batch-mean, drift and start-dependence diagnostics. Passing this screen would not prove equilibrium; failing it prevents treating the pooled chains as a settled ensemble. Curvature means below are averages over each start class and are descriptive when the

chains have not mixed.

L; k; Flat-start mean R; Rough-start mean R; Split R-hat; Screen.

3; 0.05; -1.0470; -0.6731; 1.573; fails.

3; 0.065; -0.6022; 44.1443; 13.032; fails.

4; 0.05; -1.0655; 31.2166; 5.498; fails.

4; 0.065; -0.6793; 45.4855; 17.716; fails.

All four parameter groups fail the prespecified R-hat screen. The table and trajectories distinguish initial-state dependence from an established equilibrium phase. Negative mean curvature after a flat start is insufficient to demonstrate that the ensemble has equilibrated. These finite-run tests neither locate a critical point nor establish the nonexistence of an equilibrium limit; they restrict what can be claimed from the sampled histories.

All 24 chains satisfy the scaling identity in their measurement averages to within 0.09 percent, despite the failed mixing screen. This separates a successful overall-scale consistency check from equilibration of the geometry. It provides a concrete example of an implementation check that cannot establish the sampled phase.

The next tests are normalizability of the specified ensemble and convergence from both start classes at longer durations, followed

by finite-size scaling and curvature correlations only where convergence is supported. A Newtonian potential and a Lorentzian matter-coupled theory require separate tests. The original run’s histories are kept in its archive, and this audit supplies its own complete histories and trajectory figure in the audit archive.

9. Conclusion

Among the shortest rewriting rules, under the update orders tried, none grows a four-dimensional causal order. Of the 2,935 rules that rewrite two relations into three and samples of 3,500 from three larger classes, none reads four on all three of causal-set theory’s dimension tests, in runs of up to 144,000 updates. Two-dimensional orders do appear under the first passes’ order, so the tests can find a dimension when a short rule has one; under random orders the same rules grow orders that are nearly one-dimensional on two tests, so a rule’s update order belongs in its length. The near misses read four on two tests and 1.7 to 3.0 on the third: wide like four dimensions, with chains far longer than four dimensions allow, the opposite of an expanding universe. A rule that rewrites a few relations at a time bounds the links of every update, while a random four-dimensional causal set gives

its elements ever more links. Such a rule can therefore match our spacetime only at large scales, where its small-scale frame must stay hidden. A criterion fixed in advance with noise bands puts the near misses' chains 4.7 to 21 standard deviations from four dimensions. Sequential growth escapes the link bound and still misses: transitive percolation held at four on the first test, followed to two million elements, reaches four on the third while its midpoint drifts to 5.8, and originary percolation, read against de Sitter space, runs its chains too long. The search goes on with longer rules, following the readings as the intervals grow, and from the other end, with the known laws made discrete. From that end, one family of local unitary steps already reproduces quantum fields in curved spacetime and crossing a black-hole horizon, with errors falling fourfold as the spacing halves, its single-particle Dirac step fitting in a program of 2,488 bits, and the same steps, run as that program unchanged, create pairs in an expanding universe, while a separate 3+1 lattice model drives its expansion. The proposed specification on a supplied four-dimensional lattice, with dark matter and initial data, compresses to 16,272 bits. Component tests support parts of this construction; they do not establish a single executable model covering present data. Smooth large-scale geometry, a

consistent chiral matter sector and the quantum-gravity dynamics remain separate requirements. The independent Regge audit fails the prespecified mixing screen in all four parameter groups, so the sampled regime is not established as an equilibrated smooth phase. Harder tests sharpen the search: a rule must be judged by the source of gravity it makes, since one lattice that matches every wave doubles it. What remains is the short rule that grows the geometry they need, and the dynamics it gives a black hole’s center and a black hole’s end.